

APPENDIX 1

Part 4 of 5



UNITED STATES PATENT AND TRADEMARK OFFICE

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/218,418	12/12/2018	Patrizio Vinciarelli	00614-0183001	1092
26214	7590	06/30/2020	EXAMINER	
FISH & RICHARDSON P.C. (VLT, INC)			COX, CASSANDRA F	
P.O. BOX 1022			ART UNIT	
MINNEAPOLIS, MN 55440-1022			PAPER NUMBER	
			2849	
DATE MAILED: 06/30/2020				

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 30 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability	Application No. 16/218,418	Applicant(s) Vinciarelli, Patrizio	
	Examiner CASSANDRA F COX	Art Unit 2849	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to 04/30/2020.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

3. ☒ The allowed claim(s) is/are 1-47. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) ☐ All b) ☐ Some *c) ☐ None of the:

1. ☐ Certified copies of the priority documents have been received.

2. ☐ Certified copies of the priority documents have been received in Application No. _____.

3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.
THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.

Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).

6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

1. ☒ Notice of References Cited (PTO-892) 2. ☒ Information Disclosure Statements (PTO/SB/08), Paper No./Mail Date _____. 3. ☐ Examiner's Comment Regarding Requirement for Deposit of Biological Material _____. 4. ☐ Interview Summary (PTO-413), Paper No./Mail Date. _____.	5. ☒ Examiner's Amendment/Comment 6. ☒ Examiner's Statement of Reasons for Allowance 7. ☐ Other _____.
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/CASSANDRA F COX/
Primary Examiner, Art Unit 2849

DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Information Disclosure Statement

1. The information disclosure statement filed 08/07/09 fails to comply with 37 CFR 1.98(a)(1), which requires the following: (1) a list of all patents, publications, applications, or other information submitted for consideration by the Office; (2) U.S. patents and U.S. patent application publications listed in a section separately from citations of other documents; (3) the application number of the application in which the information disclosure statement is being submitted on each page of the list; (4) a column that provides a blank space next to each document to be considered, for the examiner's initials; and (5) a heading that clearly indicates that the list is an information disclosure statement. The information disclosure statement has been placed in the application file, but the information referred to therein has not been considered.

EXAMINER'S AMENDMENT

1. An examiner's amendment to the record appears below. Should the changes and/or additions be unacceptable to applicant, an amendment may be filed as provided by 37 CFR

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1.312. To ensure consideration of such an amendment, it MUST be submitted no later than the payment of the issue fee.

The application has been amended as follows:

The abstract has been rewritten in the form of one single paragraph to place it into proper format:

-- Encapsulated electronic modules having complex contact structures may be formed by encapsulating panels containing a substrate comprising pluralities of electronic modules delineated by cut lines and having conductive interconnects buried within terminal holes and other holes drilled in the panel within the boundaries of the cut lines. Slots may be cut in the panel along the cut lines. The interior of the holes, as well as surfaces within the slots and on the surfaces of the panel may be metallized, e.g. by a series of processes including plating. Solder may be dispensed into the holes for surface mounting. Two or more panels may be stacked prior to singulation to form module stacks. Multi-cell converters having a large cell pitch may be combined with an interconnection module to provide vertical power delivery to semiconductor devices through a semiconductor power grid having a small pitch. The converters and interconnection modules may be fabricated in panels and stacked prior to singulation. --

Allowable Subject Matter

2. Claims 1-47 are allowed.

3. The following is an examiner's statement of reasons for allowance: Claims 1-16, 36-43, and 46 are allowed because the closest prior art of record fails to disclose a method of powering a semiconductor device having power input terminals, the method comprising: providing an interconnection module having x-input terminals arranged on a first side of the interconnection module in a pattern configured to align with the first pattern of converter output terminals, and x-output terminals arranged on a surface of a second opposite side of the interconnection module in a second pattern configured to align with the power

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input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals, the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals; assembling the power conversion module to the first side of the interconnection module with each converter output terminal mated with and electrically connected to a respective x-input terminal; wherein the interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern and the power input terminals of the semiconductor device arranged in the second pattern; wherein the power conversion module and the interconnection module are stacked to form a module sandwich including the power conversion module in a first layer, electrical connections between the converter output terminals and the x-input terminals in a second layer, and the interconnection module in a third layer in combination with the rest of the limitations of the base claim. Claims 17-23 are allowed because the closest prior art of record fails to disclose a method of making power converters, comprising: selectively forming a plurality of holes in the exterior surface of the panel, each hole being formed at a respective predetermined location aligned with a respective output node, spaced apart from the cut lines, and exposing within the hole the respective conductive feature; selectively forming one or more conductive metal layers on the exterior surface of the panel including within the plurality of holes, the one or more conductive metal layers within each hole being in electrical contact with the respective conductive feature, to form a plurality of electrical contacts on the exterior surface of the panel; patterning the one or more conductive metal layers on the exterior surface to electrically isolate at least one electrical contact from one or more of the plurality of electrical contacts; cutting the panel along the one

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or more cut lines to singulate the one or more power conversion modules, each singulated power conversion module having a respective plurality of the electrical contacts formed on the exterior surface of the module in combination with the rest of the limitations of the base claim. Claims 24-26 are allowed because the closest prior art of record fails to disclose a method of making electronic assemblies, the method comprising: wherein the interconnection module is configured to provide electrical and mechanical translation between the circuit terminals arranged in the first pattern and the external terminals of the third electronic device arranged in the second pattern; wherein the electronic module and the interconnection module are stacked to form a module sandwich including the electronic module in a first layer, electrical connections between the circuit terminals and the x-input terminals in a second layer, and the interconnection module in a third layer; assembling a first panel comprising a plurality of unsingulated electronic modules; assembling a second panel comprising a plurality of unsingulated interconnection modules; wherein the assembling of the electronic module and the interconnection module is performed using the first and second panels prior to singulation to form a panel sandwich and further comprises dispensing a curable adhesive compound between the first and second panels and forming solder connections between each circuit terminal and each respective x-input terminal; and cutting the panel sandwich along predetermined cut lines to singulate a plurality of the module sandwiches in combination with the rest of the limitations of the base claim. Claims 27-29 are allowed because the closest prior art of record fails to disclose a method of forming modular circuit assemblies, the method comprising: selectively forming a plurality of terminal holes in the first exterior surface of each panel through the first layer at predetermined locations within the perimeter boundaries of

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each electronic module, each terminal hole being spaced apart from the cut lines and exposing within the terminal hole a respective conductive feature; selectively forming one or more conductive metal layers on the first exterior surface of each panel including within the plurality of terminal holes, the one or more conductive metal layers within each terminal hole being in electrical contact with the respective conductive feature; patterning the one or more conductive metal layers on the exterior surface, to form a plurality of electrical contacts on the exterior surface of each panel electrically isolated from at least one other electrical contact in the plurality of electrical contacts; assembling the first and second encapsulated panels with electrical contacts on the first panel aligned and mated with respective electrical contacts on the second panel to form a panel stack, the panel stack comprising a plurality of un-singulated module stacks, each module stack including an unsingulated electronic module in the first panel and a respective unsingulated electronic module in the second panel; and cutting the panel stack along the one or more cut lines to singulate the plurality of electronic module stacks in combination with the rest of the limitations of the base claim. Claims 30-34 are allowed because the closest prior art of record fails to disclose an apparatus comprising: an interconnection module having x-input terminals arranged on a first surface of the interconnection module in a pattern configured to align with the first pattern of converter output terminals, and x-output terminals arranged on a second surface in a second pattern configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals, the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals; the converter output terminals of the power conversion module being mated

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with and electrically connected to respective x-input terminals of the interconnection module; wherein the interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern and the power input terminals of the semiconductor device arranged in the second pattern; wherein the power conversion module and the interconnection module are stacked with the power conversion module located at a first level, electrical connections between the converter output terminals and the x-input terminals at a second level, and the interconnection module at a third level in combination with the rest of the limitations of the base claim. Claims 35, 44-45, and 47 are allowed because the closest prior art of record fails to disclose a method of powering a semiconductor device having power input terminals, the method comprising: providing an interconnection module having x-input terminals arranged on a first surface of the interconnection module in a pattern configured to align with the first pattern of converter output terminals, and x-output terminals arranged on a second surface in a second pattern configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals, the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals; providing a plurality of conductive pillars for electrically connecting the converter output terminals with respective x-input terminals; assembling the power conversion module to the interconnection module with each converter output terminal aligned with and electrically connected to a respective x-input terminal by a respective one of the conductive pillars; wherein the interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern and the

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power input terminals of the semiconductor device arranged in the second pattern; wherein the power conversion module and the interconnection module are stacked to form a module sandwich including the power conversion module in a first layer, the conductive pillars connecting the converter output terminals and the x-input terminals in a second layer, and the interconnection module in a third layer; and wherein the x-output terminals are arranged with alternating polarities to reduce interconnection inductance in combination with the rest of the limitations of the base claim.

Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

Conclusion

4. The prior art made of record and not relied upon is considered pertinent to applicant's disclosure.

5. Any inquiry concerning this communication or earlier communications from the examiner should be directed to CASSANDRA F COX whose telephone number is (571)272-1741. The examiner can normally be reached on M-F 6:00-3:30; off alt Fridays.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

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If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Lincoln Donovan can be reached on 571-272-1988. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <https://ppair-my.uspto.gov/pair/PrivatePair>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/CASSANDRA F COX/
Primary Examiner, Art Unit 2849

Notice of References Cited	Application/Control No. 16/218,418		Applicant(s)/Patent Under Reexamination Vinciarelli, Patrizio	
	Examiner CASSANDRA F COX		Art Unit 2849	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	CPC Classification	US Classification
*	A	US-7245509-B1	07-2007	Chan; Chun-Kong	H02M7/538	363/17
	B					
	C					
	D					
	E					
	F					
	G					
	H					
	I					
	J					
	K					
	L					
	M					

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	CPC Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

<i>Search Notes</i> 	Application/Control No. 16/218,418	Applicant(s)/Patent Under Reexamination Vinciarelli, Patrizio
	Examiner CASSANDRA F COX	Art Unit 2849

CPC - Searched*		
Symbol	Date	Examiner

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
EAST Search (USPGPUB, USPAT, USOCR, EPO, JPO, DERWENT, IBM_TDB) see search history	05/26/2020	cfc
H02M1/08 or H01L24/26 or H05K1/144 or H05K1/0298 or H02M2001/0074 (text only - see search history)	05/26/2020	cfc
inventor search	05/26/2020	cfc

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner
H02M	1/08, 2001/0074	05/26/2020	cfc
H01L	24/26	05/26/2020	cfc
H05K	1/144, 1/0298	05/26/2020	cfc

/CASSANDRA F COX/ Primary Examiner, Art Unit 2849	
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<i>Search Notes</i> 	Application/Control No. 16/218,418	Applicant(s)/Patent Under Reexamination Vinciarelli, Patrizio
	Examiner CASSANDRA F COX	Art Unit 2849

/CASSANDRA F COX/ Primary Examiner, Art Unit 2849	
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Issue Classification 	Application/Control No. 16/218,418	Applicant(s)/Patent Under Reexamination Vinciarelli, Patrizio
	Examiner CASSANDRA F COX	Art Unit 2849

CPC						
Symbol					Type	Version
H02M	/	1	/	08	F	2013-01-01
H05K	/	1	/	144	I	2013-01-01
H01L	/	24	/	26	I	2013-01-01
H05K	/	1	/	0298	I	2013-01-01
H02M	/	2001	/	0074	A	2013-01-01

CPC Combination Sets					
Symbol				Type	Set
	/		/		

NONE (Assistant Examiner) (Date)		Total Claims Allowed: 47	
/CASSANDRA F COX/ Primary Examiner, Art Unit 2849 (Primary Examiner) (Date)		26 May 2020 O.G. Print Claim(s)	30 O.G. Print Figure

<i>Issue Classification</i> 	Application/Control No. 16/218,418	Applicant(s)/Patent Under Reexamination Vinciarelli, Patrizio
	Examiner CASSANDRA F COX	Art Unit 2849

INTERNATIONAL CLASSIFICATION			
CLAIMED			
H02M1/08	/	1	/ 08
NON-CLAIMED			
/		/	

US ORIGINAL CLASSIFICATION	
CLASS	SUBCLASS

CROSS REFERENCES(S)						
CLASS	SUBCLASS (ONE SUBCLASS PER BLOCK)					

NONE		Total Claims Allowed:	
(Assistant Examiner)	(Date)	47	
/CASSANDRA F COX/ Primary Examiner, Art Unit 2849	26 May 2020	O.G. Print Claim(s)	O.G. Print Figure
(Primary Examiner)	(Date)	30	5A

Issue Classification 	Application/Control No. 16/218,418	Applicant(s)/Patent Under Reexamination Vinciarelli, Patrizio
	Examiner CASSANDRA F COX	Art Unit 2849

☐ Claims renumbered in the same order as presented by applicant
 ☐ CPA
 ☐ T.D.
 ☐ R.1.47

CLAIMS															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
1	1	10	10	28	19	37	28	18	37	25	46				
2	2	11	11	29	20	38	29	19	38	47	47				
3	3	12	12	30	21	39	30	20	39						
4	4	13	13	31	22	40	31	21	40						
5	5	14	14	32	23	41	32	22	41						
6	6	15	15	33	24	42	33	23	42						
7	7	16	16	34	25	43	34	24	43						
8	8	26	17	35	26	44	35	45	44						
9	9	27	18	36	27	17	36	46	45						

NONE (Assistant Examiner) _____ (Date) _____		Total Claims Allowed: 47	
/CASSANDRA F COX/ Primary Examiner, Art Unit 2849 (Primary Examiner) _____ (Date) 26 May 2020		O.G. Print Claim(s) 30	O.G. Print Figure 5A

Sheet 1 of 4

Substitute Disclosure Form U.S. Department of Commerce Patent and Trademark Office	Attorney Docket No. 00614-0183001	Application No. 16/218,418
Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))	Applicant VLT, Inc.	
	Filing Date December 12, 2018	Group Art Unit

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Class	Subclass	Filing Date If Appropriate
	1.	5438294	8/1/1995	Smith			
	2.	5728600	3/17/1998	Vinciarelli et al.			
	3.	6031743	02/29/2000	Carpenter et al.			
	4.	6035261	03/07/2000	Carpenter et al.			
	5.	6118186	09/12/2000	Scott et al.			
	6.	6275958	08/14/2001	Carpenter et al.			
	7.	6300749	10/01/2001	Castelli et al.			
	8.	6403009	6/11/2002	Vinciarelli et al.			
	9.	6421262	07/16/2002	Saxelby et al.			
	10.	6466458	10/15/2002	Zhang et al.			
	11.	6603292	08/01/2003	Schouten et al.			
	12.	6700361	03/01/2004	Gregorius, Peter			
	13.	6788033	09/07/2004	Vinciarelli			
	14.	6911848	06/28/2005	Vinciarelli			
	15.	6930893	8/16/2005	Vinciarelli et al.			
	16.	6934166	8/23/2005	Vinciarelli et al.			
	17.	6969909	11/29/2005	Briere			
	18.	6975098	12/13/2005	Vinciarelli et al.			
	19.	6975098	12/13/2005	Vinciarelli			
	20.	6984965	1/10/2006	Vinciarelli et al.			
	21.	6985341	01/10/2006	Vinciarelli, et al.			
	22.	7145786	12/5/2006	Vinciarelli et al.			
	23.	7187263	3/6/2007	Vinciarelli et al.			
	24.	7245509	07/17/2007	Chan et al.			
	25.	7361844	4/22/2008	Vinciarelli et al.			
	26.	7521909	04/01/2009	Dow et al.			

Examiner Signature /CASSANDRA F COX/	Date Considered 05/26/2020
EXAMINER: Initials citation considered. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.	

Substitute Disclosure Form

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /C.F.C/

Sheet 2 of 4

Substitute Disclosure Form U.S. Department of Commerce Patent and Trademark Office	Attorney Docket No. 00614-0183001	Application No. 16/218,418
Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))	Applicant VLT, Inc.	
	Filing Date December 12, 2018	Group Art Unit

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Class	Subclass	Filing Date If Appropriate
	27.	7561446	07/14/2009	Vinciarelli			
	28.	7768807	08/01/2010	Chen, et al.			
	29.	8232783	07/01/2012	Yanagawa, Kenji			
	30.	8427269	4/23/2013	Vinciarelli et al.			
	31.	8582333	11/12/2013	Oraw et al.			
	32.	8966747	3/3/2015	Vinciarelli et al.			
	33.	9087656	7/21/2015	Vinciarelli			
	34.	9112422	08/18/2015	Vinciarelli			
	35.	9166481	10/20/2015	Vinciarelli et al.			
	36.	9402319	7/26/2016	Vinciarelli et al.			
	37.	9516761	12/6/2016	Vinciarelli et al.			
	38.	9660537	05/23/2017	Vinciarelli			
	39.	9765750	09/19/2017	Ferguson et al.			
	40.	9936580	4/3/2018	Vinciarelli et al.			
	41.	10014798	7/3/2018	Vinciarelli et al.			
	42.	10020752	7/10/2018	Vinciarelli et al.			
	43.	10158357	12/18/2018	Vinciarelli et al.			
	44.	10264664	4/16/2019	Vinciarelli et al.			
	45.	10277105	4/30/2019	Vinciarelli et al.			
	46.	14/731,287	06/04/2015	Vinciarelli et al.			
	47.	14/822,561	08/10/2015	Vinciarelli			
	48.	14/874,054	10/02/2015	Vinciarelli			
	49.	2003/0227280	12/11/2003	Vinciarelli et al.			
	50.	2006/0291265	12/28/2006	Schrom et al.			
	51.	2007/0297198	12/27/2017	Chang			
	52.	2009/0016083	1/15/2009	Soldano et al.			

Examiner Signature /CASSANDRA F COX/	Date Considered 05/26/2020
EXAMINER: Initials citation considered. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.	

Substitute Disclosure Form

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /C.F.C/

Sheet 3 of 4

Substitute Disclosure Form Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))	U.S. Department of Commerce Patent and Trademark Office	Attorney Docket No. 00614-0183001	Application No. 16/218,418
	Applicant VLT, Inc.		
	Filing Date December 12, 2018	Group Art Unit	

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Class	Subclass	Filing Date If Appropriate
	53.	2013/0030534	1/31/2013	DeLurio et al.			
	54.	2013/0069608	03/01/2013	Gakhar et al.			
	55.	2015/0188407	07/02/2015	Golder et al.			
	56.	RE40,072	02/19/2008	Prager, et al.			

Foreign Patent Documents or Published Foreign Patent Applications								
Examiner Initial	Desig. ID	Document Number	Publication Date	Country or Patent Office	Class	Subclass	Translation	
							Yes	No
	57.	EP2863531	4/22/2015	EP				
	58.	JPH02280666	11/16/1990	JP			English Abstract	

Other Documents (include Author, Title, Date, and Place of Publication)		
Examiner Initial	Desig. ID	Document
	59.	Amendment after Allowance Pursuant to 37 C.F.R. §1.312 in U.S. Application No. 13/044,063, dated March 3, 2015, 17 pages
	60.	AN2738, Application Note AN2738, "L6390 half-bridge gate driver," STMicroelectronics, August 2009
	61.	AND8311/D (Understanding the LLC Structure in Resonant Applications," Christophe Basso, ON Semiconductor, Jan., 2008) (Year: 2008)
	62.	Brown (POINT OF LOAD CONVERTERS — The Topologies, Converters, and Switching Devices Required for Efficient Conversion," Jess Brown, Vishay Siliconix, Presented at PCIM Conference, May, 2002) (Year: 2002)
	63.	Burton et al., "FIVR - Fully Integrated Voltage Regulators on 4th Generation Intel ® Core™ SoCs," In Applied Power Electronics Conference and Exposition (APEC), 2014 Twenty-Ninth Annual IEEE, March 16, 2014, 432-439
	64.	Cao et al., "A Family of Zero Current Switching Switched-Capacitor DC-DC converters," Applied Power Electronics Conference and Exposition (APEC), 2010 Twenty-Fifth Annual IEEE, Mar. 18, 2010
	65.	Cao et al., "Zero-Current-Switching Multilevel Modular Switched-Capacitor DC-DC Converter" IEEE Transactions on Industry Applications, 46(6):2536-2544, Sep. 2010
	66.	DiBene II et al., "A 400 Amp Fully Integrated Silicon Voltage Regulator with in-die Magnetically Coupled Embedded Inductors," retrieved from http://www.psma.com/sites/default/files/uploads/tech-forums-nanotechnology/resources/400a-fully-integrated-silicon-voltage-regulator.pdf , retrieved on August 3, 2016, 25 pages.
	67.	European Office Action in European Application No. 16173089.0, dated April 10, 2017, 2 pages

Examiner Signature /CASSANDRA F COX/	Date Considered 05/26/2020
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EXAMINER: Initials citation considered. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

Substitute Disclosure Form

Sheet 4 of 4

Substitute Disclosure Form U.S. Department of Commerce Patent and Trademark Office Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))	Attorney Docket No. 00614-0183001	Application No. 16/218,418
	Applicant VLT, Inc.	
	Filing Date December 12, 2018	Group Art Unit

Other Documents (include Author, Title, Date, and Place of Publication)		
Examiner Initial	Desig. ID	Document
	68.	Extended Search Report in European Application No. 16173089.0, dated February 14, 2017, 8 pages
	69.	ISL6210, Datasheet ISL6210, "Dual Synchronous Rectified MOSFET Drivers," Intersil Americas Inc., December 2008
	70.	Miller, "12V PowerStage in Embedded Die System-in-Package," APEC 2015, 33 pages
	71.	Notice of Allowance in U.S. Application No. 13/044,063, Examiner Nguyen, dated February 20, 2015, 15 pages
	72.	Notice of Allowance in U.S. Application No. 13/044,063, Examiner Nguyen, dated June 5, 2015, 11 pages
	73.	Notice of Allowance in U.S. Application No. 14/682,187, dated January 17, 2017, 22 pages
	74.	Notice of Allowance in U.S. Application No. 15/091,346, dated August 13, 2018, 8 pages
	75.	Notice of Allowance in U.S. Application No. 15/091,346, dated May 21, 2018, 9 pages
	76.	Notice of Allowance in U.S. Application No. 15/616,288, dated December 11, 2018, 9 pages
	77.	Office Action in U.S. Application No. 13/044,063, Examiner Nguyen, dated May 23, 2014, 24 pages
	78.	Okudaira and Matsuse, "A New Quasi-Resonant Inverter With Two-way Short-circuit Switch Across A Resonant Capacitor," Power Conversion Conference 2002, 2002, 1496-1501
	79.	Oraw et al., "Load Adaptive, High Efficiency, Switched Capacitor Intermediate Bus Converter," Telecommunications Energy Conference, IEEE 2007, pp. 628-635 Sept. 30 – Oct. 4, 2007
	80.	Reply to Action of May 23, 2014 in U.S. Application No. 13/044,063, dated October 24, 2014, 39 pages
	81.	Seeman et al., "Analysis and optimization of switched-capacitor dc-dc converters," Power Electronics, IEEE Transactions on, vol. 23, pp. 841-851, March 2008
	82.	Supplemental Notice of Allowability in U.S. Application No. 13/044,063, Examiner Nguyen, dated April 20, 2015, 5 pages
	83.	TND359 ("High-Efficiency 255W ATX Power Supply Reference Design Documentation Package," ON Semiconductor, Jan. 2009) (Year: 2009)
	84.	Watson, "New Techniques in the Design of Distributed Power System," Dissertation, 1998, 32 pages
	85.	Yeung et al., "Generalised analysis of switched-capacitor step-down quasi-resonant converter," Electronics Letters, vol. 38, pp. 263-264, 2002
	86.	Yeung et al., "Zerocurrent switching switched-capacitor quasiresonant step-down converter," Electric Power Applications, IEE Proceedings -, vol. 149, pp. 111-121, 2002

Examiner Signature /CASSANDRA F COX/	Date Considered 05/26/2020
EXAMINER: Initials citation considered. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.	

Substitute Disclosure Form

EAST Search History

EAST Search History (Interference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L24	13625	(H02M1/08 or H01L24/26 or H05K1/144 or H05K1/0298 or H02M2001/0074).cpc.	US-PGPUB; USPAT	OR	ON	2020/05/26 14:25
L25	2288	power adj conversion adj module	US-PGPUB; USPAT	OR	ON	2020/05/26 14:25
L26	2853	interconnection adj module	US-PGPUB; USPAT	OR	ON	2020/05/26 14:25
L27	31	(plural\$3 or multiple) adj conversion adj cells	US-PGPUB; USPAT	OR	ON	2020/05/26 14:25
L28	0	I24 and I25 and I26 and I27	US-PGPUB; USPAT	OR	ON	2020/05/26 14:26
L29	0	I24 and I25 and I26	US-PGPUB; USPAT	OR	ON	2020/05/26 14:26
L30	71483	power adj converter	US-PGPUB; USPAT	OR	ON	2020/05/26 14:26
L31	0	I25 and I26 and I27	US-PGPUB; USPAT	OR	ON	2020/05/26 14:26
L32	15	I25 and I26	US-PGPUB; USPAT	OR	ON	2020/05/26 14:26
L33	0	I24 and I30 and I26 and I27	US-PGPUB; USPAT	OR	ON	2020/05/26 14:27
L34	2	I24 and I30 and I26	US-PGPUB; USPAT	OR	ON	2020/05/26 14:27

5/26/2020 2:55:04 PM

C:\Users\ccox\Documents\EAST\Workspaces\616288.wsp

Sheet 1 of 1

Substitute Disclosure Form U.S. Department of Commerce Patent and Trademark Office Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))	Attorney Docket No. 00614-0183001	Application No. 16/218,418
	Applicant VLT, Inc.	
	Filing Date December 12, 2018	Group Art Unit

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Filing Date If Appropriate	Class	Subclass
/C.F.C/	1.	4211603	7/8/1980	Reed			
/C.F.C/	2.	20030087538	5/8/2003	Ueno			
/C.F.C/	3.	20170013748	1/12/2017	Saji et al.			

Foreign Patent Documents or Published Foreign Patent Applications							
Examiner Initial	Desig. ID	Document Number	Publication Date	Country or Patent Office	Translation	Class	Subclass

Other Documents (include Author, Title, Date, and Place of Publication)		
Examiner Initial	Desig. ID	Document

Examiner Signature /CASSANDRA F COX/	Date Considered 05/26/2020
---	-----------------------------------

EXAMINER: Initials citation considered. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

Substitute Disclosure Form

Bibliographic Data

Application No: 16/218,418

Foreign Priority claimed: ☐ Yes ☒ No

35 USC 119 (a-d) conditions met: ☐ Yes ☒ No ☐ Met After Allowance

Verified and Acknowledged: /CASSANDRA F COX/

Examiner's Signature

Initials

Title: Delivering Power To Semiconductor Loads

FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.
12/12/2018	327	2849	00614-0183001
RULE			

APPLICANTS

VLT, Inc., Andover, MA,

INVENTORS

Patrizio Vinciarelli Boston, MA, UNITED STATES

CONTINUING DATA

This application is a CIP of 15616288 06/07/2017 PAT 10277105

15616288 is a CIP of 15091346 04/05/2016 PAT 10158357

FOREIGN APPLICATIONS

IF REQUIRED, FOREIGN LICENSE GRANTED**

01/04/2019

STATE OR COUNTRY

UNITED STATES

ADDRESS

FISH & RICHARDSON P.C. (VLT, INC)

P.O. BOX 1022

MINNEAPOLIS, MN 55440-1022

UNITED STATES

FILING FEE RECEIVED

\$5,600

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

First Named Inventor : Patrizio Vinciarelli Art Unit : Unknown
Application No. : 16/218,418 Examiner : Unknown
Filed : December 12, 2018 Conf. No. : 1092
Title : DELIVERING POWER TO SEMICONDUCTOR LOADS

MAIL STOP AMENDMENT

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

INFORMATION DISCLOSURE STATEMENT

Please consider the documents listed on the disclosure form submitted with this statement. Copies of the documents are being submitted with this statement to the extent required under 37 C.F.R. § 1.98(a)(2).

The applicant wishes to make the examiner aware of the following related application. The prosecution histories for this application are available on PAIR, and are not provided with this statement.

Application No.	Filing Date	First Named Inventor	Attorney Docket No.
16/218,395	December 12, 2018	Vinciarelli	00614-0182001

This statement is being filed before the receipt of a first action on the merits. Apply any necessary charges or credits applicable under 37 C.F.R. § 1.17(p) to deposit account 06-1050, referencing the above attorney docket number.

Please contact the undersigned if there are any questions regarding this statement.

Respectfully submitted,

Date: August 7, 2019 _____

/Rex I. Huang/ _____

Rex I. Huang
Reg. No. 57,661

Customer Number 26214
Fish & Richardson P.C.
Telephone: (617) 542-5070
Facsimile: (877) 769-7945

First Named Inventor : Patrizio Vinciarelli
Application No. : 16/218,418
Filed : December 12, 2018
Page : 2 of 2

Attorney Docket: 00614-0183001 / 0269-US-1
Gearbox

EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	4219	power adj conversion adj module	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/05/26 13:18
L2	3439	interconnection adj module	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/05/26 13:18
L3	26244	multi adj cell	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/05/26 13:19
L4	36	(plural\$3 or multiple) adj conversion adj cells	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/05/26 13:19
L5	5048940	pattern	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/05/26 13:20
L6	1713492	spacing	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/05/26 13:20

L7	25397	(H02M1/08 or H01L24/26 or H05K1/144 or H05K1/0298 or H02M2001/0074).cpc.	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/05/26 13:21
L8	0	I1 and I2 and I3 and I4 and I5 and I6 and I7	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/05/26 13:22
L9	491280	(plural\$3 or multiple) adj3 cells	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/05/26 13:22
L10	0	I1 and I2 and I3 and I4 and I5 and I6	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/05/26 13:22
L11	0	I1 and I2 and I3 and I9 and I5 and I6 and I7	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/05/26 13:23
L12	0	I1 and I2 and I3 and I9 and I5 and I6	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/05/26 13:23
L13	0	I1 and I2 and I3 and I4 and I7	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/05/26 13:23

L14	0	I1 and I2 and I3 and I9 and I7	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/05/26 13:24
L15	0	I1 and I2 and I3 and I4	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/05/26 13:24
L16	0	I1 and I2 and I3 and I9	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/05/26 13:24
L17	0	I1 and I2 and I4 and I7	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/05/26 13:24
L18	0	I1 and I2 and I7 and I9	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/05/26 13:25
L19	120709	power adj converter	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/05/26 13:25
L20	6	I4 and I19	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/05/26 13:31

L21	97	("20030087538" "20170013748" "4211603" "10231333" "10390437" "10512182" "20010018285" "20010018286" "20010023980" "20010032388" "20030162434" "20030168499" "20040207089" "20040251554" "20050048692" "20060127652" "20060180346" "20060272150" "20130083495" "20160128215" "20170115144" "20170336584" "20190080931" "3846741" "4223970" "4230385" "4394711" "4526429" "4742477" "4917526" "5168432" "5321351" "5408189" "6031253" "6110213" "6116916" "6230403" "6646886" "6847853" "7294007" "RE32212" "RE36442" "0014731" "0014822" "0014874" "10014798" "10020752" "10158357" "10264664" "10277105" "20030227280" "20060291265" "20070297198" "20090016083" "20130030534" "20130069608" "20150188407" "5438294" "5728600" "6031743" "6035261" "6118186" "6275958" "6300749" "6403009" "6421262" "6466458" "6603292" "6700361" "6788033" "6911848" "6930893" "6934166" "6969909" "6975098" "6984965" "6985341" "7145786" "7187263" "7245509" "7361844" "7521909" "7561446" "7768807" "8232783" "8427269" "8582333" "8966747" "9087656" "9112422" "9166481" "9402319" "9516761" "9660537" "9765750" "9936580" "RE40072").PN.	US-PGPUB; USPAT	OR	ON	2020/05/26 13:32
L22	4219	power adj conversion adj module	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/05/26 14:09
L23	301	vinciarelli-patrizio.in.	US-PGPUB; USPAT; USOCR; EPO; JPO;	OR	ON	2020/05/26 14:25

			DERWENT; IBM_TDB			
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5/26/2020 2:54:53 PM

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Sheet 1 of 2

Substitute Disclosure Form Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))	U.S. Department of Commerce Patent and Trademark Office	Attorney Docket No. 00614-0183001	Application No. 16/218,418
	Applicant VLT, Inc.		
	Filing Date December 12, 2018	Group Art Unit	

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Filing Date If Appropriate	Class	Subclass
	1.	3846741	11/1/1974	Kunkle et al.			
	2.	4223970	9/23/1980	Walter			
	3.	4230385	10/28/1980	Ammon et al.			
	4.	4394711	7/19/1983	Conley			
	5.	4526429	7/2/1985	Kirkman			
	6.	4742477	5/3/1988	Phillips et al.			
	7.	4917526	4/17/1990	Paterson			
	8.	5168432	3/6/2019	Murphy et al.			
	9.	5321351	6/14/1994	Swart et al.			
	10.	5408189	4/18/1995	Swart et al.			
	11.	6031253	2/29/2000	Kobayashi			
	12.	6110213	8/29/2000	Vinciarelli et al.			
	13.	6116916	9/12/2000	Kasai			
	14.	6230403	5/15/2001	Skoolicas et al.			
	15.	6646886	11/11/2003	Popovich et al.			
	16.	6847853	1/25/2005	Vinciarelli et al.			
	17.	7294007	11/13/2007	Lawlyes			
	18.	10231333	3/12/2019	Liu et al.			
	19.	10390437	8/20/2019	Kudou et al.			
	20.	10512182	12/17/2019	Suzuki			
	21.	20010018285	8/30/2001	Mizumura			
	22.	20010018286	8/30/2001	Mizumura			
	23.	20010023980	9/27/2001	Ohmori			
	24.	20010032388	10/25/2001	Morris			
	25.	20030162434	8/28/2003	Kamiya			
	26.	20030168499	9/11/2003	Tanabe et al.			
	27.	20040207089	10/21/2004	Masuda			

Examiner Signature	Date Considered
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EXAMINER: Initials citation considered. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

Substitute Disclosure Form

Sheet 2 of 2

Substitute Disclosure Form Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))	U.S. Department of Commerce Patent and Trademark Office	Attorney Docket No. 00614-0183001	Application No. 16/218,418
	Applicant VLT, Inc.		
	Filing Date December 12, 2018	Group Art Unit	

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Filing Date If Appropriate	Class	Subclass
	28.	20040251554	12/16/2004	Masuda			
	29.	20050048692	3/3/2005	Hanada et al.			
	30.	20060127652	6/15/2006	Kanaya et al.			
	31.	20060180346	8/17/2006	Knight et al.			
	32.	20060272150	12/7/2006	Eguchi et al.			
	33.	20130083495	4/4/2013	Moon et al.			
	34.	20160128215	5/5/2016	Kanzaki et al.			
	35.	20170115144	4/27/2017	Watanabe et al.			
	36.	20170336584	11/23/2017	Ariga et al.			
	37.	20190080931	3/14/2019	Ito et al.			
	38.	RE32212	7/22/1986	Walter et al.			
	39.	RE36442	12/14/1999	Kardos			

Foreign Patent Documents or Published Foreign Patent Applications							
Examiner Initial	Desig. ID	Document Number	Publication Date	Country or Patent Office	Translation	Class	Subclass

Other Documents (include Author, Title, Date, and Place of Publication)		
Examiner Initial	Desig. ID	Document

Examiner Signature /CASSANDRA F COX/	Date Considered 05/26/2020
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EXAMINER: Initials citation considered. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

Substitute Disclosure Form

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Request for Continued Examination (RCE) Transmittal

Address to:
Mail Stop RCE
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Application Number	16/218,418
Filing Date	December 12, 2018
First Named Inventor	Patrizio Vinciarelli
Art Unit	
Examiner Name	
Attorney Docket Number	00614-0183001

This is a Request for Continued Examination (RCE) under 37 CFR 1.114 of the above-identified application.

Request for Continued Examination (RCE) practice under 37 CFR 1.114 does not apply to any utility or plant application filed prior to June 8, 1995, or to any design application. See Instruction Sheet for RCEs (not to be submitted to the USPTO) on page 2.

1. **Submission required under 37 CFR 1.114** Note: If the RCE is proper, any previously filed unentered amendments and amendments enclosed with the RCE will be entered in the order in which they were filed unless applicant instructs otherwise. If applicant does not wish to have any previously filed unentered amendment(s) entered, applicant must request non-entry of such amendment(s).

a. ☐ Previously submitted. If a final Office action is outstanding, any amendments filed after the final Office action may be considered as a submission even if this box is not checked.

i. ☐ Consider the arguments in the Appeal Brief or Reply Brief previously filed on _____

ii. ☐ Other _____

b. ☒ Enclosed

i. ☐ Amendment/Reply

iii. ☒ Information Disclosure Statement (IDS)

ii. ☐ Affidavit(s)/ Declaration(s)

iv. ☐ Other _____

2. Miscellaneous

Suspension of action on the above-identified application is requested under 37 CFR 1.103(c) for a

a. ☐ period of _____ months. (Period of suspension shall not exceed 3 months; Fee under 37 CFR 1.17(i) required)

b. ☐ Other _____

3. Fees

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Electronic Patent Application Fee Transmittal

Application Number:	16218418			
Filing Date:	12-Dec-2018			
Title of Invention:	Delivering Power To Semiconductor Loads			
First Named Inventor/Applicant Name:	Patrizio Vinciarelli			
Filer:	Rex I. Huang/Rita Liston			
Attorney Docket Number:	00614-0183001			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
RCE- 1ST REQUEST	1801	1	1300	1300
Total in USD (\$)				1300

Electronic Acknowledgement Receipt

EFS ID:	39874699
Application Number:	16218418
International Application Number:	
Confirmation Number:	1092
Title of Invention:	Delivering Power To Semiconductor Loads
First Named Inventor/Applicant Name:	Patrizio Vinciarelli
Customer Number:	26214
Filer:	Rex I. Huang/Rita Liston
Filer Authorized By:	Rex I. Huang
Attorney Docket Number:	00614-0183001
Receipt Date:	30-JUN-2020
Filing Date:	12-DEC-2018
Time Stamp:	16:19:51
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	DA
Payment was successfully received in RAM	\$ 1300
RAM confirmation Number	E20206TG20133713
Deposit Account	
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Request for Continued Examination (RCE)	00614_0183001_RCE.pdf	117259	no	1
			e54d038d8cfebd45d40c97079f7099f26483fa07		

Warnings:

This is not a USPTO supplied RCE SB30 form.

Information:

2	Fee Worksheet (SB06)	fee-info.pdf	29907	no	2
			e8a9925e99c8ef5bbe54b25dbe64cef775ded13f		

Warnings:

Information:

Total Files Size (in bytes):			147166
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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

First Named Inventor :	Patrizio Vinciarelli	Art Unit :	Unknown
Application No. :	16/218,418	Examiner :	Unknown
Filed :	December 12, 2018	Conf. No. :	1092
Title :	DELIVERING POWER TO SEMICONDUCTOR LOADS		

MAIL STOP AMENDMENT

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

INFORMATION DISCLOSURE STATEMENT

Please consider the documents listed on the disclosure form submitted with this statement. Copies of the documents are being submitted with this statement to the extent required under 37 C.F.R. § 1.98(a)(2).

This statement is being filed with a Request for Continued Examination.

Apply any necessary charges or credits applicable under 37 C.F.R. § 1.17(p) to deposit account 06-1050, referencing the above attorney docket number.

Please contact the undersigned if there are any questions regarding this statement.

Respectfully submitted,

Date: June 30, 2020_____

/Rex I. Huang/_____

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Substitute Disclosure Form	U.S. Department of Commerce Patent and Trademark Office	Attorney Docket No. 00614-0183001	Application No. 16/218,418
Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))		Applicant VLT, Inc.	
		Filing Date December 12, 2018	Group Art Unit

.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Filing Date If Appropriate	Class	Subclass
	1.	7015587	03/21/2006	Poddar			
	2.	7030469	4/18/2006	Mahadevan et al.			
	3.	7198987	4/3/2007	Warren et al.			
	4.	7494843	2/24/2009	Lin et al.			
	5.	8138584	3/20/2012	Wang et al.			
	6.	20070241440	10/18/2007	Hoang et al.			
	7.	20080078572	7/26/2011	Watanabe et al.			
	8.	20140355218	12/4/2014	Vinciarelli et al.			

Foreign Patent Documents or Published Foreign Patent Applications							
Examiner Initial	Desig. ID	Document Number	Publication Date	Country or Patent Office	Translation	Class	Subclass
	9.	JPH04293293	10/16/1992	JP	English Abstract		

Other Documents (include Author, Title, Date, and Place of Publication)		
Examiner Initial	Desig. ID	Document

Examiner Signature	Date Considered
EXAMINER: Initials citation considered. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.	

**Espacenet****Bibliographic data: JPH04293293 (A) — 1992-10-16**

WIRING ALTERABLE ELECTRONIC INTEGRATED CIRCUIT**Inventor(s):** INUDA MANABU ± (INUDA MANABU)**Applicant(s):** NIYUUTRON KK ± (NIYUUTRON:KK)**Classification:** - **international:** *H05K1/14; H05K1/00; (IPC1-7): H05K1/14*
- **cooperative:** *H05K1/0286 (EP); H05K1/144 (EP)***Application number:** JP19910081880 19910320**Priority number(s):** JP19910081880 19910320**Abstract of JPH04293293 (A)**

PURPOSE:To enhance the degree of freedoms of a circuit design by connecting a printed circuit of a matrix wiring board in which lead conductors extended from a high density mounting printed wiring board of a micromodule are crossed perpendicularly via through holes of the matrix board. **CONSTITUTION:**A high density mounting printed wiring board is connected to a matrix wiring board 11. A plurality of lead conductors extended from a micromodule associated with electronic components of an integrated circuit, etc., are inserted into grooves 19 formed on the board 11, and connected to connecting conductors 15 exposed in the grooves 19. Then, desired connecting conductors 15 on printed wiring boards 12, 13 for constituting the board 11 are connected to each via necessary through holes 17 on the board 11. The boards 12, 13 can be freely connected to the conductors 15. Thus, the degree of freedoms of a circuit design can be enhanced.

(19) 日本国特許庁 (J P)

(12) 公 開 特 許 公 報 (A)

(11) 特許出願公開番号

特開平4-293293

(43) 公開日 平成4年(1992)10月16日

(51) Int.Cl.⁵

H 0 5 K 1/14

識別記号

庁内整理番号

E 8727-4E

F I

技術表示箇所

審査請求 未請求 請求項の数6(全 7 頁)

(21) 出願番号 特願平3-81880

(22) 出願日 平成3年(1991)3月20日

(71) 出願人 591076464

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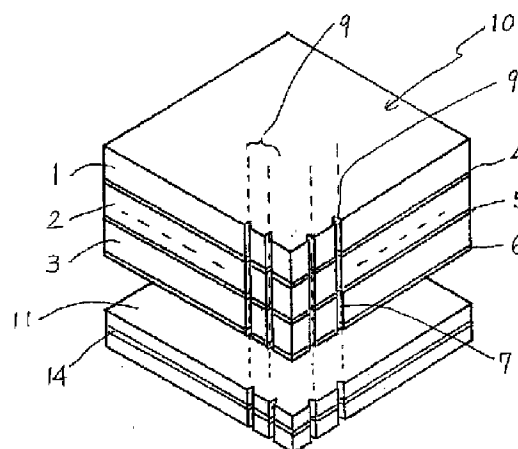
(74) 代理人 弁理士 辻 実

(54) 【発明の名称】 配線変更自在な電子集積回路

(57) 【要約】

【目的】 ブロック化されたマイクロモジュールの配線を自由に変更できるような配線変更自在な電子集積回路を得ることを目的とする。

【構成・効果】 マイクロモジュールの高密度実装プリント配線基板から引き出された引出導体をマトリクス配線板の所定のプリント回路と接続するとともに、所定のスルーホールを介して直交するマトリクス配線板のプリント回路を接続する。マトリクス配線板の接続位置を変更して、所望の機能の電子集積回路を得ることができる。



【特許請求の範囲】

【請求項1】電子部品を搭載しプリント回路と接続した引出導体を設けた高密度実装プリント配線基板からなるマイクロモジュールと、絶縁物を介して互いに直交する複数本のプリント回路を設けこれらプリント回路の交叉部分にスルーホールを有する交点端子を設けたマトリクス配線板とを有し、マイクロモジュールの高密度実装プリント配線基板から引き出された引出導体をマトリクス配線板の所定のプリント回路と接続するとともに、所定のスルーホールを介して直交するマトリクス配線板のプリント回路を接続することを特徴とする配線変更自在な電子集積回路。

【請求項2】前記マイクロモジュールは、複数層積層した高密度実装プリント配線基板を有することを特徴とする請求項1記載の配線変更自在な電子集積回路。

【請求項3】所定のスルーホールを介して直交するマトリクス配線板のプリント回路を接続する手段は、螺合自在で取り外し自在な金属ねじであることを特徴とする請求項1記載の配線変更自在な電子集積回路。

【請求項4】所定のスルーホールを介して直交するマトリクス配線板のプリント回路を接続する手段は、スルーホールを貫通し、かつ直交するマトリクス配線板のそれぞれに弾接する接点を有し、抜き差し自在なピンであることを特徴とする請求項1記載の配線変更自在な電子集積回路。

【請求項5】所定のスルーホールを介して直交するマトリクス配線板のプリント回路と接続する手段は、一方の面のプリント回路と電気的に接続しスルーホールに沿って他方の面に導出され他方の面のプリント回路の近傍に設けられた端子部と、該他方の面のプリント回路と導体にて接続したことを特徴とする請求項1記載の配線変更自在な電子集積回路。

【請求項6】前記端子部とプリント回路とを接続する導体は半田であることを特徴とする請求項5記載の配線変更自在な電子集積回路。

【発明の詳細な説明】

【0001】

【産業上の利用分野】本発明は、電子集積回路に関し、特に配線変更自在な電子集積回路に関する。

【0002】

【従来の技術】半導体装置が発明されて以来、電子回路の小型化が進行し、コンデンサ、抵抗器、コイル等の電子部品を小型化する試みがなされるとともに、小さい面積中にこれらの電子部品をでき得るかぎり多く収納する試みが多くなされた。たとえば、合成樹脂製またはセラミックス製絶縁基板の上に印刷手段により配線回路を形成し、その上に極く小さい抵抗器、コンデンサ、コイル、トランジスタ等の電子部品を接続し、各配線回路の必要部分は、その絶縁基板の周囲に設けた複数の切り欠き部分に形成した端子電極と電気的に接続する。このよ

うに構成された絶縁基板を複数枚重ねあわせた後、前記切り欠きに設けられた端子電極どうしをリード線で連結して電子集積回路を構成する。

【0003】またこの電子集積回路を合成樹脂内に封入する場合もある。このようにして構成された電子集積回路はマイクロモジュールと呼ばれている。

【0004】上述のごときマイクロモジュールは部品の集積度も低く、また大量生産に向かないため、最近では部品の集積度も大幅に向上し、かつ大量生産にも向くLSIが開発されて、今やこれが半導体製品の主流をなしている。

【0005】上述のLSIは、膨大な電子部品を小さな面積内に収納でき、かつ大量生産に向いている反面、製造のための設備に莫大な費用を投入しなければならず、また製造方法も複雑である。したがって、構成部品数も少なく、需要も限られた電子集積回路に対しては不向きである。そこで需要数も少なく、構成部品も少ない電子集積回路の場合にはマイクロモジュールの方が有利とされている。

【0006】

【発明が解決しようとする課題】ところで、マイクロモジュールは多くのカスタマーから要求される種々の回路構成に対して効率よく対応するため、予め前記絶縁基板上に基本的な回路を複数個形成しておき、カスタマーからの要求に応じてこれらの基本的な回路間を選択的に配線して所望の回路構成を得る。この時、緊急的な仕様設計変更が止むなく発生したり、また回路の設計ミスが判明した場合には、煩雑な回路中で配線を切断したりまた必要な回路間をジャンパー線で接続するというような煩雑な作業を強いられる。

【0007】一方、上述のマイクロモジュールは、組み立てられた完成品の状態で内部の回路変更を行うようなことは無く、個々のマイクロモジュールの持つ個々のスピードで処理される入出力端子が固有的在あるいは専用のに配線接続されており、このことが内部回路の設計変更に対する柔軟性を損ない、マイクロモジュールを専用機化してしまう要因のひとつでもある。

【0008】このような設計変更に対する柔軟性に乏しいマイクロモジュールを汎用化させるための一つの解決策は、電子計算機のプログラムによって対応ビットをシフトさせ信号の経路を任意に変更することであるが、プログラムの処理スピードに制約され信号のスピードで限定されることも多い。また、PLD素子の様に集積回路内部の論理回路をプログラムで固着させ得るものは、配線変更もプログラムで容易に変更することが出来、かつ高速信号を扱うことが出来るが、書き込み器が必要であり、一般的に所望の回路を構成した場合に1個のPLD素子では納まらない場合が多いので、多数のチップの書き込みが困難である。

【0009】従来のマイクロモジュールは、専用のに外

部接続ピンと配線されている。しかも固着封入される回路が複雑化し多様化しても外部接続ピンの数で制約を受けることとなり多様化できない欠点もある。つまりモールド用の合成樹脂で固めて封入されるものは、いくつかの部品を複合し、1個にするという非常に小規模なものに限定されていた。そのため配線の変更ができるようにはなっていなかった。また従来熱可塑性の合成樹脂で固める場合小さなプラスチック・ケースを使用しモールド用合成樹脂が軟化しているうちに回路部品とともに封入されるので個別的なモールド作業となるので人手を多く必要とし、生産性が非常に悪かった。

【0010】本発明はこのような従来の不都合を解決しようとするものであり、その発明の目的は、ブロック化されたマイクロモジュールの配線を自由に変更できるような配線変更自在な電子集積回路を得ることにある。

【0011】

【課題を解決するための手段】上述のような本発明の目的を達成するために、本発明は電子部品を搭載しプリント回路と接続した引出導体を設けた高密度実装プリント配線基板からなるマイクロモジュールと、絶縁物を介して互いに直交する複数本のプリント回路を設けこれらプリント回路の交叉部分にスルーホールを有する交点端子を設けたマトリクス配線板とを有し、マイクロモジュールの高密度実装プリント配線基板から引き出された引出導体をマトリクス配線板の所定のプリント回路と接続するとともに、所定のスルーホールを介して直交するマトリクス配線板のプリント回路を接続することを特徴とする配線変更自在な電子集積回路を提供する。

【0012】

【実施例】本発明の一実施例を図面を用いて詳細に説明する。

【0013】図1は、本発明にかかるマイクロモジュールを示す斜視図である。この実施例においてマイクロモジュール10は、図1から明らかなように、3層の高密度実装プリント配線基板1、2、3を有し、これら3層の高密度実装プリント配線基板1～3はそれぞれプリント回路板4、5、6を有し、これらプリント回路板4、5、6の上には、集積回路、トランジスタ、抵抗器、コンデンサなどの電子部品が取り付けられ、これら電子部品の端子はプリント回路板上に形成されたプリント回路8に接続されている。そしてこれら高密度プリント配線基板は積層されて、合成樹脂の中に埋め込まれて、ブロック化されている。

【0014】図2から分かるように、マイクロモジュール10の側面には断面V字型あるいはU字型をした凹溝7が複数本縦方向に設けられている。各プリント回路板の所定のプリント回路8は凹溝7にまで導出され、その側面が凹溝7に現われている。各凹溝7の中には引出導線9が挿入され、各プリント回路板上のプリント回路8と半田付けにより接続されており、それらの所定のもの

の上下端はマイクロモジュール10の上下面から突出しており、またあるものは所定のプリント回路板間の接続だけを行ない、さらにあるものは上方向又は下方向にだけ突出している。そしてこれらの引出導線9は各層間の電気的な接続を行なうだけでなく、マイクロモジュール10と外部回路との接続導体ともなるのであるが、マイクロモジュール10の種類によりこれらの構造はそれぞれ異なる。

【0015】図1において、11はマトリクス配線板である。マトリクス配線板11はブロック化されたマイクロチップ10の機能に応じて導出された引出導線9の所定のものどうしを接続したり、外部から入力される入力信号を、高密度実装プリント配線基板1、2、3を構成するプリント回路板4、5、6の所定のプリント回路に接続したり、また高密度実装プリント配線基板1、2、3から出力される出力信号を外部に導出する仲介を果たしたりする。なお図1では、該マトリクス配線板11がマイクロモジュール10から分離して示されているが、実際にはマイクロモジュール10とマトリクス配線板11は、接着されはしないが、積層されて一体化される。

【0016】マトリクス配線板11は、図3に示すように、2枚のプリント回路板12、13が、スペーサ14を挟んで背中合わせに接着されている。一方のプリント回路板12の表面には縦方向に平行にかつ等間隔に複数本の接続導体15がプリント配線技術によって形成されている。また、接続導体15、15・・・には、等間隔で複数個の交点端子16が設けられている。図3には示されていないが、他方のプリント回路板13の表面（図3では裏面となる）には、プリント回路板12に設けられた接続導体15と直交する方向すなわち横方向に複数本の接続導体が等間隔に設けられており、また複数個の交点端子16が等間隔に複数個設けられている。そして、これらプリント回路板12、13に設けられた交点端子16はマトリクス配線板11の厚み方向に対して重なって設けられている。これら交点端子16の中央には、マトリクス配線板を構成する2枚のプリント回路板12、13を貫通するスルーホール17が貫通している。マトリクス配線板11の縁部には、外部入力回路あるいは外部出力回路とマイクロモジュールとを接続する接続端子18が設けられている。マトリクス配線板11の側面にはマイクロモジュール10と同様な断面U字型あるいはV字型をした凹溝19が設けられており、プリント回路板12、13の所定のものは、凹溝19にまで導出され、その側面が凹溝19内に現われている。

【0017】図4はスルーホール部分を示す断面図である。図4から明らかなように、接続導体15はスルーホール17の内部にまで設けられており、各スルーホール17内にはねじが切つてある。20はプリント回路板12とプリント回路板13の接続導体を導通せしめる金属ねじである。

【0018】図5は、マイクロチップ10とマトリックス配線板11の接続関係を説明するための回路図である。なお、説明を簡単にするために、1枚のみの高密度プリント配線基板がマトリックス配線板11と接続している。また、高密度プリント配線基板には、2個の集積回路21、22と2個の入出力用のゲート回路23、24を備えているものとする。

【0019】上述のように、集積回路21、22等を組み込んだマイクロモジュール10から伸びる複数本の引出導線9をマトリックス配線板11に設けられた凹溝19に挿入し、該凹溝19内に表出している接続導体15と接続する。次いでマトリックス配線板11上の必要なスルーホール17（図5において黒ドットで示してある）に細い金属ねじ20をねじ込んで、マトリックス配線板11を構成するプリント回路板12、13上の所望の接続導体15同士を接続する。この接続で、マイクロモジュール10内に設けた集積回路21、あるいは集積回路22の入力端子は、マトリックス配線板11を通してマイクロモジュール10から導出される外部接続端子25の所定の入力端に接続され、集積回路21、あるいは集積回路22の出力端子は、マトリックス配線板11を通してマイクロモジュール10から導出される外部接続端子25の所定の出力端に接続される。

【0020】カスタマーの要求で、外部出力端子25の配置変更を要求された場合には、金属ねじ20が螺合する位置を変更するだけで、この困難な要求に添うことができる。

【0021】なお、上記実施例では、マトリックス配線板11のスルーホール17に金属ねじ20を螺合させて、プリント回路板12、13上の所望の接続導体15同士を接続しているが、図6に示すように、1枚の両面プリント回路板26を使用し、該両面プリント回路板26の表裏面にそれぞれ縦横のプリント回路30、40を設け、スルーホール27には上下面を接続する導体28を設け、該導体28はプリント回路40と反対側に導出されて端子部28'と接続される。又プリント回路30はスルーホール近傍において円形の端子部30'を形成し、小さな間隙を置いて端子部28'と対向している。そして接続を必要とするスルーホール27には図7に示すように半田29を盛って端子部28'と30'とを電気的に接続し、表裏面のプリント回路を接続する。

【0022】図8はスルーホールにおける接続方法のもう一つ他の実施例を示す断面図である。図8において、50および52は上下のプリント回路板であって、それぞれのプリント回路板50、52には縦、横方向に走るプリント回路54、56が設けられている。それらプリント回路54、56は、スルーホール58の内壁に添って設けられ、更にそれぞれ裏側に折返っている。なお、プリント回路板50と52との間には、絶縁物製のスペーサ60が介在している。又、接続を要するスルーホー

ル58には、金属性のピン62が挿入されている。該ピン62には、足部64が設けられ、該足部64はスリフト66により複数本に分割されて接点片68を形成している。接点片68にはそれぞれ、プリント回路54と接触する接点70と、プリント回路56と接触する接点72とを有する。したがって、スルーホールにピン62が挿入されたとき、プリント回路板50と52に設けられたプリント回路54と56はピン64により接続される。

【0023】また、マイクロモジュール10は、カスタマーの要求により、1個1個作成してもよいが、図9に示すように、第1層目ないし第3層目のプリント回路板の所定の位置にそれぞれ種類の異なる回路を作成してこれらの高密度実装プリント配線基板を一体に合成樹脂成型し、その後、点線部分をソーにより切断してマイクロモジュールを形成すれば、一度に複数種類のマイクロモジュールを得ることができる。

【0024】

【発明の効果】以上詳細に説明したように、本発明によれば、電子部品を搭載しプリント回路と接続した引出導体を設けた高密度実装プリント配線基板からなるマイクロモジュールと、絶縁物を介して互いに直交する複数本のプリント回路を設けこれらプリント回路の交叉部分にスルーホールを有する交点端子を設けたマトリックス配線板とを有し、マイクロモジュールの高密度実装プリント配線基板から引き出された引出導体をマトリックス配線板の所定のプリント回路と接続するとともに、所定のスルーホールを介して直交するマトリックス配線板のプリント回路を接続するものであるから、所定のスルーホールを介して直交するプリント回路を接続する部分をカスタマーの要求に応じて自由に設定でき、このため回路設計の自由度がきわめて高い電子集積回路を得ることができる。

【図面の簡単な説明】

【図1】本発明の一実施例を示す分解斜視図である。

【図2】マイクロモジュールの分解斜視図である。

【図3】マトリックス配線板の斜視図である。

【図4】スルーホールの断面図である。

【図5】マイクロチップ10とマトリックス配線板11の接続関係を説明するための回路図である。

【図6】本発明のスルーホールの他の一実施例を示す斜視図である。

【図7】図6に示すスルーホールにおいて上下面のプリント回路を半田にて接続した状態を示す断面図である。

【図8】本発明のスルーホールのもう一つ他の実施例を示す断面図である。

【図9】マイクロモジュールの製造方法を説明するための斜視図である。

【符号の説明】

1、2、3・・・高密度実装プリント配線基板

(5)

特開平4-293293

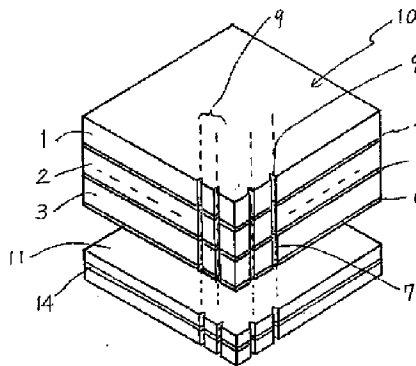
7

8

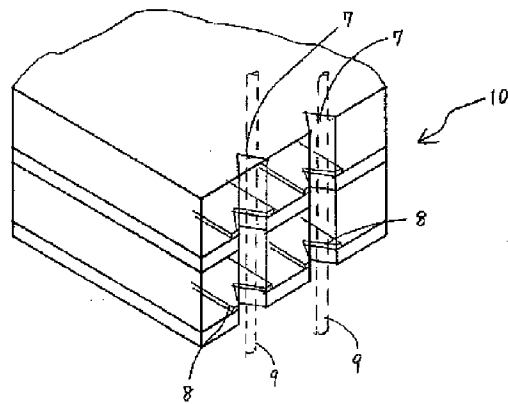
- 4、5、6・・・プリント回路板
- 7・・・凹溝
- 8・・・プリント回路
- 9・・・引出導線
- 10・・・マイクロモジュール
- 11・・・マトリクス配線板
- 12、13・・・プリント回路板
- 14・・・スペーサ
- 15・・・接続導体
- 16・・・交点端子
- 17・・・スルーホール
- 18・・・接続端子

- 19・・・凹溝
- 20・・・金属ねじ
- 21・・・集積回路
- 22・・・集積回路
- 23・・・ゲート回路
- 24・・・ゲート回路
- 25・・・外部接続端子
- 26・・・両面プリント回路板
- 27・・・スルーホール
- 10 28・・・導体
- 29・・・半田

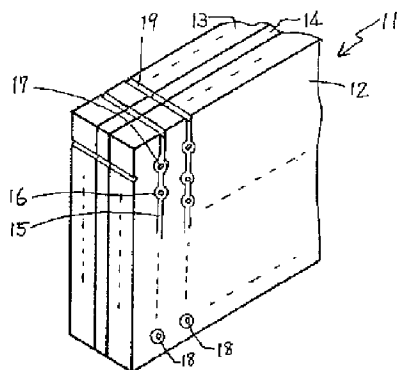
【図1】



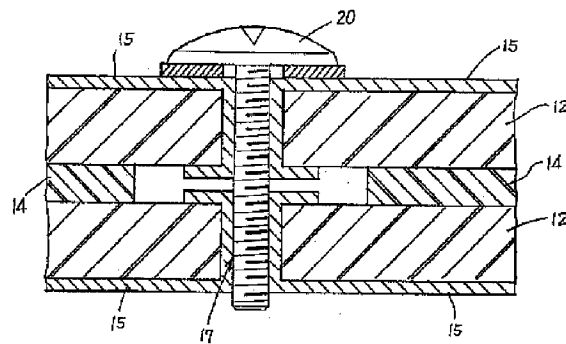
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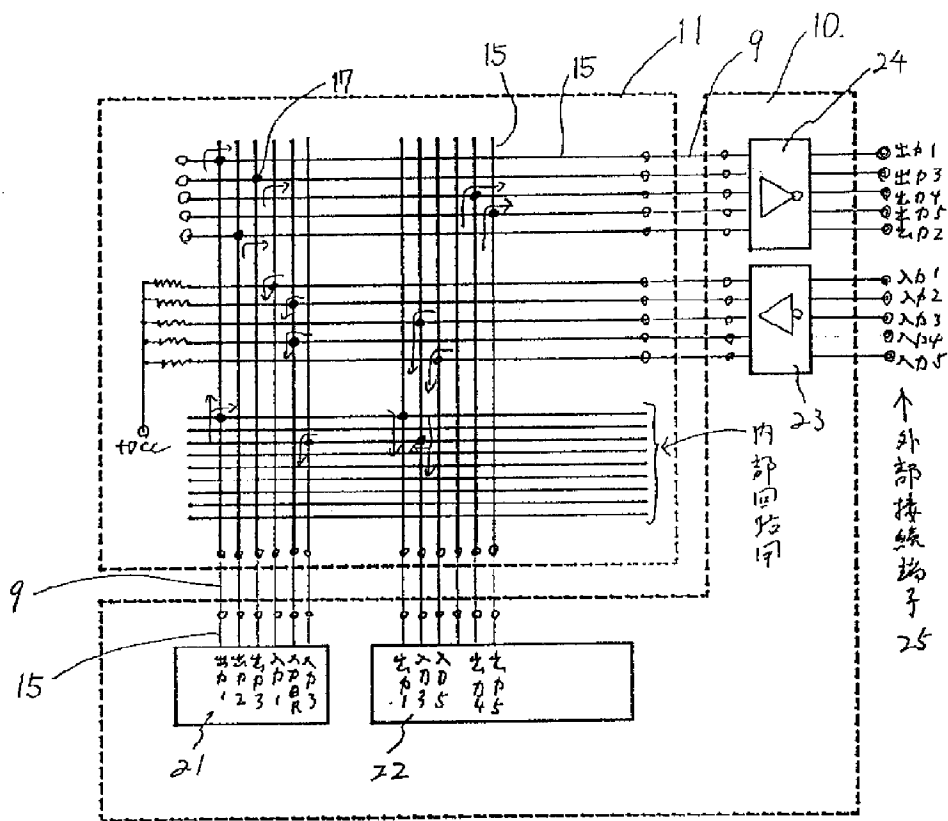
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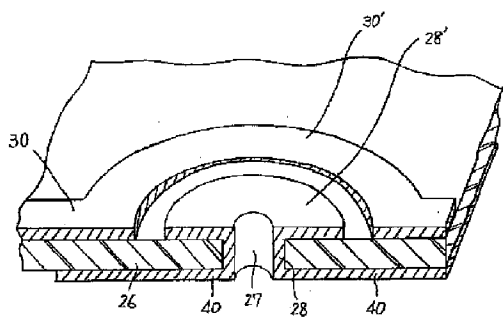
【図4】



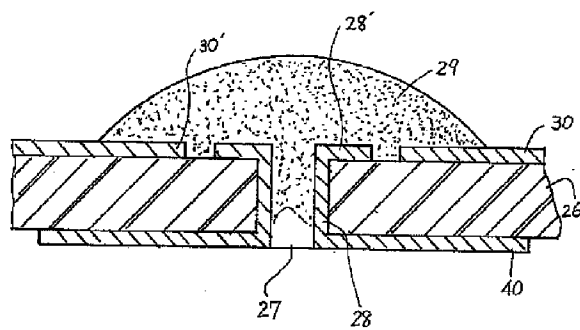
【図5】



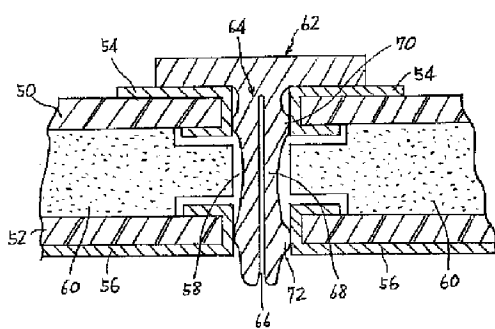
【図6】



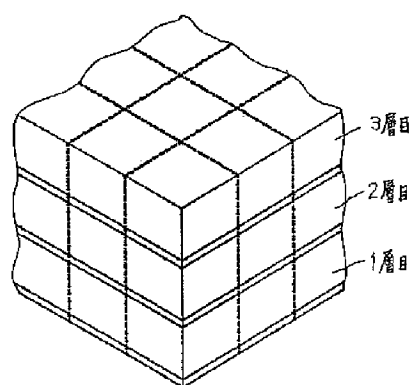
【図7】



【図8】



【図9】



Electronic Acknowledgement Receipt

EFS ID:	39875898
Application Number:	16218418
International Application Number:	
Confirmation Number:	1092
Title of Invention:	Delivering Power To Semiconductor Loads
First Named Inventor/Applicant Name:	Patrizio Vinciarelli
Customer Number:	26214
Filer:	Rex I. Huang/Mollie Kufner
Filer Authorized By:	Rex I. Huang
Attorney Docket Number:	00614-0183001
Receipt Date:	30-JUN-2020
Filing Date:	12-DEC-2018
Time Stamp:	16:59:53
Application Type:	Utility under 35 USC 111(a)

Payment information:

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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Information Disclosure Statement (IDS) Form (SB08)	00614-0183001_Transmittal.pdf	65685	no	1
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16218418	NOA	06/30/2020	00614-0183001
	892	06/30/2020	00614-0183001
	1449	06/30/2020	00614-0183001
	1449	06/30/2020	00614-0183001
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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875				Application or Docket Number 16/218,418		Filing Date 12/12/2018		☐ To be Mailed				
ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO												
APPLICATION AS FILED - PART I												
		(Column 1)	(Column 2)									
FOR		NUMBER FILED	NUMBER EXTRA		RATE (\$)		FEE (\$)					
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))		N/A	N/A		N/A							
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))		N/A	N/A		N/A							
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))		N/A	N/A		N/A							
TOTAL CLAIMS (37 CFR 1.16(i))		minus 20 = *		x \$100 =								
INDEPENDENT CLAIMS (37 CFR 1.16(h))		minus 3 = *		x \$460 =								
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))		If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).										
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))												
* If the difference in column 1 is less than zero, enter "0" in column 2.					TOTAL							
APPLICATION AS AMENDED - PART II												
		(Column 1)		(Column 2)	(Column 3)							
AMENDMENT	06/30/2020	CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)		ADDITIONAL FEE (\$)				
	Total (37 CFR 1.16(i))	* 47	Minus	** 47	= 0	x \$100 =		0				
	Independent (37 CFR 1.16(h))	* 6	Minus	*** 6	= 0	x \$460 =		0				
	☐ Application Size Fee (37 CFR 1.16(s))											
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))											
						TOTAL ADD'L FEE		0				
		(Column 1)		(Column 2)	(Column 3)							
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)		ADDITIONAL FEE (\$)				
	Total (37 CFR 1.16(i))	*	Minus	**	=	x \$0 =						
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x \$0 =						
	☐ Application Size Fee (37 CFR 1.16(s))											
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))											
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PAPER NUMBER

2849

DATE MAILED: 07/28/2020

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/218,418	12/12/2018	Patrizio Vinciarelli	00614-0183001	1092

TITLE OF INVENTION: Delivering Power To Semiconductor Loads

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1000	\$0.00	\$0.00	\$1000	10/28/2020

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If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 1/2 the amount of undiscounted fees, and micro entity fees are 1/2 the amount of small entity fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed and an extra copy of the form should be submitted. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Maintenance fees are due in utility patents issuing on applications filed on or after Dec. 12, 1980. It is patentee's responsibility to ensure timely payment of maintenance fees when due. More information is available at www.uspto.gov/PatentMaintenanceFees.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), by mail or fax, or via EFS-Web.

By mail, send to: Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450

By fax, send to: (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

26214 7590 07/28/2020
FISH & RICHARDSON P.C. (VLT, INC)
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via EFS-Web or by facsimile to (571) 273-2885, on the date below.

(Typed or printed name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/218,418	12/12/2018	Patrizio Vinciarelli	00614-0183001	1092

TITLE OF INVENTION: Delivering Power To Semiconductor Loads

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1000	\$0.00	\$0.00	\$1000	10/28/2020

EXAMINER	ART UNIT	CLASS-SUBCLASS
COX, CASSANDRA F	2849	327-109000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-09 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

1 _____

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

2 _____

3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government4a. Fees submitted: ☐ Issue Fee ☐ Publication Fee (if required) ☐ Advance Order - # of Copies _____

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☐ Electronic Payment via EFS-Web ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. _____

5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29☐ Applicant asserting small entity status. See 37 CFR 1.27☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____

Date _____

Typed or printed name _____

Registration No. _____



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/218,418	12/12/2018	Patrizio Vinciarelli	00614-0183001	1092
26214	7590	07/28/2020	EXAMINER	
FISH & RICHARDSON P.C. (VLT, INC)			COX, CASSANDRA F	
P.O. BOX 1022			ART UNIT	
MINNEAPOLIS, MN 55440-1022			PAPER NUMBER	
			2849	
DATE MAILED: 07/28/2020				

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 30 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability	Application No. 16/218,418	Applicant(s) Vinciarelli, Patrizio	
	Examiner CASSANDRA F COX	Art Unit 2849	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to 06/30/2020.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

3. ☒ The allowed claim(s) is/are 1-47. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) ☐ All b) ☐ Some *c) ☐ None of the:

1. ☐ Certified copies of the priority documents have been received.

2. ☐ Certified copies of the priority documents have been received in Application No. _____.

3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.
THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.

Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).

6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

1. ☐ Notice of References Cited (PTO-892)	5. ☐ Examiner's Amendment/Comment
2. ☒ Information Disclosure Statements (PTO/SB/08), Paper No./Mail Date _____.	6. ☒ Examiner's Statement of Reasons for Allowance
3. ☐ Examiner's Comment Regarding Requirement for Deposit of Biological Material _____.	7. ☐ Other _____.
4. ☐ Interview Summary (PTO-413), Paper No./Mail Date. _____.	

/CASSANDRA F COX/ Primary Examiner, Art Unit 2849	
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DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Allowable Subject Matter

1. Claims 1-47 are allowed.
2. The following is an examiner's statement of reasons for allowance: Claims 1-16, 36-43, and 46 are allowed because the closest prior art of record fails to disclose a method of powering a semiconductor device having power input terminals, the method comprising: providing an interconnection module having x-input terminals arranged on a first side of the interconnection module in a pattern configured to align with the first pattern of converter output terminals, and x-output terminals arranged on a surface of a second opposite side of the interconnection module in a second pattern configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals, the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals; assembling the power conversion module to the first side of the interconnection module with each converter output terminal mated with and electrically connected to a respective x-input terminal; wherein the interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern and the power input terminals of

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the semiconductor device arranged in the second pattern; wherein the power conversion module and the interconnection module are stacked to form a module sandwich including the power conversion module in a first layer, electrical connections between the converter output terminals and the x-input terminals in a second layer, and the interconnection module in a third layer in combination with the rest of the limitations of the base claim. Claims 17-23 are allowed because the closest prior art of record fails to disclose a method of making power converters, comprising: selectively forming a plurality of holes in the exterior surface of the panel, each hole being formed at a respective predetermined location aligned with a respective output node, spaced apart from the cut lines, and exposing within the hole the respective conductive feature; selectively forming one or more conductive metal layers on the exterior surface of the panel including within the plurality of holes, the one or more conductive metal layers within each hole being in electrical contact with the respective conductive feature, to form a plurality of electrical contacts on the exterior surface of the panel; patterning the one or more conductive metal layers on the exterior surface to electrically isolate at least one electrical contact from one or more of the plurality of electrical contacts; cutting the panel along the one or more cut lines to singulate the one or more power conversion modules, each singulated power conversion module having a respective plurality of the electrical contacts formed on the exterior surface of the module in combination with the rest of the limitations of the base claim. Claims 24-26 are allowed because the closest prior art of record fails to disclose a method of making electronic assemblies, the method comprising: wherein the interconnection module is configured to provide electrical and mechanical translation between the circuit terminals arranged in the first pattern and the external terminals of the third electronic device arranged

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in the second pattern; wherein the electronic module and the interconnection module are stacked to form a module sandwich including the electronic module in a first layer, electrical connections between the circuit terminals and the x-input terminals in a second layer, and the interconnection module in a third layer; assembling a first panel comprising a plurality of unsingulated electronic modules; assembling a second panel comprising a plurality of unsingulated interconnection modules; wherein the assembling of the electronic module and the interconnection module is performed using the first and second panels prior to singulation to form a panel sandwich and further comprises dispensing a curable adhesive compound between the first and second panels and forming solder connections between each circuit terminal and each respective x-input terminal; and cutting the panel sandwich along predetermined cut lines to singulate a plurality of the module sandwiches in combination with the rest of the limitations of the base claim. Claims 27-29 are allowed because the closest prior art of record fails to disclose a method of forming modular circuit assemblies, the method comprising: selectively forming a plurality of terminal holes in the first exterior surface of each panel through the first layer at predetermined locations within the perimeter boundaries of each electronic module, each terminal hole being spaced apart from the cut lines and exposing within the terminal hole a respective conductive feature; selectively forming one or more conductive metal layers on the first exterior surface of each panel including within the plurality of terminal holes, the one or more conductive metal layers within each terminal hole being in electrical contact with the respective conductive feature; patterning the one or more conductive metal layers on the exterior surface, to form a plurality of electrical contacts on the exterior surface of each panel electrically isolated from at least one other electrical contact in

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the plurality of electrical contacts; assembling the first and second encapsulated panels with electrical contacts on the first panel aligned and mated with respective electrical contacts on the second panel to form a panel stack, the panel stack comprising a plurality of un-singulated module stacks, each module stack including an unsingulated electronic module in the first panel and a respective unsingulated electronic module in the second panel; and cutting the panel stack along the one or more cut lines to singulate the plurality of electronic module stacks in combination with the rest of the limitations of the base claim. Claims 30-34 are allowed because the closest prior art of record fails to disclose an apparatus comprising: an interconnection module having x-input terminals arranged on a first surface of the interconnection module in a pattern configured to align with the first pattern of converter output terminals, and x-output terminals arranged on a second surface in a second pattern configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals, the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals; the converter output terminals of the power conversion module being mated with and electrically connected to respective x-input terminals of the interconnection module; wherein the interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern and the power input terminals of the semiconductor device arranged in the second pattern; wherein the power conversion module and the interconnection module are stacked with the power conversion module located at a first level, electrical connections between the converter output terminals and the x-input terminals at a second level, and the interconnection module at a third

level in combination with the rest of the limitations of the base claim. Claims 35, 44-45, and 47 are allowed because the closest prior art of record fails to disclose a method of powering a semiconductor device having power input terminals, the method comprising: providing an interconnection module having x-input terminals arranged on a first surface of the interconnection module in a pattern configured to align with the first pattern of converter output terminals, and x-output terminals arranged on a second surface in a second pattern configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals, the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals; providing a plurality of conductive pillars for electrically connecting the converter output terminals with respective x-input terminals; assembling the power conversion module to the interconnection module with each converter output terminal aligned with and electrically connected to a respective x-input terminal by a respective one of the conductive pillars; wherein the interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern and the power input terminals of the semiconductor device arranged in the second pattern; wherein the power conversion module and the interconnection module are stacked to form a module sandwich including the power conversion module in a first layer, the conductive pillars connecting the converter output terminals and the x-input terminals in a second layer, and the interconnection module in a third layer; and wherein the x-output terminals are arranged with alternating polarities to reduce interconnection inductance in combination with the rest of the limitations of the base claim.

Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

Conclusion

3. Any inquiry concerning this communication or earlier communications from the examiner should be directed to CASSANDRA F COX whose telephone number is (571)272-1741. The examiner can normally be reached on M-F 6:00-3:30; off alt Fridays.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Lincoln Donovan can be reached on 571-272-1988. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <https://ppair-my.uspto.gov/pair/PrivatePair>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-

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free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/CASSANDRA F COX/

Primary Examiner, Art Unit 2849

<i>Search Notes</i> 	Application/Control No. 16/218,418	Applicant(s)/Patent Under Reexamination Vinciarelli, Patrizio
	Examiner CASSANDRA F COX	Art Unit 2849

CPC - Searched*		
Symbol	Date	Examiner

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
EAST Search (USPGPUB, USPAT, USOCR, EPO, JPO, DERWENT, IBM_TDB) see search history	05/26/2020	cfc
H02M1/08 or H01L24/26 or H05K1/144 or H05K1/0298 or H02M2001/0074 (text only - see search history)	05/26/2020	cfc
inventor search	05/26/2020	cfc
update	07/20/2020	cfc

/CASSANDRA F COX/ Primary Examiner, Art Unit 2849	
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<i>Search Notes</i> 	Application/Control No. 16/218,418	Applicant(s)/Patent Under Reexamination Vinciarelli, Patrizio
	Examiner CASSANDRA F COX	Art Unit 2849

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner
H02M	1/08, 2001/0074	05/26/2020	cfc
H01L	24/26	05/26/2020	cfc
H05K	1/144, 1/0298	05/26/2020	cfc

/CASSANDRA F COX/ Primary Examiner, Art Unit 2849	
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Issue Classification 	Application/Control No. 16/218,418	Applicant(s)/Patent Under Reexamination Vinciarelli, Patrizio
	Examiner CASSANDRA F COX	Art Unit 2849

CPC						
Symbol					Type	Version
H02M	/	1	/	08	F	2013-01-01
H05K	/	1	/	144	I	2013-01-01
H01L	/	24	/	26	I	2013-01-01
H05K	/	1	/	0298	I	2013-01-01
H02M	/	2001	/	0074	A	2013-01-01

CPC Combination Sets					
Symbol				Type	Set
	/		/		

NONE (Assistant Examiner) _____ (Date) _____		Total Claims Allowed: 47	
/CASSANDRA F COX/ Primary Examiner, Art Unit 2849 (Primary Examiner) _____ (Date) _____		20 July 2020 O.G. Print Claim(s) 30	O.G. Print Figure 5A

Issue Classification 	Application/Control No. 16/218,418	Applicant(s)/Patent Under Reexamination Vinciarelli, Patrizio
	Examiner CASSANDRA F COX	Art Unit 2849

INTERNATIONAL CLASSIFICATION			
CLAIMED			
H02M1/08	/	1	/ 08
NON-CLAIMED			
/	/		

US ORIGINAL CLASSIFICATION	
CLASS	SUBCLASS

CROSS REFERENCES(S)						
CLASS	SUBCLASS (ONE SUBCLASS PER BLOCK)					

NONE		Total Claims Allowed:	
(Assistant Examiner)	(Date)	47	
/CASSANDRA F COX/ Primary Examiner, Art Unit 2849	20 July 2020	O.G. Print Claim(s)	O.G. Print Figure
(Primary Examiner)	(Date)	30	5A

Issue Classification 	Application/Control No. 16/218,418	Applicant(s)/Patent Under Reexamination Vinciarelli, Patrizio
	Examiner CASSANDRA F COX	Art Unit 2849

☐ Claims renumbered in the same order as presented by applicant
 ☐ CPA
 ☐ T.D.
 ☐ R.1.47

CLAIMS															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
1	1	10	10	28	19	37	28	18	37	25	46				
2	2	11	11	29	20	38	29	19	38	47	47				
3	3	12	12	30	21	39	30	20	39						
4	4	13	13	31	22	40	31	21	40						
5	5	14	14	32	23	41	32	22	41						
6	6	15	15	33	24	42	33	23	42						
7	7	16	16	34	25	43	34	24	43						
8	8	26	17	35	26	44	35	45	44						
9	9	27	18	36	27	17	36	46	45						

NONE (Assistant Examiner) _____ (Date) _____		Total Claims Allowed: 47	
/CASSANDRA F COX/ Primary Examiner, Art Unit 2849 (Primary Examiner) _____ (Date) 20 July 2020		O.G. Print Claim(s) 30	O.G. Print Figure 5A

Bibliographic Data

Application No: 16/218,418

Foreign Priority claimed: ☐ Yes ☒ No

35 USC 119 (a-d) conditions met: ☐ Yes ☒ No ☐ Met After Allowance

Verified and Acknowledged: /CASSANDRA F COX/

Examiner's Signature

Initials

Title: Delivering Power To Semiconductor Loads

FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.
12/12/2018	327	2849	00614-0183001
RULE			

APPLICANTS

VLT, Inc., Andover, MA,

INVENTORS

Patrizio Vinciarelli, Boston, MA, UNITED STATES

CONTINUING DATA

This application is a CIP of 15616288 06/07/2017 PAT 10277105

15616288 is a CIP of 15091346 04/05/2016 PAT 10158357

FOREIGN APPLICATIONS

IF REQUIRED, FOREIGN LICENSE GRANTED**

01/04/2019

STATE OR COUNTRY

UNITED STATES

ADDRESS

FISH & RICHARDSON P.C. (VLT, INC)

P.O. BOX 1022

MINNEAPOLIS, MN 55440-1022

UNITED STATES

FILING FEE RECEIVED

\$5,600

EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	8	("20070241440" "20080078572" "20140355218" "7015587" "7030469" "7198987" "7494843" "8138584").PN.	US-PGPUB; USPAT	OR	ON	2020/07/20 11:05

7/20/2020 12:00:38 PM

C:\Users\ccox\Documents\EAST\Workspaces\616288.wsp

Sheet 1 of 1

Substitute Disclosure Form U.S. Department of Commerce Patent and Trademark Office Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))	Attorney Docket No. 00614-0183001	Application No. 16/218,418
	Applicant VLT, Inc.	
	Filing Date December 12, 2018	Group Art Unit

.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Filing Date If Appropriate	Class	Subclass
	1.	7015587	03/21/2006	Poddar			
	2.	7030469	4/18/2006	Mahadevan et al.			
	3.	7198987	4/3/2007	Warren et al.			
	4.	7494843	2/24/2009	Lin et al.			
	5.	8138584	3/20/2012	Wang et al.			
	6.	20070241440	10/18/2007	Hoang et al.			
	7.	20080078572	7/26/2011	Watanabe et al.			
	8.	20140355218	12/4/2014	Vinciarelli et al.			

Foreign Patent Documents or Published Foreign Patent Applications							
Examiner Initial	Desig. ID	Document Number	Publication Date	Country or Patent Office	Translation	Class	Subclass
	9.	JPH04293293	10/16/1992	JP	English Abstract		

Other Documents (include Author, Title, Date, and Place of Publication)		
Examiner Initial	Desig. ID	Document

Examiner Signature /CASSANDRA F COX/	Date Considered 07/20/2020
---	-----------------------------------

EXAMINER: Initials citation considered. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

Substitute Disclosure Form

To: PATDOCTC@fr.com,,
From: PAIR_eOfficeAction@uspto.gov
Cc: PAIR_eOfficeAction@uspto.gov
Subject: Private PAIR Correspondence Notification for Customer Number 26214

Jul 28, 2020 03:47:30 AM

Dear PAIR Customer:

FISH & RICHARDSON P.C. (VLT, INC)
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022
UNITED STATES

The following USPTO patent application(s) associated with your Customer Number, 26214 , have new outgoing correspondence. This correspondence is now available for viewing in Private PAIR.

The official date of notification of the outgoing correspondence will be indicated on the form PTOL-90 accompanying the correspondence.

Disclaimer:

The list of documents shown below is provided as a courtesy and is not part of the official file wrapper. The content of the images shown in PAIR is the official record.

Application	Document	Mailroom Date	Attorney Docket No.
16218418	NOA	07/28/2020	00614-0183001
	1449	07/28/2020	00614-0183001

To view your correspondence online or update your email addresses, please visit us anytime at <https://sportal.uspto.gov/secure/myportal/privatepair>.

If you have any questions, please email the Electronic Business Center (EBC) at EBC@uspto.gov with 'e-Office Action' on the subject line or call 1-866-217-9197 during the following hours:

Monday - Friday 6:00 a.m. to 12:00 a.m.

Thank you for prompt attention to this notice,

UNITED STATES PATENT AND TRADEMARK OFFICE
PATENT APPLICATION INFORMATION RETRIEVAL SYSTEM

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Request for Continued Examination (RCE) Transmittal

Address to:
Mail Stop RCE
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Application Number	16/218,418
Filing Date	December 12, 2018
First Named Inventor	Patrizio Vinciarelli
Art Unit	
Examiner Name	
Attorney Docket Number	00614-0183001

This is a Request for Continued Examination (RCE) under 37 CFR 1.114 of the above-identified application.

Request for Continued Examination (RCE) practice under 37 CFR 1.114 does not apply to any utility or plant application filed prior to June 8, 1995, or to any design application. See Instruction Sheet for RCEs (not to be submitted to the USPTO) on page 2.

1. **Submission required under 37 CFR 1.114** Note: If the RCE is proper, any previously filed unentered amendments and amendments enclosed with the RCE will be entered in the order in which they were filed unless applicant instructs otherwise. If applicant does not wish to have any previously filed unentered amendment(s) entered, applicant must request non-entry of such amendment(s).

a. ☐ Previously submitted. If a final Office action is outstanding, any amendments filed after the final Office action may be considered as a submission even if this box is not checked.

i. ☐ Consider the arguments in the Appeal Brief or Reply Brief previously filed on _____

ii. ☐ Other _____

b. ☒ Enclosed

i. ☐ Amendment/Reply

iii. ☒ Information Disclosure Statement (IDS)

ii. ☐ Affidavit(s)/ Declaration(s)

iv. ☐ Other _____

2. Miscellaneous

Suspension of action on the above-identified application is requested under 37 CFR 1.103(c) for a

a. ☐ period of _____ months. (Period of suspension shall not exceed 3 months; Fee under 37 CFR 1.17(i) required)

b. ☐ Other _____

3. Fees

The RCE fee under 37 CFR 1.17(e) is required by 37 CFR 1.114 when the RCE is filed.

The Director is hereby authorized to charge the following fees any underpayment of fees, or credit any overpayments to

a. ☒ Deposit Account No. 06-1050

i. ☒ RCE fee required under 37 CFR 1.17(e)

ii. ☐ Extension of time fee (37 CFR 1.136 and 1.17)

iii. ☒ Other any deficiencies

b. ☐ Check in the amount of \$ _____ enclosed

c. ☐ Payment by credit card (Form PTO-2038 enclosed)

d. ☐ Payment by EFS-Web

WARNING: Information on this form may become public. Credit card information should not be included on this form. Provide credit card information and authorization on PTO-2038.

SIGNATURE OF APPLICANT, ATTORNEY, OR AGENT REQUIRED

Signature	/Rex I. Huang/	Date	August 7, 2020
Name (Print/Type)	Rex I. Huang	Registration No.	57,661

CERTIFICATE OF MAILING OR TRANSMISSION

I hereby certify that this correspondence is being EFS-Web transmitted to the United States Patent and Trademark Office (USPTO), deposited with the United States Postal Service with sufficient postage as first class mail in an envelope addressed to: Mail Stop RCE, Commissioner for Patents, P. O. Box 1450, Alexandria, VA 22313-1450 or facsimile transmitted to the USPTO on the date shown below.

Signature

Name (Print/Type)

Date

This collection of information is required by 37 CFR 1.114. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Mail Stop RCE, Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Electronic Patent Application Fee Transmittal

Application Number:	16218418			
Filing Date:	12-Dec-2018			
Title of Invention:	Delivering Power To Semiconductor Loads			
First Named Inventor/Applicant Name:	Patrizio Vinciarelli			
Filer:	Rex I. Huang/Rita Liston			
Attorney Docket Number:	00614-0183001			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
RCE- 2ND AND SUBSEQUENT REQUEST	1820	1	1900	1900
Total in USD (\$)				1900

Electronic Acknowledgement Receipt

EFS ID:	40227129
Application Number:	16218418
International Application Number:	
Confirmation Number:	1092
Title of Invention:	Delivering Power To Semiconductor Loads
First Named Inventor/Applicant Name:	Patrizio Vinciarelli
Customer Number:	26214
Filer:	Rex I. Huang/Rita Liston
Filer Authorized By:	Rex I. Huang
Attorney Docket Number:	00614-0183001
Receipt Date:	07-AUG-2020
Filing Date:	12-DEC-2018
Time Stamp:	15:35:16
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	DA
Payment was successfully received in RAM	\$ 1900
RAM confirmation Number	E202087F35316335
Deposit Account	
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Request for Continued Examination (RCE)	00614_0183001_RCE_080720.pdf	117263	no	1
			a3491f5d8b433df92bdc1b3e0aa4c0807e5cccab		
Warnings:					
This is not a USPTO supplied RCE SB30 form.					
Information:					
2	Fee Worksheet (SB06)	fee-info.pdf	30443	no	2
			772724df7dc75dc534a95f25d162efa0f597c30f		
Warnings:					
Information:					
Total Files Size (in bytes):			147706		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

First Named Inventor :	Patrizio Vinciarelli	Art Unit :	Unknown
Application No. :	16/218,418	Examiner :	Unknown
Filed :	December 12, 2018	Conf. No. :	1092
Title :	DELIVERING POWER TO SEMICONDUCTOR LOADS		

MAIL STOP RCE

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

INFORMATION DISCLOSURE STATEMENT

Please consider the documents listed on the disclosure form submitted with this statement. Copies of the documents are being submitted with this statement to the extent required under 37 C.F.R. § 1.98(a)(2).

This statement is being filed with a Request for Continued Examination. No fee is required.

Apply any necessary charges or credits applicable under 37 C.F.R. § 1.17(p) to deposit account 06-1050, referencing the above attorney docket number.

Please contact the undersigned if there are any questions regarding this statement.

Respectfully submitted,

Date: August 7, 2020 _____

/Rex I. Huang/ _____

Rex I. Huang
Reg. No. 57,661

Customer Number 26214
Fish & Richardson P.C.
Telephone: (617) 542-5070
Facsimile: (877) 769-7945

Substitute Disclosure Form	U.S. Department of Commerce Patent and Trademark Office	Attorney Docket No. 00614-0183001	Application No. 16/218,418
Information Disclosure Statement by Applicant (Use several sheets if necessary)		Applicant VLT, Inc.	
		Filing Date December 12, 2018	Group Art Unit
(37 CFR §1.98(b))			

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Filing Date If Appropriate	Class	Subclass
	1.	5027255	06/01/1991	Zeitlin			
	2.	5481436	01/01/1996	Werther			
	3.	6229216	05/08/2001	Ma et al.			
	4.	6940013	09/06/2005	Vinciarelli et al.			
	5.	7767494	08/03/2010	Sasaki et al.			
	6.	7952879	05/01/2011	Vinciarelli; Patrizio			
	7.	8240035	08/01/2012	Nishikawa			
	8.	9439297	09/06/2016	Vinciarelli			
	9.	10537015	01/14/2020	Vinciarelli			
	10.	20030058628	03/01/2003	Boylan			
	11.	20040100778	05/27/2004	Vinciarelli et al.			
	12.	20050184381	08/01/2005	Asahi Toshivuki			
	13.	20060133041	06/01/2006	Belady			
	14.	20060133042	06/01/2006	Belady			
	15.	20090140394	06/04/2009	Bathan et al.			
	16.	20090321956	12/31/2009	Sasaki et al.			
	17.	20090321957	12/31/2009	Sasaki et al.			
	18.	20090325345	12/31/2009	Sasaki et al.			

Foreign Patent Documents or Published Foreign Patent Applications							
Examiner Initial	Desig. ID	Document Number	Publication Date	Country or Patent Office	Translation	Class	Subclass

Other Documents (include Author Title Date and Place of Publication)		
Examiner Initial	Desig. ID	Document

Examiner Signature	Date Considered
--------------------	-----------------

EXAMINER: Initials citation considered. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

Electronic Acknowledgement Receipt

EFS ID:	40227324
Application Number:	16218418
International Application Number:	
Confirmation Number:	1092
Title of Invention:	Delivering Power To Semiconductor Loads
First Named Inventor/Applicant Name:	Patrizio Vinciarelli
Customer Number:	26214
Filer:	Rex I. Huang/Mollie Kufner
Filer Authorized By:	Rex I. Huang
Attorney Docket Number:	00614-0183001
Receipt Date:	07-AUG-2020
Filing Date:	12-DEC-2018
Time Stamp:	15:44:20
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
------------------------	----

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Information Disclosure Statement (IDS) Form (SB08)	00614-0183001_Transmittal.pdf	65525	no	1
			c197ce7a03d09790325330c27661dff6a7a6f391		

Warnings:

Information:					
This is not an USPTO supplied IDS fillable form					
2	Information Disclosure Statement (IDS) Form (SB08)	00614-0183001_1449.pdf	115757	no	1
			4a9e98d65da90bbf1eeb4d9435f347d23b4b197c		
Warnings:					
Information:					
This is not an USPTO supplied IDS fillable form					
Total Files Size (in bytes):				181282	
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

NOTICE OF ALLOWANCE AND FEE(S) DUE

26214 7590 08/24/2020
FISH & RICHARDSON P.C. (VLT, INC)
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022

EXAMINER

COX, CASSANDRA F

ART UNIT

PAPER NUMBER

2849

DATE MAILED: 08/24/2020

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/218,418	12/12/2018	Patrizio Vinciarelli	00614-0183001	1092

TITLE OF INVENTION: Delivering Power To Semiconductor Loads

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1000	\$0.00	\$0.00	\$1000	11/24/2020

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 1/2 the amount of undiscounted fees, and micro entity fees are 1/2 the amount of small entity fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed and an extra copy of the form should be submitted. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Maintenance fees are due in utility patents issuing on applications filed on or after Dec. 12, 1980. It is patentee's responsibility to ensure timely payment of maintenance fees when due. More information is available at www.uspto.gov/PatentMaintenanceFees.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), by mail or fax, or via EFS-Web.

By mail, send to: Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450

By fax, send to: (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

26214 7590 08/24/2020
FISH & RICHARDSON P.C. (VLT, INC)
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via EFS-Web or by facsimile to (571) 273-2885, on the date below.

(Typed or printed name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/218,418	12/12/2018	Patrizio Vinciarelli	00614-0183001	1092

TITLE OF INVENTION: Delivering Power To Semiconductor Loads

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1000	\$0.00	\$0.00	\$1000	11/24/2020

EXAMINER	ART UNIT	CLASS-SUBCLASS
COX, CASSANDRA F	2849	327-109000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-09 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

1 _____

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

2 _____

3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. Fees submitted: ☐ Issue Fee ☐ Publication Fee (if required) ☐ Advance Order - # of Copies _____

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☐ Electronic Payment via EFS-Web ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)

☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. _____

5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____

Date _____

Typed or printed name _____

Registration No. _____



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/218,418	12/12/2018	Patrizio Vinciarelli	00614-0183001	1092
26214	7590	08/24/2020	EXAMINER	
FISH & RICHARDSON P.C. (VLT, INC)			COX, CASSANDRA F	
P.O. BOX 1022			ART UNIT	
MINNEAPOLIS, MN 55440-1022			PAPER NUMBER	
			2849	
DATE MAILED: 08/24/2020				

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 30 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability	Application No. 16/218,418	Applicant(s) Vinciarelli, Patrizio	
	Examiner CASSANDRA F COX	Art Unit 2849	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to 08/07/2020.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

3. ☒ The allowed claim(s) is/are 1-47. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) ☐ All b) ☐ Some *c) ☐ None of the:

1. ☐ Certified copies of the priority documents have been received.

2. ☐ Certified copies of the priority documents have been received in Application No. _____.

3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.
THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.

Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).

6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

1. ☐ Notice of References Cited (PTO-892)	5. ☐ Examiner's Amendment/Comment
2. ☒ Information Disclosure Statements (PTO/SB/08), Paper No./Mail Date _____.	6. ☒ Examiner's Statement of Reasons for Allowance
3. ☐ Examiner's Comment Regarding Requirement for Deposit of Biological Material _____.	7. ☐ Other _____.
4. ☐ Interview Summary (PTO-413), Paper No./Mail Date. _____.	

/CASSANDRA F COX/ Primary Examiner, Art Unit 2849	
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DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Allowable Subject Matter

1. Claims 1-47 are allowed.
2. The following is an examiner's statement of reasons for allowance: Claims 1-16, 36-43, and 46 are allowed because the closest prior art of record fails to disclose a method of powering a semiconductor device having power input terminals, the method comprising: providing an interconnection module having x-input terminals arranged on a first side of the interconnection module in a pattern configured to align with the first pattern of converter output terminals, and x-output terminals arranged on a surface of a second opposite side of the interconnection module in a second pattern configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals, the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals; assembling the power conversion module to the first side of the interconnection module with each converter output terminal mated with and electrically connected to a respective x-input terminal; wherein the interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern and the power input terminals of

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the semiconductor device arranged in the second pattern; wherein the power conversion module and the interconnection module are stacked to form a module sandwich including the power conversion module in a first layer, electrical connections between the converter output terminals and the x-input terminals in a second layer, and the interconnection module in a third layer in combination with the rest of the limitations of the base claim. Claims 17-23 are allowed because the closest prior art of record fails to disclose a method of making power converters, comprising: selectively forming a plurality of holes in the exterior surface of the panel, each hole being formed at a respective predetermined location aligned with a respective output node, spaced apart from the cut lines, and exposing within the hole the respective conductive feature; selectively forming one or more conductive metal layers on the exterior surface of the panel including within the plurality of holes, the one or more conductive metal layers within each hole being in electrical contact with the respective conductive feature, to form a plurality of electrical contacts on the exterior surface of the panel; patterning the one or more conductive metal layers on the exterior surface to electrically isolate at least one electrical contact from one or more of the plurality of electrical contacts; cutting the panel along the one or more cut lines to singulate the one or more power conversion modules, each singulated power conversion module having a respective plurality of the electrical contacts formed on the exterior surface of the module in combination with the rest of the limitations of the base claim. Claims 24-26 are allowed because the closest prior art of record fails to disclose a method of making electronic assemblies, the method comprising: wherein the interconnection module is configured to provide electrical and mechanical translation between the circuit terminals arranged in the first pattern and the external terminals of the third electronic device arranged

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in the second pattern; wherein the electronic module and the interconnection module are stacked to form a module sandwich including the electronic module in a first layer, electrical connections between the circuit terminals and the x-input terminals in a second layer, and the interconnection module in a third layer; assembling a first panel comprising a plurality of unsingulated electronic modules; assembling a second panel comprising a plurality of unsingulated interconnection modules; wherein the assembling of the electronic module and the interconnection module is performed using the first and second panels prior to singulation to form a panel sandwich and further comprises dispensing a curable adhesive compound between the first and second panels and forming solder connections between each circuit terminal and each respective x-input terminal; and cutting the panel sandwich along predetermined cut lines to singulate a plurality of the module sandwiches in combination with the rest of the limitations of the base claim. Claims 27-29 are allowed because the closest prior art of record fails to disclose a method of forming modular circuit assemblies, the method comprising: selectively forming a plurality of terminal holes in the first exterior surface of each panel through the first layer at predetermined locations within the perimeter boundaries of each electronic module, each terminal hole being spaced apart from the cut lines and exposing within the terminal hole a respective conductive feature; selectively forming one or more conductive metal layers on the first exterior surface of each panel including within the plurality of terminal holes, the one or more conductive metal layers within each terminal hole being in electrical contact with the respective conductive feature; patterning the one or more conductive metal layers on the exterior surface, to form a plurality of electrical contacts on the exterior surface of each panel electrically isolated from at least one other electrical contact in

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the plurality of electrical contacts; assembling the first and second encapsulated panels with electrical contacts on the first panel aligned and mated with respective electrical contacts on the second panel to form a panel stack, the panel stack comprising a plurality of un-singulated module stacks, each module stack including an unsingulated electronic module in the first panel and a respective unsingulated electronic module in the second panel; and cutting the panel stack along the one or more cut lines to singulate the plurality of electronic module stacks in combination with the rest of the limitations of the base claim. Claims 30-34 are allowed because the closest prior art of record fails to disclose an apparatus comprising: an interconnection module having x-input terminals arranged on a first surface of the interconnection module in a pattern configured to align with the first pattern of converter output terminals, and x-output terminals arranged on a second surface in a second pattern configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals, the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals; the converter output terminals of the power conversion module being mated with and electrically connected to respective x-input terminals of the interconnection module; wherein the interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern and the power input terminals of the semiconductor device arranged in the second pattern; wherein the power conversion module and the interconnection module are stacked with the power conversion module located at a first level, electrical connections between the converter output terminals and the x-input terminals at a second level, and the interconnection module at a third

level in combination with the rest of the limitations of the base claim. Claims 35, 44-45, and 47 are allowed because the closest prior art of record fails to disclose a method of powering a semiconductor device having power input terminals, the method comprising: providing an interconnection module having x-input terminals arranged on a first surface of the interconnection module in a pattern configured to align with the first pattern of converter output terminals, and x-output terminals arranged on a second surface in a second pattern configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals, the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals; providing a plurality of conductive pillars for electrically connecting the converter output terminals with respective x-input terminals; assembling the power conversion module to the interconnection module with each converter output terminal aligned with and electrically connected to a respective x-input terminal by a respective one of the conductive pillars; wherein the interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern and the power input terminals of the semiconductor device arranged in the second pattern; wherein the power conversion module and the interconnection module are stacked to form a module sandwich including the power conversion module in a first layer, the conductive pillars connecting the converter output terminals and the x-input terminals in a second layer, and the interconnection module in a third layer; and wherein the x-output terminals are arranged with alternating polarities to reduce interconnection inductance in combination with the rest of the limitations of the base claim.

Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

Conclusion

3. Any inquiry concerning this communication or earlier communications from the examiner should be directed to CASSANDRA F COX whose telephone number is (571)272-1741. The examiner can normally be reached on M-F 6:00-3:30; off alt Fridays.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Lincoln Donovan can be reached on 571-272-1988. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <https://ppair-my.uspto.gov/pair/PrivatePair>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-

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free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/CASSANDRA F COX/

Primary Examiner, Art Unit 2849

<i>Search Notes</i> 	Application/Control No. 16/218,418	Applicant(s)/Patent Under Reexamination Vinciarelli, Patrizio
	Examiner CASSANDRA F COX	Art Unit 2849

CPC - Searched*		
Symbol	Date	Examiner

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
EAST Search (USPGPUB, USPAT, USOCR, EPO, JPO, DERWENT, IBM_TDB) see search history	05/26/2020	cfc
H02M1/08 or H01L24/26 or H05K1/144 or H05K1/0298 or H02M2001/0074 (text only - see search history)	05/26/2020	cfc
inventor search	05/26/2020	cfc
update	07/20/2020	cfc
update	08/17/2020	cfc

/CASSANDRA F COX/ Primary Examiner, Art Unit 2849	
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<i>Search Notes</i> 	Application/Control No. 16/218,418	Applicant(s)/Patent Under Reexamination Vinciarelli, Patrizio
	Examiner CASSANDRA F COX	Art Unit 2849

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner
H02M	1/08, 2001/0074	05/26/2020	cfc
H01L	24/26	05/26/2020	cfc
H05K	1/144, 1/0298	05/26/2020	cfc

/CASSANDRA F COX/ Primary Examiner, Art Unit 2849	
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Issue Classification 	Application/Control No. 16/218,418	Applicant(s)/Patent Under Reexamination Vinciarelli, Patrizio	
	Examiner CASSANDRA F COX	Art Unit 2849	

CPC						
Symbol					Type	Version
H02M	/	1	/	08	F	2013-01-01
H05K	/	1	/	144	I	2013-01-01
H01L	/	24	/	26	I	2013-01-01
H05K	/	1	/	0298	I	2013-01-01
H02M	/	2001	/	0074	A	2013-01-01

CPC Combination Sets					
Symbol			Type	Set	Ranking
/	/	/			

NONE (Assistant Examiner) (Date)		Total Claims Allowed: 47	
/CASSANDRA F COX/ Primary Examiner, Art Unit 2849 (Primary Examiner) (Date)		17 August 2020 O.G. Print Claim(s)	30 O.G. Print Figure

Issue Classification 	Application/Control No. 16/218,418	Applicant(s)/Patent Under Reexamination Vinciarelli, Patrizio	
	Examiner CASSANDRA F COX	Art Unit 2849	

INTERNATIONAL CLASSIFICATION			
CLAIMED			
H02M1/08	/	1	/ 08

NON-CLAIMED			
/	/		

US ORIGINAL CLASSIFICATION	
CLASS	SUBCLASS

CROSS REFERENCES(S)						
CLASS	SUBCLASS (ONE SUBCLASS PER BLOCK)					

NONE		Total Claims Allowed:	
(Assistant Examiner)	(Date)	47	
/CASSANDRA F COX/ Primary Examiner, Art Unit 2849 (Primary Examiner)	17 August 2020 (Date)	O.G. Print Claim(s) 30	O.G. Print Figure 5A

Issue Classification 	Application/Control No. 16/218,418	Applicant(s)/Patent Under Reexamination Vinciarelli, Patrizio
	Examiner CASSANDRA F COX	Art Unit 2849

☐ Claims renumbered in the same order as presented by applicant
 ☐ CPA
 ☐ T.D.
 ☐ R.1.47

CLAIMS															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
1	1	10	10	28	19	37	28	18	37	25	46				
2	2	11	11	29	20	38	29	19	38	47	47				
3	3	12	12	30	21	39	30	20	39						
4	4	13	13	31	22	40	31	21	40						
5	5	14	14	32	23	41	32	22	41						
6	6	15	15	33	24	42	33	23	42						
7	7	16	16	34	25	43	34	24	43						
8	8	26	17	35	26	44	35	45	44						
9	9	27	18	36	27	17	36	46	45						

NONE (Assistant Examiner) _____ (Date) _____		Total Claims Allowed: 47	
/CASSANDRA F COX/ Primary Examiner, Art Unit 2849 (Primary Examiner) _____ (Date) 17 August 2020		O.G. Print Claim(s) 30	O.G. Print Figure 5A

EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	18	("10537015" "20030058628" "20040100778" "20050184381" "20060133041" "20060133042" "20090140394" "20090321956" "20090321957" "20090325345" "5027255" "5481436" "6229216" "6940013" "7767494" "7952879" "8240035" "9439297").PN.	US-PGPUB; USPAT	OR	ON	2020/08/17 14:11
L2	4,389	power adj conversion adj module	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/17 14:13
L3	1	l1 and L2	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/17 14:13
L4	236,858	power adj conver\$4	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/17 14:13
L5	1	l1 and "l4"	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/17 14:14
L6	10	l1 and l4	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/17 14:14
L7	10,178	pattern with output adj terminals	US-PGPUB; USPAT; USOCR; EPO; JPO;	OR	ON	2020/08/17 14:22

			DERWENT; IBM_TDB			
L8	8,503	pattern with input adj terminals	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/17 14:22
L9	0	I6 and I7 and I8	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/17 14:23

8/17/2020 2:32:15 PM

C:\Users\ccox\Documents\EAST\Workspaces\616288.wsp

Sheet 1 of 1

Substitute Disclosure Form Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))	U.S. Department of Commerce Patent and Trademark Office	Attorney Docket No. 00614-0183001	Application No. 16/218,418
	Applicant VLT, Inc.		
	Filing Date December 12, 2018	Group Art Unit	

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Filing Date If Appropriate	Class	Subclass
	1.	5027255	06/01/1991	Zeitlin			
	2.	5481436	01/01/1996	Werther			
	3.	6229216	05/08/2001	Ma et al.			
	4.	6940013	09/06/2005	Vinciarelli et al.			
	5.	7767494	08/03/2010	Sasaki et al.			
	6.	7952879	05/01/2011	Vinciarelli; Patrizio			
	7.	8240035	08/01/2012	Nishikawa			
	8.	9439297	09/06/2016	Vinciarelli			
	9.	10537015	01/14/2020	Vinciarelli			
	10.	20030058628	03/01/2003	Boylan			
	11.	20040100778	05/27/2004	Vinciarelli et al.			
	12.	20050184381	08/01/2005	Asahi Toshivuki			
	13.	20060133041	06/01/2006	Belady			
	14.	20060133042	06/01/2006	Belady			
	15.	20090140394	06/04/2009	Bathan et al.			
	16.	20090321956	12/31/2009	Sasaki et al.			
	17.	20090321957	12/31/2009	Sasaki et al.			
	18.	20090325345	12/31/2009	Sasaki et al.			

Foreign Patent Documents or Published Foreign Patent Applications							
Examiner Initial	Desig. ID	Document Number	Publication Date	Country or Patent Office	Translation	Class	Subclass

Other Documents (include Author Title Date and Place of Publication)		
Examiner Initial	Desig. ID	Document

Examiner Signature /CASSANDRA F COX/	Date Considered 08/17/2020
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EXAMINER: Initials citation considered. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

Substitute Disclosure Form

Bibliographic Data

Application No: 16/218,418

Foreign Priority claimed: ☐ Yes ☒ No

35 USC 119 (a-d) conditions met: ☐ Yes ☒ No ☐ Met After Allowance

Verified and Acknowledged: /CASSANDRA F COX/

Examiner's Signature

Initials

Title: Delivering Power To Semiconductor Loads

FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.
12/12/2018	327	2849	00614-0183001
RULE			

APPLICANTS

VLT, Inc., Andover, MA,

INVENTORS

Patrizio Vinciarelli, Boston, MA, UNITED STATES

CONTINUING DATA

This application is a CIP of 15616288 06/07/2017 PAT 10277105

15616288 is a CIP of 15091346 04/05/2016 PAT 10158357

FOREIGN APPLICATIONS

IF REQUIRED, FOREIGN LICENSE GRANTED**

01/04/2019

STATE OR COUNTRY

UNITED STATES

ADDRESS

FISH & RICHARDSON P.C. (VLT, INC)

P.O. BOX 1022

MINNEAPOLIS, MN 55440-1022

UNITED STATES

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Aug 24, 2020 03:38:23 AM

Dear PAIR Customer:

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UNITED STATES

The following USPTO patent application(s) associated with your Customer Number, 26214 , have new outgoing correspondence. This correspondence is now available for viewing in Private PAIR.

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Application	Document	Mailroom Date	Attorney Docket No.
16218418	NOA	08/24/2020	00614-0183001
	1449	08/24/2020	00614-0183001

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Request for Continued Examination (RCE) Transmittal

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Application Number	16/218,418
Filing Date	December 12, 2018
First Named Inventor	Patrizio Vinciarelli
Art Unit	2849
Examiner Name	Cassandra F. Cox
Attorney Docket Number	00614-0183001

This is a Request for Continued Examination (RCE) under 37 CFR 1.114 of the above-identified application.

Request for Continued Examination (RCE) practice under 37 CFR 1.114 does not apply to any utility or plant application filed prior to June 8, 1995, or to any design application. See Instruction Sheet for RCEs (not to be submitted to the USPTO) on page 2.

1. **Submission required under 37 CFR 1.114** Note: If the RCE is proper, any previously filed unentered amendments and amendments enclosed with the RCE will be entered in the order in which they were filed unless applicant instructs otherwise. If applicant does not wish to have any previously filed unentered amendment(s) entered, applicant must request non-entry of such amendment(s).

a. ☐ Previously submitted. If a final Office action is outstanding, any amendments filed after the final Office action may be considered as a submission even if this box is not checked.

i. ☐ Consider the arguments in the Appeal Brief or Reply Brief previously filed on _____

ii. ☐ Other _____

b. ☒ Enclosed

i. ☒ Amendment/Reply

iii. ☐ Information Disclosure Statement (IDS)

ii. ☐ Affidavit(s)/ Declaration(s)

iv. ☐ Other _____

2. Miscellaneous

Suspension of action on the above-identified application is requested under 37 CFR 1.103(c) for a

a. ☐ period of _____ months. (Period of suspension shall not exceed 3 months; Fee under 37 CFR 1.17(i) required)

b. ☐ Other _____

3. Fees

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a. ☒ Deposit Account No. 06-1050

i. ☒ RCE fee required under 37 CFR 1.17(e)

ii. ☐ Extension of time fee (37 CFR 1.136 and 1.17)

iii. ☒ Other any deficiencies

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SIGNATURE OF APPLICANT, ATTORNEY, OR AGENT REQUIRED

Signature	/Rex I. Huang/	Date	September 25, 2020
Name (Print/Type)	Rex I. Huang	Registration No.	57,661

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IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

First Named Inventor :	Patrizio Vinciarelli	Art Unit :	2849
Application No. :	16/218,418	Examiner :	Cassandra F. Cox
Filed :	December 12, 2018	Conf. No. :	1092
Title :	DELIVERING POWER TO SEMICONDUCTOR LOADS		

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

PRELIMINARY AMENDMENT

Prior to examination, please amend the application as indicated on the following pages.

Amendments to the Claims:

This listing of claims replaces all prior versions and listings of claims in the application:

Listing of Claims:

1. (Original) A method of powering a semiconductor device having power input terminals, the method comprising:

providing a power conversion module having conversion circuitry arranged in a multi-cell structure having a plurality of conversion cells, each conversion cell including one or more converter output terminals, the conversion circuitry being configured to convert power received at a first voltage for delivery via the converter output terminals of the plurality of conversion cells at a second lower voltage, the converter output terminals being arranged in a first pattern having a first spacing between the converter output terminals;

providing an interconnection module having x-input terminals arranged on a first side of the interconnection module in a pattern configured to align with the first pattern of converter output terminals, and x-output terminals arranged on a surface of a second opposite side of the interconnection module in a second pattern configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals, the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals;

assembling the power conversion module to the first side of the interconnection module with each converter output terminal mated with and electrically connected to a respective x-input terminal;

wherein the interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern and the power input terminals of the semiconductor device arranged in the second pattern;

wherein the power conversion module and the interconnection module are stacked to form a module sandwich including the power conversion module in a first layer, electrical connections between the converter output terminals and the x-input terminals in a second layer, and the interconnection module in a third layer.

2. (Currently Amended) The method of claim 1 or 48 wherein the interconnection module further comprises filter circuitry connected to the x-input terminals and the x-output terminals.

3. (Original) The method of claim 2 wherein the interconnection module further comprises a multilayer printed circuit board (“PCB”), and the filter circuitry comprises a plurality of capacitors mounted to a first surface of the PCB on the first side of the interconnection module.

4. (Original) The method of claim 2 wherein the second spacing is less than the first spacing.

5. (Original) The method of claim 2 wherein each conversion cell further comprises a first dimension and the first spacing is configured as a function of the first dimension.

6. (Original) The method of claim 5 wherein each conversion cell further comprises a second dimension, the first spacing further comprises a latitudinal spacing and a longitudinal spacing, and the latitudinal spacing is a function of the first dimension and the longitudinal spacing is a function of the second dimension.

7. (Currently Amended) The method of claim 1 or 48, further comprising configuring the conversion circuitry as a buck converter having a plurality of phases with each conversion cell configured to operate as a selected one of the phases.

8. (Currently Amended) The method of claim 1 or 48, further comprising configuring the conversion circuitry to convert power received from an input to the power conversion module at an input voltage, V_{IN} , and input current, I_{IN} , for delivery to the converter output terminals at an output voltage, V_{OUT} , and an output current, I_{OUT} , using a transformation ratio, K , where $K = V_{OUT} / V_{IN}$, that is essentially constant, subject to an equivalent output resistance.

9. (Original) The method of claim 8, further comprising encapsulating the conversion circuitry to form a first surface of the power conversion module; encapsulating the interconnection module to form a first surface on the first side of the interconnection module; affixing the first surface of the power conversion module to the first surface of the interconnection module with the converter output terminals and the x-input terminals aligned using an adhesive, and forming solder connections between respective one of the output terminals and the x-input terminals.

10. (Currently Amended) The method of claim 1 or 48 wherein the interconnection module further comprises a multilayer printed circuit board ("PCB") and the method further comprises mounting at least one solder ball to the PCB for each x-output terminal.

11. (Currently Amended) The method of claim 1 or 48 wherein the interconnection module further comprises a multilayer printed circuit board ("PCB") and the method further comprises mounting at least one conductive pin to the PCB for each x-output terminal.

12. (Currently Amended) The method of claim 1 or 48 wherein the interconnection module further comprises a multilayer printed circuit board ("PCB") and the method further comprises providing the x-output terminals in a land grid array on the PCB.

13. (Currently Amended) The method of claim 1 or 48, further comprising attaching the semiconductor device to the interconnection module with the x-output terminals aligned with and electrically connected to respective ones of the power input terminals; and wherein the

connections between the x-output terminals and the power input terminals form a fourth layer in the module sandwich.

14. (Currently Amended) The method of claim 1 or 48, further comprising assembling a first panel comprising a plurality of unsingulated power conversion modules;

assembling a second panel comprising a plurality of unsingulated interconnection modules;

wherein the assembling of the power conversion module and the interconnection module is performed using the first and second panels prior to singulation to form a panel sandwich and further comprises dispensing epoxy between the first and second panels and forming solder connections between each converter output terminal and each respective x-input terminal; and further comprising:

cutting the panel sandwich along predetermined cut lines to singulate a plurality of the module sandwiches.

15. (Original) The method of claim 14 wherein the dispensing epoxy further comprises dispensing epoxy on the surface of at least one of the panels along the cut lines.

16. (Original) The method of claim 15, further comprising reflowing solder and curing the epoxy.

17. (Original) A method of making power converters, the method comprising:
assembling an encapsulated panel including a printed circuit board ("PCB") having electronic components mounted to the PCB and a plurality of conductive features enclosed within the panel and unexposed to an exterior surface of the panel, the panel comprising one or more power conversion modules having boundaries defined by one or more predetermined cut lines;

configuring each of the one or more power conversion modules in a multi-cell structure having a plurality of conversion cells, each conversion cell including one or more converter

output nodes arranged in a first pattern having a first spacing between the output nodes, and wherein each output node is electrically connected to a respective conductive feature;

selectively forming a plurality of holes in the exterior surface of the panel, each hole being formed at a respective predetermined location aligned with a respective output node, spaced apart from the cut lines, and exposing within the hole the respective conductive feature;

selectively forming one or more conductive metal layers on the exterior surface of the panel including within the plurality of holes, the one or more conductive metal layers within each hole being in electrical contact with the respective conductive feature, to form a plurality of electrical contacts on the exterior surface of the panel;

patterning the one or more conductive metal layers on the exterior surface to electrically isolate at least one electrical contact from one or more of the plurality of electrical contacts;

cutting the panel along the one or more cut lines to singulate the one or more power conversion modules, each singulated power conversion module having a respective plurality of the electrical contacts formed on the exterior surface of the module.

18. (Original) The method of claim 17 further comprising selectively filling the plurality of holes with solder.

19. (Original) The method of claim 18, further comprising planarizing the solder to a predetermined height relative to the exterior surface.

20. (Original) The method of claim 17 wherein each conversion cell further comprises a first dimension and a second dimension, the first spacing further comprises a latitudinal spacing and a longitudinal spacing, and the latitudinal spacing is a function of the first dimension and the longitudinal spacing is a function of the second dimension.

21. (Original) The method of claim 17, further comprising:
providing an interconnection module having x-input terminals arranged on a first surface of the interconnection module in a pattern configured to align with the first pattern of converter output terminals, and x-output terminals arranged on a second surface in a second pattern, the

second pattern having a second spacing between the x-output terminals, the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals;

assembling the first surface of the interconnection module to the power conversion module with each converter output terminal mated with and electrically connected to a respective x-input terminal,

wherein the power conversion module and the interconnection module are stacked to form a sandwich structure including the power conversion module in a first layer, electrical connections between the converter output terminals and the x-input terminals in a second layer, and the interconnection module in a third layer.

22. (Original) The method of claim 21, further comprising assembling the sandwich structure to a substrate with the x-output terminals aligned with and electrically connected to respective terminals on a surface of the substrate.

23. (Original) The method of claim 22 wherein the substrate includes a semiconductor device electrically connected to receive power from the power conversion module.

24. (Original) A method of making electronic assemblies, the method comprising:
providing an electronic module having electronic circuitry including circuit terminals arranged in a first pattern having a first spacing between the circuit terminals;

providing an interconnection module having x-input terminals arranged on a first surface of the interconnection module in a pattern configured to align with the first pattern of circuit terminals, and x-output terminals arranged on a second surface in a second pattern configured to align with external terminals of a third electronic device, the second pattern having a second spacing between the x-output terminals, the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals;

assembling the electronic module to the first surface of the interconnection module with each circuit terminal mated with and electrically connected to a respective x-input terminal;

wherein the interconnection module is configured to provide electrical and mechanical translation between the circuit terminals arranged in the first pattern and the external terminals of the third electronic device arranged in the second pattern;

wherein the electronic module and the interconnection module are stacked to form a module sandwich including the electronic module in a first layer, electrical connections between the circuit terminals and the x-input terminals in a second layer, and the interconnection module in a third layer;

assembling a first panel comprising a plurality of unsingulated electronic modules;

assembling a second panel comprising a plurality of unsingulated interconnection modules;

wherein the assembling of the electronic module and the interconnection module is performed using the first and second panels prior to singulation to form a panel sandwich and further comprises dispensing a curable adhesive compound between the first and second panels and forming solder connections between each circuit terminal and each respective x-input terminal; and

cutting the panel sandwich along predetermined cut lines to singulate a plurality of the module sandwiches.

25. (Original) The method of claim 24 wherein the dispensing the curable adhesive compound further comprises dispensing the curable adhesive compound on the surface of at least one of the panels along the cut lines.

26. (Original) The method of claim 24, further comprising reflowing solder and curing the adhesive compound.

27. (Original) A method of forming modular circuit assemblies, the method comprising:

assembling first and second encapsulated panels each including a first multilayer printed circuit board ("PCB") having a plurality of conductive layers, a first plurality of electronic components mounted to a first surface of the PCB, and a first layer of cured encapsulant

covering the first plurality of electronic components and the surface of the PCB, the first layer of cured encapsulant forming a first exterior surface of the encapsulated panel, the encapsulated panel comprising a plurality of unsingulated electronic modules, each electronic module having perimeter boundaries defined by one or more predetermined cut lines;

selectively forming a plurality of terminal holes in the first exterior surface of each panel through the first layer at predetermined locations within the perimeter boundaries of each electronic module, each terminal hole being spaced apart from the cut lines and exposing within the terminal hole a respective conductive feature;

selectively forming one or more conductive metal layers on the first exterior surface of each panel including within the plurality of terminal holes, the one or more conductive metal layers within each terminal hole being in electrical contact with the respective conductive feature;

patterning the one or more conductive metal layers on the exterior surface, to form a plurality of electrical contacts on the exterior surface of each panel electrically isolated from at least one other electrical contact in the plurality of electrical contacts;

assembling the first and second encapsulated panels with electrical contacts on the first panel aligned and mated with respective electrical contacts on the second panel to form a panel stack, the panel stack comprising a plurality of un-singulated module stacks, each module stack including an unsingulated electronic module in the first panel and a respective unsingulated electronic module in the second panel; and

cutting the panel stack along the one or more cut lines to singulate the plurality of electronic module stacks.

28. (Original) The method of claim 27, further comprising dispensing a curable adhesive compound on the surface of at least one of the panels along the cut lines.

29. (Original) The method of claim 28, further comprising reflowing solder and curing the adhesive compound.

30. (Original) An apparatus comprising:

a semiconductor device having power input terminals;

a power conversion module having conversion circuitry arranged in a multi-cell structure having a plurality of conversion cells, each conversion cell including one or more converter output terminals, the conversion circuitry being configured to convert power received at a first voltage for delivery via the converter output terminals of the plurality of conversion cells at a second lower voltage, the converter output terminals being arranged in a first pattern having a first spacing between the converter output terminals; and

an interconnection module having x-input terminals arranged on a first surface of the interconnection module in a pattern configured to align with the first pattern of converter output terminals, and x-output terminals arranged on a second surface in a second pattern configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals, the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals;

the converter output terminals of the power conversion module being mated with and electrically connected to respective x-input terminals of the interconnection module;

wherein the interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern and the power input terminals of the semiconductor device arranged in the second pattern;

wherein the power conversion module and the interconnection module are stacked with the power conversion module located at a first level, electrical connections between the converter output terminals and the x-input terminals at a second level, and the interconnection module at a third level.

31. (Currently Amended) The ~~method~~ apparatus of claim 30 wherein the semiconductor device comprises a die.

32. (Currently Amended) The ~~method~~ apparatus of claim 30 wherein the semiconductor device comprises a wafer.

33. (Currently Amended) The ~~method~~ apparatus of claim 30 wherein the semiconductor device comprises a substrate.

34. (Currently Amended) The ~~method~~ apparatus of claim 30 wherein the semiconductor device comprises a semiconductor die or package mounted to a substrate.

35. (Original) A method of powering a semiconductor device having power input terminals, the method comprising:

providing a power conversion module having conversion circuitry arranged in a multi-cell structure having a plurality of conversion cells, each conversion cell including one or more converter output terminals, the conversion circuitry being configured to convert power received at a first voltage for delivery via the converter output terminals of the plurality of conversion cells at a second lower voltage, the converter output terminals being arranged in a first pattern having a first spacing between the converter output terminals;

providing an interconnection module having x-input terminals arranged on a first surface of the interconnection module in a pattern configured to align with the first pattern of converter output terminals, and x-output terminals arranged on a second surface in a second pattern configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals, the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals;

providing a plurality of conductive pillars for electrically connecting the converter output terminals with respective x-input terminals;

assembling the power conversion module to the interconnection module with each converter output terminal aligned with and electrically connected to a respective x-input terminal by a respective one of the conductive pillars;

wherein the interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern and the power input terminals of the semiconductor device arranged in the second pattern;

wherein the power conversion module and the interconnection module are stacked to form a module sandwich including the power conversion module in a first layer, the conductive pillars connecting the converter output terminals and the x-input terminals in a second layer, and the interconnection module in a third layer; and

wherein the x-output terminals are arranged with alternating polarities to reduce interconnection inductance.

36. (Original) The method of claim 1 further comprising arranging the x-output terminals with alternating polarities.

37. (Original) The method of claim 1 further comprising arranging the x-output terminals with polarities that alternate between adjacent columns.

38. (Original) The method of claim 1 further comprising arranging the x-output terminals with polarities that alternate between adjacent rows.

39. (Original) The method of claim 1 further comprising arranging the x-output terminals with polarities that alternate between adjacent x-output terminals.

40. (Original) The method of claim 1 further comprising arranging the converter output terminals with alternating polarities.

41. (Original) The method of claim 1 further comprising arranging the converter output terminals with polarities that alternate between rows.

42. (Original) The method of claim 1 further comprising arranging the converter output terminals with polarities that alternate between columns.

43. (Original) The method of claim 1 further comprising arranging the converter output terminals with polarities that alternate between adjacent converter output terminals.

44. (Original) The method of claim 35 further comprising arranging the x-output terminals with alternating polarities.

45. (Original) The method of claim 35 further comprising arranging the converter output terminals with alternating polarities.

46. (Previously Presented) The method of claim 39, further comprising assembling the sandwich structure to a substrate with the x-output terminals aligned with and electrically connected to respective terminals on a surface of the substrate, the substrate including a semiconductor device electrically connected to receive power from the power conversion module; and

further comprising conducting heat vertically from the converter module through the sandwich structure including through the interconnection module and substrate.

47. (Previously Presented) The method of claim 35, further comprising assembling the sandwich structure to a substrate with the x-output terminals aligned with and electrically connected to respective terminals on a surface of the substrate, the substrate including a semiconductor device electrically connected to receive power from the power conversion module; and

further comprising conducting heat vertically from the converter module through the sandwich structure and out the top of the substrate.

48. (New) A method of powering a semiconductor device, including at least one semiconductor die having a plurality of die power input terminals arranged on a surface of the die and having edges of the semiconductor die defining a die footprint, the semiconductor device having power input terminals arranged within the footprint of the die, the method comprising:

providing a power conversion module having conversion circuitry arranged in a multi-cell structure having a plurality of conversion cells, each conversion cell including one or more converter output terminals, the conversion circuitry being configured to convert power received at a first voltage for delivery via the converter output terminals of the plurality of conversion

cells at a second lower voltage, the converter output terminals being arranged in a first pattern having a first spacing between the converter output terminals;

providing an interconnection module having x-input terminals arranged on a first side of the interconnection module in a pattern configured to align with the first pattern of converter output terminals, and x-output terminals arranged on a surface of a second opposite side of the interconnection module in a second pattern arranged within the die footprint and configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals, the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals;

assembling the power conversion module to the first side of the interconnection module with each converter output terminal mated with and electrically connected to a respective x-input terminal;

wherein the interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern and the power input terminals of the semiconductor device arranged in the second pattern;

wherein the power conversion module and the interconnection module are stacked to form a module sandwich including the power conversion module in a first layer, electrical connections between the converter output terminals and the x-input terminals in a second layer, and the interconnection module in a third layer.

49. (New) An apparatus comprising:

a semiconductor device, including at least one semiconductor die having a plurality of die power input terminals arranged on a surface of the die and having edges of the semiconductor die defining a die footprint, the semiconductor device having power input terminals arranged within the footprint of the die;

a power conversion module having switching power conversion circuitry arranged in a multi-cell structure having a plurality of conversion cells, each conversion cell including one or more converter output terminals, the conversion circuitry being configured to convert power received at a first voltage for delivery via the converter output terminals of the plurality of

conversion cells at a second lower voltage, the converter output terminals being arranged in a first pattern having a first spacing between the converter output terminals; and

an interconnection module having x-input terminals arranged on a first surface of the interconnection module in a pattern configured to align with the first pattern of converter output terminals, and x-output terminals arranged on a second surface in a second pattern with alternating polarities configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals, the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals;

the converter output terminals of the power conversion module being mated with and electrically connected to respective x-input terminals of the interconnection module;

wherein the interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern and the power input terminals of the semiconductor device arranged in the second pattern;

wherein the power conversion module and the interconnection module are stacked with the power conversion module located at a first level, electrical connections between the converter output terminals and the x-input terminals at a second level, and the interconnection module at a third level.

50. (New) The apparatus of claim 49 wherein the semiconductor device comprises a wafer.

51. (New) The apparatus of claim 50 wherein the wafer comprises a multiplicity of un-singulated dice on the wafer.

52. (New) The apparatus of claim 49 wherein the semiconductor device further comprises a substrate, and the semiconductor die is mounted to the substrate within the die footprint.

53. (New) An apparatus for delivering power vertically to a semiconductor die, the

die having a plurality of semiconductor power input terminals arranged on a surface of the semiconductor die and having edges of the semiconductor die defining a die footprint, the apparatus comprising:

switching power conversion circuitry arranged on a first substrate having a plurality of switching power conversion cells and a plurality of converter output terminals, the switching power conversion circuitry being configured to convert power received at a first voltage for delivery via the converter output terminals at a second lower voltage;

translation interconnections including X-input terminals configured and arranged to align with and electrically connect to the converter output terminals, a plurality of X-output terminals configured and arranged with alternating polarities to align with respective power input terminals of the semiconductor die, and a plurality of vertical conductors to make electrical connections to the X-output terminals;

a capacitive array arranged between the power conversion circuitry and the semiconductor die and electrically connected to the translation interconnections to provide energy storage;

wherein the translation interconnections are configured to provide electrical and mechanical translation between the converter output terminals and the power input terminals of the semiconductor die;

wherein the switching power conversion circuitry, the translation interconnections, and the capacitive array, are each configured and arranged within a respective footprint configured to occupy substantially all of the die footprint, and are together configured and arranged to form a vertical stack configured to occupy substantially all of the die footprint.

54. (New) The apparatus of claim 53 wherein the power conversion circuitry further comprises converter input terminals arranged on a first surface of the first substrate.

55. (New) The apparatus of claim 53, further comprising the semiconductor die and wherein the semiconductor die is mounted to the stack with the power input terminals of the semiconductor die aligned with and electrically connected to the X-output terminals.

56. (New) The apparatus of claim 55 wherein the power flows vertically through the stack essentially within the die footprint.

57. (New) The apparatus of claim 55 wherein the capacitive array comprises a plurality of discrete capacitors arranged to occupy spaces between the X-input terminals.

58. (New) The apparatus of claim 55 wherein the X-output terminals use a spacing between terminals of 1mm or less.

59. (New) The apparatus of claim 53 wherein the capacitive array comprises a plurality of discrete capacitors arranged to occupy spaces between the X-input terminals.

60. (New) The apparatus of claim 59 wherein the capacitors are arranged on a second substrate.

61. (New) The apparatus of claim 60 wherein the power conversion circuitry comprises a fixed-ratio conversion topology.

62. (New) The apparatus of claim 60 wherein the power conversion circuitry comprises a multi-phase buck conversion topology.

63. (New) The apparatus of claim 60 wherein the power conversion circuitry comprises a switched capacitor conversion topology.

64. (New) The apparatus of claim 53 wherein the power conversion circuitry further comprises converter input terminals arranged along an edge of a surface of the first substrate.

65. (New) The apparatus of claim 53 further comprising one or more conductive terminals passing through the vertical stack configured to carry signals through the vertical stack.

66. (New) The apparatus of claim 60 further comprising one or more conductive terminals passing through the vertical stack configured to carry signals through the vertical stack.

67. (New) A vertical power delivery apparatus for a semiconductor die, the semiconductor die having a plurality of semiconductor power input terminals arranged on a surface of the semiconductor die and having edges of the semiconductor die defining a die footprint;

the vertical power delivery apparatus comprising in a vertical stack configured to occupy, in whole or in part, the die footprint, the vertical stack comprising:

(a) point-of-load (“POL”) switching power conversion circuitry (i) arranged on a substrate within a conversion footprint, (ii) including a plurality of switching power conversion cells connected in parallel to convert power from an input at an input voltage for delivery to the semiconductor power input terminals at a die voltage, (iii) the input voltage being at least five times higher than the die voltage, and (iv) the conversion footprint being configured to occupy, in whole or in part, the die footprint;

(b) interdigitated output terminals (i) electrically connected to receive power from the switching power conversion cells, (ii) configured within a terminal footprint to match the plurality of semiconductor power input terminals, (iii) the terminal footprint being configured to occupy, in whole or in part, the die footprint; and

(c) a capacitive array (i) arranged between the power conversion circuitry and the semiconductor die, (ii) within a filter footprint configured to occupy, in whole or in part, the die footprint and (iii) electrically connected to the interdigitated output terminals.

68. (New) A vertical power delivery apparatus for a semiconductor die, the semiconductor die having a plurality of semiconductor power input terminals arranged on a surface of the semiconductor die and having edges of the semiconductor die defining a die footprint;

the vertical power delivery apparatus comprising in a vertical stack configured to occupy, in whole or in part, the die footprint, the vertical stack comprising:

(a) point-of-load (“POL”) switching power conversion circuitry (i) arranged on a substrate within a conversion footprint, (ii) including a plurality of switching power conversion cells connected in parallel to convert power from an input at an input voltage to the semiconductor power input terminals at a die voltage, (iii) the input voltage being at least five times higher than the die voltage, (iv) the conversion footprint being configured to occupy, in whole or in part, the die footprint;

(b) interdigitated output terminals (i) electrically connected to receive power from the switching power conversion cells, (ii) configured within a terminal footprint to match the plurality of semiconductor power input terminals, (iii) the terminal footprint being configured to occupy, in whole or in part, the die footprint;

(c) a capacitive array (i) arranged between the power conversion circuitry and the semiconductor die, (ii) within a filter footprint configured to occupy, in whole or in part, the die footprint, and (iii) electrically connected to the interdigitated output terminals; and

(d) a plurality of vertical conductors arranged within the die footprint to connect the interdigitated output terminals to the switching power conversion cells.

69. (New) The vertical power delivery apparatus of claim 68 wherein the POL switching power conversion circuitry further comprises a plurality of power conversion output terminals including a space between adjacent ones of the power conversion output terminals and wherein the interdigitated output terminals are electrically connected to the plurality of the power conversion output terminals by the vertical conductors.

70. (New) The vertical power delivery apparatus of claim 69 wherein multiple ones of the plurality of vertical conductors are electrically connected together by lateral electrical connections within the die footprint.

71. (New) The vertical power delivery apparatus of claim 70 wherein the power conversion output terminals are connected to the vertical conductors by the lateral electrical connections.

72. (New) The vertical power delivery apparatus of claim 67 or 68 wherein the interdigitated output terminals further comprise a space of one millimeter or less between adjacent ones of the interdigitated output terminals.

73. (New) The vertical power delivery apparatus of claim 67 or 68 wherein the POL switching power conversion circuitry further comprises a plurality of power conversion output terminals including a space between adjacent ones of the power conversion output terminals and wherein the capacitive array comprises a plurality of discrete capacitors arranged in the spaces between the power conversion output terminals.

74. (New) The apparatus of claim 67 or 68 wherein the switching power conversion circuitry comprises a plurality of power switches, an inductance, a fixed-ratio conversion topology, and is configured to use a current flowing in the inductance to charge and discharge capacitances in the switching power conversion circuitry and reduce a voltage across the one or more power switches prior to turn ON or turn OFF of the one or more of the power switches.

75. (New) The apparatus of claim 74 wherein the switching power conversion circuitry comprises a switched capacitor conversion topology.

76. (New) The apparatus of claim 74 wherein the switching power conversion circuitry further comprises a transformer, and wherein power is converted via the transformer from an input at an input voltage to the semiconductor power input terminals.

77. (New) The apparatus of claim 74 wherein the power conversion circuitry comprises a multi-phase buck conversion topology.

78. (New) The apparatus of claim 75 wherein the power conversion circuitry comprises a multi-phase buck conversion topology.

79. (New) The apparatus of claim 75 wherein the interdigitated output terminals, the plurality of semiconductor power input terminals, and the vertical power connections are configured to provide the shortest electrical path between the semiconductor die and the switching power conversion circuitry.

80. (New) The apparatus of claim 67 or 68 wherein the switching power conversion circuitry comprises a plurality of power switches and an inductance and is configured to use a current flowing in the inductance to charge and discharge capacitances in the switching power conversion circuitry and reduce a current through one or more one of the power switches prior to turn ON or turn OFF of the one or more of the power switches.

81. (New) The apparatus of claim 67 or 68 further comprising first stage switching power conversion circuitry for converting power from a first stage input at a first stage input voltage for delivery to the input of the POL switching power conversion circuitry; wherein the first stage switching power conversion circuitry and the POL switching power conversion circuitry comprise a fixed-ratio conversion topology and a regulating topology.

82. (New) The apparatus of claim 81 wherein the first stage switching power conversion circuitry comprises a regulating topology.

83. (New) The apparatus of claim 81 wherein the POL switching power conversion circuitry comprises a regulating topology.

84. (New) The apparatus of claim 68 wherein the vertical conductors are arranged to align with the interdigitated output terminals.

85. (New) The apparatus of claim 84 wherein the interdigitated output terminals and the vertical conductors occupy the entire die footprint.

86. (New) The apparatus of claim 85 wherein the interdigitated output terminals and

the vertical conductors extend beyond the die footprint in at least one lateral direction.

87. (New) The apparatus of claim 85 wherein the interdigitated output terminals and the vertical conductors extends beyond the die footprint in at least two lateral directions.

88. (New) The apparatus of claim 67 or 68 wherein at least a portion of the stack is encapsulated.

89. (New) The apparatus of claim 88 wherein the POL switching power conversion circuitry is encapsulated.

90. (New) The apparatus of claim 84 wherein the matrix of vertical conductors are formed in a second substrate and the interdigitated output terminals are formed on a surface of the second substrate.

91. (New) The apparatus of claim 67 or 68 wherein the stack is provided as a self-contained module adapted to be installed as a unit.

92. (New) The apparatus of claim 67 or 68 wherein the capacitive array comprises a multiplicity of discrete capacitors.

93. (New) The apparatus of claim 67 or 68 wherein the input of the POL switching power conversion circuitry further comprises terminals on a surface of the substrate.

94. (New) The apparatus of claim 67 wherein the matrix of interdigitated output terminals further comprise conductors electrically connecting the interdigitated output terminals to the switching power conversion circuitry, the conductors being substantially arranged to occupy the die footprint.

95. (New) The apparatus of claim 67 wherein the conductors are arranged in a matrix

with alternating polarities to align with the interdigitated output terminals.

96. (New) The apparatus of claim 95 wherein the matrix of interdigitated output terminals and the matrix of conductors occupy the entire die footprint.

97. (New) The apparatus of claim 96 wherein the matrix of interdigitated output terminals and the matrix of conductors extend beyond the die footprint in at least one lateral direction.

98. (New) The apparatus of claim 96 wherein the matrix of interdigitated output terminals and the matrix of conductors extend beyond the die footprint in at least two lateral directions.

99. (New) The apparatus of claim 95 wherein the matrix of conductors are formed in a second substrate and the interdigitated output terminals are formed on a surface of the second substrate.

100. (New) The apparatus of claim 32, 49, 50, or 51, further comprising a capacitive array vertically arranged between the power conversion module and the semiconductor device, within a filter footprint configured to occupy, in whole or in part, the die footprint, and electrically connected to the X-output terminals.

101. (New) The apparatus of claim 100 wherein the capacitive array comprises a plurality of discrete capacitors arranged between the x-input terminals.

102. (New) The apparatus of claim 100 wherein the interconnection module further comprises a second substrate, the x-output terminals are formed on a surface of the second substrate, and further comprising a matrix of vertical conductors formed in the second substrate within the die footprint.

103. (New) The apparatus of claim 102 wherein the power conversion module and the interconnection module are provided stacked together as a self-contained module adapted to be installed as a unit.

104. (New) The apparatus of claim 101 further comprising first stage switching power conversion circuitry for converting power from a first stage input at a first stage input voltage for delivery to the input of the power conversion module; wherein the first stage switching power conversion circuitry and the power conversion module comprise a fixed-ratio conversion topology and a regulating topology.

105. (New) The apparatus of claim 104 wherein the first stage switching power conversion circuitry comprises a regulating topology.

106. (New) The apparatus of claim 104 wherein the power conversion module comprises a regulating topology.

107. (New) The apparatus of claim 30, 32, 33, 49, 50, or 51 wherein the second spacing between x-output terminals is one millimeter or less.

108. (New) The apparatus of claim 30, 32, 33, 49, 50, or 51 wherein the power conversion circuitry comprises a plurality of power switches, an inductance, a fixed-ratio conversion topology, and is configured to use a current flowing in the inductance to charge and discharge capacitances in the switching power conversion circuitry and reduce a voltage across the one or more power switches prior to turn ON or turn OFF of the one or more of the power switches.

109. (New) The apparatus of claim 108 wherein the switching power conversion circuitry comprises a switched capacitor conversion topology.

110. (New) The apparatus of claim 108 wherein the switching power conversion

circuitry further comprises a transformer, and wherein power is converted power from an input at an input voltage to the semiconductor power input terminals.

111. (New) The apparatus of claim 30, 32, 33, 49, 50, or 51 wherein the switching power conversion circuitry comprises a multi-phase buck conversion topology.

112. (New) The apparatus of claim 32, 49, 50, or 51 wherein the x-output terminals are electrically connected to the x-input terminals by vertical power conductors arranged with the die footprint.

113. (New) The apparatus of claim 30, 32, 33, 49, 50, or 51 wherein the switching power conversion circuitry comprises a plurality of power switches and an inductance and is configured to use a current flowing in the inductance to charge and discharge capacitances in the switching power conversion circuitry and reduce a current through one or more one of the power switches prior to turn ON or turn OFF of the one or more of the power switches.

First Named Inventor : Patrizio Vinciarelli
Application No. : 16/218,418
Filed : December 12, 2018
Page : 26 of 26

Attorney Docket 00614-0183001 / 0269-US-1

REMARKS

Claims 2, 7, 8, 10-14, and 31-34 have been amended.

New claims 48-113 have been added. Support for the new claims can be found throughout the specification, e.g., FIGS. 3-6, 10A, and 10B and their corresponding description.

Applicant asks that all claims be examined in view of the amendment to the claims.

The fees in the amount of \$1,900 for the second Request for Continued Examination and \$18,320 for the excess claim fees are being paid with this reply on the Electronic Filing System. Please apply any other necessary charges or credits to Deposit Account 06-1050, referencing the above attorney docket number.

Respectfully submitted,

Date: September 25, 2020_____

/Rex I. Huang/

Rex I. Huang
Reg. No. 57,661

Customer Number 26214
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Telephone: (617) 542-5070
Facsimile: (877) 769-7945

Electronic Patent Application Fee Transmittal

Application Number:	16218418			
Filing Date:	12-Dec-2018			
Title of Invention:	Delivering Power To Semiconductor Loads			
First Named Inventor/Applicant Name:	Patrizio Vinciarelli			
Filer:	Rex I. Huang/Lisa Gray			
Attorney Docket Number:	00614-0183001			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
CLAIMS IN EXCESS OF 20	1202	152	100	15200
INDEPENDENT CLAIMS IN EXCESS OF 3	1201	5	460	2300
MULTIPLE DEPENDENT CLAIMS	1203	1	820	820
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				
Miscellaneous:				
RCE- 2ND AND SUBSEQUENT REQUEST	1820	1	1900	1900
Total in USD (\$)				20220

Electronic Acknowledgement Receipt

EFS ID:	40677441
Application Number:	16218418
International Application Number:	
Confirmation Number:	1092
Title of Invention:	Delivering Power To Semiconductor Loads
First Named Inventor/Applicant Name:	Patrizio Vinciarelli
Customer Number:	26214
Filer:	Rex I. Huang/Lisa Gray
Filer Authorized By:	Rex I. Huang
Attorney Docket Number:	00614-0183001
Receipt Date:	25-SEP-2020
Filing Date:	12-DEC-2018
Time Stamp:	18:31:16
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	DA
Payment was successfully received in RAM	\$ 20220
RAM confirmation Number	E202090I31309778
Deposit Account	
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Request for Continued Examination (RCE)	00614-0183001rce.pdf	117293	no	1
			ce72065a7ab5262ee45fe9e98a36bedeab9e2df3		
Warnings:					
This is not a USPTO supplied RCE SB30 form.					
Information:					
2		00614-0183001prelimamend.pdf	163875	yes	26
			09f3fc1b5e1430b268e78c3ef82b610a72cae867		
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Amendment Submitted/Entered with Filing of CPA/RCE		1	1	
	Claims		2	25	
	Applicant Arguments/Remarks Made in an Amendment		26	26	
Warnings:					
Information:					
3	Fee Worksheet (SB06)	fee-info.pdf	37271	no	2
			a7beec72dc99ad32826121aa3afbc07edaf8ee9a		
Warnings:					
Information:					
Total Files Size (in bytes):			318439		

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875				Application or Docket Number 16/218,418		Filing Date 12/12/2018		☐ To be Mailed		
ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO										
APPLICATION AS FILED - PART I										
		(Column 1)	(Column 2)							
FOR		NUMBER FILED	NUMBER EXTRA	RATE (\$)		FEE (\$)				
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))		N/A	N/A	N/A						
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))		N/A	N/A	N/A						
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))		N/A	N/A	N/A						
TOTAL CLAIMS (37 CFR 1.16(i))		minus 20 =	*	x \$100 =						
INDEPENDENT CLAIMS (37 CFR 1.16(h))		minus 3 =	*	x \$460 =						
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))		If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).								
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))										
* If the difference in column 1 is less than zero, enter "0" in column 2.				TOTAL						
APPLICATION AS AMENDED - PART II										
		(Column 1)		(Column 2)	(Column 3)					
AMENDMENT	09/25/2020	CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)		ADDITIONAL FEE (\$)		
	Total (37 CFR 1.16(i))	* 199	Minus	** 47	= 152	x \$100 =		15200		
	Independent (37 CFR 1.16(h))	* 11	Minus	*** 6	= 5	x \$460 =		2300		
	☐ Application Size Fee (37 CFR 1.16(s))									
	☒ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))							820		
						TOTAL ADD'L FEE		18320		
		(Column 1)		(Column 2)	(Column 3)					
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)		ADDITIONAL FEE (\$)		
	Total (37 CFR 1.16(i))	*	Minus	**	=	x \$0 =				
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x \$0 =				
	☐ Application Size Fee (37 CFR 1.16(s))									
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))									
						TOTAL ADD'L FEE				
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.						LIE				
** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".						/DORIAN EVANS/				
*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".										
The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.										

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.



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Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

NOTICE OF ALLOWANCE AND FEE(S) DUE

26214 7590 10/20/2020
FISH & RICHARDSON P.C. (VLT, INC)
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022

EXAMINER

COX, CASSANDRA F

ART UNIT

PAPER NUMBER

2849

DATE MAILED: 10/20/2020

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/218,418	12/12/2018	Patrizio Vinciarelli	00614-0183001	1092

TITLE OF INVENTION: DELIVERING POWER TO SEMICONDUCTOR LOADS

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1200	\$0.00	\$0.00	\$1200	01/20/2021

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 1/2 the amount of undiscounted fees, and micro entity fees are 1/2 the amount of small entity fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed and an extra copy of the form should be submitted. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Maintenance fees are due in utility patents issuing on applications filed on or after Dec. 12, 1980. It is patentee's responsibility to ensure timely payment of maintenance fees when due. More information is available at www.uspto.gov/PatentMaintenanceFees.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), by mail or fax, or via EFS-Web.

By mail, send to: Mail Stop ISSUE FEE
 Commissioner for Patents
 P.O. Box 1450
 Alexandria, Virginia 22313-1450

By fax, send to: (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

26214 7590 10/20/2020
FISH & RICHARDSON P.C. (VLT, INC)
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via EFS-Web or by facsimile to (571) 273-2885, on the date below.

(Typed or printed name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/218,418	12/12/2018	Patrizio Vinciarelli	00614-0183001	1092

TITLE OF INVENTION: DELIVERING POWER TO SEMICONDUCTOR LOADS

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1200	\$0.00	\$0.00	\$1200	01/20/2021

EXAMINER	ART UNIT	CLASS-SUBCLASS
COX, CASSANDRA F	2849	327-109000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-09 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

1 _____

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

2 _____

3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. Fees submitted: ☐ Issue Fee ☐ Publication Fee (if required) ☐ Advance Order - # of Copies _____

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☐ Electronic Payment via EFS-Web ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)

☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. _____

5. **Change in Entity Status** (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____

Date _____

Typed or printed name _____

Registration No. _____



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/218,418	12/12/2018	Patrizio Vinciarelli	00614-0183001	1092
26214	7590	10/20/2020	EXAMINER	
FISH & RICHARDSON P.C. (VLT, INC)			COX, CASSANDRA F	
P.O. BOX 1022			ART UNIT	
MINNEAPOLIS, MN 55440-1022			PAPER NUMBER	
			2849	
DATE MAILED: 10/20/2020				

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 30 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability	Application No. 16/218,418	Applicant(s) Vinciarelli, Patrizio	
	Examiner CASSANDRA F COX	Art Unit 2849	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to 09/25/2020.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on ____.

2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on ____; the restriction requirement and election have been incorporated into this action.

3. ☒ The allowed claim(s) is/are 1-113. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) ☐ All b) ☐ Some *c) ☐ None of the:

1. ☐ Certified copies of the priority documents have been received.

2. ☐ Certified copies of the priority documents have been received in Application No. ____.

3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: ____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.
THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date ____.

Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).

6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

1. ☐ Notice of References Cited (PTO-892) 2. ☐ Information Disclosure Statements (PTO/SB/08), Paper No./Mail Date ____. 3. ☐ Examiner's Comment Regarding Requirement for Deposit of Biological Material ____. 4. ☒ Interview Summary (PTO-413), Paper No./Mail Date _____.	5. ☐ Examiner's Amendment/Comment 6. ☒ Examiner's Statement of Reasons for Allowance 7. ☐ Other _____.
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/CASSANDRA F COX/
Primary Examiner, Art Unit 2849

DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Allowable Subject Matter

1. Claims 1-113 are allowed.
2. The following is an examiner's statement of reasons for allowance: Claims 1-16, 36-43, and 46 are allowed because the closest prior art of record fails to disclose a method of powering a semiconductor device having power input terminals, the method comprising: providing an interconnection module having x-input terminals arranged on a first side of the interconnection module in a pattern configured to align with the first pattern of converter output terminals, and x-output terminals arranged on a surface of a second opposite side of the interconnection module in a second pattern configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals, the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals; assembling the power conversion module to the first side of the interconnection module with each converter output terminal mated with and electrically connected to a respective x-input terminal; wherein the interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern and the power input terminals of

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the semiconductor device arranged in the second pattern; wherein the power conversion module and the interconnection module are stacked to form a module sandwich including the power conversion module in a first layer, electrical connections between the converter output terminals and the x-input terminals in a second layer, and the interconnection module in a third layer in combination with the rest of the limitations of the base claim. Claims 17-23 are allowed because the closest prior art of record fails to disclose a method of making power converters, comprising: selectively forming a plurality of holes in the exterior surface of the panel, each hole being formed at a respective predetermined location aligned with a respective output node, spaced apart from the cut lines, and exposing within the hole the respective conductive feature; selectively forming one or more conductive metal layers on the exterior surface of the panel including within the plurality of holes, the one or more conductive metal layers within each hole being in electrical contact with the respective conductive feature, to form a plurality of electrical contacts on the exterior surface of the panel; patterning the one or more conductive metal layers on the exterior surface to electrically isolate at least one electrical contact from one or more of the plurality of electrical contacts; cutting the panel along the one or more cut lines to singulate the one or more power conversion modules, each singulated power conversion module having a respective plurality of the electrical contacts formed on the exterior surface of the module in combination with the rest of the limitations of the base claim. Claims 24-26 are allowed because the closest prior art of record fails to disclose a method of making electronic assemblies, the method comprising: wherein the interconnection module is configured to provide electrical and mechanical translation between the circuit terminals arranged in the first pattern and the external terminals of the third electronic device arranged

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in the second pattern; wherein the electronic module and the interconnection module are stacked to form a module sandwich including the electronic module in a first layer, electrical connections between the circuit terminals and the x-input terminals in a second layer, and the interconnection module in a third layer; assembling a first panel comprising a plurality of unsingulated electronic modules; assembling a second panel comprising a plurality of unsingulated interconnection modules; wherein the assembling of the electronic module and the interconnection module is performed using the first and second panels prior to singulation to form a panel sandwich and further comprises dispensing a curable adhesive compound between the first and second panels and forming solder connections between each circuit terminal and each respective x-input terminal; and cutting the panel sandwich along predetermined cut lines to singulate a plurality of the module sandwiches in combination with the rest of the limitations of the base claim. Claims 27-29 are allowed because the closest prior art of record fails to disclose a method of forming modular circuit assemblies, the method comprising: selectively forming a plurality of terminal holes in the first exterior surface of each panel through the first layer at predetermined locations within the perimeter boundaries of each electronic module, each terminal hole being spaced apart from the cut lines and exposing within the terminal hole a respective conductive feature; selectively forming one or more conductive metal layers on the first exterior surface of each panel including within the plurality of terminal holes, the one or more conductive metal layers within each terminal hole being in electrical contact with the respective conductive feature; patterning the one or more conductive metal layers on the exterior surface, to form a plurality of electrical contacts on the exterior surface of each panel electrically isolated from at least one other electrical contact in

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the plurality of electrical contacts; assembling the first and second encapsulated panels with electrical contacts on the first panel aligned and mated with respective electrical contacts on the second panel to form a panel stack, the panel stack comprising a plurality of un-singulated module stacks, each module stack including an unsingulated electronic module in the first panel and a respective unsingulated electronic module in the second panel; and cutting the panel stack along the one or more cut lines to singulate the plurality of electronic module stacks in combination with the rest of the limitations of the base claim. Claims 30-34, and 100-113 are allowed because the closest prior art of record fails to disclose an apparatus comprising: an interconnection module having x-input terminals arranged on a first surface of the interconnection module in a pattern configured to align with the first pattern of converter output terminals, and x-output terminals arranged on a second surface in a second pattern configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals, the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals; the converter output terminals of the power conversion module being mated with and electrically connected to respective x-input terminals of the interconnection module; wherein the interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern and the power input terminals of the semiconductor device arranged in the second pattern; wherein the power conversion module and the interconnection module are stacked with the power conversion module located at a first level, electrical connections between the converter output terminals and the x-input terminals at a second level, and the interconnection module at a third

level in combination with the rest of the limitations of the base claim. Claims 35, 44-45, and 47 are allowed because the closest prior art of record fails to disclose a method of powering a semiconductor device having power input terminals, the method comprising: providing an interconnection module having x-input terminals arranged on a first surface of the interconnection module in a pattern configured to align with the first pattern of converter output terminals, and x-output terminals arranged on a second surface in a second pattern configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals, the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals; providing a plurality of conductive pillars for electrically connecting the converter output terminals with respective x-input terminals; assembling the power conversion module to the interconnection module with each converter output terminal aligned with and electrically connected to a respective x-input terminal by a respective one of the conductive pillars; wherein the interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern and the power input terminals of the semiconductor device arranged in the second pattern; wherein the power conversion module and the interconnection module are stacked to form a module sandwich including the power conversion module in a first layer, the conductive pillars connecting the converter output terminals and the x-input terminals in a second layer, and the interconnection module in a third layer; and wherein the x-output terminals are arranged with alternating polarities to reduce interconnection inductance in combination with the rest of the limitations of the base claim. Claims 48, 2, 7-8, and 10-14 are allowed because the closest prior

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art of record fails to disclose a method of powering a semiconductor device, including at least one semiconductor die having a plurality of die power input terminals arranged on a surface of the die and having edges of the semiconductor die defining a die footprint, the semiconductor device having power input terminals arranged within the footprint of the die, the method comprising: providing an interconnection module having x-input terminals arranged on a first side of the interconnection module in a pattern configured to align with the first pattern of converter output terminals, and x-output terminals arranged on a surface of a second opposite side of the interconnection module in a second pattern arranged within the die footprint and configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals, the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals; assembling the power conversion module to the first side of the interconnection module with each converter output terminal mated with and electrically connected to a respective x-input terminal; wherein the interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern and the power input terminals of the semiconductor device arranged in the second pattern; wherein the power conversion module and the interconnection module are stacked to form a module sandwich including the power conversion module in a first layer, electrical connections between the converter output terminals and the x-input terminals in a second layer, and the interconnection module in a third layer in combination with the rest of the limitations of the base claim. Claims 49-52, and 100-113 are allowed because the closest prior art of record fails to disclose an apparatus comprising: a semiconductor device,

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including at least one semiconductor die having a plurality of die power input terminals arranged on a surface of the die and having edges of the semiconductor die defining a die footprint, the semiconductor device having power input terminals arranged within the footprint of the die; an interconnection module having x-input terminals arranged on a first surface of the interconnection module in a pattern configured to align with the first pattern of converter output terminals, and x-output terminals arranged on a second surface in a second pattern with alternating polarities configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals, the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals; the converter output terminals of the power conversion module being mated with and electrically connected to respective x-input terminals of the interconnection module; wherein the interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern and the power input terminals of the semiconductor device arranged in the second pattern; wherein the power conversion module and the interconnection module are stacked with the power conversion module located at a first level, electrical connections between the converter output terminals and the x-input terminals at a second level, and the interconnection module at a third level in combination with the rest of the limitations of the base claim. Claims 53-66 are allowed because the closest prior art of record fails to disclose apparatus for delivering power vertically to a semiconductor die, the die having a plurality of semiconductor power input terminals arranged on a surface of the semiconductor die and having edges of the semiconductor die defining a die footprint, the apparatus comprising: translation

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interconnections including X-input terminals configured and arranged to align with and electrically connect to the converter output terminals, a plurality of X-output terminals configured and arranged with alternating polarities to align with respective power input terminals of the semiconductor die, and a plurality of vertical conductors to make electrical connections to the X-output terminals; a capacitive array arranged between the power conversion circuitry and the semiconductor die and electrically connected to the translation interconnections to provide energy storage; wherein the translation interconnections are configured to provide electrical and mechanical translation between the converter output terminals and the power input terminals of the semiconductor die; wherein the switching power conversion circuitry, the translation interconnections, and the capacitive array, are each configured and arranged within a respective footprint configured to occupy substantially all of the die footprint, and are together configured and arranged to form a vertical stack configured to occupy substantially all of the die footprint in combination with the rest of the limitations of the base claim. Claims 67, 72-83, 88-89, 91-99 are allowed because the closest prior art of record fails to disclose a vertical power delivery apparatus for a semiconductor die, the semiconductor die having a plurality of semiconductor power input terminals arranged on a surface of the semiconductor die and having edges of the semiconductor die defining a die footprint;

3.

4. the vertical power delivery apparatus comprising in a vertical stack configured to occupy, in whole or in part, the die footprint, the vertical stack comprising:

5.

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6. (a) point-of-load ("POL") switching power conversion circuitry; (b) interdigitated output terminals (i) electrically connected to receive power from the switching power conversion cells, (ii) configured within a terminal footprint to match the plurality of semiconductor power input terminals, (iii) the terminal footprint being configured to occupy, in whole or in part, the die footprint; and (c) a capacitive array (i) arranged between the power conversion circuitry and the semiconductor die, (ii) within a filter footprint configured to occupy, in whole or in part, the die footprint and (iii) electrically connected to the interdigitated output terminals in combination with the rest of the limitations of the base claim. Claims 68-93 are allowed because the closest prior art of record fails to disclose a vertical power delivery apparatus for a semiconductor die, the semiconductor die having a plurality of semiconductor power input terminals arranged on a surface of the semiconductor die and having edges of the semiconductor die defining a die footprint; the vertical power delivery apparatus comprising in a vertical stack configured to occupy, in whole or in part, the die footprint, the vertical stack comprising: (a) point-of-load ("POL") switching power conversion circuitry; (b) interdigitated output terminals (i) electrically connected to receive power from the switching power conversion cells, (ii) configured within a terminal footprint to match the plurality of semiconductor power input terminals, (iii) the terminal footprint being configured to occupy, in whole or in part, the die footprint; (c) a capacitive array (i) arranged between the power conversion circuitry and the semiconductor die, (ii) within a filter footprint configured to occupy, in whole or in part, the die footprint and (iii) electrically connected to the interdigitated output terminals; and (d) a plurality of vertical conductors arranged within the die footprint to

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connect the interdigitated output terminals to the switching power conversion cells in combination with the rest of the limitations of the base claim.

Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

Conclusion

7. Any inquiry concerning this communication or earlier communications from the examiner should be directed to CASSANDRA F COX whose telephone number is (571)272-1741. The examiner can normally be reached on M-F 6:00-3:30; off alt Fridays.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Lincoln Donovan can be reached on 571-272-1988. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR

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system, see <https://ppair-my.uspto.gov/pair/PrivatePair>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/CASSANDRA F COX/

Primary Examiner, Art Unit 2849

Summary of Record of Interview Requirements

Manual of Patent Examining Procedure (MPEP), Section 713.04, Substance of Interview Must be Made of Record

A complete written statement as to the substance of any face-to-face, video conference, or telephone interview with regard to an application must be made of record in the application whether or not an agreement with the examiner was reached at the interview.

Title 37 Code of Federal Regulations (CFR) 1.133 Interviews

Paragraph (b)

In every instance where reconsideration is requested in view of an interview with an examiner, a complete written statement of the reasons presented at the interview as warranting favorable action must be filed by the applicant. An interview does not remove the necessity for reply to Office action as specified in §§ 1.111, 1.135. (35 U.S.C. 132)

37 CFR §1.2 Business to be transacted in writing.

All business with the Patent or Trademark Office should be transacted in writing. The personal attendance of applicants or their attorneys or agents at the Patent and Trademark Office is unnecessary. The action of the Patent and Trademark Office will be based exclusively on the written record in the Office. No attention will be paid to any alleged oral promise, stipulation, or understanding in relation to which there is disagreement or doubt.

The action of the Patent and Trademark Office cannot be based exclusively on the written record in the Office if that record is itself incomplete through the failure to record the substance of interviews.

It is the responsibility of the applicant or the attorney or agent to make the substance of an interview of record in the application file, unless the examiner indicates he or she will do so. It is the examiners responsibility to see that such a record is made and to correct material inaccuracies which bear directly on the question of patentability.

Examiners must complete an Interview Summary Form for each interview held where a matter of substance has been discussed during the interview by checking the appropriate boxes and filling in the blanks. Discussions regarding only procedural matters, directed solely to restriction requirements for which interview recordation is otherwise provided for in Section 812.01 of the Manual of Patent Examining Procedure, or pointing out typographical errors or unreadable script in Office actions or the like, are excluded from the interview recordation procedures below. Where the substance of an interview is completely recorded in an Examiners Amendment, no separate Interview Summary Record is required.

The Interview Summary Form shall be given an appropriate Paper No., placed in the right hand portion of the file, and listed on the "Contents" section of the file wrapper. In a personal interview, a duplicate of the Form is given to the applicant (or attorney or agent) at the conclusion of the interview. In the case of a telephone or video-conference interview, the copy is mailed to the applicants correspondence address either with or prior to the next official communication. If additional correspondence from the examiner is not likely before an allowance or if other circumstances dictate, the Form should be mailed promptly after the interview rather than with the next official communication.

The Form provides for recordation of the following information:

- Application Number (Series Code and Serial Number)
- Name of applicant
- Name of examiner
- Date of interview
- Type of interview (telephonic, video-conference, or personal)
- Name of participant(s) (applicant, attorney or agent, examiner, other PTO personnel, etc.)
- An indication whether or not an exhibit was shown or a demonstration conducted
- An identification of the specific prior art discussed
- An indication whether an agreement was reached and if so, a description of the general nature of the agreement (may be by attachment of a copy of amendments or claims agreed as being allowable). Note: Agreement as to allowability is tentative and does not restrict further action by the examiner to the contrary.
- The signature of the examiner who conducted the interview (if Form is not an attachment to a signed Office action)

It is desirable that the examiner orally remind the applicant of his or her obligation to record the substance of the interview of each case. It should be noted, however, that the Interview Summary Form will not normally be considered a complete and proper recordation of the interview unless it includes, or is supplemented by the applicant or the examiner to include, all of the applicable items required below concerning the substance of the interview.

A complete and proper recordation of the substance of any interview should include at least the following applicable items:

- 1) A brief description of the nature of any exhibit shown or any demonstration conducted,-
- 2) an identification of the claims discussed,
- 3) an identification of the specific prior art discussed,
- 4) an identification of the principal proposed amendments of a substantive nature discussed, unless these are already described on the Interview Summary Form completed by the Examiner,
- 5) a brief identification of the general thrust of the principal arguments presented to the examiner,
(The identification of arguments need not be lengthy or elaborate. A verbatim or highly detailed description of the arguments is not required. The identification of the arguments is sufficient if the general nature or thrust of the principal arguments made to the examiner can be understood in the context of the application file. Of course, the applicant may desire to emphasize and fully describe those arguments which he or she feels were or might be persuasive to the examiner.)
- 6) a general indication of any other pertinent matters discussed, and
- 7) if appropriate, the general results or outcome of the interview unless already described in the Interview Summary Form completed by the examiner.

Examiners are expected to carefully review the applicants record of the substance of an interview. If the record is not complete and accurate, the examiner will give the applicant an extendable one month time period to correct the record.

Examiner to Check for Accuracy

If the claims are allowable for other reasons of record, the examiner should send a letter setting forth the examiners version of the statement attributed to him or her. If the record is complete and accurate, the examiner should place the indication, Interview Record OK on the paper recording the substance of the interview along with the date and the examiners initials.

Issue Classification 	Application/Control No. 16/218,418	Applicant(s)/Patent Under Reexamination Vinciarelli, Patrizio
	Examiner CASSANDRA F COX	Art Unit 2849

CPC						
Symbol					Type	Version
H02M	/	1	/	08	F	2013-01-01
H05K	/	1	/	144	I	2013-01-01
H01L	/	24	/	26	I	2013-01-01
H05K	/	1	/	0298	I	2013-01-01
H02M	/	2001	/	0074	A	2013-01-01

CPC Combination Sets					
Symbol				Type	Version
/	/	/	/		

NONE (Assistant Examiner) (Date)		Total Claims Allowed: 113	
/CASSANDRA F COX/ Primary Examiner, Art Unit 2849 (Primary Examiner) (Date)		10 October 2020 O.G. Print Claim(s)	30 O.G. Print Figure

Issue Classification 	Application/Control No. 16/218,418	Applicant(s)/Patent Under Reexamination Vinciarelli, Patrizio
	Examiner CASSANDRA F COX	Art Unit 2849

INTERNATIONAL CLASSIFICATION			
CLAIMED			
H02M1/08	/	1	/ 08

NON-CLAIMED			
/		/	

US ORIGINAL CLASSIFICATION	
CLASS	SUBCLASS

CROSS REFERENCES(S)						
CLASS	SUBCLASS (ONE SUBCLASS PER BLOCK)					

NONE		Total Claims Allowed:	
(Assistant Examiner)	(Date)	113	
/CASSANDRA F COX/ Primary Examiner, Art Unit 2849 (Primary Examiner)	10 October 2020 (Date)	O.G. Print Claim(s) 30	O.G. Print Figure 5A

Issue Classification 	Application/Control No. 16/218,418	Applicant(s)/Patent Under Reexamination Vinciarelli, Patrizio
	Examiner CASSANDRA F COX	Art Unit 2849

☐ Claims renumbered in the same order as presented by applicant
 ☐ CPA
 ☐ T.D.
 ☐ R.1.47

CLAIMS															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
1	1	16	16	40	31	25	46	61	61	76	76	91	91	106	106
2	2	26	17	41	32	47	47	62	62	77	77	92	92	107	107
3	3	27	18	42	33	48	48	63	63	78	78	93	93	108	108
4	4	28	19	43	34	49	49	64	64	79	79	94	94	109	109
5	5	29	20	44	35	50	50	65	65	80	80	95	95	110	110
6	6	30	21	17	36	51	51	66	66	81	81	96	96	111	111
7	7	31	22	18	37	52	52	67	67	82	82	97	97	112	112
8	8	32	23	19	38	53	53	68	68	83	83	98	98	113	113
9	9	33	24	20	39	54	54	69	69	84	84	99	99		
10	10	34	25	21	40	55	55	70	70	85	85	100	100		
11	11	35	26	22	41	56	56	71	71	86	86	101	101		
12	12	36	27	23	42	57	57	72	72	87	87	102	102		
13	13	37	28	24	43	58	58	73	73	88	88	103	103		
14	14	38	29	45	44	59	59	74	74	89	89	104	104		
15	15	39	30	46	45	60	60	75	75	90	90	105	105		

NONE (Assistant Examiner) _____ (Date) _____		Total Claims Allowed: 113	
/CASSANDRA F COX/ Primary Examiner, Art Unit 2849 (Primary Examiner) _____ (Date) 10 October 2020		O.G. Print Claim(s) 30	O.G. Print Figure 5A

<i>Search Notes</i> 	Application/Control No. 16/218,418	Applicant(s)/Patent Under Reexamination Vinciarelli, Patrizio
	Examiner CASSANDRA F COX	Art Unit 2849

CPC - Searched*		
Symbol	Date	Examiner

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
EAST Search (USPGPUB, USPAT, USOCR, EPO, JPO, DERWENT, IBM_TDB) see search history	05/26/2020	cfc
H02M1/08 or H01L24/26 or H05K1/144 or H05K1/0298 or H02M2001/0074 (text only - see search history)	05/26/2020	cfc
inventor search	05/26/2020	cfc
update	07/20/2020	cfc
update	08/17/2020	cfc
update	10/10/2020	cfc

/CASSANDRA F COX/ Primary Examiner, Art Unit 2849	
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<i>Search Notes</i> 	Application/Control No. 16/218,418	Applicant(s)/Patent Under Reexamination Vinciarelli, Patrizio
	Examiner CASSANDRA F COX	Art Unit 2849

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner
H02M	1/08, 2001/0074	05/26/2020	cfc
H01L	24/26	05/26/2020	cfc
H05K	1/144, 1/0298	05/26/2020	cfc

/CASSANDRA F COX/ Primary Examiner, Art Unit 2849	
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EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	288,227	semiconductor adj chip	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:58
L2	170,766	driver adj circuit	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:58
L3	5,153	output adj rail	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:58
L4	242	rail adj selection	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:58
L5	4,898	rail adj switch\$3	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:58
L6	3	L1 and L2 and L3 and L4 and L5	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:58
L7	3	L1 and L2 and L3 and L5	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:58

L8	3	L2 and L3 and L4 and L5	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:58
L9	6	L2 and L3 and L5	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:58
L10	30	("8582333" "9660537" "6031743" "6035261" "6275958" "6300749" "6421262" "6466458" "6603292" "6700361" "6788033" "6911848" "6934166" "6969909" "6985341" "7145786" "7521909" "7561446" "7768807" "8232783" "RE40072" "0014731" "0014822" "0014874" "20130069608" "6930893" "6975098" "6984965" "9112422" "9166481").PN.	US-PGPUB; USPAT	OR	ON	2020/10/10 18:58
L11	44,885	(H02M1/08 or H02M3/155 or H01L23/645 or H01L23/5385 or H01L23/5383 or H01L23/5386 or H01L25/18 or H01L23/5384).cpc.	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:58
L12	5,153	output adj rail	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:58
L13	242	rail adj selection	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:58
L14	4,898	rail adj switch\$3	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:58
L15	90	L12 and L14	US-PGPUB; USPAT;	OR	ON	2020/10/10 18:58

			USOCR; EPO; JPO; DERWENT; IBM_TDB			
L16	243	point adj load adj circuit	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:58
L17	5,153	output adj rail	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:58
L18	242	rail adj selection	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:58
L19	14	L17 and L18	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:58
L20	77,593	rail\$1 with switch\$3	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:58
L21	10,664	rail adj voltage	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:58
L22	288,227	semiconductor adj chip	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:58
L23	170,766	driver adj circuit	US-PGPUB; USPAT; USOCR;	OR	ON	2020/10/10 18:58

			EPO; JPO; DERWENT; IBM_TDB			
L24	100	L22 and L23 and L20	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:58
L25	33	("20130030534" "5438294" "9087656" "8582333" "9660537" "6031743" "6035261" "6275958" "6300749" "6421262" "6466458" "6603292" "6700361" "6788033" "6911848" "6934166" "6969909" "6985341" "7145786" "7521909" "7561446" "7768807" "8232783" "RE40072" "0014731" "0014822" "0014874" "20130069608" "6930893" "6975098" "6984965" "9112422" "9166481").PN.	US-PGPUB; USPAT	OR	ON	2020/10/10 18:58
L26	39,853	power adj bus	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:58
L27	77,593	rail\$1 with switch\$3	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:58
L28	288,227	semiconductor adj chip	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:58
L29	170,766	driver adj circuit	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:58
L30	100	L28 and L29 and L27	US-PGPUB; USPAT; USOCR;	OR	ON	2020/10/10 18:58

			EPO; JPO; DERWENT; IBM_TDB			
L31	2	L26 and L30	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:58
L32	44,885	(H02M1/08 or H02M3/155 or H01L23/645 or H01L23/5385 or H01L23/5383 or H01L23/5386 or H01L25/18 or H01L23/5384).cpc.	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:58
L33	28	L32 and L30	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:58
L34	19,212	voltage adj driver	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:58
L35	10,664	rail adj voltage	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:58
L36	492	(dc or direct adj current) adj rail adj voltage	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:58
L37	51	plural\$3 adj3 rail adj voltage	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:58
L38	122	(plural\$3 or multiple) adj3 rail adj voltage	US-PGPUB; USPAT; USOCR; EPO; JPO;	OR	ON	2020/10/10 18:58

			DERWENT; IBM_TDB			
L39	0	(plural\$3 or multiple) adj3 rail adj voltage16 and L38	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:58
L40	170,766	driver adj circuit	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:58
L41	122	(plural\$3 or multiple) adj3 rail adj voltage	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:58
L42	13	(plural\$3 or multiple) adj3 rail adj voltage and L40 and L41	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:58
L43	25	generat\$3 near3 (plural\$3 or multiple) adj3 rail adj voltage	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:58
L44	5	L40 and L41 and L43	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:58
L45	122	(plural\$3 or multiple) adj3 rail adj voltages	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:58
L46	61	driver and L45	US-PGPUB; USPAT; USOCR; EPO; JPO;	OR	ON	2020/10/10 18:58

			DERWENT; IBM_TDB			
L47	1,921	(generat\$3 or output\$4) near3 rail adj voltage\$1	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:58
L48	30	L46 and L47	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:58
L49	1,753	half adj bridge adj driver	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:58
L50	36,868	switch\$3 with capacitor with transformer	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:59
L51	154	L49 and L50	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:59
L52	22	L49 same L50	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:59
L53	44,885	(H02M1/08 or H02M3/155 or H01L23/645 or H01L23/5385 or H01L23/5383 or H01L23/5386 or H01L25/18 or H01L23/5384).cpc.	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:59
L54	5	L51 and L53	US-PGPUB; USPAT; USOCR; EPO; JPO;	OR	ON	2020/10/10 18:59

			DERWENT; IBM_TDB			
L55	309	vinciarelli-patrizio.in.	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:59
L56	18	damico-andrew-t.in.	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:59
L57	36,868	switch\$3 with capacitor with transformer	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:59
L58	309	vinciarelli-patrizio.in.	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:59
L59	36,868	switch\$3 with capacitor with transformer	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:59
L60	83	L58 and L59	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:59
L72	4,518	power adj conversion adj module	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:59
L73	3,548	interconnection adj module	US-PGPUB; USPAT; USOCR; EPO; JPO;	OR	ON	2020/10/10 18:59

			DERWENT; IBM_TDB			
L74	27,240	multi adj cell	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:59
L75	37	(plural\$3 or multiple) adj conversion adj cells	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:59
L76	5,161,715	pattern	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:59
L77	1,746,305	spacing	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:59
L78	26,479	(H02M1/08 or H01L24/26 or H05K1/144 or H05K1/0298 or H02M2001/0074).cpc.	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:59
L79	0	L72 and L73 and L74 and L75 and L76 and L77 and L78	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:59
L80	507,745	(plural\$3 or multiple) adj3 cells	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:59
L81	0	L72 and L73 and L74 and L75 and L76 and L77	US-PGPUB; USPAT; USOCR; EPO; JPO;	OR	ON	2020/10/10 18:59

			DERWENT; IBM_TDB			
L82	0	L72 and L73 and L74 and L80 and L76 and L77 and L78	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:59
L83	0	L72 and L73 and L74 and L80 and L76 and L77	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:59
L84	0	L72 and L73 and L74 and L75 and L78	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:59
L85	0	L72 and L73 and L74 and L80 and L78	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:59
L86	0	L72 and L73 and L74 and L75	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:59
L87	0	L72 and L73 and L74 and L80	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:59
L88	0	L72 and L73 and L75 and L78	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:59
L89	0	L72 and L73 and L78 and L80	US-PGPUB; USPAT; USOCR; EPO; JPO;	OR	ON	2020/10/10 18:59

			DERWENT; IBM_TDB			
L90	124,553	power adj converter	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:59
L91	6	L75 and L90	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:59
L92	97	("20030087538" "20170013748" "4211603" "10231333" "10390437" "10512182" "20010018285" "20010018286" "20010023980" "20010032388" "20030162434" "20030168499" "20040207089" "20040251554" "20050048692" "20060127652" "20060180346" "20060272150" "20130083495" "20160128215" "20170115144" "20170336584" "20190080931" "3846741" "4223970" "4230385" "4394711" "4526429" "4742477" "4917526" "5168432" "5321351" "5408189" "6031253" "6110213" "6116916" "6230403" "6646886" "6847853" "7294007" "RE32212" "RE36442" "0014731" "0014822" "0014874" "10014798" "10020752" "10158357" "10264664" "10277105" "20030227280" "20060291265" "20070297198" "20090016083" "20130030534" "20130069608" "20150188407" "5438294" "5728600" "6031743" "6035261" "6118186" "6275958" "6300749" "6403009" "6421262" "6466458" "6603292" "6700361" "6788033" "6911848" "6930893" "6934166" "6969909" "6975098" "6984965" "6985341" "7145786" "7187263" "7245509" "7361844" "7521909" "7561446")	US-PGPUB; USPAT	OR	ON	2020/10/10 18:59

		"7768807" "8232783" "8427269" "8582333" "8966747" "9087656" "9112422" "9166481" "9402319" "9516761" "9660537" "9765750" "9936580" "RE40072").PN.				
L93	4,518	power adj conversion adj module	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:59
L94	309	vinciarelli-patrizio.in.	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:59
L106	8	("20070241440" "20080078572" "20140355218" "7015587" "7030469" "7198987" "7494843" "8138584").PN.	US-PGPUB; USPAT	OR	ON	2020/10/10 18:59
L107	18	("10537015" "20030058628" "20040100778" "20050184381" "20060133041" "20060133042" "20090140394" "20090321956" "20090321957" "20090325345" "5027255" "5481436" "6229216" "6940013" "7767494" "7952879" "8240035" "9439297").PN.	US-PGPUB; USPAT	OR	ON	2020/10/10 18:59
L108	4,518	power adj conversion adj module	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:59
L109	1	L107 and L108	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:59
L110	239,871	power adj conver\$4	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:59

L111	1	L107 and "I4"	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:59
L112	10	L107 and L110	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:59
L113	10,230	pattern with output adj terminals	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:59
L114	8,546	pattern with input adj terminals	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:59
L115	0	L112 and L113 and L114	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:59
L116	436	switch\$3 adj power adj conversion adj circuit\$2	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 18:59
L117	68,715	capacit\$3 near5 array	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 19:00
L118	1,218	die adj voltage	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 19:00

L119	34,072	vertical adj stack\$3	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 19:00
L120	0	switch\$3 adj power adj conversion adj cells	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 19:01
L121	0	l116 and l117 and l118 and l119	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 19:25
L122	2	l117 and l118 and l119	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/10 19:26

10/10/2020 8:14:03 PM

C:\Users\ccox\Documents\EAST\Workspaces\616288.wsp

Bibliographic Data

Application No: 16/218,418

Foreign Priority claimed: ☐ Yes ☒ No

35 USC 119 (a-d) conditions met: ☐ Yes ☒ No ☐ Met After Allowance

Verified and Acknowledged: /CASSANDRA F COX/

Examiner's Signature

Initials

Title: Delivering Power To Semiconductor Loads

FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.
12/12/2018	327	2849	00614-0183001
RULE			

APPLICANTS

VLT, Inc., Andover, MA,

INVENTORS

Patrizio Vinciarelli, Boston, MA, UNITED STATES

CONTINUING DATA

This application is a CIP of 15616288 06/07/2017 PAT 10277105

15616288 is a CIP of 15091346 04/05/2016 PAT 10158357

FOREIGN APPLICATIONS

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01/04/2019

STATE OR COUNTRY

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FILING FEE RECEIVED

\$5,600

EAST Search History

EAST Search History (Interference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L61	26,597	(H02M1/08 or H02M3/155 or H01L23/645 or H01L23/5385 or H01L23/5383 or H01L23/5386 or H01L25/18 or H01L23/5384).cpc.	US-PGPUB; USPAT	OR	ON	2020/10/10 18:59
L62	23,045	switch\$3 with capacitor with transformer	US-PGPUB; USPAT	OR	ON	2020/10/10 18:59
L63	9,824	rail adj voltage	US-PGPUB; USPAT	OR	ON	2020/10/10 18:59
L64	32,218	rail\$1 with switch\$3	US-PGPUB; USPAT	OR	ON	2020/10/10 18:59
L65	34,124	power adj bus	US-PGPUB; USPAT	OR	ON	2020/10/10 18:59
L66	1,068	L61 and L62	US-PGPUB; USPAT	OR	ON	2020/10/10 18:59
L67	2	L63 and L65 and L66	US-PGPUB; USPAT	OR	ON	2020/10/10 18:59
L68	357	L63 and L65	US-PGPUB; USPAT	OR	ON	2020/10/10 18:59
L69	37	L63 and L66	US-PGPUB; USPAT	OR	ON	2020/10/10 18:59
L70	2	L63 and L65 and L61	US-PGPUB; USPAT	OR	ON	2020/10/10 18:59
L71	154	L64 and L68	US-PGPUB; USPAT	OR	ON	2020/10/10 18:59
L95	14,285	(H02M1/08 or H01L24/26 or H05K1/144 or H05K1/0298 or H02M2001/0074).cpc.	US-PGPUB; USPAT	OR	ON	2020/10/10 18:59
L96	2,390	power adj conversion adj module	US-PGPUB; USPAT	OR	ON	2020/10/10 18:59

L97	2,926	interconnection adj module	US-PGPUB; USPAT	OR	ON	2020/10/10 18:59
L98	31	(plural\$3 or multiple) adj conversion adj cells	US-PGPUB; USPAT	OR	ON	2020/10/10 18:59
L99	0	L95 and L96 and L97 and L98	US-PGPUB; USPAT	OR	ON	2020/10/10 18:59
L100	0	L95 and L96 and L97	US-PGPUB; USPAT	OR	ON	2020/10/10 18:59
L101	74,134	power adj converter	US-PGPUB; USPAT	OR	ON	2020/10/10 18:59
L102	0	L96 and L97 and L98	US-PGPUB; USPAT	OR	ON	2020/10/10 18:59
L103	17	L96 and L97	US-PGPUB; USPAT	OR	ON	2020/10/10 18:59
L104	0	L95 and L101 and L97 and L98	US-PGPUB; USPAT	OR	ON	2020/10/10 18:59
L105	2	L95 and L101 and L97	US-PGPUB; USPAT	OR	ON	2020/10/10 18:59

10/10/2020 8:14:22 PM

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Application	Document	Mailroom Date	Attorney Docket No.
16218418	NOA	10/20/2020	00614-0183001
	INTV.SUM.APP	10/20/2020	00614-0183001

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Request for Continued Examination (RCE) Transmittal

Address to:
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P.O. Box 1450
Alexandria, VA 22313-1450

Application Number	16/218,418
Filing Date	December 12, 2018
First Named Inventor	Patrizio Vinciarelli
Art Unit	2849
Examiner Name	Cassandra F. Cox
Attorney Docket Number	00614-0183001

This is a Request for Continued Examination (RCE) under 37 CFR 1.114 of the above-identified application.

Request for Continued Examination (RCE) practice under 37 CFR 1.114 does not apply to any utility or plant application filed prior to June 8, 1995, or to any design application. See Instruction Sheet for RCEs (not to be submitted to the USPTO) on page 2.

1. **Submission required under 37 CFR 1.114** Note: If the RCE is proper, any previously filed unentered amendments and amendments enclosed with the RCE will be entered in the order in which they were filed unless applicant instructs otherwise. If applicant does not wish to have any previously filed unentered amendment(s) entered, applicant must request non-entry of such amendment(s).

a. ☐ Previously submitted. If a final Office action is outstanding, any amendments filed after the final Office action may be considered as a submission even if this box is not checked.

i. ☐ Consider the arguments in the Appeal Brief or Reply Brief previously filed on _____

ii. ☐ Other _____

b. ☒ Enclosed

i. ☐ Amendment/Reply

iii. ☒ Information Disclosure Statement (IDS)

ii. ☐ Affidavit(s)/ Declaration(s)

iv. ☐ Other _____

2. Miscellaneous

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a. ☐ period of _____ months. (Period of suspension shall not exceed 3 months; Fee under 37 CFR 1.17(i) required)

b. ☐ Other _____

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a. ☒ Deposit Account No. 06-1050

i. ☒ RCE fee required under 37 CFR 1.17(e)

ii. ☐ Extension of time fee (37 CFR 1.136 and 1.17)

iii. ☒ Other any deficiencies

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Signature	/Rex I. Huang/	Date	October 28, 2020
Name (Print/Type)	Rex I. Huang	Registration No.	57,661

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Electronic Patent Application Fee Transmittal

Application Number:	16218418			
Filing Date:	12-Dec-2018			
Title of Invention:	DELIVERING POWER TO SEMICONDUCTOR LOADS			
First Named Inventor/Applicant Name:	Patrizio Vinciarelli			
Filer:	Rex I. Huang/Mollie Kufner			
Attorney Docket Number:	00614-0183001			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
RCE- 2ND AND SUBSEQUENT REQUEST	1820	1	2000	2000
Total in USD (\$)				2000

Electronic Acknowledgement Receipt

EFS ID:	40975928
Application Number:	16218418
International Application Number:	
Confirmation Number:	1092
Title of Invention:	DELIVERING POWER TO SEMICONDUCTOR LOADS
First Named Inventor/Applicant Name:	Patrizio Vinciarelli
Customer Number:	26214
Filer:	Rex I. Huang/Mollie Kufner
Filer Authorized By:	Rex I. Huang
Attorney Docket Number:	00614-0183001
Receipt Date:	28-OCT-2020
Filing Date:	12-DEC-2018
Time Stamp:	17:49:12
Application Type:	Utility under 35 USC 111(a)

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Submitted with Payment	yes
Payment Type	DA
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RAM confirmation Number	E20200RH49364886
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IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

First Named Inventor :	Patrizio Vinciarelli	Art Unit :	2849
Application No. :	16/218,418	Examiner :	Cassandra F. Cox
Filed :	December 12, 2018	Conf. No. :	1092
Title :	DELIVERING POWER TO SEMICONDUCTOR LOADS		

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INFORMATION DISCLOSURE STATEMENT

Please consider the documents listed on the disclosure form submitted with this statement. Copies of the documents are being submitted with this statement to the extent required under 37 C.F.R. § 1.98(a)(2).

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Apply any necessary charges or credits applicable under 37 C.F.R. § 1.17(p) to deposit account 06-1050, referencing the above attorney docket number.

Please contact the undersigned if there are any questions regarding this statement.

Respectfully submitted,

Date: October 28, 2020 _____

/Rex I. Huang/ _____

Rex I. Huang
Reg. No. 57,661

Customer Number 26214
Fish & Richardson P.C.
Telephone: (617) 542-5070
Facsimile: (877) 769-7945

Substitute Disclosure Form U.S. Department of Commerce Patent and Trademark Office Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))	Attorney Docket No. 00614-0183001	Application No. 16/218,418
	Applicant VL T, Inc.	
	Filing Date December 12, 2018	Group Art Unit 2849

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Filing Date If Appropriate	Class	Subclass
	1.	4326765	4/27/1982	Brancaleone			
	2.	5471366	11/28/1995	Ozawa			
	3.	6147876	11/1/2000	Yamaguchi; Masayoshi			
	4.	6184585	2/6/2001	Martinez et al.			
	5.	6391685	5/21/2002	Hikita et al.			
	6.	6838754	1/4/2005	Kim			
	7.	7298038	11/20/2007	Filoteo et al.			
	8.	8288846	10/16/2012	Narendra et al.			
	9.	8896111	11/1/2014	Vinciarelli			
	10.	9107290	8/11/2015	Chen			
	11.	9571084	2/14/2017	Vinciarelli et al.			
	12.	9584026	2/28/2017	Vinciarelli et al.			
	13.	9660537	5/23/2017	Vinciarelli			
	14.	10701828	6/30/2020	Vinciarelli			
	15.	10757816	8/25/2020	Vinciarelli et al.			
	16.	20030227280	12/11/2003	Vinciarelli et al.			
	17.	20040157410	8/1/2004	Yamaguchi			
	18.	20050167814	8/4/2005	Beroz et al.			
	19.	20070158799	7/1/2007	Chiu; Chin-Tien			
	20.	20080116589	5/22/2008	Li et al.			
	21.	20090251873	10/1/2009	Chenq; Sun-Wen Cyrus			
	22.	20100072604	3/13/2010	Komatsu et al.			
	23.	20100172116	7/1/2010	Yorita			
	24.	20100246152	9/30/2010	Lin et al.			
	25.	20120287582	11/15/2012	Vinciarelli et al.			
	26.	20140152350	6/5/2014	Ikriannikov et al.			

Examiner Signature	Date Considered
EXAMINER: Initials citation considered. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.	

Substitute Disclosure Form U.S. Department of Commerce Patent and Trademark Office	Attorney Docket No. 00614-0183001	Application No. 16/218,418
Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))	Applicant VLT, Inc.	
	Filing Date December 12, 2018	Group Art Unit 2849

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Filing Date If Appropriate	Class	Subclass
	27.	20140218155	8/1/2014	Akre			
	28.	20150181719	6/25/2015	Vinciarelli et al.			
	29.	20150181727	6/25/2015	Vinciarelli			
	30.	20160302312	10/13/2016	Vinciarelli et al.			
	31.	20200253060	8/6/2020	Vinciarelli et al.			

Foreign Patent Documents or Published Foreign Patent Applications							
Examiner Initial	Desig. ID	Document Number	Publication Date	Country or Patent Office	Translation	Class	Subclass
	32.	WO 1995/027308	10/12/1995	WIPO			
	33.	WO 2012/155036	11/15/2012	WIPO			

Other Documents (include Author, Title, Date, and Place of Publication)		
Examiner Initial	Desig. ID	Document

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INTERNATIONAL APPLICATION PUBLISHED UNDER THE PATENT COOPERATION TREATY (PCT)

(51) International Patent Classification ⁶ : H01L 23/02, 23/28	A1	(11) International Publication Number: WO 95/27308 (43) International Publication Date: 12 October 1995 (12.10.95)
(21) International Application Number: PCT/US95/03468 (22) International Filing Date: 20 March 1995 (20.03.95) (30) Priority Data: 223,207 5 April 1994 (05.04.94) US (71) Applicant: OLIN CORPORATION [US/US]; 350 Knotter Drive, P.O. Box 586, Cheshire, CT 06410-0586 (US). (72) Inventors: PARTHASARATHI, Arvind; 142 Linsley Lake Road, North Branford, CT 06471 (US). HOFFMAN, Paul, R.; 5542 American Avenue, Modesto, CA 95346 (US). LIANG, Dexin; 3708 Glendower Court, Modesto, CA 95356 (US). MAHULIKAR, Deepak; 20 Martleshamheath Lane, Madison, CT 06443 (US). PASQUALONI, Anthony, M.; 219 Fairview Avenue, Hamden, CT 06514 (US). (74) Agents: ROSENBLATT, Gregory, S. et al.; Wiggin & Dana, One Century Tower, New Haven, CT 06508-1832 (US).		(81) Designated States: AM, AU, BB, BG, BR, BY, CA, CN, CZ, EE, FI, GE, HU, JP, KE, KG, KP, KR, KZ, LK, LR, LT, LV, MD, MG, MN, MW, MX, NO, NZ, PL, RO, RU, SD, SG, SI, SK, TJ, TT, UA, UZ, VN, European patent (AT, BE, CH, DE, DK, ES, FR, GB, GR, IE, IT, LU, MC, NL, PT, SE), OAPI patent (BF, BJ, CF, CG, CI, CM, GA, GN, ML, MR, NE, SN, TD, TG). Published <i>With international search report.</i>
(54) Title: CAVITY FILLED METAL ELECTRONIC PACKAGE (57) Abstract There is provided an electronic package (70) where the package components (52, 54) define a cavity (56). A semiconductor device (16) and a portion (18) of a leadframe (20) occupy part of the cavity (56). Substantially the remainder of the cavity (56) is filled with a polymer (26) so that an outermost surface (74) of the polymer (26) is coplanar with a surface (76) of one of the package components (54).		

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-1-

CAVITY FILLED METAL ELECTRONIC PACKAGE

This invention relates to an electronic package for housing at least one semiconductor device. More particularly, by filling the package cavity with a compliant polymer the seal width of an adhesive sealant is reduced. When the package cavity is at least partially filled with a polymer, the requirement of a separate cover component is eliminated. Also, the rigidity of certain of electronic packages is increased.

Integrated circuit devices, such as silicon based semiconductor circuits, are encased within a package to protect the device from mechanical shock. The packages are manufactured from a variety of materials, such as plastics, ceramics and metals.

U.S. Patent No. 4,939,316 to Mahulikar et al discloses an anodized aluminum alloy base and cover defining a cavity. A leadframe is disposed between the base and cover and bonded to both by an adhesive seal ring. An integrated circuit device is electrically interconnected to the leadframe and encased within the cavity. The assembled package is light weight, durable and efficiently dissipates heat generated by the device.

The aluminum alloy package components have a coefficient of thermal expansion significantly lower than the adhesive sealants, typically an epoxy, used to seal the package. When the packages are heated, thermally generated mechanical stresses can cause delamination between the leadframe and the adhesive. As the packages are heated or cooled, air trapped within the package cavity expands and contracts. The cavity has a fixed volume so the expansion and contraction of the trapped air is compensated as changes in air pressure. The pressure changes apply

-2-

a stress on the seal ring which could eventually lead to delamination.

Seal ring delamination results in a gross leak failure. One way to identify gross leak failure is
5 by immersing the package in an inert liquid fluorocarbon heated to 125°C typically for about 1 minute. A stream of bubbles emanating from the package indicates seal delamination and a passageway to the package cavity. Water vapor and other
10 contaminants can enter the package cavity leading to corrosion of the device or electric circuitry.

One method to prevent seal delamination is to increase the width of the seal path. This solution is contrary to a primary electronic package
15 manufacturing objective, namely to maximize the area available within the package cavity for housing the electronic device. One continuing goal of electronic packaging is to encapsulate the largest device possible within a given package outline.

20 As discussed below, Applicants increase the area available for the semiconductor device by filling the package cavity with a compliant polymer, such as a silicone gel. In U.S. Patent No. 4,961,106 to Butt et al a package cavity is filled
25 with a fluid or gel having good thermal conductivity, such as helium or hydrogen gas and silicone or silicate gels.

U.S. Patent No. 5,060,114 to Feinberg et al discloses a conformable gel pad inserted within the
30 cavity of a package. The gel pad has a compressible matrix made from silicone or polyurethane with a thermally conductive material dispersed throughout.

-3-

"Glop topping" is utilized in molded plastic packages as disclosed in U.S. Patent No. 4,965,227 to Chang et al. The package cavity is filled with a soft gel, for example silicone, to provide environmental protection. The glop top is used in molded packages because the molding resin is permeable to moisture. The glop top prevents corrosion of package components by inhibiting moisture diffusion.

While the addition of a compliant polymer to a package cavity has been disclosed for improving thermal conductivity and for providing a moisture barrier, compliant polymers have not, until now, been used to reduce the seal path width thereby increasing the size of the electronic device which can be encapsulated within a fixed package outline.

Additionally, the polymers have not been used to eliminate the need for a separate cover component in electronic packages having a discrete base component. Further, the use of the polymers to increase the rigidity of a metallic package has not been recognized.

Accordingly, it is a first object of the invention to provide an electronic package having an adhesive seal ring with a reduced width. A second object of the invention is to provide an electronic package in which the cavity defined by a base component and window frame is filled with a polymer such that a discrete cover component is not required. A further object of the invention is to provide an electronic package having metallic base and cover components with increased rigidity.

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It is a feature of the invention that the package cavity is filled with a compliant polymer to prevent the ingress of contaminants. It is a benefit of the invention that larger integrated circuit devices or multiple devices can be housed within a fixed package outline. It is a further benefit of the invention that the cavity is essentially contaminant free, facilitating the use of aluminum based bond wires rather than gold based bond wires. By filling the package cavity with a gel, the bond wires are held firmly in place. The bond wires do not move during mechanical shock or vibration reducing the chance of adjacent wires touching and creating an electrical short circuit. Also, vibration of the bond wires can cause the wire to work harden, become brittle and break.

Yet another benefit of the invention is that thermoplastic polymer sealants may be used in addition to thermosetting adhesives to assemble the package. By sealing the package lid with a thermoplastic adhesive, a sealed package may be subsequently opened for replacement or repair. As another benefit of the invention, the compliant gel may be selected to be ultraviolet light transparent and the package sealed with a UV transmitting glass lid for a package to house an erasable-programmable (EPROM) device.

In accordance with the invention there is provided an adhesively sealed electronic package. The package has a base and a cover defining a cavity. A leadframe electrically interconnected to an integrated circuit device is disposed between the base and cover with the device and a portion of the

-5-

leadframe positioned within the cavity. A compliant polymer fills substantially the remainder of the cavity. An adhesive having a reduced seal width bonds the leadframe to both the base and to the
5 cover.

In accordance with a second embodiment of the invention, there is provided an electronic package having a base and a window frame defining a cavity. A leadframe is disposed between and bonded to both
10 the base and to the window frame with a first adhesive. A semiconductor device is electrically interconnected to the leadframe such that the device and a portion of the leadframe occupy a portion of the cavity defined by the base and the window frame.
15 A polymer occupies substantially the remainder of the cavity.

In accordance with yet another embodiment of the invention, there is provided an electronic package having a base and a cover defining a cavity.
20 A leadframe is disposed between and bonded to both the base and the cover with a first adhesive. A semiconductor device is electrically interconnected to the leadframe and the device and a portion of the leadframe occupy a portion of said cavity defined by
25 the base and cover. A second adhesive is disposed within the cavity and bonds the semiconductor device to the cover.

The objects, features and benefits described above will become more apparent from the
30 specification and drawings which follow.

-6-

Figure 1 shows in cross sectional representation an adhesively sealed metal package as known from the prior art.

Figure 2 shows in cross sectional
5 representation an adhesively sealed metal package in accordance with a first embodiment of the invention.

Figure 3 shows in cross sectional representation an adhesively sealed window frame package in accordance with a second embodiment of
10 the invention.

Figure 4 shows in cross sectional representation a cavity filled window frame package in accordance with an embodiment of the invention.

Figure 5 shows in cross sectional
15 representation the window frame package of claim 4 further including a die attach paddle.

Figure 6 shows in cross sectional representation a thin metallic electronic package having increased rigidity in accordance with another
20 embodiment of the invention.

Figure 7 shows in cross sectional representation an embodiment of the electronic package of Figure 6 in which the bond wires are partially insulated to prevent contact with the
25 metallic cover.

Figure 1 shows in cross sectional representation a metal electronic package 10 as known from the prior art. The package 10 has a metallic base 12 and a cover 14. The cover 14 is
30 typically formed from the same material as the base 12. To maximize thermal conductivity and minimize package weight, the base 12 and cover 14 can be manufactured from aluminum or an aluminum alloy.

-7-

One suitable aluminum alloy is designated by the American Society of Metals (ASM) as A3003 and has the nominal composition by weight of 0.12% copper, 1.2% manganese and the balance aluminum. A3003 can
5 be imparted with a black color by integral color anodization as disclosed in U.S. Patent No. 5,066,368 to Pasqualoni et al.

An electronic device 16 is electrically interconnected to the inner leads 18 of a leadframe
10 20. A first adhesive 22 bonds the leadframe 20 to both the base 12 and to the cover 14. Tenacious adhesion between the first adhesive 22 and the other package components is required to withstand the stresses generated by the coefficient of thermal
15 expansion mismatch between the package components and the adhesive as well as by variations in the pressure of the air trapped within the package cavity 24. To prevent gross leak failures, the first adhesive is selected to be a strongly adherent
20 thermosetting polymer such as an epoxy.

The peripheral dimension of the package 10, designated "D", is dictated by the available space on a printed circuit board (not shown). This space is limited and due to the trend in the electronics
25 industry for standardization, "D" is usually fixed. The maximum size of the device 16 is limited by the size of the cavity 24. The only way to increase the size of the cavity 24 is to reduce the seal width, designated "W". However, when "W" is reduced below
30 a critical value, the mechanical stress on the adhesive induced by the combination of thermal expansion mismatch and the cavity pressure variations causes seal delamination. When the base

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12 and cover 14 are an aluminum alloy, the leadframe unplated copper and the sealant a novalac based epoxy, this critical value is about 3.8mm (0.15 inches).

5 Figure 2 illustrates in cross sectional representation an electronic package 30 in accordance with the invention. The package 30 has a base 12 and a cover 14. While both the base 12 and the cover 14 may be manufactured from any suitable
10 material, such as a plastic, ceramic, metal or composite, metals are preferred. Aluminum and aluminum alloys are most preferred due to the low weight and good thermal conductivity of the metals. Portions of the base 12 and cover 14 may be anodized
15 to increase corrosion resistance and to improve adhesion to the first adhesive 22. It is most preferred that the base 12 and cover 14 be formed from aluminum alloys of the ASM 3xxx and 6xxx series. 3xxx series aluminum alloys contain up to
20 about 1.5% by weight manganese along with other alloying elements while 6xxx alloys contain magnesium and silicon in an approximate proportion to form Mg_2Si . Both alloy series are capable of forming a gray to black color by integral color
25 anodization.

 The package 30 has a reduced seal width designated "RW". As used throughout the specification and claims herein, "RW", or reduced seal width, means a seal ring width less than the
30 critical value. The seal ring width is less than the width required to prevent thermally induced and pressure variation induced adhesive delamination in a package having an unfilled cavity. Since the

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peripheral dimension "D" does not change, reducing the seal width "RW" increases the area of the cavity 24. The mechanism by which "W" may be decreased to "RW" is by filling the cavity 24 with a compliant polymer 26.

5 Disposed between the base 12 and cover 14 is a leadframe 20. The leadframe is usually manufactured from copper or a copper alloy to maximize electrical conductivity. An integrated circuit device 16 is
10 electrically interconnected to inner lead portions 18 of the leadframe 20. Electrical interconnection may be by means of thin bond wires 28 or thin copper foil strips (not shown) as used in tape automated bonding (TAB). The thin bond wires may be
15 manufactured from any suitable metal such as gold, aluminum, copper or alloys thereof. The bond wires are protected from environmental corrosion by the compliant polymer 26 so it is not necessary to use expensive gold based bond wires. Similarly, it is
20 not necessary to coat the TAB leads with gold or another barrier layer when the compliant polymer 26 is utilized.

 The reliability of the bond wires 28 is further protected by the compliant polymer 26 by
25 being held rigidly in place. The bond wires do not move due to mechanical shock or vibration. Movement of the wires may cause two bond wires to touch creating an electrical short circuit. Vibration can work harden a wire causing the wire to become
30 brittle and break. The compliant polymer yields when stressed so there is not the problem of bond wire breakage due to coefficient of thermal expansion mismatch as occurs in molded plastic

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packages such as PDIPs (plastic dual in line packages) and PQFPs (plastic quad flat packages).

The integrated circuit device 16 may be free floating within the cavity 24 or bonded to the package base 12 by a compliant adhesive 31. For ease of assembly, the integrated circuit device 16 is usually bonded to a die attach paddle 32. That portion of leadframe 20 which includes the inner leads 18 and die attach paddle 32 as well as the semiconductor device 16 occupies a portion of the cavity 24. The first adhesive 22 bonds the leadframe 20 to both the base 12 and cavity 24.

Substantially the remainder of the cavity 24 is occupied by the compliant polymer 26. The compliant polymer 26 may be any suitable material such as a viscous liquid, a gel or a compliant plastic. Gelatinizable materials which are introduced into the cavity 24 in liquid form and subsequently coagulated into a gelatinous solid are preferred. While in the liquid phase, the gelatinizable compliant polymer 26 should have a sufficiently low viscosity to fill all corners and hidden features of the cavity 24. After coagulating to a gel, the viscosity should be sufficiently high to avoid cavity leakage through the seal ring. The coagulated gel should further be chemically inert with respect to the leadframe 20, integrated circuit device 16 and bond wires 28. Ideally, the compliant material 26 should have good thermal conductivity to aid in the removal of heat from the integrated circuit device 16. However, in metallic packages good thermal conductivity is not critical because most thermal dissipation is through the base 12.

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Preferred compliant polymers include liquids, gels and plastics such as gelatinazable silicones, fluorocarbons and epoxies. A most preferred polymer is a gelatinazable silicone manufactured by Dow
5 Corning, Midland, Michigan.

The compliant polymer while in the liquid phase is introduced to the cavity 24 through a first vent port 34 by any suitable means such as a large diameter hypodermic needle. Sufficient compliant
10 polymer is introduced to the cavity 24 such that substantially the entire cavity is filled. To remove air displaced from the cavity 24, a second vent port 36 is provided. The compliant polymer is added until there is sufficient compliant polymer
15 that both the first 34 and second 36 vent ports are at least partially filled. The compliant polymer, if gelatinizable, is coagulated into a gel and the vent ports sealed.

Once the cavity 24 is filled with the gel,
20 seal ring delamination is not a problem. Delamination is less likely because the air in the package has been displaced. The pressure variation induced stress on the first adhesive 22 during package heating and cooling has been eliminated.
25 Beneficially, should a portion of the first adhesive 22 delaminate, the result is not a gross leak failure. The flow of contaminants into the package cavity 24 is inhibited by the compliant polymer 26. Contaminants are unable to reach bond wires 28 or
30 semiconductor device 16.

One additional benefit realized is that because the bond wires 28 are fully protected from chlorides and other contaminants, aluminum or

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aluminum alloy bond wires may be utilized rather than the more chemically inert, but much more costly gold or gold alloys.

A second additional benefit which is realized is that the adhesive strength of the first adhesive 22 is less critical. When the package cavity is not filled, or only partially filled, any delamination of the first adhesive 22 leads to a gross leak failure. When the package cavity 24 is substantially filled, in accordance with the invention, delamination can be tolerated. The primary function of the first adhesive 22 changes from maintaining package hermeticity to merely holding the package components together. As a result, rather than a tenacious adhesive such as an epoxy being required, less tenacious adhesives such as thermoplastic polymer may be utilized.

Suitable thermoplastic polymers include those commonly referred to as "Hot-Melt" Adhesives. These adhesives, with the addition of suitable fillers, will remain solid up to about 175°C and can withstand momentary exposure to higher temperatures the package may achieve during soldering to a printed circuit board. As the temperature is increased above the melting temperature, the adhesives rapidly melt to a low viscosity fluid which rapidly sets when the temperature is lowered again. Because the adhesives are thermoplastic, the melting/resolidification process is repeatable. Hot melt adhesives include ethylene and vinyl acetate copolymers (EVA), polyvinyl acetate (PVA), polyethylene, amorphous polypropylene, block copolymers such as those based on styrene and

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elastomeric segments or ether and amide segments (ie. thermoplastic elastomers), polyamides and polyesters. EVA is one preferred hot melt adhesive.

5 A primary advantage of using a thermoplastic polymer as the first adhesive 22 is that the package can be reopened after sealing to perform device repair or replacement. Repair is not possible with a thermosetting polymer such as an epoxy.

10 Figure 3 illustrates in cross sectional representation a window frame package 50 in accordance with the invention. The package has a base which may be a metal plate 52, optionally containing a depressed central region for receiving the semiconductor device. The metallic plate 52 may
15 be manufactured from any suitable material and is preferably metallic, and most preferably aluminum or an aluminum alloy. The leadframe 20 with an electrically interconnected semiconductor device 16 is disposed between the metallic plate 52 and a
20 window frame 54. A first adhesive 22 bonds the leadframe 20 to both metallic plate 52 and the window frame 54. The window frame 54 is manufactured from a suitable material, such as plastic, metal or ceramic. Preferably, the window
25 frame 54 has a coefficient of thermal expansion about equal to that of metallic plate 52. Most preferably, the window frame 54 is manufactured from the same material as the metallic base plate 52.

30 After bonding the leadframe and electrically interconnecting the semiconductor device 16, the metallic plate 52 and window frame 54 define a cavity 56. The cavity 56 is filled with a compliant polymer 26 up to the sealing surface 58 of the

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window frame 54. Because the center of the window frame 54 is a large aperture, introduction of the compliant material may be by any suitable means such as injection from a small diameter tube. After the compliant polymer coagulates to a gel, a cover 60 is bonded to the seal surface 58 by a lidding adhesive 62. The lidding adhesive 62 is any suitable material such as a thermosetting polymer, a thermoplastic polymer or a solder. A thermoplastic polymer such as the hot melt adhesives (ie EVA) is preferred so that the cover 60 is removable. The cover 60 may be any suitable material such as a plastic, glass, ceramic or metal. To prevent thermally induced flexure of the package, in one preferred embodiment, the cover 60 is manufactured from the same material as the metallic plate 52.

In a second embodiment of the invention, the cover 60 is selected to be a glass capable of transmitting a certain wavelength of electromagnetic radiation. Exposure of certain devices to electromagnetic radiation erases the device memory, these devices are commonly referred to as EPROM devices. A preferred cover would be capable of transmitting ultraviolet (UV) light. One UV transparent cover is quartz (silicon dioxide). The selected compliant polymer should also be transparent to UV radiation. This package is superior to other EPROM packages. By filling the package cavity with the compliant polymer, changes in temperature and the corresponding pressure changes will not flex the glass cover which may distort the UV light path. The cover can be sealed

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with a thermoplastic adhesive permitting replacement in the event the glass is scratched.

The window frame package 50 of Figure 3 can encapsulate a larger semiconductor device 16 or a plurality of devices within a fixed package profile than was possible with prior art packages. The coagulated compliant polymer mitigates the destructive effect of delamination. The bond wires 28 may be manufactured from a less inert material than gold and the cover 14 may be sealed with a removable lidding adhesive 62 such as a thermosetting polymer.

In another embodiment of the invention, the cover 60 can be omitted. With reference to Figure 4, a window frame package 70 has a base 52 and a window frame 54 defining a cavity 56. A leadframe 20 is bonded to both the base 52 and a first surface 72 of the window frame 54. Typically, the leadframe is bonded with a first adhesive 22, such as a thermosetting epoxy.

A semiconductor device 16 is bonded to a central portion of the base 52. Bond wires 28 electrically interconnect the leadframe 20 to the semiconductor device 16. A portion of the cavity 56 is therefore occupied by the combination of the inner leads 18 of the leadframe 20, the bond wires 28 and the semiconductor device 16. Substantially the remainder of the cavity 56 is filled with a polymer 26. The polymer 26 fills the cavity 56 to sufficient height that the outermost surface 74 of the polymer 26 is substantially coplanar with the second surface 76 of the window frame. The combination of the outermost surface 74 and second

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surface 76 form the top surface of the electronic package 70.

While any polymer satisfying the requirements detailed above is suitable for this embodiment of the invention, namely, initially the polymer has a low viscosity to fill the package cavity 56 and the viscosity subsequently increases during a cure, epoxy potting compounds are preferred. The epoxies are more rigid after curing than silicones or similar materials and better resist deformation or penetration.

Preferably, the polymer 26 is filled with a thermally conducting, electrically insulating material, such as silicon nitride, to enhance the dissipation of heat from the semiconductor device 16. Additionally, the polymer 26 preferably has a black color to enhance the absorption of infrared radiation during soldering of the outer portions of the leadframe 20.

Figure 5 illustrates in cross sectional representation, another electronic package 80 of the invention. This package is similar to the electronic package 70 illustrated in Figure 4, but further includes a die attach paddle 82. The die attach paddle 82 is generally formed from the same material as the leadframe 20, typically copper or a copper alloy, and is located within the aperture defined by the inner leads 18. The semiconductor device 16 is bonded to the die attach paddle 82 by a die attach adhesive 84. Following electrical interconnection to the inner leads 18, the device/leadframe assembly is bonded to the package base 12 by pad attach adhesive 31. Preferably, the

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pad attach adhesive is cured at essentially the same time as the leadframe 20 is bonded to the base 12 and first surface 72 of the window frame 54. The cavity 56 defined by the base 12 and window frame 54 is then filled with a polymer 26, such as an epoxy, such that the outermost surface of the polymer 26 and the second surface 76 of the window frame 54 define the package 80 cover.

An ongoing goal of electronic packages is to reduce the thickness of the package. A target thickness is a maximum of 1.4mm (0.055 inch). This target thickness requires the package base and cover to have a thickness, in the region overlying or underlying the semiconductor device, of at most about .25mm (0.01 inch). When the base and cover are formed from metal, the thin package is susceptible to deflection. Exposure of the package to temperature variations, either during assembly or during use, leads to changes in the air pressure within the package cavity. Rather than these pressure changes leading to a gross leak through the first adhesive as occurs with conventional packages, in the thin packages, these pressure excursions are manifest as inward and outward deflection of the base and cover. Outward deflection can cause the base (or cover in a cavity down format) to contact a printed circuit board. Inward deflection can cause the lid to contact the bond wires.

The magnitude of deflection is highly dependent on the cavity size. The larger the cavity, the more severe the problem. Since increasing the cavity volume for a given package outline is a primary packaging goal, for thin metal

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packages to be successful, the rigidity must be increased.

Figure 6 illustrates in cross sectional representation a metal electronic package 90 having increased rigidity. The package has a metallic base 12 and cover 14. While the cover may be any material, preferably the cover has a coefficient of thermal expansion about equal to that of the base to prevent flexure of the package during heating and cooling. Preferably the base and cover are formed from copper, aluminum or alloys thereof, and most preferably from an aluminum alloy coated with an anodization layer.

A leadframe 20 is bonded to the base 12 and cover 14 by a first adhesive 22. The base 12 and the cover 14 define a cavity 24. A semiconductor device 16 is bonded to the base 12, either directly, or by means of a die attach paddle 32. The semiconductor device 16 is also bonded to the package cover 14 by a second adhesive 92. The second adhesive is any suitable polymer adhesive which is electrically non-conducting to avoid creating an electrical short circuit on the electrically active face of the semiconductor device 16. Also, the second adhesive should be noncorrosive to the electrically active face. A preferred polymer adhesive is an epoxy potting compound.

A controlled volume of the second adhesive is dispensed onto the semiconductor device prior to cover 14 sealing. The cover contacts the second adhesive 92 during sealing and is bonded both to the leadframe 20 by means of the first adhesive 22 and

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to the semiconductor device 16 by means of the second adhesive 92.

To egress the reaction products of the polymer cure, a vent hole 94 is provided. Since the central portion of the base 12 and cover 14 is quite thin, the vent hole is positioned over a thicker peripheral portion to facilitate assembly and sealing. The vent hole can be sealed with a drop of a polymer adhesive.

10 The electronic package 90 of Figure 6 eliminates deflection of the package base and cover by forming a rigid beam through the center of the package. However, due to the reduced thickness, the bond wires 28 are close to the cover 14. Contact with the cover could cause an electrical short circuit or cause a bond wire to deflect against an adjoining bond wire or break. These problems are eliminated by the embodiment illustrated in Figure 7. The package 100, illustrated in cross-sectional representation in Figure 7, has the same features as the package 90 of Figure 6, except that the second adhesive 92 extends beyond the perimeter of the semiconductor device 16 and encapsulates the apex 102 of the loop formed by wire bond 28. Preferably, the second adhesive encapsulates the entire wire bond 28, terminating somewhat past the point where the wire bond is joined to the inner lead 18. With this embodiment, the wire bonds are protected from corrosion, deflection and permanently electrically isolated from the cover 14.

The advantage of the electronic package of the invention will become more apparent from the Example

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which follows. The Example is intended to be exemplary and not limiting.

EXAMPLE

A series of metal electronic package were
 5 assembled with a novalac based epoxy film seal ring
 of a width as specified in Table 1. The packages
 had 28mm x 28mm (1.1 inch x 1.1 inch) anodized
 aluminum base and cover components sealed to a 208
 lead copper lead frame. The assembled packages were
 10 placed in a pressure cooker at 121°C, 100% relative
 humidity for 200 hours and then tested for gross
 leaks. As shown in Table 1, the packages had a
 critical value of about 3.8mm (.150 inches).

Table 1

15	Seal Ring Width	Gross leak Results Failures/Parts Tested
	1.27mm (.050 in.)	8/8
	1.91mm (.075 in.)	7/8
	2.54mm (.100 in.)	8/8
20	3.18mm (.125 in.)	6/8
	3.81mm (.150 in.)	0/8

By filling the package cavity with the
 compliant polymer in accordance with the invention,
 gross leak failures were eliminated permitting the
 25 assembly of packages with a seal ring width of less
 than 2.54mm.

To illustrate the advantage of reducing the
 seal ring width, the portion of the exemplary bases
 (unfilled cavity), seal ring dimensions (28mm x
 30 3.81mm Outside dimension; 20.4mm x 3.81mm Inside

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dimension) available for mounting a semiconductor device is:

$$\begin{array}{rclcl} \text{Area of Base} & \text{Area of Seal Ring} & = & \text{Available Area} & \\ (1) \quad 784\text{mm}^2 & - \quad 369\text{mm}^2 & & = & 415\text{mm}^2 \end{array}$$

5 By filling the cavity with a compliant polymer in accordance with the invention, the adhesive seal width can be reduced to below about 2.54mm and more preferably to the range of about 1.25mm to about 2.0mm. At the upper end of the preferred range,
10 seal width of 2.0mm, the area available for mounting the semiconductor device is:

$$(2) \quad 784\text{mm}^2 \quad - \quad 246\text{mm}^2 \quad = \quad 538\text{mm}^2$$

an increase of 123mm^2 or about 30% more area available for mounting the semiconductor device.

15 It is apparent that there has been provided in accordance with this invention an adhesively sealed electronic package having a reduced seal ring width which fully satisfies the objects, means and advantages set forth herein above. While the
20 invention has been described in combination with the embodiments thereof. It is evident that many alternatives, modifications and variations will be apparent to those skilled in the art in light of the foregoing description. Accordingly, it is intended
25 to embrace all such alternatives, modifications and variations as fall within the spirit and broad scope of the appended claims.

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IN THE CLAIMS

1. An electronic package (70), characterized by:

a base (52) and a window frame (54) having first (72) and second (76) opposing surfaces

5 defining a cavity (56);

a leadframe (20) disposed between said base (52) and said window frame (54);

a first adhesive (22) bonding said leadframe (20) both to said base (52) and to said
10 first surface (72) of said window frame (54);

a semiconductor device (16) electrically interconnected to said leadframe (20), wherein said semiconductor device (16) and a portion (18) of said leadframe (20) occupy a portion of said cavity (56);
15 and

a polymer (26) occupying substantially the remainder of said cavity (56) wherein an outermost surface (74) of said polymer (26) is substantially coplanar with said second surface (76) of said
20 window frame (54).

2. The electronic package (70) of claim 1 characterized in that said polymer (26) is selected from the group consisting of viscous liquids, gels and compliant plastics.

25 3. The electronic package (70) of claim 2 characterized in that said polymer (26) is an epoxy.

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4. The electronic package (70) of claim 3 characterized in that said epoxy (26) is filled with a thermal conductive, electrically insulating material.

5 5. The electronic package (70) of claim 4 characterized in that said base (52) and said window frame (54) are both formed from an aluminum alloy and at least partially coated with an anodization layer.

10 6. An electronic package (90, 100), characterized by:
 a base (12) and a cover (14) defining a cavity (24);
 a leadframe (20) disposed between said base
15 (12) and said cover (14);
 a first adhesive (22) bonding said leadframe (20) both to said base (12) and to said cover (14);
 an adhesive 31 bonding a semiconductor device (16) to said base (12), said semiconductor device
20 (16) electrically interconnected to said leadframe (20), wherein said semiconductor device (16) and a portion (18) of said leadframe (20) occupy a portion of said cavity (24); and
 a second adhesive (92) disposed within said
25 cavity (24) bonding the semiconductor device (16) to the cover (14) forming a rigid beam through the center of said package (90, 100).

 7. The electronic package (90, 100) of claim 6 characterized in that a vent hole (94) extends
30 through a peripheral portion of said cover (14).

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8. The electronic package (90, 100) of claim 7 characterized in that said second adhesive (92) is an epoxy.

9. The electronic package (90, 100) of claim 5 7 characterized in that said second adhesive (92) extends over at least a portion of one face of said semiconductor device (16).

10. The electronic package (90, 100) of claim 9 characterized in that said second adhesive (92) 10 extends beyond the perimeter of said semiconductor device (16) and encapsulates a bond wire (28) electrically interconnecting said semiconductor device (16) to said leadframe (20) through the apex (102) of the loop formed by said wire bonds (28) 15 terminating at a point where the wire bond (28) is joined to the inner lead (18).

11. The electronic package (90, 100) of claim 10 characterized in that said second adhesive (92) extends to encapsulate said portion (18) of said 20 leadframe (20).

12. The electronic package (90, 100) of claim 11 characterized in that the thickness of said electronic package (90, 100) is less than or equal to about 1.4mm.

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13. The electronic package (90, 100) of claim
12 characterized in that the thickness of the
portion of said base (12) underlying said
semiconductor device (16) is less than or equal to
5 about 0.25mm.

14. The electronic package (90, 100) of claim
12 characterized in that said base (12) and said
cover (14) are both formed from an aluminum alloy
and at least partially coated with an anodization
10 layer.

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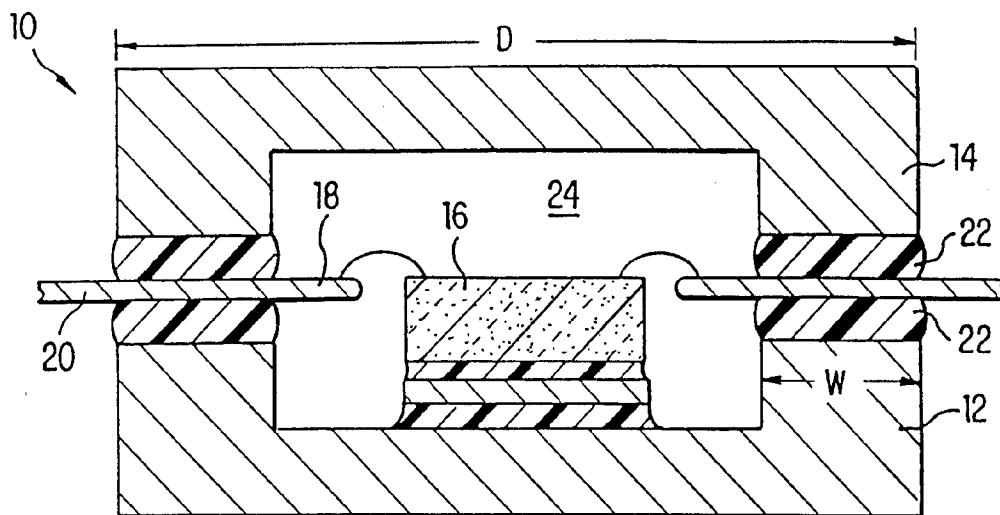


FIG. 1 PRIOR ART

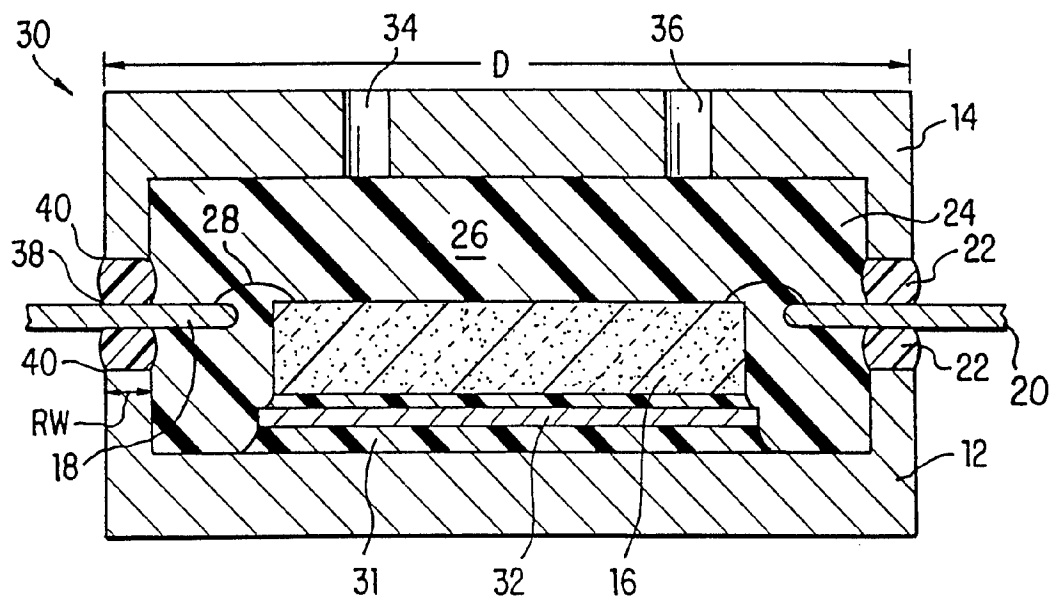


FIG. 2

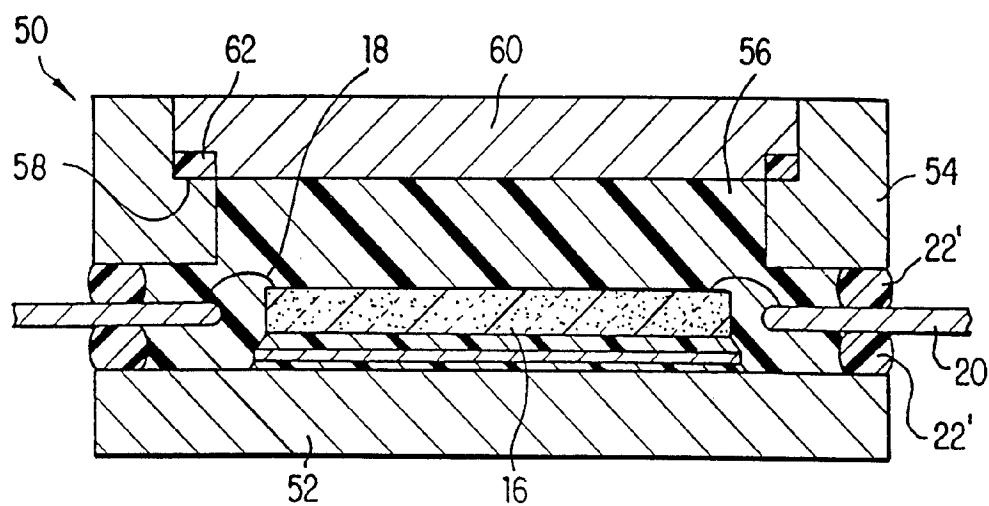


FIG. 3

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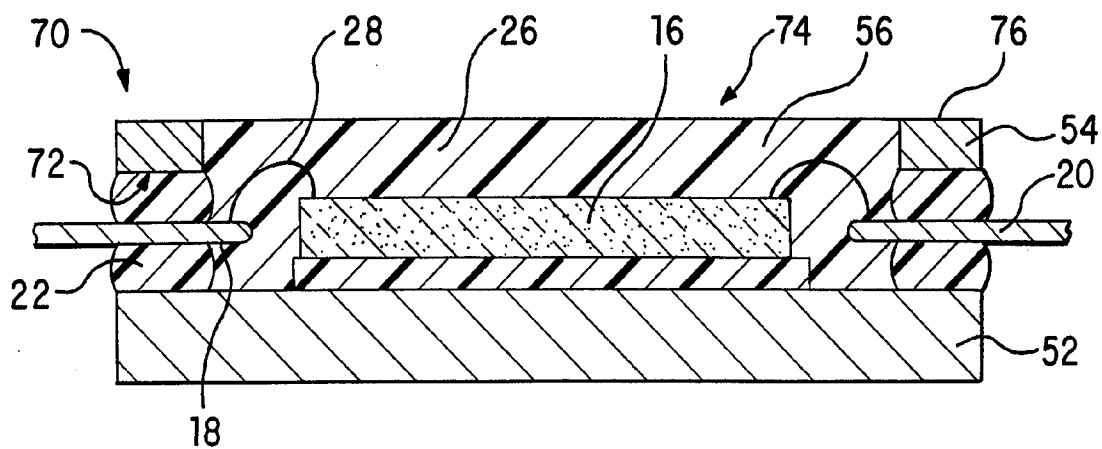


FIG. 4

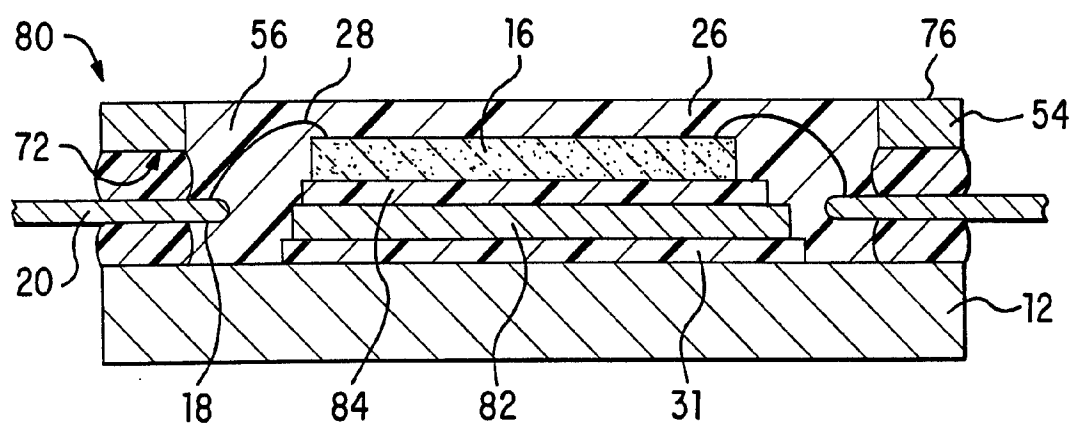


FIG. 5

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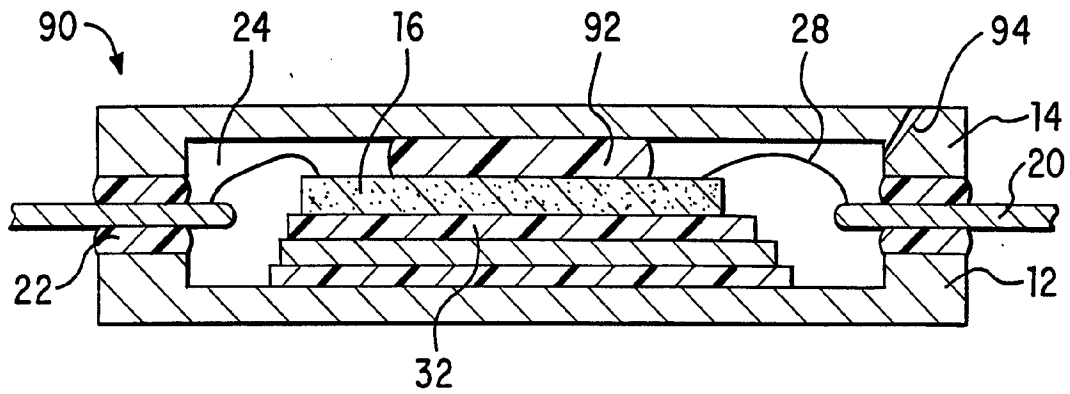


FIG. 6

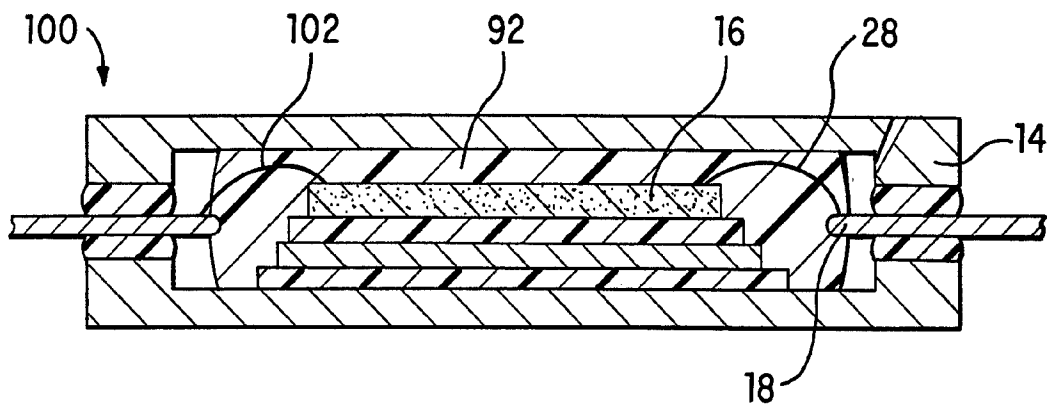


FIG. 7

INTERNATIONAL SEARCH REPORT

International application No.
PCT/US95/03468**A. CLASSIFICATION OF SUBJECT MATTER**

IPC(6) :HO1L 23/02, HO1L 23/28

US CL :257/687,792

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

U.S. : 257/668,687,676,693,698,709,710,787,793,791,792,433,434

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

NONE

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

APS

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
Y	US,A, 5,239,131, (HOFFMAN ET AL.), 24 AUGUST 1993, FIGURE 1	1-14
Y	US,A, 4,961,105 (YAMAMOTO), 02 OCTOBER 1990, FIGURE 3	1-14
Y	US,A, 5,098,864, (MAHULIKAR), 24 MARCH 1992, FIGURE 4	1-14
Y	JP,A, 5-267486, (MORI), 15 OCTOBER 1993, FIGURE 9	1-14
Y	US,A,4, 897,508, (MAHULIKAR ET AL), 30 JANUARY 1990 FIGURE 4	1-14

☐ Further documents are listed in the continuation of Box C.
 ☐ See patent family annex.

* Special categories of cited documents:	"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention
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"P" document published prior to the international filing date but later than the priority date claimed	

Date of the actual completion of the international search

07 JUNE 1995

Date of mailing of the international search report

19 JUN 1995

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(51) International Patent Classification:

H05K 7/00 (2006.01) *H05K* 5/00 (2006.01)
H01R 43/00 (2006.01) *B29C* 33/20 (2006.01)

(21) International Application Number:

PCT/US2012/037495

(22) International Filing Date:

11 May 2012 (11.05.2012)

(25) Filing Language:

English

(26) Publication Language:

English

(30) Priority Data:

13/105,696 11 May 2011 (11.05.2011) US

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(81) Designated States (unless otherwise indicated, for every kind of national protection available): AE, AG, AL, AM, AO, AT, AU, AZ, BA, BB, BG, BH, BR, BW, BY, BZ, CA, CH, CL, CN, CO, CR, CU, CZ, DE, DK, DM, DO, DZ, EC, EE, EG, ES, FI, GB, GD, GE, GH, GM, GT, HN, HR, HU, ID, IL, IN, IS, JP, KE, KG, KM, KN, KP, KR, KZ, LA, LC, LK, LR, LS, LT, LU, LY, MA, MD, ME, MG, MK, MN, MW, MX, MY, MZ, NA, NG, NI, NO, NZ, OM, PE, PG, PH, PL, PT, QA, RO, RS, RU, RW, SC, SD, SE, SG, SK, SL, SM, ST, SV, SY, TH, TJ, TM, TN, TR, TT, TZ, UA, UG, US, UZ, VC, VN, ZA, ZM, ZW.

(84) Designated States (unless otherwise indicated, for every kind of regional protection available): ARIPO (BW, GH, GM, KE, LR, LS, MW, MZ, NA, RW, SD, SL, SZ, TZ, UG, ZM, ZW), Eurasian (AM, AZ, BY, KG, KZ, RU, TJ, TM), European (AL, AT, BE, BG, CH, CY, CZ, DE, DK, EE, ES, FI, FR, GB, GR, HR, HU, IE, IS, IT, LT, LU, LV,

[Continued on next page]

(54) Title: PANEL-MOLDED ELECTRONIC ASSEMBLIES

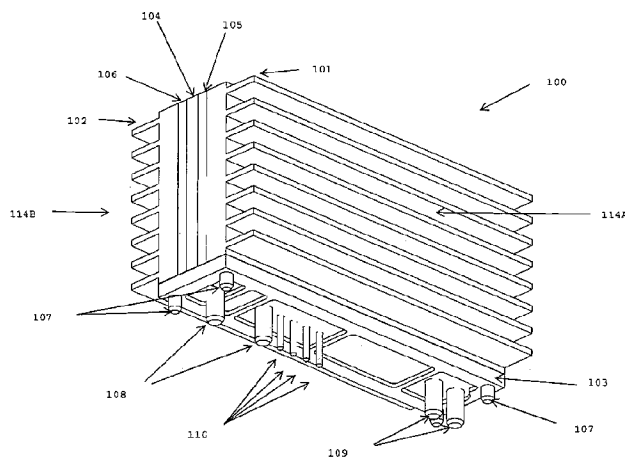


FIG. 1

(57) Abstract: A method of encapsulating a panel of electronic components such as power converters reduces wasted printed circuit board area. The panel, which may include a plurality of components, may be cut into one or more individual pieces after encapsulation. The mold may be used to form part of the finished product, e.g. providing heat sink fins or a surface mount solderable surface. Interconnection features provided along boundaries of individual circuits are exposed during the singulation process providing electrical connections to the components without wasting valuable PCB surface area. The molds may include various internal features such as registration features accurately locating the circuit board within the mold cavity, interlocking contours for structural integrity of the singulated module, contours to match component shapes and sizes enhancing heat removal from internal components and reducing the required volume of encapsulant, clearance channels providing safety agency spacing and setbacks for the interconnects. Wide cuts may be made in the molds after encapsulation reducing thermal stresses and reducing the thickness of material to be cut during subsequent singulation. External mold features can include various fin configurations for heat sinks, flat surfaces for surface mounting or soldering, etc. Blank mold panels may be machined to provide some or all of the above features in an on-demand manufacturing system. Connection adapters may be provided to use the modules in vertical or horizontal mounting positions in connector, through-hole, surface-mount solder

[Continued on next page]



MC, MK, MT, NL, NO, PL, PT, RO, RS, SE, SI, SK, SM, TR), OAPI (BF, BJ, CF, CG, CI, CM, GA, GN, GQ, GW, ML, MR, NE, SN, TD, TG).

Published:

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variations. The interconnects may be plated to provide a connectorized module that may be inserted into a mating connector. Re-useable plates may be used instead of the heat sink panels. Alternatively the panel may be encapsulated in and separated from a re-useable mold after curing.

PANEL-MOLDED ELECTRONIC ASSEMBLIES

CROSS-REFERENCE TO RELATED APPLICATION

[001] This is a continuation-in-part of U.S. Patent Application 13/105,696, filed on May 11, 2011, the entire disclosure of which is incorporated herein by reference.

FIELD OF THE INVENTION

[002] This invention relates to the field of encapsulating electronic assemblies and more particularly to encapsulated power converters.

BACKGROUND

[003] Contemporary electronic power systems require power converters capable of deployment at the point of load. Competing considerations require increasing power density, decreasing mounting area on customer motherboard, and lower cost.

[004] An encapsulated electronic module, such as an electronic power converter module for example, may comprise a printed circuit assembly over-molded with an encapsulant to form some or all of the package and exterior structure or surfaces of the module. Encapsulation in this manner may aid in conducting heat out of the over-molded components, i.e., components that are mounted on the printed circuit assembly and covered with encapsulant. In the case of an electronic power converter module, the printed circuit assembly may include one or more inductive components, such as inductors and transformers. Encapsulated electronic power converters capable of being surface mount soldered to a customer motherboard are described in Vinciarelli et al., *Power Converter Package and Thermal Management*, U.S. Patent No. 7,361,844, issued Apr. 22, 2008, (the "SAC Package Patent") (assigned to VLT, Inc. of Sunnyvale, CA, the entire disclosure of which is incorporated herein by reference). Encapsulated electronic modules having at least one surface of a magnetic core structure exposed and methods for manufacturing the same are described in Vinciarelli et al., *Encapsulation Method and Apparatus for Electronic Modules*, U.S. Patent Application Ser. No. 12/493,773, filed Jun.

29, 2009, (the “Exposed Core Application”) (assigned to VI Chip Inc. of Andover, MA, the entire disclosure of which is incorporated herein by reference).

[005] Methods of over-molding both sides of a printed circuit board assembly while leaving opposing regions on both sides of the printed circuit board free of encapsulant are described in Saxelby, et al., *Circuit Encapsulation Process*, U.S. Patent No. 5,728,600, issued Mar. 17, 1998 and Saxelby, et al., *Circuit Encapsulation*, U.S. Pat. No. 6,403,009, issued June 11, 2002 (collectively the “Molding Patents”) (both assigned to VLT, Inc. of Sunnyvale, CA and incorporated by reference in their entirety).

[006] Leads for connecting the encapsulated power converter substrate to the customer motherboard are described in Vinciarelli et al., *Surface Mounting A Power Converter*, U.S. Patent No. 6,940,013, issued Sep. 6, 2005 (the “J-Lead Patent”) (assigned to VLT, Inc. of Sunnyvale, CA, the entire disclosure of which is incorporated herein by reference).

SUMMARY

[007] In general, in one aspect, a method of making a plurality of electronic devices is provided. The method includes providing a plurality of mold panels, which when assembled, form an internal chamber; providing a substrate having a plurality of conductive traces and a plurality of components electrically connected to form at least one circuit, the substrate having at least one contact for making an electrical connection to the circuit; forming a panel assembly including the mold panels assembled with the substrate in the internal chamber and an encapsulant filling spaces between the substrate and interior surfaces of the chamber; curing the encapsulant; and cutting the panel assembly to expose at least a portion of the at least one contact and form a respective exposed contact.

[008] Implementations of the method may include one or more of the following features. The cutting can include making a first cut in at least one of the mold panels, and in the first cut, making a second narrower cut through the panel assembly. The curing

can include raising the temperature of the panel assembly and the first cut is made before the panel assembly cools after the curing. The cutting can divide the panel assembly producing at least one module having a plurality of layers including respective portions of each of the panel molds, the substrate, and the encapsulant, and the module can contain the at least one circuit. The cutting can define at least one side of the module. The cutting can define two or more sides of the module. The method can further include treating the exposed contact to protect against oxidation. Treating the exposed contact can include applying a removable conformal coating to the exposed contact. Treating the exposed contact can include applying a layer of metal to the exposed contact. The metal applied to the exposed contact can include solder or a precious metal. Applying a layer of metal can include plating. The method can further include providing an adapter having at least one electrical terminal; and attaching the at least one electrical terminal to the exposed contact. The at least one contact can include a plurality of contacts; the module can include a plurality of exposed contacts; the adapter can have a plurality of electrical terminals arranged to match respective ones of the plurality of exposed contacts; and the adapter can be mechanically secured to the module. The mold panels can be metal. The respective portions of the mold panels can provide heat sink surfaces for the module. Contours can be provided in an internal surface of one or more of the mold panels. Providing the contours can include matching depths of portions of the internal surface to heights of one or more selected components. The selected components can include at least one magnetically permeable core and at least one semiconductor device. The traces and components can be electrically connected to form a plurality of separable circuits arranged in a pattern on the substrate; and the cutting can divide the panel assembly along spaces between the separable circuits into a plurality of modules each containing at least one respective circuit. The spaces between the separable circuits can have dimensions approximately matching a width of cuts produced by equipment used to cut the panel assembly. The at least one contact can include a plurality of contacts located in the spaces between the separable circuits. The plurality of contacts can be formed in the substrate. The plurality of contacts can be formed and buried below at least one surface of the substrate. The method can include treating at least one exterior surface of at least one of the mold panels for solderability. Forming the panel assembly can include

dispensing encapsulant into a bottom panel mold; assembling the substrate into the bottom panel mold; dispensing encapsulant onto a top of the substrate; and assembling a top panel mold onto the substrate. The method can include centrifuging the assembly before curing the encapsulant. Forming the panel assembly can include assembling the substrate with a first substrate surface facing into a first panel mold, closing a second panel mold onto the substrate covering a second substrate surface, providing one or more conduits to the internal chamber, and forcing encapsulant through the one or more conduits into the chamber. The method can further include centrifuging the assembly before curing the encapsulant. The method can further include providing a connector for removably mating with and providing electrical connection to the metal. The method can further include providing a center plate having an opening to accommodate the substrate. Forming a panel assembly can include positioning the substrate in the opening of the center plate and closing the mold panels against the center plate. The method can further include providing at least one opening in the center plate connected to the internal chamber by at least one conduit; and forcing the encapsulant through the at least one opening and at least one conduit into the internal chamber. Forming a panel assembly can include securing the mold panels together prior to curing the encapsulant.

[009] In general, in another aspect, a method of forming an electrical contact is provided. The method includes assembling a panel including a substrate having one or more conductive features enclosed within the panel and unexposed to an exterior surface of the panel, the one or more conductive features being located along a cut line; cutting the panel along the cut line exposing portions of the one or more conductive features for use as electrical connections to the substrate; and treating the portions of the one or more conductive features exposed from the cutting for preservation as electrical connections.

[010] Implementations of the method may include one or more of the following features. The treating can include applying solder to the portions of the one or more conductive features exposed from the cutting. The treating can include applying a conformal coating to the portions of the one or more conductive features exposed from the cutting to protect against oxidation. The treating can include applying a metal layer to the portions of the one or more conductive features exposed from the cutting. The

treating can include soldering a lead of an adapter to the portions of the one or more conductive features exposed from the cutting. The method can further include covering at least one surface of the substrate in an area including the cut line and the conductive features prior to cutting the substrate. The covering can include encapsulating the substrate with a molding compound. The method can further include providing a registration feature having a predetermined relationship to the substrate, and using the registration feature to align the cutting relative to the cut line. The method can further include establishing a pattern including at least one conductive layer in the substrate along the cut line to form the conductive features. The substrate can include a multilayer printed circuit board, and the pattern can include a plurality of conductive layers that are established along the cut line to form the conductive features. The method can further include establishing a pattern including at least one conductive via in the substrate along the cut line to form the conductive features. The conductive via can be filled with a conductive material. The conductive via can be buried in the substrate. The conductive via can be a through hole contacting the surfaces of the substrate. The through hole can be filled with a conductive material. The conductive features can be covered at the surfaces of the substrate by an insulative layer.

[011] In general, in another aspect, an apparatus including a first mold panel is provided. The first mold panel includes an exterior surface, an interior surface defining an internal cavity, a clamp region located at points along a circumference of the internal cavity, and an opening. The first mold panel is adapted to (a) be engaged by pressure in the clamp region, (b) receive in the cavity a circuit panel containing a plurality of components and mold compound to fill empty spaces in the internal cavity, and (c) be cut after curing of the mold compound.

[012] Implementations of the apparatus may include one or more of the following features. The interior surface in the region of the internal cavity can be adapted to adhere to the mold compound. The apparatus can further include a second mold panel. The second mold panel can include an exterior surface, an interior surface, and a clamp region, in which the second mold panel can be adapted to (a) close against and mate with the first mold panel, (b) be engaged by pressure in the clamp region, and (c) be cut after

curing of the mold compound. The interior surface of the second mold can further define a second internal cavity, the clamp region of the second mold panel can be located at points along a circumference of the second internal cavity, and the internal cavity of the first mold panel can be adapted to receive a first side of the circuit panel and the second internal cavity can be adapted to receive a second opposite side of the circuit panel. The apparatus can further include contours formed in the interior surface of at least one of the mold panels. The contours can be adapted to match predetermined characteristics of selected ones of the plurality of components. The internal surface of the mold panel can be adapted to adhere to the molding compound. The contours can form interlocking features with cured mold compound. One or more of the mold panels can include a non-ferrous metal, aluminum, or a thermally conductive material. One or more of the mold panels can include a non-metallic substance. The apparatus can further include at least one channel connected to the internal cavity for allowing expansion of the mold compound. The apparatus can further include at least one channel connected to the internal cavity for injecting mold compound into the internal cavity. The exterior surface of the mold panel can include a plurality of fins. The exterior surface of the mold panel can include a flat surface. The flat surface can be adapted for a solder joint. The internal cavity can include features formed in the interior surface and arranged in a predetermined pattern. The internal cavity can include features formed in the interior surface along lines through which the mold panel maybe cut to establish a setback from a cut edge of the mold panel. The mold panel can include at least one registration feature adapted to engage and establish a predetermined relationship with the circuit panel. One or more selected portions of the mold panel can be adapted for incorporation into one or more products, the products being formed by a process which uses the mold panel to contain the molding compound. The exterior surface of the mold panel can include a flat surface with one or more pins protruding from the flat surface. The exterior surface of the mold panel can include a flat surface having one or more holes in the flat surface. The apparatus can further include a pin inserted into a respective one of the one or more holes.

[013] In general, in another aspect, an apparatus including a panel assembly having external surfaces defined by a first mold panel and a second mold panel is

provided. The first and second mold panels form an internal cavity enclosing an internal circuit board, the internal circuit board having a first surface and second surface and a plurality of components in an active circuit area on at least one of the surfaces, the components being electrically connected to interconnects contained within the internal cavity. The internal cavity is filled with mold compound in spaces unoccupied by the circuit board and components, and the panel assembly is adapted to have first selected portions of the first and second mold panels cut away to expose the interconnects and to have second selected portions of the respective mold panels situated near the active circuit area remain attached to the assembly following the cut.

[014] Implementations of the apparatus may include one or more of the following features. The panel assembly can include a clamp region located at points along a circumference of the first and second internal cavities, and the first selected portions can include the clamp region.

[015] In general, in another aspect, an apparatus including a panel assembly having external surfaces defined by a first mold panel and a second mold panel is provided. The first and second mold panels form an internal cavity enclosing an internal circuit board, the internal circuit board having a first surface and second surface and a plurality of components electrically connected to form a plurality of individual circuits, each individual circuit being electrically connected to respective interconnects located along a respective circuit perimeter, the interconnects being contained within the cavity.

[016] Implementations of the apparatus may include one or more of the following features. The panel assembly can be adapted to be cut along the circuit perimeter separating the individual circuits, dividing the panel assembly into individual circuit modules, and exposing selected portions of the interconnects. The panel assembly can be constructed to retain an integral layered structure after being filled with mold compound which is subsequently cured, the layered structure comprising a first layer including a portion of the first mold, a second layer including a portion of the mold compound, a third layer including a portion of the circuit board, a fourth layer including a portion of the mold compound, and a fifth layer including a portion of the second mold. The

internal circuit board can include a plurality of circuits having the same functionality. The apparatus can further include one or more conduits connecting the internal cavity to an external opening. Spaces unoccupied by the circuit board, components, and interconnects in the internal cavity can be filled with mold compound. The mold compound has been cured. Selected portions of at least one of the mold panels have been removed, the selected portions being located near the circuit perimeters.

[017] In general, in another aspect, an apparatus including a modular package is provided. The modular package has a first external surface, a second external surface opposite the first external surface, and a side wall extending along the perimeter of and connecting with the first and second external surfaces. The modular package includes a first layer defining the first external surface and a second layer defining the second external surface, the first and second layers being separated by and in contact with cured mold compound. An electrical circuit is located between the first and second layers and within the cured mold compound and including at least one electrical component electrically connected to a plurality of interconnects. The side wall includes a strip formed by the first layer, a strip formed by the second layer, and a strip formed by the cured mold compound. The interconnects are disposed within the side wall.

[018] Implementations of the apparatus may include one or more of the following features. The first and second layers can include a non-ferrous metal or aluminum. The first external surface can include a plurality of fins. The first external surface can include an essentially flat area. The essentially flat area can be adapted for a solder joint. The electrical circuit can include a circuit board, and the side wall can include a strip formed by the circuit board and an additional strip formed by cured mold compound. The interconnects can include conductive features in the circuit board. The circuit board can include a multilayer printed circuit board ("PCB") and each of the interconnects can include a plurality of conductive layers in the PCB. Each of the interconnects can include a plurality of conductive vias in the circuit board. The apparatus can further include interlocking features. The interlocking features can include a contour formed in an interior surface of the first layer, the contour being filled with cured mold compound. The circuit board can include a top surface and a bottom surface and the at least one

electrical component can include a set of top-side components mounted on the top surface and a set of bottom-side components mounted on the bottom surface. The top-side components can include a number, T, of large-footprint components, the bottom-side components can include a number, B, of large-footprint components, and the number T can be approximately equal to the number B. Each of most of the top-side large-footprint components can share a respective set of conductive vias with a corresponding one of the bottom-side large-footprint components. Each of most of the top-side large-footprint components is located in a respective footprint shared by a corresponding one of the bottom-side large-footprint components. The circuit board can include a top surface and a bottom surface, the at least one electrical component can include a number, T, of large-footprint components mounted on the top surface, and most of the large-footprint components can be distributed symmetrically in relation to an axis on the top surface. The axis can be along a midline of the top surface. The axis can be defined in relation to a predetermined component. The apparatus can further include an adapter for providing mechanical and electrical connections between the modular package and an external mounting surface, the adapter having a body and a plurality of electrical terminals supported by the body; and electrical connections formed between the adapter terminals and respective interconnects on the modular package. The adapter body can be mechanically secured to the modular package, and the adapter terminals can be arranged to mate with the external mounting surface. The external mounting surface can be a circuit board and the terminals can include ends constructed and arranged to be inserted into conductive holes in the circuit board. The external mounting surface can be a circuit board and the terminals can include ends constructed and arranged to be surface mount soldered to the circuit board. The external mounting surface can include a connector and the terminals can include ends constructed and arranged to mate with the connector. The interconnects can be disposed along a long edge of the modular package, the adapter can be secured to the long edge of the modular package, and the first and second layers of the modular package can be oriented perpendicular to the mounting surface. The interconnects can be disposed along opposite edges of the modular package, the adapter can be secured to the opposite edges, and the first and second layers of the modular package can be oriented essentially parallel to the mounting surface. The adapter can be

constructed and arranged to maintain one of the first or second layers in contact with the mounting surface. The interconnects can include a surface constructed and arranged for engagement with a connector terminal. The interconnects can include a layer of metal plating.

[019] In general, in another aspect, an apparatus includes a modular package having a first external surface, a second external surface opposite the first external surface, and a side wall extending along the perimeter of and connecting with the first and second external surfaces, the modular package including a first layer of metal defining the first external surface and a second layer of metal defining the second external surface, the first and second layers being separated by and in contact with cured mold compound; and an electrical circuit for converting power, including a printed circuit board ("PCB"), a plurality of components including semiconductors mounted to the PCB, and a plurality of interconnects electrically connected to the components, the electrical circuit being located between the first and second layers and within the cured mold compound. The side wall includes a strip formed by the first layer, a strip formed by the second layer, a strip formed by the PCB, and one or more strips formed by the cured mold compound. The interconnects are disposed within the side wall.

[020] Implementations of the apparatus may include one or more of the following features. The first and second layers can include aluminum, and the first external surface can include a plurality of fins. The first and second layers can include aluminum, and the first external surface can include an essentially flat area. The essentially flat area can be adapted for a solder joint. The PCB can include a top surface and a bottom surface, the semiconductors can include large-footprint switches. A number, T , of top-side large-footprint switches can be mounted on the top surface, a number, B , of bottom-side large-footprint switches can be mounted on the bottom surface, and the number T can be approximately equal to the number B . Each large-footprint switch can be connected to one or more other components by a respective set of conductive vias in the PCB, and each of a plurality of the top-side large-footprint switches can share its respective set of conductive vias with a corresponding one of the bottom-side large-footprint switches. Most of the large-footprint switches can be positioned on one surface in a location

substantially overlapping a location on the other surface occupied by another large-footprint switch. The apparatus of claim 151 wherein the PCB includes a top surface and a bottom surface, the semiconductors include large-footprint switches mounted on the top surface, most of the large-footprint components are distributed symmetrically between a left side of the top surface and an opposite right side of the top surface. The PCB can include a top surface and a bottom surface, the semiconductors can include large-footprint switches mounted on the top surface, and most of the large-footprint components can be distributed symmetrically between quadrants on the top surface. The components can include a transformer core. The quadrants can surround the transformer core. The PCB can include a top surface and a bottom surface, and the components can include a set of high-profile components having similar heights. A number, T, of the high-profile components can be mounted on the top surface, a number, B, of the high-profile components can be mounted on the bottom surface, and the number T can be approximately equal to the number B. A spatial distribution of the high-profile components on the top surface can roughly match a spatial distribution of the high-profile components on the bottom surface. The apparatus can further include an adapter for providing mechanical and electrical connections between the modular package and an external mounting surface, the adapter having a body and a plurality of electrical terminals supported by the body; and electrical connections formed between the adapter terminals and respective interconnects on the modular package. The adapter body can be mechanically secured to the modular package, and the adapter terminals can be arranged to mate with the external mounting surface. The external mounting surface can be a circuit board and the terminals can include ends constructed and arranged to be inserted into conductive holes in the circuit board. The external mounting surface can be a circuit board and the terminals can include ends constructed and arranged to be surface mount soldered to the circuit board. The external mounting surface can include a connector and the terminals can include ends constructed and arranged to mate with the connector. The interconnects can be disposed along a long edge of the modular package, the adapter can be secured to the long edge of the modular package, and the first and second layers of the modular package can be oriented perpendicular to the mounting surface. The interconnects can be disposed along opposite edges of the modular package, the adapter

can be secured to the opposite edges, and the first and second layers of the modular package can be oriented essentially parallel to the mounting surface. The adapter can be constructed and arranged to maintain one of the first or second layers in contact with the mounting surface. The interconnects can include a surface constructed and arranged for engagement with a connector terminal. The interconnects can include a layer of metal plating. The first layer can include contours formed in an interior surface of the first layer, the contours including a first feature having a shape and an elevation to accommodate a first component on the PCB, the first component having a height greater than or less than other components on the PCB. The first component can include a magnetic core structure and the elevation can be a recess in the interior surface. The first component can include a semiconductor switch and the elevation can be a protrusion from the interior surface. The apparatus can further include interlocking features having a contour formed in an interior surface of the first layer, the contour being filled with cured mold compound. The modular package can include a recess formed in the first layer adjacent one or more of the interconnects providing a setback between the first layer and the one or more interconnects.

[021] In general, in another aspect, an apparatus includes a power converter is provided. The power converter includes a printed circuit board ("PCB") having a plurality of conductive layers and having a top surface and a bottom surface; a magnetic core structure magnetically coupled to a winding formed by traces in one or more of the conductive layers in the PCB; and a plurality of power semiconductor devices. A first set of the power semiconductor devices is mounted on the top surface and electrically connected to dissipate power at a level, P_t , during operation of the converter, and a second set of the power semiconductor devices is mounted on the bottom surface and electrically connected to dissipate power at a level, P_b , during operation of the converter. The power semiconductor devices are distributed between the first and second sets to distribute heat generation during operation of the converter such that each level P_t , P_b is less than 150% of the other level P_b , P_t .

[022] Implementations of the apparatus may include one or more of the following features. A plurality of the power semiconductor devices in the first set can each be

positioned in a location on the top surface substantially overlapping a location on the bottom surface occupied by a power semiconductor device in the second set. The power semiconductor devices can be electrically connected using a respective set of conductive vias in the PCB, and a plurality of the power semiconductor devices in the first set can share their respective sets of conductive vias with corresponding power semiconductor devices in the second set. The power converter can include circuitry having a pair of cells that have a common circuit topology and each including power semiconductor switches from each of the first and second sets. Each cell can have its respective components arranged in a pattern, in which the pattern of components of one cell is substantially a mirror image of the pattern of components in the other cell. A component from one of the cells can be located on an opposite surface of a respective component from the other one of the cells. The cells can include input cells. The power semiconductor devices can include output switches.

[023] In general, in another aspect, a method of manufacturing a plurality of products is provided. The method includes inserting a plurality of components into a cavity formed by one or more molds; closing the one or more molds to form a seal around the cavity; filling the cavity with mold compound; curing the mold compound in the cavity to secure the components, cured mold compound, and molds together into an assembly; and cutting the assembly to separate the plurality of products from the assembly, the products each including a respective section of the one or more molds which remains as an integral part of each respective product.

[024] Implementations of the method may include one or more of the following features. The method can further include maintaining a predetermined alignment between the plurality of components and the one or more molds. The plurality of components can include a substrate having conductive features. The cutting can include exposing portions of the conductive features in each respective product.

[025] In general, in another aspect, a method of making a plurality of electronic devices is provided. The method includes providing a PCB panel including a PCB having a plurality of conductive traces and a plurality of components electrically

connected to form a plurality of circuits, the PCB having a plurality of contacts for making an electrical connections to the plurality of circuits; enclosing the PCB panel, including the PCB, components, and contacts in a mold cavity; filling the mold cavity with mold compound; curing the mold compound; removing the encapsulated PCB panel from the mold; and cutting the encapsulated PCB panel to expose at least a portion of the plurality of contacts and form respective exposed contacts.

[026] Implementations of the method may include one or more of the following features. The method can include mating the PCB panel with a center plate having an opening for receiving the PCB panel; covering the opening of the center plate with at least one plate; and wherein the enclosing comprises closing the mold against the center plate and exerting pressure to force the at least one plate against the center plate; wherein the removing comprises removing the at least one plate from the encapsulated assembly. The method can include mating the PCB panel with a center plate; and wherein the enclosing comprises closing the mold against the center plate; wherein the curing bonds the PCB panel to the center plate; wherein the removing comprises removing the bonded centerplate and PCB panel. The method can include providing one or more registration features for positioning the panel assembly in the mold cavity; creating voids in the mold compound with one or more of the registration features by displacing uncured mold compound for reference after the encapsulated PCB panel is removed from the mold. The method can include making holes in the cured mold compound to expose portions of the PCB. The method can include removing a layer of material from a first surface of the encapsulated panel assembly. The method can include removing a layer of material from a second surface of the encapsulated PCB panel. The components can include a magnetic core and the removal of a layer of material may expose a surface of the magnetic core. The removal of a layer of material from the top surface of the encapsulated panel assembly can provide finished surfaces and reduce the encapsulated panel assembly to a controlled thickness. The cutting can divide the substrate producing at least one module having a plurality of layers including respective portions of the substrate and the encapsulant on each side of the substrate, and the module can contain at least one of the circuits. The method can include providing an adapter having at least one electrical terminal, and attaching the at least one electrical terminal to the exposed contact. The at

least one contact can include a plurality of contacts; the module can include a plurality of exposed contacts; the adapter can have a plurality of electrical terminals arranged to match respective ones of the plurality of exposed contacts; and the adapter can be mechanically secured to the module. The traces and components can be electrically connected to form a plurality of separable circuits arranged in a pattern on the PCB; and the cutting can divide the PCB panel along spaces between the separable circuits into a plurality of modules each containing at least one respective circuit.

[027] In general, in another aspect, an apparatus includes a modular package having a first external surface, a second external surface opposite the first external surface, and a side wall extending along the perimeter of and connecting with the first and second external surfaces, the first and second external surfaces each being defined at least in part by a respective layer of cured mold compound; and an electrical circuit for converting power, including a printed circuit board ("PCB"), a plurality of components including semiconductors mounted to the PCB and a plurality of interconnects electrically connected to the components, the electrical circuit being located within the cured mold compound. The side wall includes a strip formed by the PCB between strips formed by the layers of cured mold compound defining the first and second external surfaces; and the interconnects are disposed within the side wall.

[028] Implementations of the apparatus may include one or more of the following features. The electrical circuit can include a magnetic core, and at least one of the first and second external surfaces can also be defined by a surface of the magnetic core. The apparatus can include an adapter for providing mechanical and electrical connections between the modular package and an external mounting substrate, the adapter having a body and a plurality of electrical terminals; and electrical connections formed between the adapter terminals and respective interconnects on the modular package. The adapter body can be mechanically secured to the modular package and covering the electrical connections; and the adapter terminals can be arranged to mate with the external mounting substrate.

[029] In general, in another aspect, a method of manufacturing a plurality of products is provided. The method includes inserting a printed circuit board including a plurality of components into a mold cavity; closing the mold to form a seal around the cavity; filling the cavity with mold compound; curing the mold compound in the cavity to form an encapsulated panel assembly; and cutting the encapsulated panel assembly to separate the plurality of products from the encapsulated panel assembly, the products each including a respective section of the printed circuit board.

[030] In general, in another aspect, an apparatus includes a panel assembly having external surfaces defined at least in part by cured mold compound; the panel assembly including an internal circuit board, the internal circuit board having a first surface and second surface and a plurality of components electrically connected to form a plurality of individual circuits, each individual circuit being electrically connected to respective interconnects located along a respective circuit perimeter, the interconnects being contained within the cured mold compound.

[031] Implementations of the apparatus may include the following feature. The panel assembly can be adapted to be cut along the circuit perimeter separating the individual circuits, dividing the panel assembly into individual circuit modules, and exposing selected portions of the interconnects.

[032] The details of one or more embodiments of the invention are set forth in the accompanying drawings and description below. Other features, objects, and advantages of the invention will be apparent from the description and drawings, and from the claims.

DESCRIPTION OF DRAWINGS

[033] FIG. 1 shows an electronic module 100.

[034] FIG. 2 shows a panel assembly 120 comprising a plurality of electronic modules before singulation.

[035] FIG. 3 shows an exploded view of the panel assembly revealing the heat sink panels 121 and 122 and internal PCB panel 124.

[036] FIG. 4 shows the panel assembly 120 with the top heat sink panel 121 removed.

[037] FIG. 4A shows an exploded view of the PCB panel and lower heat sink panel in a modified panel assembly 120.

[038] FIG. 4B shows a cross-section of the panel assembly 120A along lines 4B-4B in Fig. 4A.

[039] FIG. 5 shows a plan view of the PCB panel 124 assembled with the bottom heat sink panel 122.

[040] FIG. 6 shows a cross-section of the panel assembly 120.

[041] FIG. 7 shows a partial cross section of the panel assembly closed in a mold.

[042] FIG. 8 shows another partial cross section of the panel assembly closed in a mold.

[043] FIG. 9A shows a plan view of the panel assembly 120 following singulation.

[044] FIGs. 9B, 9C show optional channels formed in the panel before singulation.

[045] FIGs. 10A, 10B show single modules 115, 115B after singulation.

[046] FIG. 11 shows an enlarged portion of the module 115 revealing details of exposed buried-embedded interconnects.

[047] FIG. 12 shows a cross section of the singulated module 115 of Fig. 10A.

[048] FIG. 13 shows a connector assembly.

[049] FIG. 14 shows an exploded view of the connector assembly.

[050] FIG. 15 shows a horizontal through-hole mount module 200.

[051] FIG. 16 shows an exploded view of the through-hole mount module 200.

[052] FIG. 17 shows a horizontal surface-mount module 300.

[053] FIG. 18 shows an exploded view of the surface-mount module 300.

[054] FIG. 19 shows an alternative horizontal surface-mount module 400.

[055] FIG. 20 shows a top view of the surface-mount module 400.

[056] FIG. 21 shows an exploded top view of module-connector set 500.

[057] FIG. 22 shows an exploded view of the connector set 503.

[058] FIG. 23 shows an exploded bottom view of module-connector set 500.

[059] FIG. 24 shows an isometric view of the module-connector set 500 assembled.

[060] FIG. 25 shows an alternative horizontal through-hole flush-mount module 600 exploded from a customer PCB.

[061] FIG. 25A shows an alternative horizontal through-hole flush-mount module 650 exploded from a customer PCB.

[062] FIG. 26 shows the horizontal through-hole flush-mount module 600 assembled onto a customer PCB.

[063] FIG. 27 shows top and bottom plan views of a section of a PCB illustrating symmetry of component layouts.

[064] FIG. 28 shows an exploded perspective view of a panel assembly 720 including a manifold plate.

[065] FIG. 29 is an exploded side view of the panel assembly 720 showing the PCB mated with the manifold plate.

[066] FIG. 30 shows a top plan view of the panel assembly 720.

[067] FIG. 31 shows a top plan view of the panel assembly 720 with the top heat sink panel removed.

[068] FIG. 32 shows a side view of the panel assembly 720 closed in a mold.

[069] FIG. 33 shows a top perspective view of a singulated module.

[070] FIG. 34 shows a bottom perspective view of a singulated module.

[071] FIG. 35 shows a top exploded perspective view of a singulated module and connectors.

[072] FIG. 36 shows a bottom exploded perspective view of a singulated module and connectors.

[073] FIG. 37 shows an exploded perspective view of a PCB panel assembly including a manifold plate.

[074] FIG. 38 is an exploded side view of the PCB panel assembly with the PCB mated with the manifold plate.

[075] FIG. 39 shows a top plan view of the PCB panel assembly.

[076] FIG. 40 shows a side view of the PCB panel assembly closed in a mold.

[077] FIG. 41 shows a top plan view of the encapsulated PCB panel assembly and manifold plate.

[078] FIG. 42 shows a cross-sectional view of the encapsulated panel assembly.

[079] FIG. 43 shows a bottom (as molded) isometric view of an encapsulated PCB panel.

[080] FIG. 44 shows a top (as molded) isometric view of an encapsulated PCB panel.

[081] Like references symbols in the various drawings indicate like elements.

DETAILED DESCRIPTION

I. Vertical PCB Package.

[082] Referring to Fig. 1, an electronic module 100, e.g. a power converter, is shown having a generally rectangular form factor with two large faces 114A, 114B covered by heat sinks 101, 102. As shown between the heat sinks 101 and 102, the module 100 includes a printed circuit board ("PCB") 104 having its large faces arranged generally coplanar to the two large faces 114A, 114B of the electronic module 100. Electronic components (Figs. 3, 5, 6) may be mounted to one or both sides of the PCB 104 and electrically interconnected, e.g. by conductive traces on or in the PCB 104 to form the module circuitry. Using a power converter as an example, the electronic components may include power transistors, control ICs, and discrete resistors and capacitors. One or more magnetic core structures may be provided, which in combination with conductive traces on PCB 104, may form planar magnetic components such as inductors and transformers.

[083] The electronic components may protrude from one or both sides of the PCB 104 to varying degrees depending upon component size. Spaces between the faces of PCB 104 and the components on the PCB on one hand and the interior surfaces of the heat sinks on the other hand may be filled with molding compound, which when cured may form integral structural layers 105, 106 as shown in Fig. 1 and further provides a thermally conductive medium in which heat may be readily conducted away from the PCB and components to the heat sinks 101, 102. The interior surfaces of the heat sinks may be contoured to match the height of one or more of the components while maintaining an appropriate clearance for insulation and safety agency requirements. Contouring the heat sinks 101, 102 in this way: (1) to match the height of the magnetic

core structure may be used as an alternate approach to the exposed core encapsulation method described in Vinciarelli, *Encapsulation Method and Apparatus for Electronic Modules*, U.S. Patent Application, Serial No. 12/493,773 filed June 29, 2009 (assigned to VI Chip Corp. of Andover, MA, the entire disclosure of which is incorporated herein by reference); (2) to match the height of lower profile components, such as power semiconductors, may (a) increase thermal performance in the case of heat dissipating components by replacing molding compound with heat sink metal; and (b) reduce cost generally by reducing the volume of molding compound required; and (c) further reduce cost by allowing less expensive molding compound to be used because of reduced thermal pathways through the encapsulant, easing the thermal conductivity requirements of the encapsulant (e.g., an encapsulant having a 1 degree Celsius per watt thermal resistance may be used with the contoured heat sink instead of an encapsulant having a 3 degrees Celsius per watt thermal resistance used without the contoured heat sink).

[084] A connector 103, including terminals 108, 109, 110 and standoffs 107, may be provided as shown along an edge of the PCB 104 to make electrical connections between the electronic module 100 and external circuitry. As shown in Fig. 1 with the connector 103 situated along one edge of the module 100, preferably one of the longest edges, the module 100 may be mounted vertically, i.e. with its internal PCB 104 perpendicular to a chassis or another circuit board such as a motherboard. Using the vertical mount module construction illustrated in Fig. 1 for a power converter may provide advantages over the more conventional horizontal mounting technique. For example, using the vertical PCB arrangement may allow use of a magnetic core structure that is thicker than in a horizontal PCB configuration, e.g. because of height restrictions, enabling increased power throughput, as compared to a similar converter using a horizontal PCB orientation. The length of the magnetic path may be also reduced in the vertical PCB configuration further reducing losses in the magnetic components. Shorter windings may also be used further reducing transformer or inductor losses. Further details and variations of, and a process for making, the electronic module will be discussed below in connection with a panel molding process.

II. Panel Molding Process Integrated Mold

A. Overview

[085] The electronic module 100 shown in Fig. 1 may be fabricated using a panel molding process described with reference to Figs. 2- 9. The panel molding process may be used to produce a multiplicity of modules at a time. A PCB panel 124 may be provided with a plurality of individual circuits for building the electronic modules. Figs. 3, 4, and 5 show the PCB panel 124 populated with electronic components revealing a 3-by-4 pattern of twelve circuits to make twelve individual modules 115 (labeled 115A through 115L in Fig. 5). The illustrative example of Figs. 3-5, being for power converters, includes magnetic core structures 131 (Fig. 3) in addition to electronic components 132. As shown in Fig. 5, the pattern of individual circuits 115A-115L are arranged close together and separated by small spaces 135 preferably sufficient to allow the PCB panel to be cut during the singulation process without necessitating two cuts between modules or unnecessary waste of PCB material. The spacing may be adjusted based upon the cut dimensions produced by the equipment used to make the cuts.

[086] The PCB panel 124 containing the multiplicity of the electronic circuits (115A-115L in Fig. 5) may be assembled with matching heat sink panels 121, 122 as shown in Fig. 3 to form a panel assembly 120 (Fig. 2). As shown in Figs. 3 and 6, the two heat sink panels 121 and 122 when assembled together may form an internal cavity 146, which completely encloses the populated PCB panel 124. Figs. 2 and 6 show that the heat sink panels 121 and 122 may be pressed together (e.g. by a mold press 161, 162 as shown in Fig. 7) to form a seal 123 (Figs. 2, 6-8) around the perimeter of the internal cavity 146. In this example, the heat sink panels 121 and 122 also function as mold panels by forming a mold cavity (e.g., the internal cavity 146) that may be filled at least in part by an encapsulant encapsulating the surfaces of the PCB panel 124 and the electronic components on the PCB panel 124.

B. Heat Sink Panels

[087] Referring to Fig. 2, the heat sink panels 121, 122 may include fins on the exterior surfaces as shown. The fins may be arranged in any direction relative to the panel or in any pattern and may vary in height, thickness, and spacing as required for the particular application. For example, modules 115 and 115B shown in Figs. 10A and 10B respectively illustrate longitudinal and transverse fin orientations. Alternatively, one or both of the heat sink panels may have a generally flat exterior surface omitting the fins altogether. For example, Figs. 17 and 25 show modules 315 and 615 produced when both heat sink panel exterior surfaces are flat and one is flat and one is finned, respectively. The thickness of the panels between the internal cavity and the external surface may be varied to suit the particular requirements of the application.

C. Heat Sink Internal Contours

[088] Referring to Fig. 3, the contoured interior surface of the bottom heat sink panel 122 is shown including a 3-by-4 repeating pattern of twelve (A through L) prominent recesses 141 and shallower recesses 142 and 143. Recesses 141 may be matched to the downward protruding portion of magnetic core structures 131. Similarly, recesses 142 and 143 may be matched to other downwardly protruding components on the PCB panel 124. Note that in the example of Fig. 3, because the core structure 131 protrudes from the PCB 124 more than the other components, the recesses 141 are deeper than recesses 142, 143. Although not visible in Fig. 3, the interior surface of heat sink panel 121 may similarly include contour features to match the components and core structure on the upward facing side of the PCB panel 124 (e.g. as shown in the cross-section of Figs. 6, 7). Although the interior contour of panel 122 in Fig. 3 is shown including three recesses 141, 142, 143 repeated for each circuit (A through L in the 3-by-4 pattern), the interior contours of the heat sink panels 121 and 122 may range from simple flat surfaces (accommodating the height of the tallest component) to a complex arrangement of a multiplicity of recessed and protruding features (accommodating a multiplicity of component heights) which at the extreme could match every component individually.

[089] Additional features may be provided in the heat sink panels to facilitate the panel molding process, to enhance the mechanical integrity or performance of the finished module 100, or to satisfy safety agency clearance requirements for the finished product. By way of example undercut features, such as undercuts 148 shown in Figs. 6, 7, and 12 may be provided at each circuit site (i.e. within each individual module location) in the heat sink panels 121, 122. As shown the undercuts 148 may be provided in selected recesses, such as recesses 143-1 and 143-2, and may be dispersed along one or more of the boundary lines of each circuit 115. During encapsulation, molding compound fills the recesses and trenches and their respective undercuts 148. When cured, the hardened molding compound in the undercuts forms a dovetail-like joint securing the heat sinks to the encapsulated PCB 124. When provided at each circuit site, the undercuts secure the heat sinks 101, 102 to the PCB 104 in the individual module 115 providing mechanical integrity after singulation. Referring to the cross-section of a singulated module 115 in Fig. 12, dovetail interfaces 149-2 and 149-1 are shown securing the top and bottom heat sinks 101, 102 to the encapsulated PCB assembly.

[090] Additionally, clearance features may be provided in the heat sink panels to satisfy minimum safety agency clearances between electrical contacts on the PCB 124 and the metal heat sinks 121, 122. As shown in Figs. 6 and 7, trenches 147 may be provided in heat sink panels 121 and 122 along the side of each module 115 (the long side as shown in Figs. 6 and 7) where electrical contacts (discussed in more detail below) are exposed in or on the PCB 124. The trenches 147 may also include the undercut features 148 discussed above. For example, cut trench 147 in Fig. 12 results in a recess 150 of the heat sink 101 away from the edge 104E of the PCB 104 after singulation.

D. PCB-Heat Sink Registration

[091] Registration features may be provided in one or both of the heat sink panels 121, 122 helping to correctly position the PCB panel 124 in the cavity 146 relative to the heat sink panels 121, 122 (which is particularly important when the panel is cut during the singulation process) and to correctly position the heat sink panels relative to each other during assembly and during subsequent molding processes. Referring to Figs. 3

and 4, beveled corners 144 may be provided in heat sink panel 122 to interface with matching indentations 133 which may be provided in PCB panel 124 for registration when assembled together (Figs. 4 and 5). Fig. 5, which is a top plan view of the PCB panel 124 assembled with the bottom heat sink 122, shows the beveled corners 144 interfacing with the indentations 133.

[092] Referring to Fig. 4A, a modified version of the assembly is shown. As shown, a registration pin 151 is press fit into a registration hole 152 in the lower heat sink panel 122 and a matching registration hole 134 is provided in the PCB panel 124. The completed panel assembly, including the lower heat sink panel 122, PCB panel 124, and upper heat sink panel 121, is shown in Fig. 4B in cross-section taken along the broken lines 4B-4B in Fig. 4A. As shown in Fig. 4B, the registration pin 151, which fits snugly in registration holes 152 and 134, provides registration for the PCB panel 124 relative to the lower heat sink panel 122. An indentation 153 may be provided in the opposite heat sink panel (121) to accommodate protrusion of the pin 151 past the PCB 124. As shown generally at 154 in Fig. 4B, the heat sink panels may include additional features to provide registration between the top and bottom heat sink panels. Although a registration pin is shown at one corner of the panel 122 in Fig. 4A, it will be appreciated that additional pins may be used at other locations. For example, Fig. 4 shows two registration holes 134, one on a corner and another in the middle of the opposite side of the PCB panel 124.

E. Encapsulation

[093] Fig. 6 shows a cross-sectional view (through lines 6-6 in Fig. 5) of the panel 120 through an opening 125 and a conduit 145 in the heat sink panels. The openings 125, which may be slot shaped as shown (Figs. 2, 3, 4, 5, 9), may be connected to the interior cavity 146 of the panel assembly 120 by conduits 145 for conveying molding compound or venting during the panel molding process. The openings may be formed in one of the heat sink panels, e.g. openings 125 in the top heat sink panel 121 as shown in Figs. 2, 3, 4, 5, and 9, or in both heat sink panels, e.g. openings 125B in the top and bottom heat sink panels 121, 122 as shown in Figs. 6 and 7. Referring to Fig. 6, the conduits 145 may

be formed by recesses in the interior surfaces of heat sink panels 121 and 122, connecting the openings 125 to the interior cavity 146. The recesses forming the conduit 145 may be situated near the edge 124C of PCB panel 124 to allow the molding compound to flow over both top 124A and bottom 124B surfaces of the PCB 124.

[094] Fig. 7 shows an enlarged cross-section of one end of the panel assembly 120 closed between an upper mold press 161 and lower mold press 162 taken through some of the smaller components, e.g. components 132-1, 132-2, i.e. through lines 6-6 in Fig. 5. A channel 163 may be provided, e.g. between the upper mold press 161 and lower mold press 162 as shown, to interface with openings 125. Molding compound may be forced through the channel 163 under pressure after the panel assembly 120 is closed in the mold presses 161, 162. The dashed line 167 with directional arrows in Figs. 7 and 8 illustrates the flow of molding compound through the channel 163 into openings 125 through conduits 145 over the top 124A and bottom 124B surfaces of the PCB panel 124 during encapsulation. The molding compound may be forced into the internal cavity 146 to fill all of the unoccupied spaces between the heat sinks 121, 122 and the PCB panel 124.

[095] Fig. 8 shows an enlarged cross-section of one end of the panel assembly 120 closed between an upper mold press 161 and lower mold press 162 taken through the magnetic core structures, i.e. through lines 8-8 in Fig. 5. Each magnetic core 131 is shown having an upper E-core 131-2 and a lower E-core 131-1 separated by a gap 131-3. The cores have openings 131-4 to accommodate windings formed on the PCB 124. As shown, the lower core 131-1 and upper core 131-2 are accommodated by recesses 141-1 and 141-2 in the lower heat sink 122 and upper heat sink 121, respectively. As shown, compliant pads 136 may be provided on the surfaces of the cores 131-1 and 131-2 or in recesses 141-1, 141-2 to accommodate dimensional differences in the components. The compliant pads may be chosen for good thermal conduction and optionally adhesive properties facilitating heat removal from the cores into the heat sink while optionally providing structural integrity to the assembly. Gap Pad A2000 available from the Bergquist Company, 18930 West 78th St, Chanhassen, MN 55317 is one example of the type of compliant pad that may be used. Alternatively and perhaps depending upon the

tolerances involved, a phase change material may accommodate the dimensional difference during assembly and when cured afterward provide structural integrity between the heat sinks and the cores.

[096] Molding compound may be deposited in one or more of the recesses, e.g. recesses 141-1 or 141-2, in the internal cavity prior to assembly of the PCB panel 124 into one or both of the heat sink panels 121, 122 to ensure that the molding compound fills narrow spaces between the heat sink and the components, e.g. the cores 131-1, 131-2. One or more vent openings (not shown) may be provided in the heat sink panels, preferably at an end opposite the fill openings 125, to allow the molding compound to flow completely through the internal cavity 146. The cavity may be filled with encapsulant either (1) by transfer through one or more conduits, (2) by measured deposition of encapsulant during assembly of the panels, or (3) by both measured deposition during assembly and transfer through conduits.

[097] As shown in Figs. 4, 5, and 6, the border areas between the modules and around the periphery of the PCB panel 124 may be minimized to avoid wasted PCB material. Because the heat sink panels 121 and 122 close against each other rather than the PCB panel 124, areas on the PCB normally reserved for closing the mold may be eliminated further reducing PCB waste and thus overall cost.

[098] After the panels are assembled together and the interior cavities are filled with molding compound, the panel assembly 120 may be cured, e.g. by elevating the temperature.

F. Singulation

[099] Singulation is the process by which individual modules, e.g. singulated module 115 in Fig. 10A, are separated from the panel assembly 120, e.g. by cutting. The panel assembly 120 may be singulated after the molding compound is cured. The panel assembly 120 is separated from the upper and lower mold presses 161, 162 and may be cut, e.g. along lines 128, 129 as shown in Fig. 9A, to singulate the modules 115A-115L. For example, a narrow saw may be used to cut through the layers (e.g. as shown in Figs.

1, 10A, 10B, 12) of panel 120, which may include heat sink 101, cured molding compound 105, PCB 104, cured molding compound 106, and heat sink panel 102. For example, a 0.025 inch thick saw such as model number EAD-3350 available from Disco Corp., Ota-ku, Tokyo, Japan may be used. Referring to the cross-sectional views of Figs. 6 and 7, dashed lines 129 illustrate the lines along which the panel 120 may be cut to singulate modules 115D, 115E, and 115F. As shown, the cuts 129 (along the long side of the individual modules 115) go through the middle of the trenches 147. The trenches 147 may be made wide enough and the cuts 129 may be made at an appropriate distance from the edges of the trench 147 to ensure that a minimum thickness of cured molding compound remains in the trench after singulation to be mechanically robust. The trenches 147 may also be used to reduce the thickness of metal through which the saw must cut during singulation, e.g. by optionally providing trenches along the perimeter of each module.

G. Electrical Connections

[0100] Interconnection features may be embedded in the PCB panel 124, preferably along the boundaries of the individual circuits, so that electrical contacts are at least in part formed by or exposed during singulation. To maximize the area on a PCB panel 124 available for circuitry, the interconnection features may preferably be buried in the PCB panel 124 below the top and bottom surface layers. For example, the interconnection features may be formed in the inner conductive layers but not occupying valuable area on the surface conductive layers, potentially reducing setback and other spacing requirements. The interconnect features therefore are preferably formed in the PCB panel 124 before the panel 120 is assembled and exposed when the panels are cut, e.g. during singulation. The interconnection features may comprise a pattern of conductive layers or buried vias (frequently used to form connections between internal conductive layers) or both in the PCB situated along the circuit boundary, i.e. lying in the cut line, e.g. cut line 129 (Fig. 9A).

[0101] Referring to Fig. 10A, a module 115 is shown after singulation. Exposed interconnects 111, 112, and 113 are shown embedded in the edge of the PCB 104 along

one of the edges 116 of the module 115. As shown in Fig. 10A, longer connections 111 and 113 provide greater interconnect surface area and higher current carrying capacity, making them amenable for use as power connections, than shorter connections 112 which have less interconnect area and lower current carrying capacity, making them useful for control signal connections. Fig. 11 shows an enlarged view of a portion of the module 115 along edge 116 revealing detail of the two interconnections 111 after singulation. The width along the boundary and number of the conductive features arranged vertically through the PCB layers may be adjusted to provide the requisite contact area for each connection. In the example of Fig. 11, conductive features, e.g. conductive lands 111A through 111L, formed along the cut line on a plurality of the conductive layers of the PCB form a stack of conductive strips resembling a bar code after singulation. The stack of conductive lands may provide more contact area than possible with a single buried via. In Fig. 11, each interconnect 111 is made up of twelve lands, each land being formed on a respective one of twelve inner conductive layers. Note that Fig. 11 shows the twelve lands (111A through 111L) buried below the top 124A and bottom 124B surfaces of the PCB, i.e. not occupying surface area on those layers. During singulation, the saw cuts through the buried embedded lands, exposing the edges of the remaining conductive material forming the interconnects 111A through 111L shown in Fig. 11. Note that the buried embedded lands may be shared between two adjacent modules 115 on the panel 120, e.g. where the PCB patterns are laid out in mirror image to each other, allowing the modules to be laid out on the PCB panel close to a singulation cut width apart. Also, the interconnect features may be aligned along a single module boundary as shown or may occupy two or more boundaries of each individual module in the panel. The exposed edges may then be used to form electrical connections immediately after singulation, i.e. before the cut interconnects oxidize or may be protected against oxidation, e.g. with a conformal coating, such as an organic solderability preservative ("OSP"), applied after singulation, to ensure subsequent ability to form electrical connections to the edges. For example, Entek Plus HT available from Enthone, Inc., a Division of Cookson Electronics, West Haven, CT 06516 may be used as an OSP to protect the interconnects.

[0102] The number of lands, i.e. conductive layers, used to form each interconnect may be increased for better electrical connections or reduced for less critical connections.

Although embedded conductive lands are shown in the example of Figs. 10A, 11, additional or alternative conductive features may be used to form the interconnects. For example, buried conductive vias located along the cut line may be used, either alone or in combination with the lands. The buried vias may be located so that the singulation cut leaves the walls of the conductive vias exposed resulting in vertical conductive strips, i.e. generally perpendicular to the PCB top and bottom surfaces 124A, 124B, in the PCB edge. Buried conductive vias may tend to fill with adhesive during fabrication of the PCB panel 124. Empty conductive vias, i.e. free of non-conductive adhesives, may be preferable for the resulting concave conductive features, i.e. embedded conductive half cylinders. Similarly, solid buried conductive vias or conductive vias filled with conductive material during the PCB manufacturing process may be preferable for the resulting continuous flat conductive surface.

[0103] In the alternative to buried conductive features, conductive through-holes, which are generally free of adhesives following fabrication of the PCB panel 124 may be used along the cut line to form interconnects extending through the thickness of the PCB from top surface to bottom surface providing generally half-cylindrical interconnects. A penalty of using through-holes for the interconnects is the loss of surface area on the top and bottom of the PCB which may otherwise be used for setback or other safety and agency approval requirements. Preferably, the through-holes may be filled, for example with a conductive material such as solder or silver paste, or conductive pins may be inserted into through holes and soldered to the through holes, to prevent molding compound from filling the through hole during the encapsulation process and to provide a greater contact surface area yielding generally flat conductive interconnects following singulation.

[0104] The exposed interconnect features, e.g. 111A through 111L may be used to make a variety of electrical connections. For example, the exposed interconnects may be solder plated and then subsequently soldered to a motherboard, e.g. using surface mount soldering techniques. The interconnects may be soldered to a connector, such as connector 103 shown in Figs. 1, 13, 14. Alternatively, a lead frame or PCB may be soldered to the exposed interconnects 111, 112, 113 of the module quickly after

singulation. Yet another alternative includes a precious metal such as gold or silver or other suitable conductive material deposited to the exposed contacts, e.g. by plating, to build up the contacts into a larger area, e.g. continuous conductive contacts, well suited to connectorized modules described in more detail below.

G1. Vertical Mount Connector

[0105] Referring to Figs. 1, 13 and 14, a connector 103 suitable for attachment to the edge 116 of the module 115 is shown having a connector body 118 with a plurality of recesses 117 and a plurality of holes 119A, 119B, and 119C. A plurality of connector pins 108, 109, 110 may be inserted into holes 119A, 119B, 119C, respectively, from the top 118A. As shown, the holes may be contoured to provide a pressure fit having a gripping force suitable for retaining the pins. The broad top surfaces 108A, 109A, 110A of the pins are suitable for making solder connections to the exposed interconnects 111, 113, 112, respectively on the edge 116 of the PCB 104 of module 115. As shown, the recesses 117 provide countersinking for the top portions of the pins allowing the connector body to mount flush to the edge 116 of module 115 and allowing space for a solder joint between the pin and the respective interconnect. The interconnection features may preferably be provided along a long edge of the module 115 yielding a more stable vertical mount module 100 such as shown in Fig. 1. However, the interconnection features may be deployed along any or all of the edges of the module 115 for different mounting configurations as discussed more fully below.

G2. Horizontal Through-Hole Mount

[0106] Referring to Figs. 15 and 16, a horizontal mount component 200 suitable for through hole mounting in a motherboard is shown including a singulated module 215 and through-hole adapters 203A, 203B. The horizontal mount module 200 may be constructed in the same manner as described above for the module 100, except that the interconnects are preferably disposed along two edges of the module PCB and connectors adapted for horizontal mounting may be used. As shown in Fig. 16, the interconnects are disposed along the two shorter edges of the PCB 204 in module 215. However, the longer edges may be used instead of, or in addition to, the shorter edges for the

interconnects. Although only one set of the interconnects 211, 212 is visible in the perspective view of Fig. 16, it will be understood that a second set, including two additional power interconnects, is disposed along the opposite hidden edge of the PCB.

[0107] Through-hole adapters 203A and 203B, suitable for attachment to the edges of the singulated module 215 are shown having adapter bodies 218 supporting conductive terminals 208, 210 and 209, respectively. A portion of each terminal may be exposed on an internal surface via an opening in the adapter body optionally providing a small recess. In Fig. 16 for example, adapter 203B is shown having two power terminals 209, each having an exposed areas 209A recessed in openings in surface 218A of the adapter body 218. The exposed areas 209A align with their respective interconnects when the adapters 203A, 203B are assembled onto the module 215. The recesses provide countersinking for the exposed terminals allowing the internal surface 218A of the adapter body 218 to mount flush to the edge of module 215 and allowing space for a solder joint between the terminal and the respective interconnect. The adapter body 218 may include a flange 218B which may form a pressure fit with the adjacent edges of the modules 25. Additional features may be provided for maintaining the structural integrity of the module and connectors.

[0108] As shown in Fig. 16, epoxy may be deposited along an internal edge of the connector body, e.g. in the shaded area 219 preferably aligned with the encapsulant layer 206 to secure the adapters 203A, 203B to the module 215. The horizontal mount module 200 may be readily adapted to match industry standard brick footprints for power converters, in particular the module 200 may fit within the standard 1/8th brick footprint.

G3. Horizontal Surface-Mount

[0109] Referring to Figs. 17 and 18, a horizontal-mount component 300 suitable for surface-mount soldering to a motherboard is shown including a singulated module 315 and surface-mount adapters 303A and 303B. The horizontal-mount module 300 may be constructed in the same manner as described above for the module 200 (Figs. 15, 16) substituting surface mount adapters 303A, 303B for through-hole adapters 203A, 203B. As shown in Fig. 17, the singulated module 315 may have flat heat sinks 301, 302 instead

of the finned heat sinks (shown in Figs. 1-4, 6-12, 15-16). Although shown disposed along the two shorter edges in Figs. 17 and 18, the interconnects may be deployed along the longer edges of the PCB 304 in the singulated module 315 instead of, or in addition to, the longer edges.

[0110] In Figs. 17 and 18, the surface-mount adapters are shown with smaller bodies 318 than the through-hole adapters (218: Fig. 15, 16) exposing the connections between the terminals and their respective interconnects during assembly and for post assembly inspection. The terminals 308, 309, 310 each may include a portion, e.g. solder pads 308A, 310A, 309A, adapted for connection, such as a solder joint, to a respective interconnect on the module, e.g. interconnect 311, 312, and 313 (not visible in Figs. 17 and 18). Holes 308B, 309B, 310B may be provided in solder pads 308A, 309A, 310A for better solder joints. Each terminal 308, 309, 310 may include a bend (e.g. 308C) to produce a surface-mount pad (e.g. 308D) for attachment, e.g. by surface-mount soldering, to a customer motherboard.

[0111] The adapter bodies 318 may include flanges 318B, preferably along two or more sides to form a pressure fit with the adjacent edges of the modules 315. Additional features may be provided for maintaining the structural integrity of the module and adapters. As shown in Fig. 18, epoxy may be deposited along an internal edge of the connector body, e.g. in the shaded area 319 preferably aligned with the encapsulant layer 306 to secure the adapters 303A, 303B to the module 315.

G4. Surface-Mount Lead Frame

[0112] An alternate embodiment of a horizontal-mount component 400 suitable for surface-mount soldering to a motherboard is shown in Figs. 19 and 20 including a singulated module 415. The horizontal-mount module 400 may be constructed in the same manner as described above for the module 300 (Figs. 17, 18) substituting lead frame adapter 403 for the surface-mount adapters 303A, 303B. Like module 315 (Fig. 17), the singulated module 415 may have flat heat sinks 301, 302 instead of the finned heat sinks of the previous examples. However, in the example of Figs. 19 and 20, the interconnects are shown deployed along the longer sides of the singulated module 415.

However, as noted above the interconnects may be deployed along any edges of the PCB 404 as desired in the singulated module 415.

[0113] The surface-mount adapter is shown in Figs. 19 and 20 having a unitary rectangular frame-like body 418 supporting a plurality of terminals 408. The profile of the frame body 418 may as shown leave a portion of the terminals exposed for making connections to their respective interconnects during assembly and for post assembly inspection. The terminals 408 each may include a portion, e.g. solder pad 408A, adapted for connection, such as a solder joint, to a respective interconnect on the module (not visible in Figs. 19, 20). Although shown without holes in Fig. 19, the solder pads may optionally include holes such as those shown in Figs. 17 and 18. Each terminal 408 may include a bend (e.g. 408C) to produce a surface-mount pad (e.g. 408D) for attachment, e.g. by surface-mount soldering, to a customer motherboard.

[0114] The opening in the frame body 418 may be sized to accommodate the perimeter edges of the singulated module 415 and optionally form a pressure fit. The frame body 418 may include recesses 417 for accommodating the terminals 408, allowing the interior surface 418A of the frame body 418 to rest flush against the module 415 surface. Additional features may be provided for maintaining the structural integrity of the module and adapters. Gaps may be provided in the interior surface 418A to allow the application of epoxy to secure the frame body 418 to the module 415.

G5. Connectorized Module

[0115] The modules 100, 200, 300, and 400 discussed above in connection with Figs. 1, and 15-20 are all examples in which connectors or adapters are mechanically and electrically connected to the interconnects on the singulated modules forming an integral modular component. Yet another option is to adapt the module to be removably mated with a connector that may be mounted on a customer circuit board. For example, the module interconnects may be plated up with an appropriate conductive material, such as silver or gold, to form contacts that may be reliably engaged with connector contacts, i.e. "connectorized." Referring to Figs. 21 and 22, a module-connector set 500 is shown

including a connectorized module 515 in exploded view with a mating connector 503 into which the module 515 may be removably inserted.

[0116] The connector 503 as shown includes a body 518 having side walls 518B creating an opening 518C adapted to receive the connectorized module 515. Terminals 508 formed, e.g. bends 508B (Fig. 22), to provide a pressure fit between a contact area 508A of each terminal 508 and a respective interconnect 511. The terminals as shown may be retained in recesses 517 in the interior surface 518A of the side walls 518B. The recesses 517 may provide support to keep the terminals 508 in place and allow the interior surfaces to engage the surfaces of the module 515. The terminals 508 may include a flat portion 508D (Figs. 22, 23) adapted for making a solder connection to surface contacts on a customer circuit board (not shown). The connector body may include a bottom surface 518D enhancing the structural integrity of the connector walls 518B which are subjected to the forces exerted by the terminals 508 against the interconnects 511. The bottom 518D may include openings as shown in Fig. 23 through which the terminals may be inserted during assembly of the connector 503. The bottom 518D may provide electrical insulation between the metal heat sink 502 and the customer circuit board (not shown) on which the connector 503 may be mounted. Alternatively, the bottom may be partially or completely removed to allow better conduction of heat from the module 515 out through the customer circuit board. Yet another alternative is to use a thermally conductive material in the bottom 518D of the connector 503. Fig. 24 shows the connectorized module 515 inserted into the connector 503.

[0117] As shown, the connector terminals 508 exert inward pressure from opposing ends of the module, however, the contacts may be arranged along a single side of the module with the connector body providing the necessary resistive force for the pressure fit. Although the connectorized module is shown having plated interconnects 511 forming contacts for engagement with the connector terminals, it should be appreciated by those of skill in the art that many variations are possible. For example, adapters of the type illustrated in connection with Figs. 15-20 may be used to provide contacts for a hybrid connectorized module allowing other orientations of the module relative to the connector and to the customer circuit board.

G6. Flush Mount

[0118] A flush-mount technique may be used with the horizontal PCB-mounting techniques discussed above in connection with Figs. 15-24 to allow the bottom heat sink to come into contact with the customer PCB, e.g. for heat removal. As shown in Figs. 25 and 26, a through-hole mount module 600 is shown adapted for flush-mounting to a customer PCB 900. The module 600 as shown includes two through-hole adapters 603A, 603B attached to the singulated module 615. The singulated module 615 may, as shown, have a finned top heat sink 601 and a generally flat bottom heat sink 602 for flush mounting against the PCB 900. Similar to the adapters discussed above in connections with Figs. 16-20, through-hole adapters 603A, 603B have terminals, 608, 609, 610 which include features, such as solder pad 608A, adapted to be attached, e.g. by solder, to respective interconnects on the module 615. The terminals 608, 609, 610 may be adapted to be soldered into through holes 908, 909, 910, respectively, in the customer PCB 900. As shown in Figs. 25 and 26, the generally flat heat sink 602 may include recesses 602B to accommodate flanges 618B of the adapter bodies 618 allowing most of the surface of heat sink 602 to rest flush against the surface of the PCB 900. Epoxy or other adhesive may be used in the recess to secure the adapter body to the module. The recesses may be an integral feature of the heat sink panel or may be added at an appropriate point during the manufacturing process, preferably before singulation.

[0119] A thermally conductive material 901, e.g. thermal adhesive, may be applied between the PCB 900 and the module heat sink 602 to facilitate removal of heat through the PCB 900. Additionally, the PCB surface may include thermally conductive features to conduct heat away from the module 615. For some applications particularly involving smaller module sizes, it may be desirable to solder the bottom heat sink 602 to one or more pads on the PCB 900, in which case the heat sink may include a solderable finish, applied for example by plating. Threaded holes may be provided, preferably in the flush mount heat sink panel, allowing the module to be secured using screws to a customer board or cold plate. The flush-mount modification may allow taller heat sink fins to be used on the top of the module without increasing the module profile above the customer PCB which may provide better thermal management in some environments.

Additionally, the flush-mount may provide a more robust shock and vibration resistant mechanical solution.

[0120] Another flush mount module 650 may include a plurality of pins 661 protruding from the bottom heat sink 652 for engagement in through holes 911 in the customer mother board 900 as shown in Fig. 25A. Similar to the flush mount module 600 of Figs. 25 and 26, the flush mount module 650 may include adapters 653A and 653B adapted for making electrical connections with through holes on the customer mother board 900. Instead of fins, the top heat sink 651 may include a flat surface to create a low-profile package as shown in Fig. 25A. The pins 661 may be formed as an integral part of the bottom heat sink panel instead of fins. Alternatively, blind holes may be provided in the heat sink panel into which the pins may be press fit at any suitable stage of the fabrication process. The pins 661 may be used to electrically connect the bottom heat sink 652 to the customer board, e.g. to ground, conduct heat out of the module into the customer board 900, and provide mechanical support. The through holes 908, 909, 910, 911 in the customer board may be sized to provide clearance between the hole and the respective pin to compensate for any dimensional variations. The pins 661 may optionally protrude beyond the bottom surface of the customer board 900 into forced air along the bottom surface of the board for additional heat removal. Additionally, a heat sink component (not shown) may be fitted onto the protruding pins to help dissipate heat.

H. Heat Sink Setback

[0121] As internal components are reduced in height, e.g. reducing the thickness of the magnetic core, the depth of the interior cavity may be decreased bringing the heat sink panels closer together, reducing the encapsulant thickness and the resulting module thickness. However, reduction of the encapsulant thickness has the potentially undesirable effect of reducing the spacing between the electrical interconnects and the edges of the heat sink panels in the finished module. When desirable, e.g. to satisfy safety agency requirements, the separation between the exposed interconnects, e.g. interconnects 111, 112, 113, and the edge of the heat sink, e.g. heat sink 101B, may be

increased using a setback, e.g. setback 155, between the edges of the heat sink panels and the edges of the module 115B as shown in Fig. 10B. The setback 155 may be created by making wide cuts through the heat sink panels 121, 122 along the singulation lines prior to singulation. The wide cuts preferably extend through the heat sink material, e.g. aluminum, and partially into the encapsulation material to form channels 128A and 129A in the panel assembly 120A as shown in Figs. 9B and 9C. If machined while the assembly is still hot from the encapsulation process, the channels 128A and 129A may be used to divide the heat sink panels into singulated module dimensions reducing stresses due to differential contraction between the heat sink and the encapsulant due to differences in thermal coefficients of expansion while the panel assembly cools. Stresses on the narrow saw may also be reduced by eliminating the heat sink metal through which the saw must cut during singulation as a result of the channels 128A, 129A.

I. Process Efficiencies

[0122] Using interconnection features that may be exposed during singulation allows the PCB panel 124, containing a plurality of modules, to be molded as a single unit. Providing embedded interconnects along the perimeter of the circuit that occupy little or no PCB surface area help reduce wasted PCB area that would otherwise be cut away, allowing close to full utilization of the PCB for product which may save on cost. Encapsulating the PCB panel with the heat sink panels simplifies the structural aspects of the modules. Using interior contours in the heat sink to match component heights helps reduce the amount of molding compound required for encapsulation. Furthermore, controlling the distance between the magnetic cores and the internal surface of the heat sink can be used as an alternative to and eliminating the complications of the exposed core molding process described in the Exposed Core Application.

[0123] Furthermore, using the mold panels to form the mold cavity for encapsulating the PCB panel helps free the molding equipment from product specific requirements that may otherwise require customized molds, allowing a single piece of molding equipment to be used for a wide variety of product mixes. The finished products, e.g. modules 115 made using a standard panel size, may have diverse dimensions not only in the lateral

(length and width) directions, but also in the vertical (thickness) direction (e.g. due to heat sink fin height or component thickness). However, because the lateral panel dimensions remain the same, and variations in thickness from panel to panel may be accommodated by the molding press, the same general purpose molding equipment may be used for a wide variety of products of diverse dimensions. Using power converters as an example, the same mold press may be used to encapsulate panels of power converters ranging in (1) footprint size from full size, to half, to quarter, to eighth size (or any other size), and in (2) thickness (height), and in (3) topology, e.g. isolated DC-DC regulating converter, non-isolated buck regulator, DC transformer, etc. to produce a large mix of products.

[0124] A panel molding manufacturing process for a mix of products may include some or all of the following steps. Select a specific product to build. Select the requisite blank heat sink panels, e.g. based upon fin orientation, spacing, and height for the specific product. Alternatively, machine the exterior of the heat sink blank panels to produce the requisite external surface (heat sink surface, mounting features such as threaded holes, fin orientation, thickness, and spacing). Machine the interior surfaces of the heat sink blanks to form the recesses and other features (i.e. the contours of interior cavity to match some or all component locations, size, and height), of the finished heat sink panels required for the specific product, preferably under computer control. Select the appropriate PCB panel for the specific product. Select and assemble the magnetic cores and other components onto the PCB panel, e.g. by surface mount soldering, etc. Dispense a measured quantity of molding compound into the bottom heat sink panel. Press the bottom side of PCB panel up against bottom heat sink panel. Dispense a measured quantity of molding compound on the top side of the PCB panel. Press top heat sink panel into place on the PCB panel. Place the panel assembly on a rotary table away from the axis of rotation, preferably a large distance from the axis, and spin the rotary table and panel assembly to evacuate air bubbles in the interior cavity to achieve essentially void free fill of panel assembly with molding compound. Cure the molding compound. Cut the panel along the cut lines for singulation. Apply a conformal coating to protect the interconnects, or plate the interconnects, or attach a lead frame, motherboard, or connector to the exposed interconnects.

J. PCB Symmetries

[0125] The components may be symmetrically arranged on the PCB such as shown in the power converter example of Fig. 27. The top 104-2 and bottom 104-1 faces of a populated PCB 104 from an individual power converter module are shown in plan view in Fig. 27. The populated PCB 104 is shown rotated along the vertical axis 27 in Fig. 27 to show the symmetry of the components.

J1. Symmetrical Distribution between PCB Surfaces

[0126] Many of the larger components may be distributed equally between both faces of PCB 104 as shown in Fig. 27. For example, the four input field effect transistors (FETs) 132-2D, 132-2E, 132-1D, 132-1E are shown equally distributed between the top 104-2 and bottom 104-1 surfaces with two FETs on each surface. Similar equal distribution between the top 104-2 and bottom 104-1 surfaces of the PCB 104 are shown for the eight output FETs 132-2B, 132-2C, 132-1B, 132-1C with four output FETs on each surface; the twelve input capacitors, 132-2F, 132-2G, 132-1F, 132-1G, with six input capacitors on each surface; and twenty four output capacitors 132-2A, 132-1A with twelve output capacitors on each surface. Some of the FETs can function as switches. Distributing larger components between the two surfaces of the PCB may decrease stresses on the PCB, e.g. due to differences in the coefficient of expansion of the encapsulant, e.g. while curing, which may improve the co-planarity and mechanical integrity of the device.

J2. Symmetrical Distribution on a PCB Surface

[0127] On each surface of the PCB, components having similar characteristics, such as size or in-circuit power dissipation, may be arranged symmetrically for example as shown in Fig. 27 with respect to horizontal axis 28. It can be seen that relative to the horizontal axis 28, which is drawn longitudinally through the midline of the top and bottom surfaces of PCB 104, many of the components are arranged symmetrically. For example, the six input capacitors 132-1F and 132-1G on the bottom surface 104-1 are symmetrically distributed in a mirror-image relationship to each other relative to

longitudinal midline axis 28. The same basic mirror image relationship is true for the six input capacitors 132-2F and 132-2G on the top 104-2 surface of the PCB. Similarly, the mirror image relationship is shown for the following pairs of components: bottom-side input FETs 132-1D and 132-1E, topside input FETs 132-2D and 132-2E, bottom-side output FETs 132-1B and 132-1C, topside output FETs 132-2B and 132-2C and also within the bottom-side and top-side banks of output capacitors 132-1A and 132-2A.

[0128] Distributing larger components symmetrically on a surface especially with respect to the longitudinal axis of the PCB may also decrease stresses on the PCB, e.g. due to differences in the coefficient of expansion of the encapsulant, e.g. while curing, which may also improve the mechanical integrity of the device. Additionally, spreading the components out symmetrically on each surface helps to spread the heat produced by power dissipating devices using a greater surface area for heat extraction improving the thermal performance.

J3. Symmetrical Footprints between PCB Surfaces

[0129] In addition to being equally distributed between the top and bottom surfaces and being symmetrically distributed on each PCB surface, the components may also be situated such that pairs of components (wherein each component on one surface has a respective counterpart on the other surface) may be arranged to occupy essentially the same space on the PCB, i.e. a component may occupy a space on one surface that substantially overlaps with the footprint of a component on the other surface. For example, input capacitors 132-1F on the bottom surface are in the same position as their counterparts 132-2F on the top surface, i.e. they share the same footprint on the PCB. The same relationship is generally true for: input capacitors 132-1G and 132-2G; output capacitors 132-1A and 132-2A; input FETs 132-1D and 132-2D; input FETs 132-1E and 132-2E; output FETs 132-1B and 132-2B; output FETs 132-1C and 132-2C; in which the pairs of components occupy the same basic footprint, albeit on opposite surfaces, of the PCB. One benefit of sharing footprints allows the pair of components to share a common set of conductive vias used to electrically connect the components on the PCB surfaces to internal conductive layers, e.g. used to form the windings of the transformer. Because

each via is used for both components in the pair, the total number of vias for making connections to the pair of components may be reduced (by as much as a factor of two) increasing the area of conductive layers useable for making connections and thus reducing resistance. For example, assuming 6 vias are required for each output FET (a total of 12 vias for two FETs), using symmetrical footprint approach, the pair of FETs can share the same 6 vias (without increasing the via resistance) and because the number has been reduced the useable area for conductors may be increased. Alternatively, while reducing the total number of vias from 12 to some intermediate number, e.g. 8, the resistance of the vias may be decreased because of the increase in effective vias per FET while still increasing the area useable for conductors.

J4. Symmetrical Power Dissipation between PCB Surfaces

[0130] The components may be arranged between the PCB surfaces according to heat dissipated during operation. For example, the heat dissipative components may be arranged in a manner that distributes the heat evenly between the two PCB surfaces allowing heat produced by power dissipating devices to be extracted from both surfaces of the PCB improving the thermal performance. This type of heat dissipation symmetry is also factored into the component layout shown in the power converter of Fig. 27. For example, two input cells each using the same basic circuit topology are shown, one above and another below axis 28. As shown, the components of each input cell occupy both sides of the PCB in observance of other factors influencing component layout such as winding locations, etc. In this example, the upper input cell includes the two input FETs 132-1D and 132-2D, and the six input capacitors 132-1F and 132-2F. The lower input cell includes the two input FETs 132-1E and 132-2E, and the six input capacitors 132-1G and 132-2G. To ensure heat dissipation symmetry between the two surfaces, the cells may be arranged in mirror image layouts as shown. In Fig. 27, input FET 132-2E (top surface) in the lower cell corresponds to input FET 132-1D (bottom surface) in the upper cell. Similarly, lower cell input FET 132-1E (bottom surface) corresponds to upper cell input FET 132-2D (top surface). As can be seen, the significant power dissipative components of the input cells are arranged to have one component of one cell mounted on one surface with the respective component from the other cell mounted on the other

surface. This type of symmetry may be seen in Fig. 27 with lower cell component 132-2H mounted on the top surface and the respective upper cell component 132-1H mounted on the bottom surface. In some examples, the FETs are arranged such that during operation, the power FETs on the top surface dissipate power at a level that is comparable to the level of power dissipated by the power FETs on the bottom surface. Also, the level of power dissipated by the power FETs in the upper input cell is comparable to the level of power dissipated by the power FETs in the lower input cell. For example, the level of power dissipated by the power FETs on one surface is less than 150% of the level of power dissipated by the power FETs on the other surface. The level of power dissipated by the power FETs in the upper cell is less than 150% of the level of power dissipated by the power FETs in the lower cell, and vice versa.

[0131] Laying out the components using any or all of the above symmetries produces several key benefits including, enhanced thermal performance, reducing top to bottom and side to side imbalances during encapsulation caused by asymmetrical distribution of components may enhance the co-planarity and structural integrity, and shared component footprints on top and bottom PCB surfaces may help reduce conduction losses and increase efficiency.

K. Center Plate Panel Assembly

[0132] In an alternate embodiment, an optional center plate 727 may be used between the top 721 and bottom 722 heat sink panels as illustrated in Fig. 28 through Fig. 32. The center plate 727, which may be made from aluminum, a molded high temperature plastic or any other material suitable for the molding process, includes an opening 729 in which the populated PCB panel 724 may sit during the panel molding process. As shown in the side view of Fig. 29 and the cross-sectional view of Fig. 32, the PCB panel 724 may sit entirely within the opening 729 and may have some high profile components, such as magnetic cores 131 and capacitors 132 (continuing with the power converter example), extending beyond the planar surfaces of the center plate. Alternatively, the center plate may be made thick enough, or include a ridge around the periphery tall enough, such that the PCB panel and all components sit entirely within the

opening allowing a flat surface (heatsink panel or mold panel) to close against the center plate. The recesses formed in the interior surfaces of the heat sink panels 721, 722 (described above) may accommodate portions of the components extending beyond the center plate surface. Conversely, protrusions from the interior surface of the heat sink panels may be used to reduce the distance between the heat sink and lower profile components. However, it may be preferable for ease of fabrication and tolerance control to avoid protrusions of the heat sink beyond the surface of the center plate which may put an upper limit on the thickness of the center plate in some embodiments.

[0133] As shown in the exploded perspective view of Fig. 28 and the top plan view of Fig. 31, registration features may be provided in the center plate 727. For example, registration pins 728 may mate with corresponding holes 734 in the PCB 724 to establish the horizontal position, i.e. in the X and Y directions, of the PCB relative to the center plate 727. Additional registration features such as the illustrated horizontal shelf 728A (Fig. 28), may be provided to establish the vertical position, i.e. in the Z direction, of the PCB relative to the center plate. The registration pins 728 may be long enough to extend beyond the upper surface of the PCB panel 724 in the upward direction and beyond the horizontal shelf in the downward direction to mate with holes (analogous to holes 152 and 153 in Fig. 4B) which may be provided in the top and bottom heat sink panels 721, 722 establishing the horizontal positions of the mold panels relative to the center plate 727. Provision of the registration features in the center plate may help relax certain precision requirements, the complexity, and thus the cost of the heat sink panels.

[0134] A cross-section of the panel assembly 720 closed in a mold press taken through line 32-32 in Figs. 30 and 31 is shown in Fig. 32. As shown, the upper mold press 761 engages the center plate 727 directly along its perimeter in regions 768 and engages the heat sink panels directly along their perimeters in regions 769. Preferably the mold press includes recessed surfaces 766 providing cavities 765 large enough to accommodate a full range of fin heights (or other heat sink panel features) supporting a diverse range of products. To compensate for dimensional differences between the thickness of the heat sink panels in regions 769 and the difference in elevation between interface regions 768 and 769 in the mold press, one or more compressible features may

be provided at the interface between the heat sink panels 721, 722 and the center plate 727. For example, a small crushable feature 723 may be formed along the perimeter of and as an integral part of the heat sink panels as illustrated in Fig. 32. Alternatively a gasket may be used between the center plate and one or both of the heat sink panels. As the press closes on the panel assembly 720 the crushable features 723 are compressed until the press is closed securely against both the center plate 727 and the heat sink panels 721, 722. As shown in Fig. 31, the crushable features may extend along the perimeter of the interior cavity forming a seal between each heat sink panel and the center plate.

[0135] The center plate may preferably include an extension, e.g. extension 730, to at least one side of opening 729 providing space for one or more chambers 725 as shown in Figs. 30, 31, and 32. During the transfer molding process, encapsulation material may be forced from the chambers through one or more channels 726 (as shown in Figs 28 and 32) in the center plate into opening 729 and thus the cavity in which the populated PCB panel 724 is enclosed. An example of encapsulant flow through a chamber 725, channel 726, and into the interior cavity 746 is illustrated by the arrows 767 shown in Fig. 32.

[0136] In the center plate panel mold assembly, the top and bottom mold panels (i.e., heat sink panels 721, 722) close against the center plate instead of each other, reducing the thickness of the top and bottom mold panels, increasing the symmetry between, and reducing the complexity of, the top and bottom heat sink panels 721, 722, potentially simplifying the molding press, eliminating critical tolerance accumulations in the assembly, simplifying the process and reducing cost. For example, provision of the chambers 725 and conduits 726 in the center plate 727 eliminates the need for sealing along a second axis, e.g. in a horizontal direction and allows for use of a simpler cull-on-plate molding press. Critical tolerances are reduced to the one vertical dimension of the heat sink panels which can be relaxed using a crushable feature or a compliant material on the surface that interfaces with the center plate. Additionally, the center plate 727 may be standardized allowing a single configuration to be used with a large variety of heat sink panels such that the center plate may be cost-effectively molded.

III. Molding Process Non-Integrated Re-Useable Mold

A. Components without Heatsinks

[0137] In some applications it may be desirable to use the panel-molding process to produce components, such as power converters, without the flat heat sink surfaces (Figs. 17-24, 25, 26), finned heat sinks (Figs. 1, 10, 12, 15-16, 25, 26), or pinned heat sink (Fig. 25A) features described in the above embodiments. Referring to Figs. 33 and 34, a horizontal-mount component 840 is shown including a singulated module 815 and two adapters 803A, 803B, suitable for through-hole mounting to a motherboard. Comparing the singulated module 815 (Fig. 33) with the singulated modules 215 (Fig. 16) and 315 (Fig. 17) reveals that the integral heat sink surfaces of the previous examples are omitted in the singulated module 815. As shown in Figs. 33 and 34 and described in greater detail below, the top 815A and bottom 815B surfaces of the singulated module 815 are defined by the cured encapsulant 805, 806 in which the magnetic core surfaces 815C, 815D are exposed.

[0138] Like the singulated modules with heat sinks, various types of connectors or adapters may be coupled to the exposed interconnects 811, 812 at the edges of the singulated module 815. All of the variations described above in connection with the heat-sink versions may be adapted for use with the singulated module 815. As shown in Figs. 33 and 34, adapters 803 having terminal portions 808 and interconnect portions 808A may be formed for example by a precision metal stamping process. Initially, the terminal portions 808 may be coupled to a lead frame 808B as shown in Figs. 35 and 36 forming a unitary unit for ease of assembly. The interconnect portions 808A may be soldered to the exposed interconnects 811, 812 in the manner described above, and protective caps 818 may be attached, e.g. with epoxy, covering the interconnect portions 808A, to provide structural integrity, and exposing only the terminals 808. The lead frame 808B may be separated from the terminals 808 after the terminals are attached to the exposed contacts.

B. Center Plate Molding

[0139] The above-described panel molding process may be adapted to produce the heat-sinkless modules 815 shown in Figs. 33 through 36. Referring to the exploded view of Fig. 37, re-useable flat top and bottom plates 834 may be used in place of the top and bottom heat sink panels of the previous examples. The center plate 827, which like the center plate 727 in Fig. 28 may be made from aluminum, a molded high temperature plastic or any other material suitable for the molding process, includes an opening 829 in which the populated PCB panel 824 may sit during the molding process. Like the example of Fig. 28, the center plate 827 may include an extension, e.g. extension 830 in Fig. 38, to at least one side of opening 829 providing space for one or more chambers 825 (shown in Figs. 37, 39, and 40).

[0140] The center plate 827 may include a sealing ridge 823 along the periphery of the opening 829 to provide a total thickness of the center plate 827 sufficient to accommodate the PCB panel 824 and all of the components, including for example the magnetic cores 131, while minimizing waste material in the center plate. Referring to the side view of Fig. 38, the thickness of the manifold provides adequate clearance for all of the components on the PCB panel 824. As shown in the cut away section 827A-827A in Fig. 38, the space between the cores and the plates 821, 822 may preferably be minimized, reducing the volume of encapsulant required. The plate registration pins 828 are shown in Fig. 38 protruding above and below the centerplate sufficiently to engage mating openings 834 provided in the top and bottom plates, 821, 822.

[0141] The PCB panel 824 may be seated on registration shelves 828B (Fig. 37) for establishing the correct vertical positioning of the PCB panel with respect to the manifold 827. The horizontal position of the PCB panel may be established using the registration pins shown in the previous example (e.g. pins 728 in Fig. 28), or may be accurately placed in the opening, e.g. using pick and place equipment or a fixture, and affixed to the center plate, e.g. using epoxy, e.g. deposited on the registration shelves 828B prior to placement of the PCB panel. The use of an adhesive such as epoxy helps to prevent bowing of the PCB panel as mold compound is forced into the cavity during the molding

process. Venting features 823A may be provided in the sealing ridges 823 along the side opposite where the mold compound is injected into the cavity as shown in Figs. 37 and 39.

[0142] Referring to the Fig. 40, a section of the panel assembly 820 (taken along lines 40-40 shown in Fig. 39) is shown closed in a mold press. The mold press includes an upper press 861 and a lower press 862 which close directly against the center plate 827 along its perimeter forming seals 868 as shown. The mold presses 861, 862 may include recessed surfaces 864 creating cavities 865 large enough to accommodate a full range of PCB panel heights, supporting a diverse range of products. Shim plates 863 may be used between the recessed surfaces 864 of the mold presses 861, 862 and the top and bottom plates 821, 822 to force the plates against and form a seal with the center plate 827, e.g. along sealing ridges 823. For example, mold-to-shim-plate interface 869A and shim-plate-to-plate interface 869B are shown in Fig. 40. Variations in thickness of the center plates 827 are accommodated by the mold press closing directly against the center plate 827. Variations in the height of the top or bottom plate from the respective center plate surface may be accommodated by varying the thickness of the respective shim plate. In this way, components of varying thicknesses may be produced using an center plate 827 having the appropriate thickness for the desired component height and using thicker or thinner shim plates 863 as necessary to adjust for any difference in height between the center plate 827 and the top and bottom plates 821, 822. Alternatively, one or both of the shim plates may be omitted completely allowing the mold presses to press directly against the respective top and/or bottom plates.

[0143] During the transfer molding process, encapsulation material may be forced from the chambers 825 through one or more channels 826 (Fig. 40) in the center plate into opening 829 (Fig. 37) and thus the cavity in which the populated PCB panel 824 is enclosed. An example of encapsulant flow through a chamber 825, channel 826, and into the cavity is illustrated by the arrows 867 shown in Fig. 40. After the encapsulant is cured, e.g., by heating, the manifold plate 827 and the encapsulated PCB panel may be separated from the upper and lower mold presses 861, 862. During or after the cooling process, the top and bottom plates may be removed from the panel assembly. The

surfaces of the top and bottom plates 821, 822 are relatively smooth and separate easily from the encapsulant as the panel assembly cools due to differences in thermal expansion coefficients of the materials.

[0144] Fig. 41 shows a top plan view of the encapsulated panel assembly 820A separated from the upper and lower mold presses after removal of the top and bottom plates 821, 822. The center plate 827 may include registration features 828A (Figs. 37, 39, 41) for positional accuracy in pre- and post-encapsulation processing of the panel. For example, the registration features 828A may be used to accurately position the PCB panel with respect to the center plate during installation or may be engaged by pins in the mold to accurately position the center plate in the mold. Greater positional accuracy may be achieved by making one or more holes 871 through the encapsulant in predetermined regions of the encapsulated PCB to expose corresponding fiducial marks 872 incorporated in the PCB panel. The center plate 827 and its registration features 828A thus may be used to establish an approximate horizontal position (of the PCB panel) for making the holes 871 which preferably would be larger than the fiducial marks 872. The exposed fiducial marks may be used to register the position of the PCB panel with greater accuracy for precise singulation. Fig. 42 shows a cross-sectional view of the encapsulated panel assembly through one of the holes 871 in the encapsulant exposing the fiducial mark 872 on the PCB panel.

[0145] Preferably before singulation, the encapsulated panel assembly may be lapped to expose the magnetic cores, finish the surface, and accurately establish the thickness of the panel-molded components. After singulation, the interconnects may be processed and connectors, terminals or adapters may be attached to the components as described above.

C. Direct Molding

[0146] Alternatively, the panel molding process may be further adapted to encapsulate the PCB panel directly within the mold cavity without using a center plate (such as center plates 727 in Fig. 28 and 827 in Fig. 37), heat sink panels (such as panels 721, 722 in Fig. 28) or top and bottom plates (such as plates 821, 822 in Fig. 37). In the direct-mold embodiment, the PCB panel may be placed in the bottom mold, with the top

mold closing against the bottom (instead of each mold closing against the center plate). The depth of the mold cavity may be set to match the overall height of the PCB panel and components with a custom mold for each product thickness. Alternatively, the mold cavity may have moveable interior surfaces that may be adjusted for the selected product or changeable shim plates designed for each product may be used with a standardized mold to adjust the depth of the mold cavity and thus the encapsulation thickness.

[0147] The mold may include fixed or movable registration supports (e.g. similar to registration shelves 828A provided in the center plate 827 in Fig. 37) preferably along the periphery of the mold cavity. The PCB panel may be seated on the supports to establish the appropriate vertical position in the mold cavity and which may also be used to reduce (displace) the volume of encapsulant in selected regions of the panel. Referring to Figs. 43 and 44 for example, an encapsulated PCB panel 890 produced using the direct mold technique is shown after removal from the mold cavity. Fig. 43 shows a bottom view as molded (component top view) of the panel revealing indentations in the cured encapsulant 805, indentations 891 along the sides, indentations 891B at the corners, a large indentation 891A at one end, and circular indentations 891C and 891D at each end of the panel 890. The large indentation 891A may be used to expose identifying features on the PCB such as a bar code used during manufacture to identify the panel. Similarly, the circular indentations 891C and 891D may be used to expose positional information on the PCB, such as the fiducial markings described above.

[0148] The PCB panel may be placed in the bottom mold cavity, and the upper mold press may be pressed against the lower mold forming a seal around the cavity. As the upper mold is closed against the lower mold, preferably moveable pins within the upper mold may make contact with the PCB in the regions of the supports to bias the PCB against the registration supports in the lower mold and securing the PCB panel. For example, the top view as molded (component bottom view) of Fig. 44 shows, additional circular indentations 891E in the cured encapsulant 806 located within the indentations 891 and 891B revealing potential locations at which moveable pins may be deployed in the top mold section. It may be advantageous to locate the movable pins within other support features to limit the exposed length of the pin.

[0149] With the mold closed and the PCB panel secured, molding compound may be injected into the mold cavity and cured, e.g. by elevating the temperature. After the molding compound is cured, the encapsulated PCB panel may be separated from the upper and lower mold presses, and any encapsulant remaining on the PCB in the region of holes 891A, 891C, 891D may be removed as described above (e.g., by drilling or laser ablation) to expose the identifying features (bar code and fiducial marks) on the PCB panel 124. The PCB panel may be lapped, and singulated as required.

[0150] In some implementations, after singulation, a portion of the top encapsulant 822 can be removed by mechanical lapping to expose a top surface of a magnetic core structure 826. Similarly, a portion of the bottom encapsulant material 824 can be removed by mechanical lapping to expose a bottom surface of the magnetic core structure 826. The lapping process provides great dimensional control of the finished product.

[0151] As described above in connection with the heat sink panel versions, the finished products, e.g. modules made using a standard panel size, may have diverse dimensions not only in the lateral (length and width) directions, but also in the vertical (thickness) direction (e.g. due to component thickness). However, because the lateral PCB panel dimensions remain the same, and variations in component thickness from panel to panel may be accommodated by the molding press, the same general purpose molding equipment may be used for a wide variety of products of diverse dimensions to produce a large mix of products.

[0152] In some implementations, a panel molding manufacturing process for a mix of products may include some or all of the following steps. Select a specific product to build. Select the requisite mold presses, e.g. based upon spacing and height for the specific product. Select the appropriate PCB panel for the specific product. Select and assemble the magnetic cores and other components onto the PCB panel, e.g. by surface mount soldering, etc. Dispense a measured quantity of molding compound into the bottom mold press. Press the bottom side of PCB panel up against bottom mold press. Dispense a measured quantity of molding compound on the top side of the PCB panel. Press top mold press into place on the PCB panel. Inject additional molding compound

into the cavity between the top and bottom mold presses and the PCB panel through a channel formed between the top and bottom mold presses. Cure the molding compound. Remove the encapsulated PCB panel from the mold presses. Form holes in the encapsulant to expose fiducial marks to facilitate alignment. Cut the encapsulated PCB panel along the cut lines for singulation. Apply a conformal coating to protect the interconnects, or plate the interconnects, or attach a lead frame, motherboard, or connector to the exposed interconnects.

[0153] In some implementations, a panel molding manufacturing process for a mix of products may include some or all of the following steps. Select a specific product to build. Select the requisite mold presses, e.g. based upon spacing and height for the specific product. Select the appropriate PCB panel for the specific product. Select and assemble the magnetic cores and other components onto the PCB panel, e.g. by surface mount soldering, etc. Position the PCB panel in a manifold plate (e.g., 827 in Figs. 37-42). Press top and bottom mold presses against the manifold plate. Inject molding compound into the cavity between the top and bottom mold presses and the PCB panel through chambers and channels formed in the manifold until the PCB panel is entirely enclosed by the molding compound. Cure the molding compound. Remove the manifold plate and the encapsulated PCB panel from the mold presses. Form holes in the encapsulant to expose fiducial marks to facilitate alignment. Mechanically lap one or both surfaces of the panel. Cut the encapsulated PCB panel along the cut lines for singulation. Apply a conformal coating to protect the interconnects, or plate the interconnects, or attach a lead frame, motherboard, or connector to the exposed interconnects.

[0154] A number of embodiments of the invention have been described. Nevertheless, it will be understood that various modifications may be made without departing from the spirit and scope of the invention. For example, non-metallic mold panels may be used. The center plate may be provided as a single-use consumable or may be modified to be used as a reusable fixture in the molding process. The center plate may be provided with or without the encapsulation channels. The registration holes in the heat sink panels may extend completely through the heat sink panels similar to hole 152

shown in Fig. 4B and the registration pins, e.g. pins 728 (Figs. 28 and 31), may extend completely through each of the heat sink panels allowing fasteners to be used in conjunction with the pins to hold the panel assembly together before and after the encapsulation process. The fasteners may function as or be used in conjunction with a seal around the pin to contain any encapsulant. Throughput through the mold press may be increased by using panel assemblies that are secured together and removed from the mold press before the panel has cooled or the encapsulant has set or both. The registration pins 728 and corresponding holes 734 may be used to align the panel assembly during the singulation process. In some examples, a panel assembly (e.g., 120) may be formed by dispensing encapsulant into a bottom panel mold (e.g., heat sink panel 122), assembling a substrate (e.g., PCB panel 124) into the bottom panel mold, dispensing encapsulant onto a top of the substrate, and assembling a top panel mold (e.g., heat sink panel 121) onto the substrate. In some examples, the surfaces of the PCB panel 124 may have conductive features that are covered by an insulative layer. Blank mold panels may be machined to provide some or all of the various features described above in an on-demand manufacturing system.

[0155] In some examples, the upper and lower heat sinks 121, 122 are clamped together by the upper and lower mold presses 161, 162 at respective clamp regions of the upper and lower heat sinks 121, 122. The clamp region of the upper heat sink 121 can be located at points along a circumference of an internal cavity defined by the interior surface of the upper heat sink 121. The clamp region of the lower heat sink 122 can be located at points along a circumference of an internal cavity defined by the interior surface of the lower heat sink 122. In some examples, the clamp regions are cut away from the panel assembly 120 to expose the interconnects 111, 112, and 113. After the cut, portions of the upper and lower heat sinks 121, 122 near an active circuit area remain attached to the panel assembly 120, allowing heat from the active circuit area during operation to be dissipated through the remaining portions of the upper and lower heat sinks 121, 122. The active circuit area can be, e.g., an area of the PCB panel 124 having active components, such as magnetic core structures 131 and electronic components 132. Interlocking contours, other than the undercuts 148 shown in Figs. 6, 7, and 12, can also be formed in the interior surface of the mold panel, the contour being filled with cured

mold compound enhancing the structural integrity of the singulated module. In some examples, most of the large-footprint components (e.g., 132-2D, 132-2E, 132-2B, 132-2C) are distributed substantially symmetrically between quadrants surrounding the transformer core (e.g., 131-2) on a surface of the PCB panel 124. For example, in FIG. 27, the input FETs 132-2D and 132-2E are distributed substantially symmetrically between the upper-right and lower-right quadrants surrounding the transformer core 131-2. The output FETs 132-2B and 132-2C are distributed substantially symmetrically between the upper-left and lower-left quadrants surrounding the transformer core 131-2.

[0156] Accordingly, other embodiments are within the scope of the following claims.

WHAT IS CLAIMED IS:

1. A method of making a plurality of electronic devices, comprising:
providing a plurality of mold panels, which when assembled, form an internal chamber;
providing a substrate having a plurality of conductive traces and a plurality of components electrically connected to form at least one circuit, the substrate having at least one contact for making an electrical connection to the circuit;
forming a panel assembly including the mold panels assembled with the substrate in the internal chamber and an encapsulant filling spaces between the substrate and interior surfaces of the chamber;
curing the encapsulant; and
cutting the panel assembly to expose at least a portion of the at least one contact and form a respective exposed contact.
2. The method of claim 1 wherein the cutting comprises making a first cut in at least one of the mold panels, and in the first cut, making a second narrower cut through the panel assembly.
3. The method of claim 2 wherein the curing comprises raising the temperature of the panel assembly and the first cut is made before the panel assembly cools after the curing.
4. The method of claim 1 wherein the cutting divides the panel assembly producing at least one module having a plurality of layers including respective portions of each of the panel molds, the substrate, and the encapsulant, and the module contains the at least one circuit.
5. The method of claim 4 wherein the cutting defines at least one side of the module.

6. The method of claim 4 wherein the cutting defines two or more sides of the module.

7. The method of claim 1 further comprising treating the exposed contact to protect against oxidation.

8. The method of claim 7 wherein the treating comprises applying a removable conformal coating to the exposed contact.

9. The method of claim 7 wherein the treating comprises applying a layer of metal to the exposed contact.

10. The method of claim 9 wherein the applying a layer of metal comprises applying solder.

11. The method of claim 9 wherein the applying a layer of metal comprises applying a precious metal.

12. The method of claim 9 wherein the applying a layer of metal comprises plating.

13. The method of claim 4 further comprising
providing an adapter having at least one electrical terminal; and
attaching the at least one electrical terminal to the exposed contact.

14. The method of claim 13 wherein
the at least one contact comprises a plurality of contacts;
the module comprises a plurality of exposed contacts;
the adapter has a plurality of electrical terminals arranged to match respective ones of the plurality of exposed contacts; and
the adapter is mechanically secured to the module.

15. The method of claim 4 wherein the mold panels are metal.

16. The method of claim 15 wherein the respective portions of the mold panels provide heat sink surfaces for the module.

17. The method of claim 1 wherein the providing mold panels further comprises providing contours in an internal surface of one or more of the mold panels.

18. The method of claim 17 wherein the providing contours further comprises matching depths of portions of the internal surface to heights of one or more selected components.

19. The method of claim 18 wherein the selected components include at least one magnetically permeable core and at least one semiconductor device.

20. The method of claim 4, wherein the traces and components are electrically connected to form a plurality of separable circuits arranged in a pattern on the substrate; and the cutting divides the panel assembly along spaces between the separable circuits into a plurality of modules each containing at least one respective circuit.

21. The method of claim 20 wherein the spaces between the separable circuits have dimensions approximately matching a width of cuts produced by equipment used to cut the panel assembly.

22. The method of claim 21 wherein the at least one contact comprises a plurality of contacts located in the spaces between the separable circuits.

23. The method of claim 22 wherein the plurality of contacts are formed in the substrate.

24. The method of claim 23 wherein the plurality of contacts are formed and buried below at least one surface of the substrate.

25. The method of claim 15 further comprising treating at least one exterior surface of at least one of the mold panels for solderability.

26. The method of claim 1 wherein forming the panel assembly comprises:
dispensing encapsulant into a bottom panel mold;
assembling the substrate into the bottom panel mold;
dispensing encapsulant onto a top of the substrate; and
assembling a top panel mold onto the substrate.

27. The method of claim 26 further comprising centrifuging the assembly before curing the encapsulant.

28. The method of claim 1 wherein forming the panel assembly comprises:
assembling the substrate with a first substrate surface facing into a first panel mold,
closing a second panel mold onto the substrate covering a second substrate surface,
providing one or more conduits to the internal chamber, and
forcing encapsulant through the one or more conduits into the chamber.

29. The method of claim 28 further comprising centrifuging the assembly before curing the encapsulant.

30. The method of claim 9 further comprising providing a connector for removably mating with and providing electrical connection to the metal.

31. The method of claim 1 further comprising:

providing a center plate having an opening to accommodate the substrate; and
wherein the forming a panel assembly includes positioning the substrate in the opening of the center plate and closing the mold panels against the center plate.

32. The method of claim 31 further comprising providing at least one opening in the center plate connected to the internal chamber by at least one conduit; and
forcing the encapsulant through the at least one opening and at least one conduit into the internal chamber.

33. The method of claim 32 wherein the forming a panel assembly further comprises securing the mold panels together prior to curing the encapsulant.

34. A method of forming an electrical contact comprising:

assembling a panel including a substrate having one or more conductive features enclosed within the panel and unexposed to an exterior surface of the panel, the one or more conductive features being located along a cut line;

cutting the panel along the cut line exposing portions of the one or more conductive features for use as electrical connections to the substrate; and

treating the portions of the one or more conductive features exposed from the cutting for preservation as electrical connections.

35. The method of claim 34 wherein the treating further comprises applying solder to the portions of the one or more conductive features exposed from the cutting.

36. The method of claim 34 wherein the treating further comprises applying a conformal coating to the portions of the one or more conductive features exposed from the cutting to protect against oxidation.

37. The method of claim 34 wherein the treating further comprises applying a metal layer to the portions of the one or more conductive features exposed from the cutting.

38. The method of claim 34 wherein the treating further comprises soldering a lead of an adapter to the portions of the one or more conductive features exposed from the cutting.

39. The method of claim 34 further comprising covering at least one surface of the substrate in an area including the cut line and the conductive features prior to cutting the substrate.

40. The method of claim 39 wherein the covering comprises encapsulating the substrate with a molding compound; and further comprising:

providing a registration feature having a predetermined relationship to the substrate, and
using the registration feature to align the cutting relative to the cut line.

41. The method of claim 34 further comprising establishing a pattern including at least one conductive layer in the substrate along the cut line to form the conductive features.

42. The method of claim 41 wherein the substrate comprises a multilayer printed circuit board and the establishing a pattern includes establishing a plurality of conductive layers along the cut line to form the conductive features.

43. The method of claim 34 further comprising establishing a pattern including at least one conductive via in the substrate along the cut line to form the conductive features.

44. The method of claim 43 wherein the conductive via is filled with a conductive material.

45. The method of claim 43 wherein the conductive via is buried in the substrate.

46. The method of claim 43 wherein the conductive via is a through hole contacting the surfaces of the substrate.

47. The method of claim 46 wherein the through hole is filled with a conductive material.

48. The method of claim 34 wherein the conductive features are covered at the surfaces of the substrate by an insulative layer.

49. An apparatus comprising:
a first mold panel comprising
 an exterior surface,
 an interior surface defining an internal cavity,
 a clamp region located at points along a circumference of the internal
cavity, and
 an opening,
the first mold panel being adapted to
 (a) be engaged by pressure in the clamp region,
 (b) receive in the cavity a circuit panel containing a plurality of
components and mold compound to fill empty spaces in the internal cavity, and
 (c) be cut after curing of the mold compound.

50. The apparatus of claim 49 wherein the interior surface in the region of the internal cavity is adapted to adhere to the mold compound.

51. The apparatus of claim 49 further comprising

a second mold panel comprising

an exterior surface,

an interior surface, and

a clamp region,

wherein the second mold panel is adapted to

(a) close against and mate with the first mold panel,

(b) be engaged by pressure in the clamp region, and

(c) be cut after curing of the mold compound.

52. The apparatus of claim 51 wherein the interior surface of the second mold further defines a second internal cavity, the clamp region of the second mold panel is located at points along a circumference of the second internal cavity, and the internal cavity of the first mold panel is adapted to receive a first side of the circuit panel and the second internal cavity is adapted to receive a second opposite side of the circuit panel.

53. The apparatus of claim 52 further comprising contours formed in the interior surface of at least one of the mold panels.

54. The apparatus of claim 53 wherein the contours are adapted to match predetermined characteristics of selected ones of the plurality of components.

55. The apparatus of claim 53 wherein the internal surface of the mold panel is adapted to adhere to the molding compound.

56. The apparatus of claim 55 wherein the contours form interlocking features with cured mold compound.

57. The apparatus of claim 53 wherein at least one of the mold panels comprises a non-ferrous metal.

58. The apparatus of claim 57 wherein at least one of the mold panels comprises aluminum.

59. The apparatus of claim 51 wherein at least one of the mold panels comprises a thermally conductive material.

60. The apparatus of claim 53 wherein at least one of the mold panels comprises a non-metallic substance.

61. The apparatus of claim 51 further comprising at least one channel connected to the internal cavity for allowing expansion of the mold compound.

62. The apparatus of claim 51 further comprising at least one channel connected to the internal cavity for injecting mold compound into the internal cavity.

63. The apparatus of claim 59 wherein the exterior surface of the mold panel comprises a plurality of fins.

64. The apparatus of claim 59 wherein the exterior surface of the mold panel comprises a flat surface.

65. The apparatus of claim 64 wherein the flat surface is adapted for a solder joint.

66. The apparatus of claim 49 wherein the internal cavity includes features formed in the interior surface and arranged in a predetermined pattern.

67. The apparatus of claim 49 wherein the internal cavity includes features formed in the interior surface along lines through which the mold panel maybe cut to establish a setback from a cut edge of the mold panel.

68. The apparatus of claim 49 wherein the mold panel further comprises at least one registration feature adapted to engage and establish a predetermined relationship with the circuit panel.

69. The apparatus of claim 49 wherein one or more selected portions of the mold panel are adapted for incorporation into one or more products, the products being formed by a process which uses the mold panel to contain the molding compound.

70. The apparatus of claim 59 wherein the exterior surface of the mold panel comprises a flat surface with one or more pins protruding from the flat surface.

71. The apparatus of claim 59 wherein the exterior surface of the mold panel comprises a flat surface having one or more holes in the flat surface.

72. The apparatus of claim 71 further comprising a pin inserted into a respective one of the one or more holes.

73. An apparatus comprising:

a panel assembly having external surfaces defined by a first mold panel and a second mold panel;

the first and second mold panels forming an internal cavity enclosing an internal circuit board, the internal circuit board having a first surface and second surface and a plurality of components in an active circuit area on at least one of the surfaces, the components being electrically connected to interconnects contained within the internal cavity;

the internal cavity being filled with mold compound in spaces unoccupied by the circuit board and components;

the panel assembly being adapted to have first selected portions of the first and second mold panels cut away to expose the interconnects and to have second selected portions of the respective mold panels situated near the active circuit area remain attached to the assembly following the cut.

74. The apparatus of claim 73 wherein the panel assembly includes a clamp region located at points along a circumference of the first and second internal cavities, and the first selected portions comprise the clamp region.

75. An apparatus comprising:

a panel assembly having external surfaces defined by a first mold panel and a second mold panel;

the first and second mold panels forming an internal cavity enclosing an internal circuit board, the internal circuit board having a first surface and second surface and a plurality of components electrically connected to form a plurality of individual circuits, each individual circuit being electrically connected to respective interconnects located along a respective circuit perimeter, the interconnects being contained within the cavity.

76. The apparatus of claim 75 wherein the panel assembly is adapted to be cut along the circuit perimeter separating the individual circuits, dividing the panel assembly into individual circuit modules, and exposing selected portions of the interconnects.

77. The apparatus of claim 75 wherein the panel assembly is constructed to retain an integral layered structure after being filled with mold compound which is subsequently cured, the layered structure comprising a first layer including a portion of the first mold, a second layer including a portion of the mold compound, a third layer including a portion of the circuit board, a fourth layer including a portion of the mold compound, and a fifth layer including a portion of the second mold.

78. The apparatus of claim 75 wherein the internal circuit board comprises a plurality of circuits having the same functionality.

79. The apparatus of claim 75 further comprising one or more conduits connecting the internal cavity to an external opening.

80. The apparatus of claim 75 wherein spaces unoccupied by the circuit board, components, and interconnects in the internal cavity are filled with mold compound.

81. The apparatus of claim 80 wherein the mold compound has been cured.

82. The apparatus of claim 81 wherein selected portions of at least one of the mold panels have been removed, the selected portions being located near the circuit perimeters.

83. An apparatus comprising:

a modular package having a first external surface, a second external surface opposite the first external surface, and a side wall extending along the perimeter of and connecting with the first and second external surfaces, the modular package including a first layer defining the first external surface and a second layer defining the second external surface, the first and second layers being separated by and in contact with cured mold compound; and

an electrical circuit located between the first and second layers and within the cured mold compound and including at least one electrical component electrically connected to a plurality of interconnects;

the side wall comprising a strip formed by the first layer, a strip formed by the second layer, and a strip formed by the cured mold compound; and

the interconnects being disposed within the side wall.

84. The apparatus of claim 83 wherein the first and second layers further comprise a non-ferrous metal.

85. The apparatus of claim 84 wherein the first and second layers further comprise aluminum.

86. The apparatus of claim 85 wherein the first external surface further comprises a plurality of fins.

87. The apparatus of claim 85 wherein the first external surface further comprises an essentially flat area.

88. The apparatus of claim 87 wherein the essentially flat area is adapted for a solder joint.

89. The apparatus of claim 83 wherein the electrical circuit further comprises a circuit board and the side wall further comprises a strip formed by the circuit board and an additional strip formed by cured mold compound.

90. The apparatus of claim 89 wherein the interconnects further comprise conductive features in the circuit board.

91. The apparatus of claim 90 wherein the circuit board comprises a multilayer printed circuit board ("PCB") and each of the interconnects comprises a plurality of conductive layers in the PCB.

92. The apparatus of claim 90 wherein each of the interconnects comprises a plurality of conductive vias in the circuit board.

93. The apparatus of claim 83 further comprising interlocking features comprising a contour formed in an interior surface of the first layer, the contour being filled with cured mold compound.

94. The apparatus of claim 91 wherein the circuit board includes a top surface and a bottom surface and the at least one electrical component comprises a set of top-side components mounted on the top surface and a set of bottom-side components mounted on the bottom surface.

95. The apparatus of claim 94 wherein
the top-side components include a number, T, of large-footprint components, the
bottom-side components include a number, B, of large-footprint components, and the
number T is approximately equal to the number B.

96. The apparatus of claim 95 wherein each of most of the top-side large-footprint
components shares a respective set of conductive vias with a corresponding one of the
bottom-side large-footprint components.

97. The apparatus of claim 95 wherein each of most of the top-side large-footprint
components is located in a respective footprint shared by a corresponding one of the
bottom-side large-footprint components.

98. The apparatus of claim 89 wherein the circuit board includes a top surface and
a bottom surface, the at least one electrical component comprises a number, T, of large-
footprint components mounted on the top surface, and most of the large-footprint
components are distributed symmetrically in relation to an axis on the top surface.

99. The apparatus of claim 98 wherein the axis is along a midline of the top
surface.

100. The apparatus of claim 98 wherein the axis is defined in relation to a
predetermined component.

101. The apparatus of claim 83 further comprising:
an adapter for providing mechanical and electrical connections between the
modular package and an external mounting surface, the adapter having a body and a
plurality of electrical terminals supported by the body; and
electrical connections formed between the adapter terminals and respective
interconnects on the modular package;
the adapter body being mechanically secured to the modular package;

the adapter terminals being arranged to mate with the external mounting surface.

102. The apparatus of claim 101 wherein the external mounting surface is a circuit board and the terminals comprise ends constructed and arranged to be inserted into conductive holes in the circuit board.

103. The apparatus of claim 101 wherein the external mounting surface is a circuit board and the terminals comprise ends constructed and arranged to be surface mount soldered to the circuit board.

104. The apparatus of claim 101 wherein the external mounting surface includes a connector and the terminals comprise ends constructed and arranged to mate with the connector.

105. The apparatus of claim 101 wherein the interconnects are disposed along a long edge of the modular package, the adapter is secured to the long edge of the modular package, and the first and second layers of the modular package are oriented perpendicular to the mounting surface.

106. The apparatus of claim 101 wherein the interconnects are disposed along opposite edges of the modular package, the adapter is secured to the opposite edges, and the first and second layers of the modular package are oriented essentially parallel to the mounting surface.

107. The apparatus of claim 106 wherein the adapter is constructed and arranged to maintain one of the first or second layers in contact with the mounting surface.

108. The apparatus of claim 83 wherein the interconnects further comprise a surface constructed and arranged for engagement with a connector terminal.

109. The apparatus of claim 83 wherein the interconnects further comprise a layer of metal plating.

110. An apparatus comprising:

a modular package having a first external surface, a second external surface opposite the first external surface, and a side wall extending along the perimeter of and connecting with the first and second external surfaces, the modular package including a first layer of metal defining the first external surface and a second layer of metal defining the second external surface, the first and second layers being separated by and in contact with cured mold compound; and

an electrical circuit for converting power, including a printed circuit board ("PCB"), a plurality of components including semiconductors mounted to the PCB, and a plurality of interconnects electrically connected to the components, the electrical circuit being located between the first and second layers and within the cured mold compound;

the side wall comprising a strip formed by the first layer, a strip formed by the second layer, a strip formed by the PCB, and one or more strips formed by the cured mold compound;

the interconnects being disposed within the side wall.

111. The apparatus of claim 110 wherein the first and second layers further comprise aluminum, and the first external surface further comprises a plurality of fins.

112. The apparatus of claim 110 wherein the first and second layers further comprise aluminum, and the first external surface comprises an essentially flat area.

113. The apparatus of claim 112 wherein the essentially flat area is adapted for a solder joint.

114. The apparatus of claim 110 wherein the PCB includes a top surface and a bottom surface, the semiconductors include large-footprint switches, a number, T, of top-side large-footprint switches are mounted on the top surface, a number, B, of bottom-side large-footprint switches are mounted on the bottom surface, and the number T is approximately equal to the number B.

115. The apparatus of claim 114 wherein each large-footprint switch is connected to one or more other components by a respective set of conductive vias in the PCB, and each of a plurality of the top-side large-footprint switches shares its respective set of conductive vias with a corresponding one of the bottom-side large-footprint switches.

116. The apparatus of claim 114 wherein each of most of the large-footprint switches is positioned on one surface in a location substantially overlapping a location on the other surface occupied by another large-footprint switch.

117. The apparatus of claim 110 wherein the PCB includes a top surface and a bottom surface, the semiconductors include large-footprint switches mounted on the top surface, most of the large-footprint components are distributed symmetrically between a left side of the top surface and an opposite right side of the top surface.

118. The apparatus of claim 110 wherein the PCB includes a top surface and a bottom surface, the semiconductors include large-footprint switches mounted on the top surface, and most of the large-footprint components are distributed symmetrically between quadrants on the top surface.

119. The apparatus of claim 118 wherein the components further comprise a transformer core, and the quadrants surround the transformer core.

120. The apparatus of claim 110 wherein the PCB includes a top surface and a bottom surface, the components include a set of high-profile components having similar heights, a number, T, of the high-profile components are mounted on the top surface, a number, B, of the high-profile components are mounted on the bottom surface, and the number T is approximately equal to the number B.

121. The apparatus of claim 120 wherein a spatial distribution of the high-profile components on the top surface roughly matches a spatial distribution of the high-profile components on the bottom surface.

122. The apparatus of claim 110 further comprising:

an adapter for providing mechanical and electrical connections between the modular package and an external mounting surface, the adapter having a body and a plurality of electrical terminals supported by the body; and

electrical connections formed between the adapter terminals and respective interconnects on the modular package,

the adapter body being mechanically secured to the modular package;

the adapter terminals being arranged to mate with the external mounting surface.

123. The apparatus of claim 122 wherein the external mounting surface is a circuit board and the terminals comprise ends constructed and arranged to be inserted into conductive holes in the circuit board.

124. The apparatus of claim 122 wherein the external mounting surface is a circuit board and the terminals comprise ends constructed and arranged to be surface mount soldered to the circuit board.

125. The apparatus of claim 122 wherein the external mounting surface includes a connector and the terminals comprise ends constructed and arranged to mate with the connector.

126. The apparatus of claim 122 wherein the interconnects are disposed along a long edge of the modular package, the adapter is secured to the long edge of the modular package, and the first and second layers of the modular package are oriented perpendicular to the mounting surface.

127. The apparatus of claim 122 wherein the interconnects are disposed along opposite edges of the modular package, the adapter is secured to the opposite edges, and the first and second layers of the modular package are oriented essentially parallel to the mounting surface.

128. The apparatus of claim 127 wherein the adapter is constructed and arranged to maintain one of the first or second layers in contact with the mounting surface.

129. The apparatus of claim 110 wherein the interconnects further comprise a surface constructed and arranged for engagement with a connector terminal.

130. The apparatus of claim 110 wherein the interconnects further comprise a layer of metal plating.

131. The apparatus of claim 110 wherein the first layer further comprises contours formed in an interior surface of the first layer, the contours including a first feature having a shape and an elevation to accommodate a first component on the PCB, the first component having a height greater than or less than other components on the PCB.

132. The apparatus of claim 131 wherein the first component includes a magnetic core structure and the elevation is a recess in the interior surface.

133. The apparatus of claim 131 wherein the first component includes a semiconductor switch and the elevation is a protrusion from the interior surface.

134. The apparatus of claim 110 further comprising interlocking features comprising a contour formed in an interior surface of the first layer, the contour being filled with cured mold compound.

135. The apparatus of claim 110 wherein the modular package further comprises a recess formed in the first layer adjacent one or more of the interconnects providing a setback between the first layer and the one or more interconnects.

136. An apparatus comprising:
a power converter including
a printed circuit board ("PCB") comprising a plurality of conductive layers and having a top surface and a bottom surface;
a magnetic core structure magnetically coupled to a winding formed by traces in one or more of the conductive layers in the PCB; and
a plurality of power semiconductor devices, a first set of the power semiconductor devices being mounted on the top surface and electrically connected to dissipate power at a level, P_t , during operation of the converter, a second set of the power semiconductor devices being mounted on the bottom surface and electrically connected to dissipate power at a level, P_b , during operation of the converter;
wherein the power semiconductor devices are distributed between the first and second sets to distribute heat generation during operation of the converter such that each level P_t , P_b is less than 150% of the other level P_b , P_t .

137. The apparatus of claim 136 wherein a plurality of the power semiconductor devices in the first set are each positioned in a location on the top surface substantially overlapping a location on the bottom surface occupied by a power semiconductor device in the second set.

138. The apparatus of claim 136 wherein the power semiconductor devices are electrically connected using a respective set of conductive vias in the PCB, and a plurality of the power semiconductor devices in the first set share their respective sets of conductive vias with corresponding power semiconductor devices in the second set.

139. The apparatus of claim 137 wherein the power converter further comprises circuitry including a pair of cells having a common circuit topology and each including power semiconductor switches from each of the first and second sets; each cell having its respective components arranged in a pattern, wherein the pattern of components of one cell is substantially a mirror image of the pattern of components in the other cell.

140. The apparatus of claim 139 wherein a component from one of the cells is located on an opposite surface of a respective component from the other one of the cells.

141. The apparatus of claim 140 wherein the cells comprise input cells.

142. The apparatus of claim 138 wherein the power semiconductor devices comprise output switches.

143. A method of manufacturing a plurality of products comprising:
inserting a plurality of components into a cavity formed by one or more molds;
closing the one or more molds to form a seal around the cavity;
filling the cavity with mold compound;
curing the mold compound in the cavity to secure the components, cured mold compound, and molds together into an assembly; and
cutting the assembly to separate the plurality of products from the assembly, the products each including a respective section of the one or more molds which remains as an integral part of each respective product.

144. The method of claim 143 further comprising maintaining a predetermined alignment between the plurality of components and the one or more molds.

145. The method of claim 143 wherein the plurality of components include a substrate having conductive features; and the cutting comprises exposing portions of the conductive features in each respective product.

146. A method of making a plurality of electronic devices, comprising:
providing a PCB panel including a PCB having a plurality of conductive traces and a plurality of components electrically connected to form a plurality of circuits, the PCB having a plurality of contacts for making an electrical connections to the plurality of circuits;
enclosing the PCB panel, including the PCB, components, and contacts in a mold cavity;
filling the mold cavity with mold compound;
curing the mold compound;
removing the encapsulated PCB panel from the mold; and
cutting the encapsulated PCB panel to expose at least a portion of the plurality of contacts and form respective exposed contacts.

147. The method of claim 146 further comprising:
mating the PCB panel with a center plate having an opening for receiving the PCB panel;
covering the opening of the center plate with at least one plate; and
wherein the enclosing comprises closing the mold against the center plate and exerting pressure to force the at least one plate against the center plate; and
wherein the removing comprises removing the at least one plate from the encapsulated assembly.

148. The method of claim 146 further comprising:
mating the PCB panel with a center plate; and
wherein the enclosing comprises closing the mold against the center plate;
wherein the curing bonds the PCB panel to the center plate;

wherein the removing comprises removing the bonded centerplate and PCB panel.

149. The method of claim 146 further comprising:
providing one or more registration features for positioning the panel assembly in the mold cavity;
creating voids in the mold compound with one or more of the registration features by displacing uncured mold compound for reference after the encapsulated PCB panel is removed from the mold.

150. The method of claim 146 further comprising:
making holes in the cured mold compound to expose portions of the PCB.

151. The method of claim 146 further comprising removing a layer of material from a first surface of the encapsulated panel assembly.

152. The method of claim 151 further comprising removing a layer of material from a second surface of the encapsulated PCB panel.

153. The method of claim 151 wherein the components include a magnetic core and the removing a layer of material exposes a surface of the magnetic core.

154. The method of claim 153 wherein the removing a layer of material from the top surface of the encapsulated panel assembly provides finished surfaces and reduces the encapsulated panel assembly to a controlled thickness.

155. The method of claim 146 wherein the cutting divides the substrate producing at least one module having a plurality of layers including respective portions of the substrate and the encapsulant on each side of the substrate, and the module contains at least one of the circuits.

156. The method of claim 155 further comprising

providing an adapter having at least one electrical terminal; and
attaching the at least one electrical terminal to the exposed contact.

157. The method of claim 156 wherein the at least one contact comprises a plurality of contacts;

the module comprises a plurality of exposed contacts;

the adapter has a plurality of electrical terminals arranged to match respective ones of the plurality of exposed contacts; and

the adapter is mechanically secured to the module.

158. The method of claim 155, wherein the traces and components are electrically connected to form a plurality of separable circuits arranged in a pattern on the PCB; and the cutting divides the PCB panel along spaces between the separable circuits into a plurality of modules each containing at least one respective circuit.

159. An apparatus comprising:

a modular package having a first external surface, a second external surface opposite the first external surface, and a side wall extending along the perimeter of and connecting with the first and second external surfaces, the first and second external surfaces each being defined at least in part by a respective layer of cured mold compound; and

an electrical circuit for converting power, including a printed circuit board ("PCB"), a plurality of components including semiconductors mounted to the PCB and a plurality of interconnects electrically connected to the components, the electrical circuit being located within the cured mold compound;

the side wall comprising a strip formed by the PCB between strips formed by the layers of cured mold compound defining the first and second external surfaces;

the interconnects being disposed within the side wall.

160. The apparatus of claim 159 wherein the electrical circuit further comprises a magnetic core, and at least one of the first and second external surfaces is also defined by a surface of the magnetic core.

161. The apparatus of claim 160 further comprising:
an adapter for providing mechanical and electrical connections between the modular package and an external mounting substrate, the adapter having a body and a plurality of electrical terminals; and
electrical connections formed between the adapter terminals and respective interconnects on the modular package,
the adapter body being mechanically secured to the modular package and covering the electrical connections;
the adapter terminals being arranged to mate with the external mounting substrate.

162. A method of manufacturing a plurality of products comprising:
inserting a printed circuit board including a plurality of components into a mold cavity;
closing the mold to form a seal around the cavity;
filling the cavity with mold compound;
curing the mold compound in the cavity to form an encapsulated panel assembly;
and
cutting the encapsulated panel assembly to separate the plurality of products from the encapsulated panel assembly, the products each including a respective section of the printed circuit board.

163. An apparatus comprising:
a panel assembly having external surfaces defined at least in part by cured mold compound;
the panel assembly including an internal circuit board, the internal circuit board having a first surface and second surface and a plurality of components electrically connected to form a plurality of individual circuits, each individual circuit being

electrically connected to respective interconnects located along a respective circuit perimeter, the interconnects being contained within the cured mold compound.

164. The apparatus of claim 163 wherein the panel assembly is adapted to be cut along the circuit perimeter separating the individual circuits, dividing the panel assembly into individual circuit modules, and exposing selected portions of the interconnects.

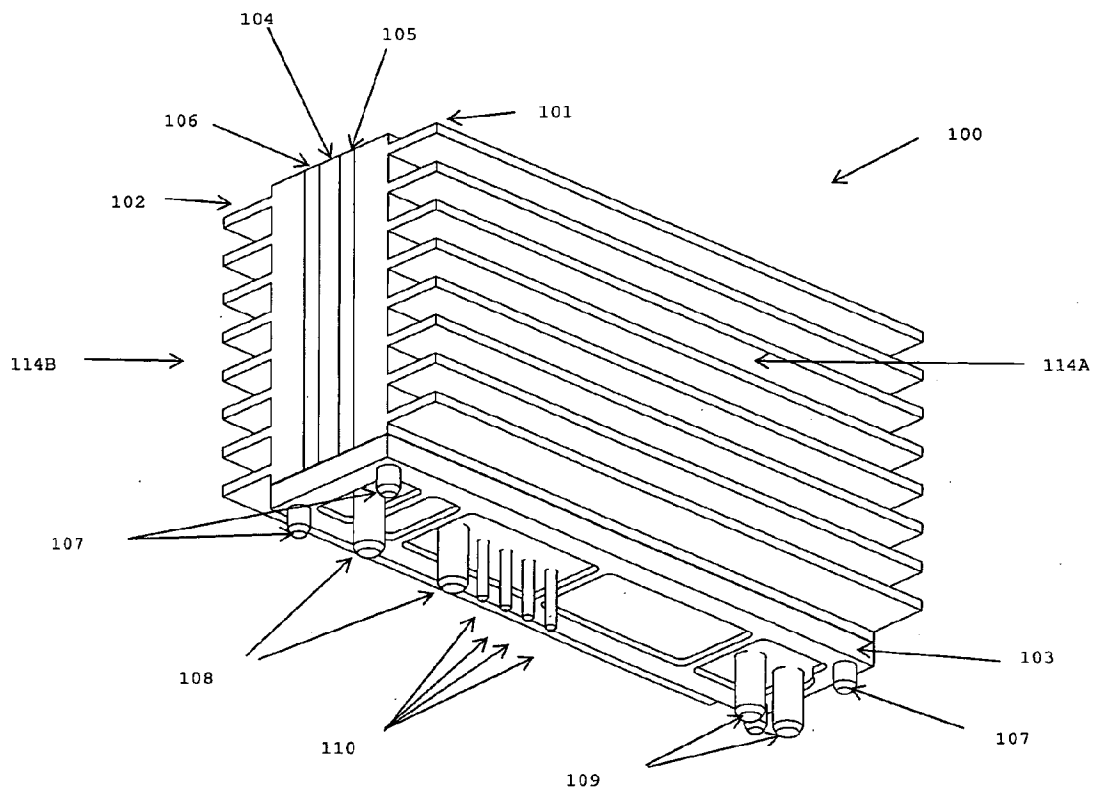


FIG. 1

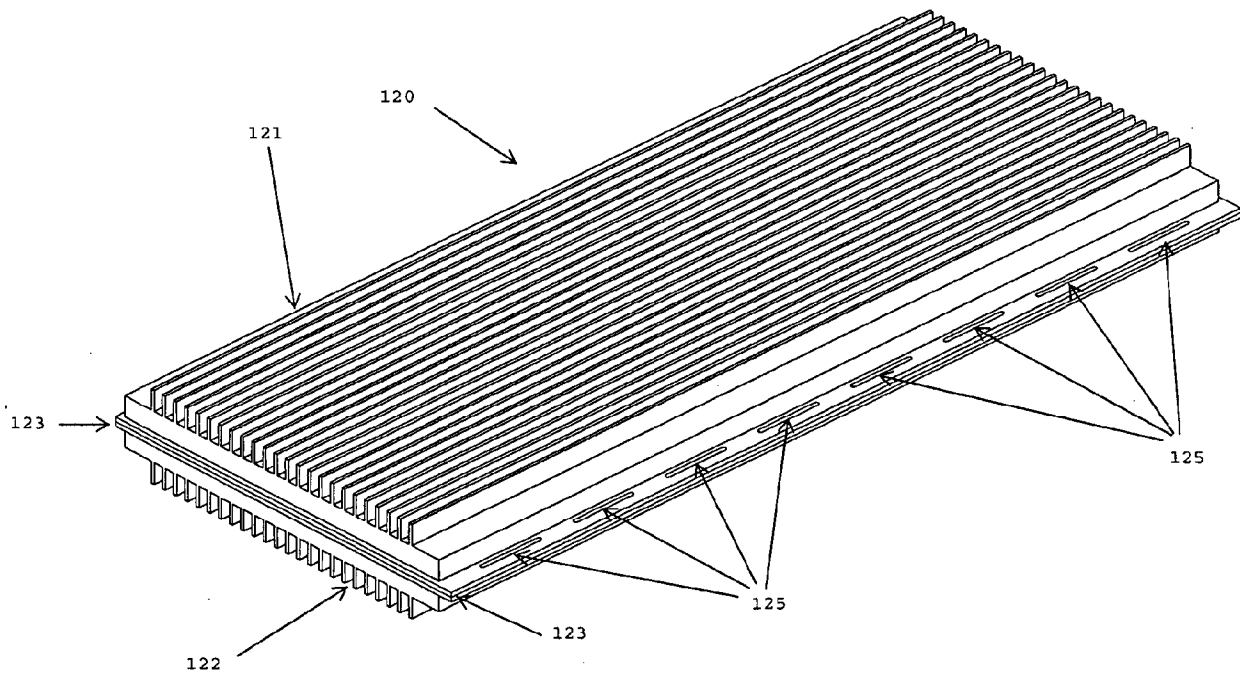


FIG. 2

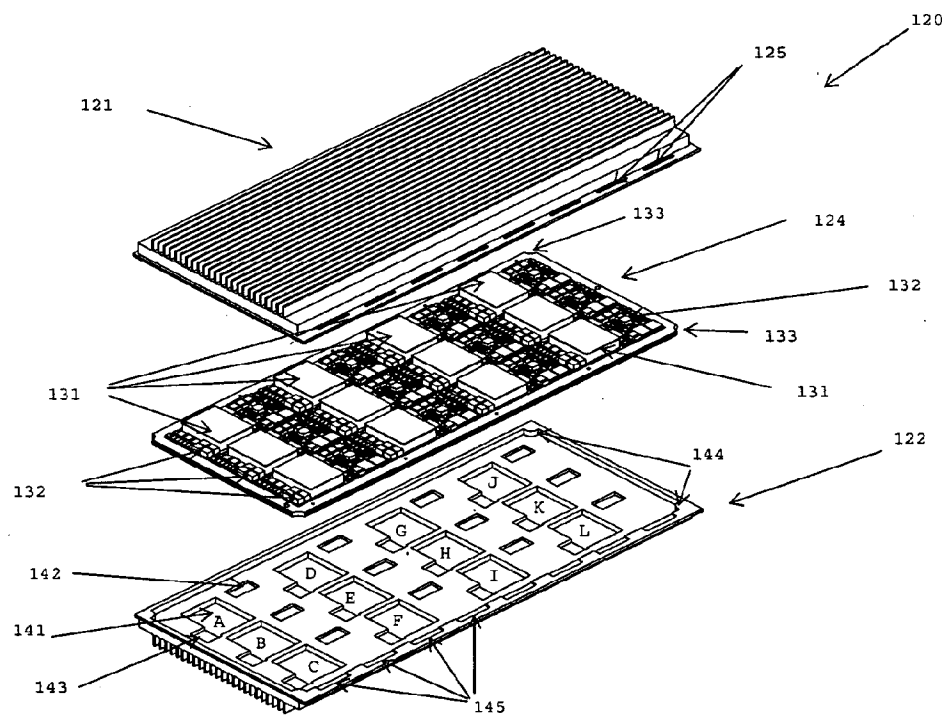


FIG. 3

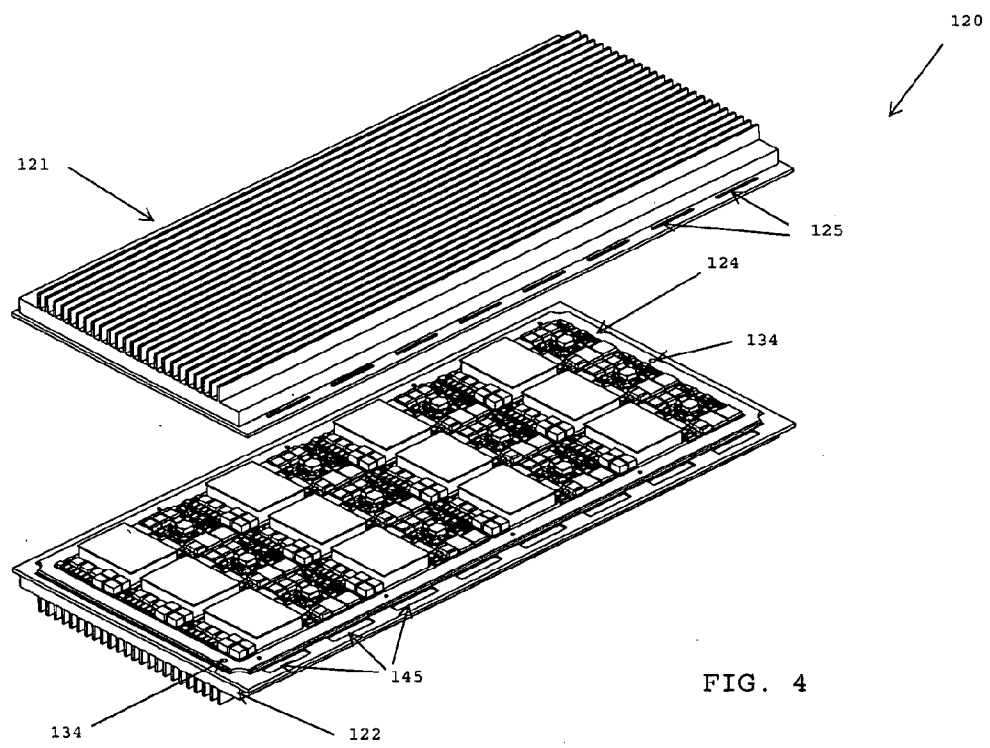
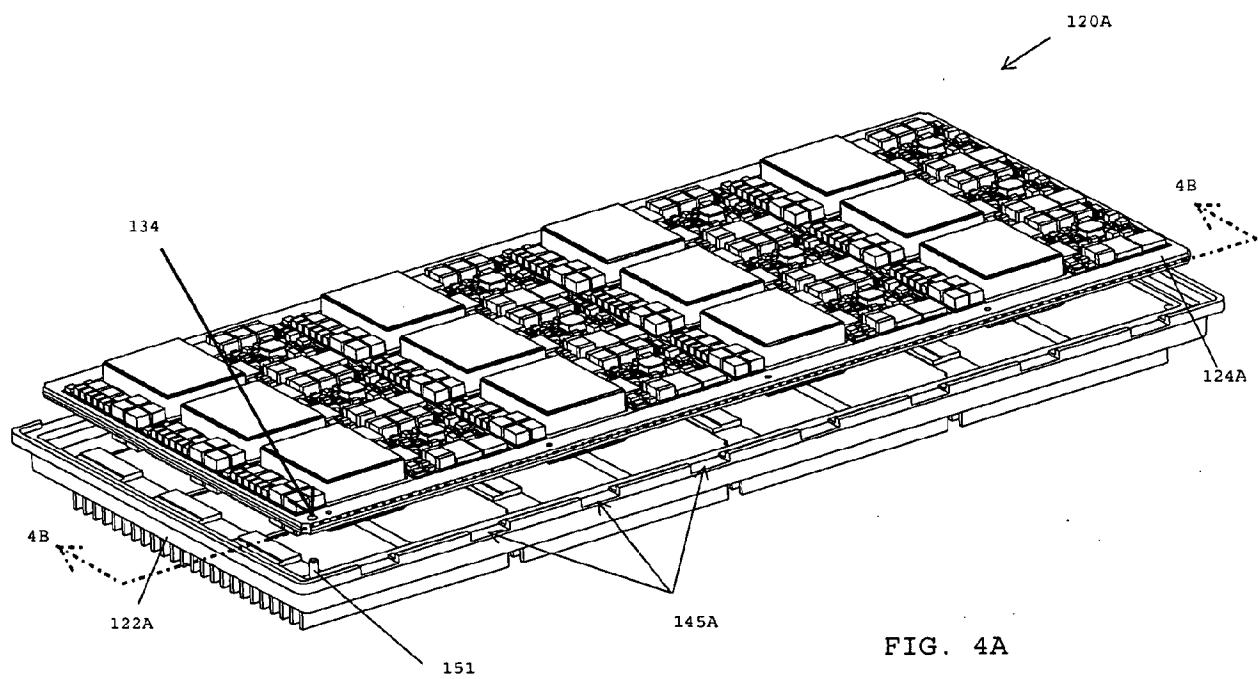


FIG. 4



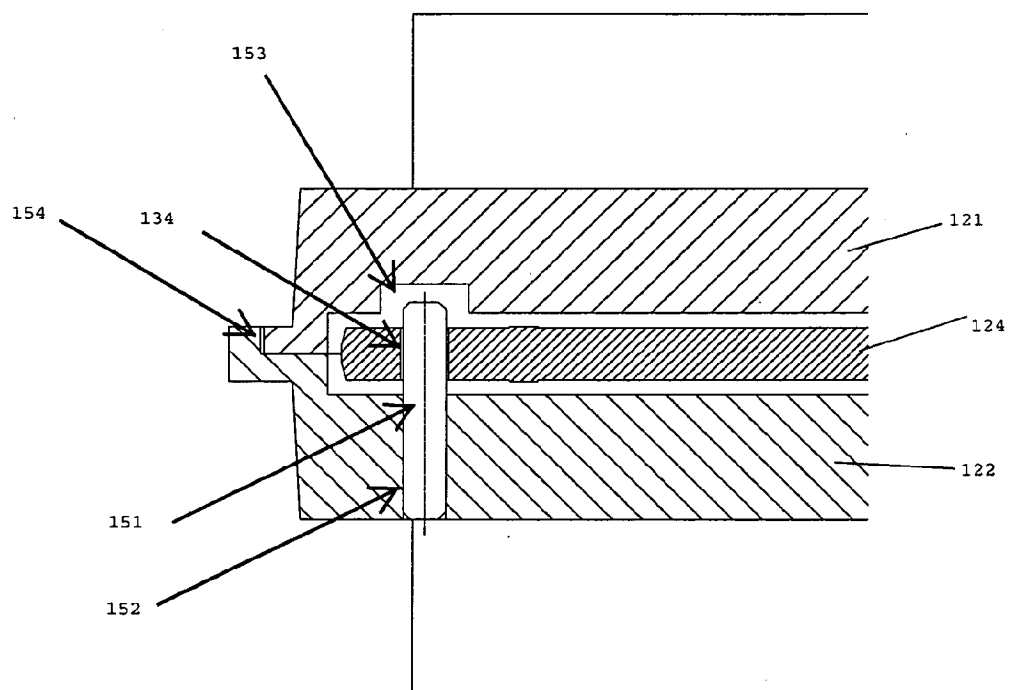


FIG. 4B

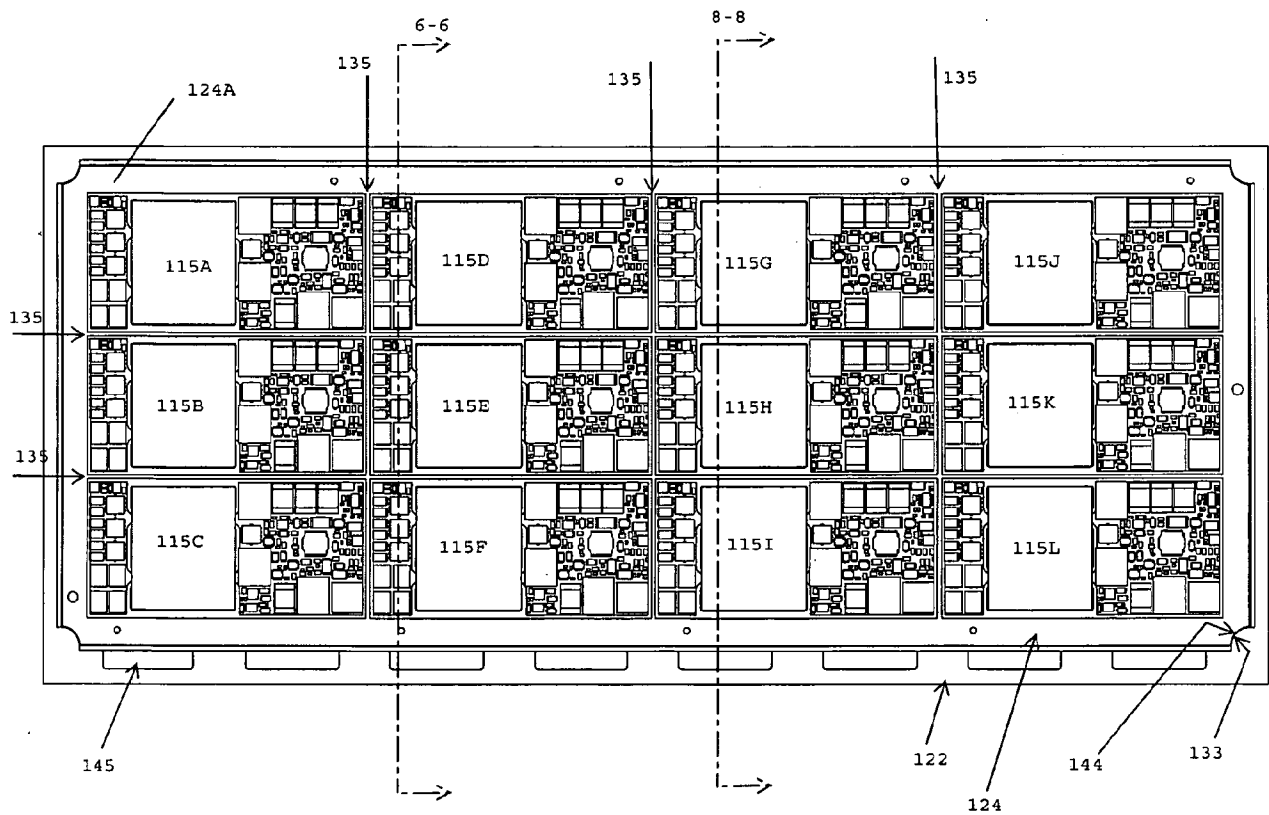


FIG. 5

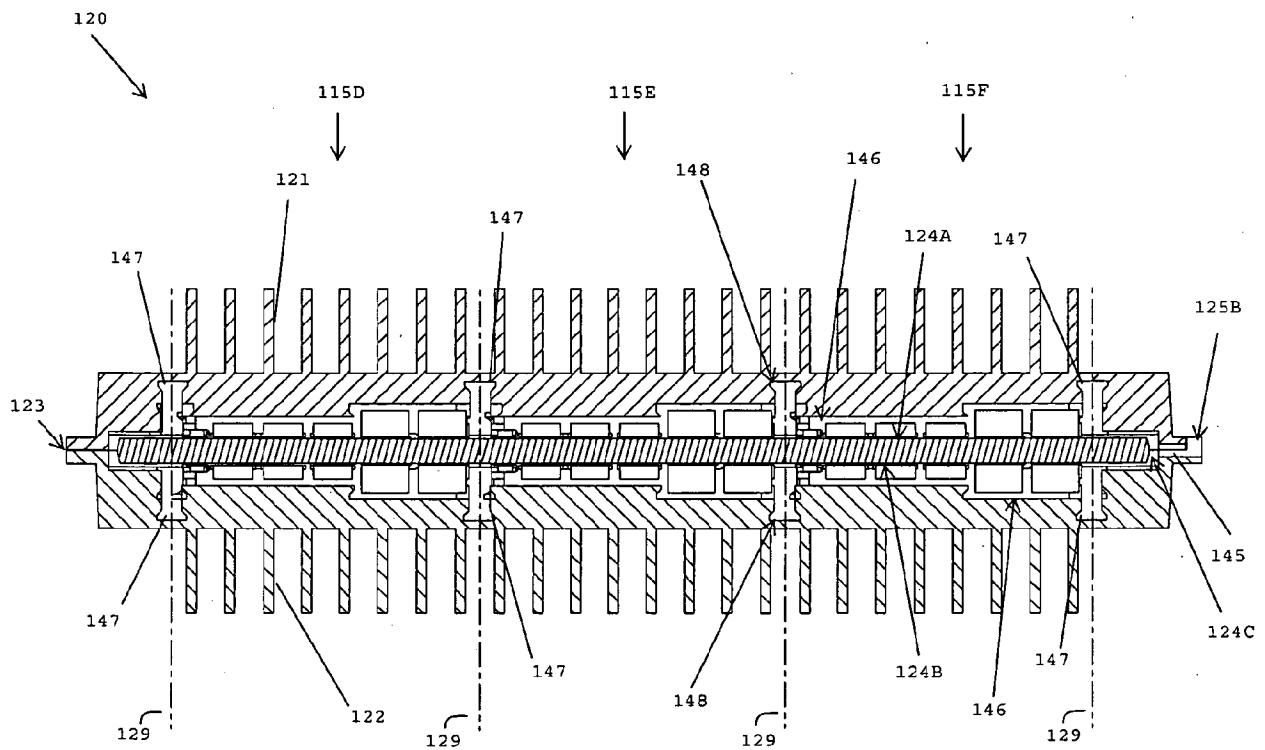


FIG. 6

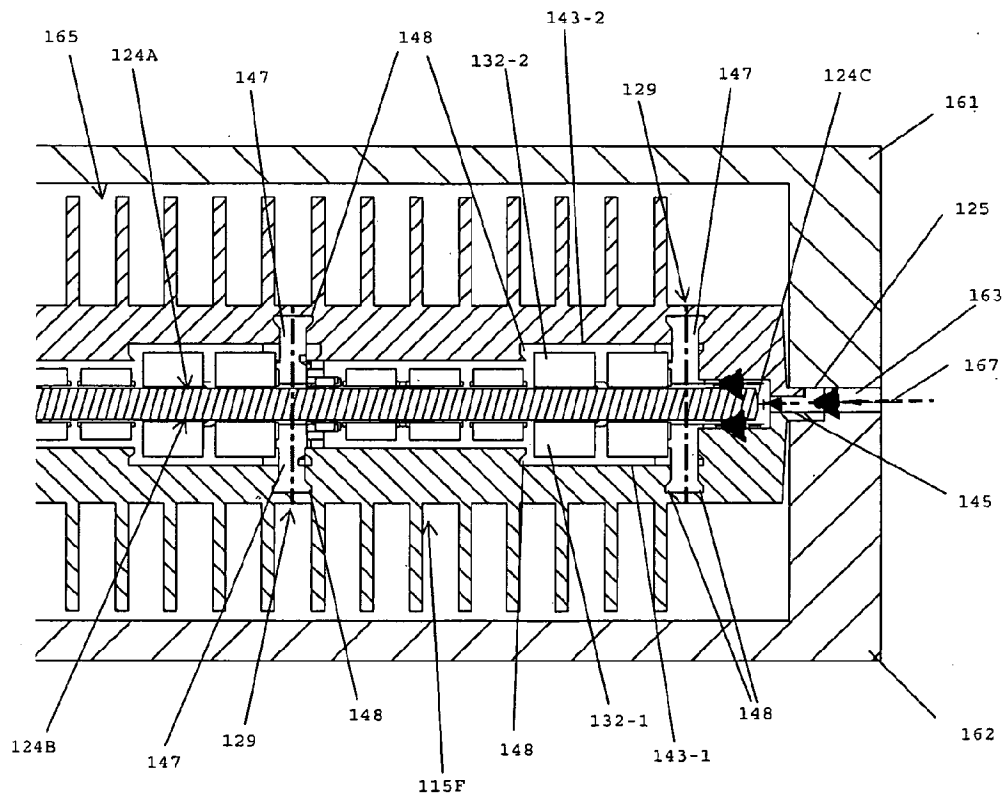
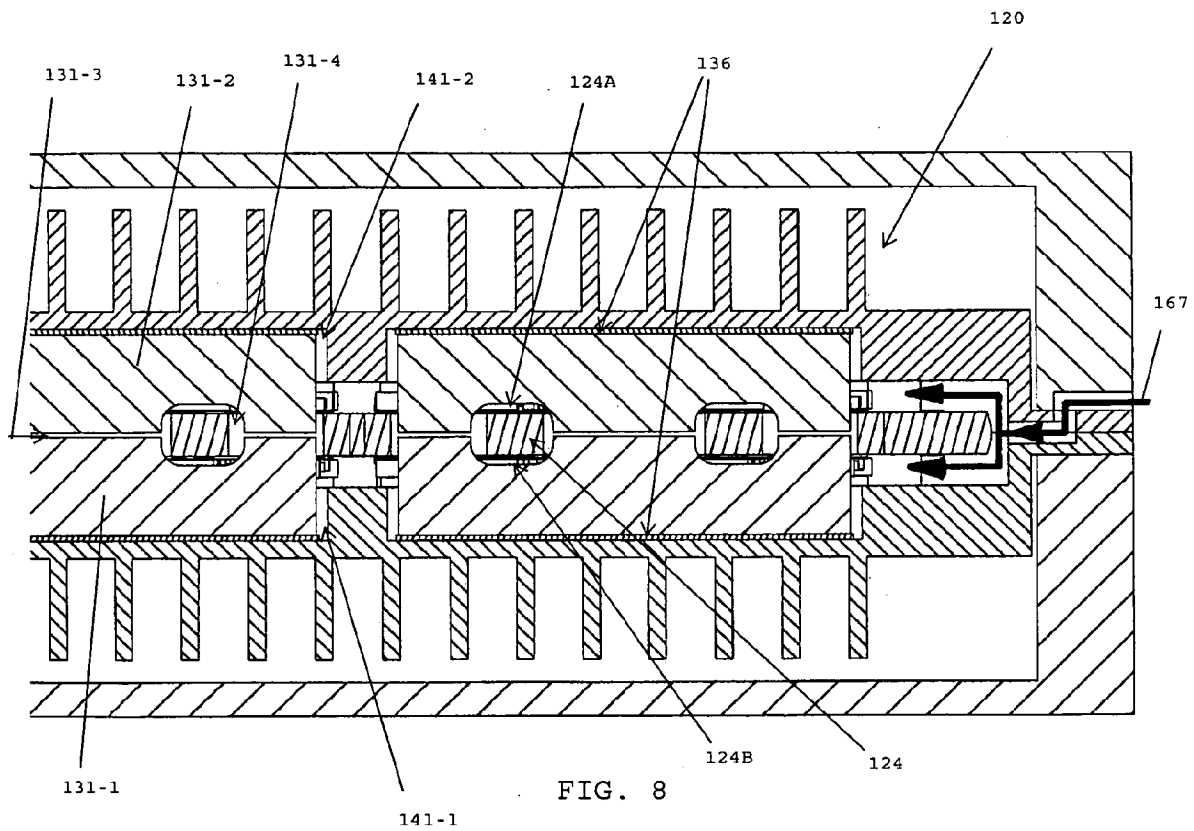


FIG. 7



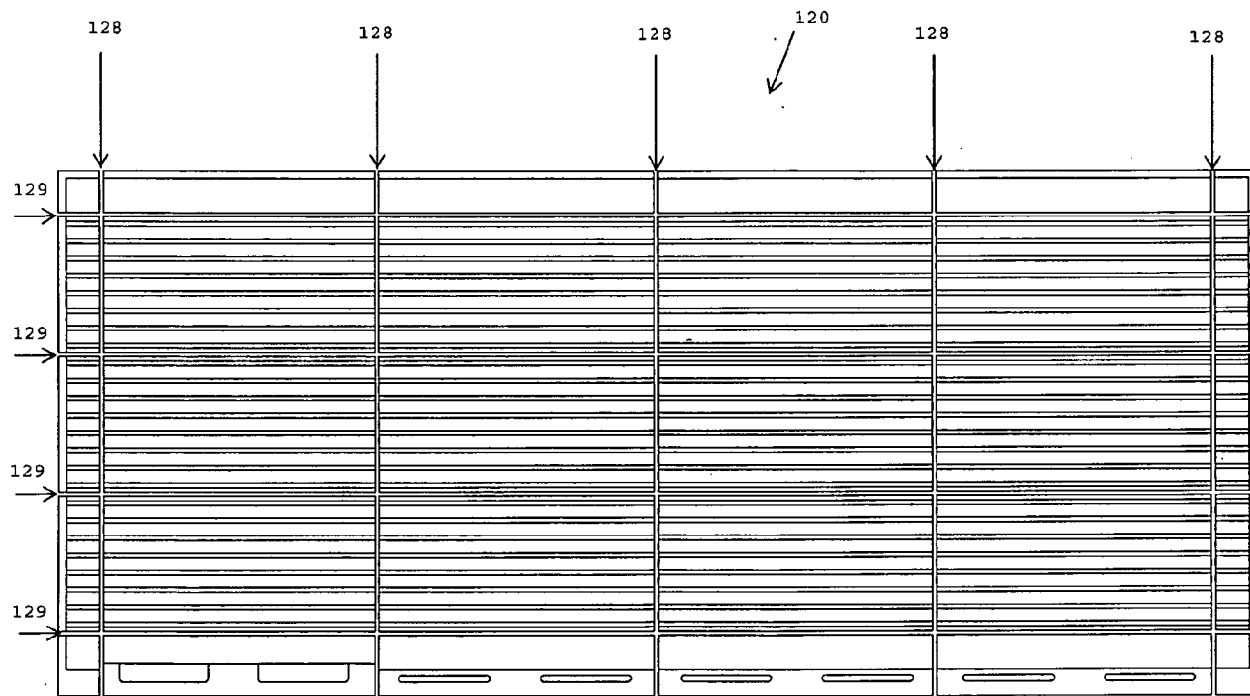


FIG. 9A

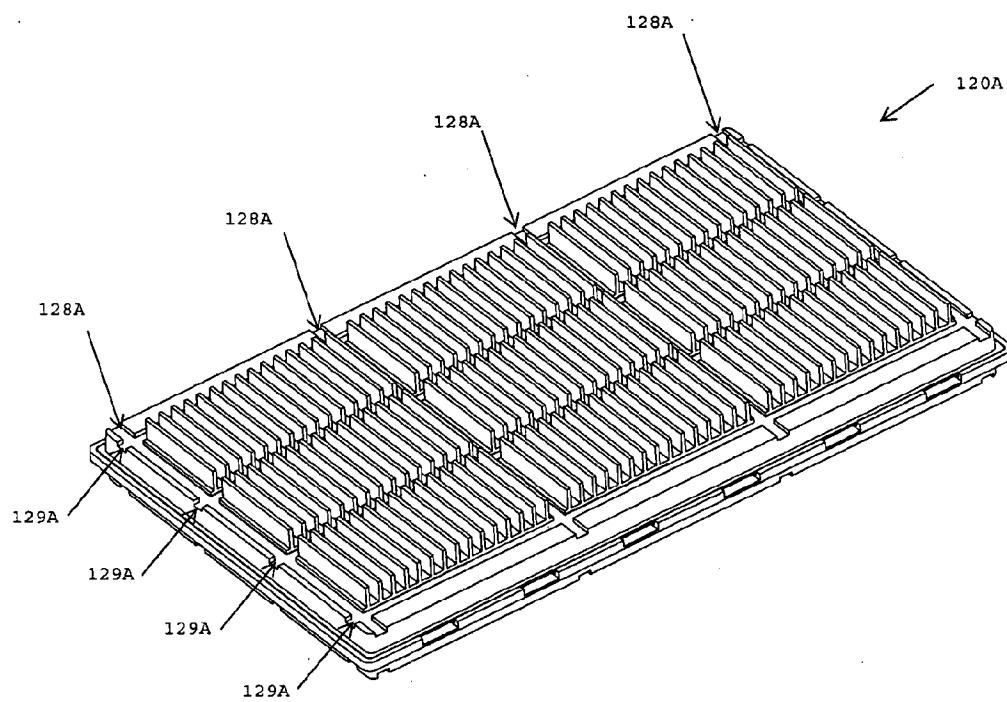


FIG. 9B

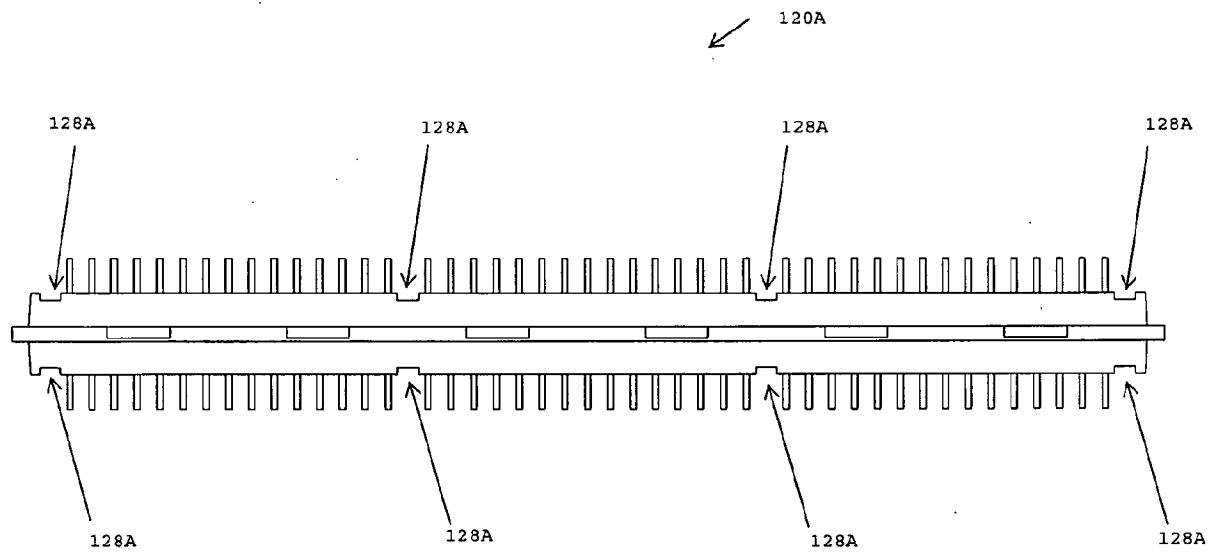


FIG. 9C

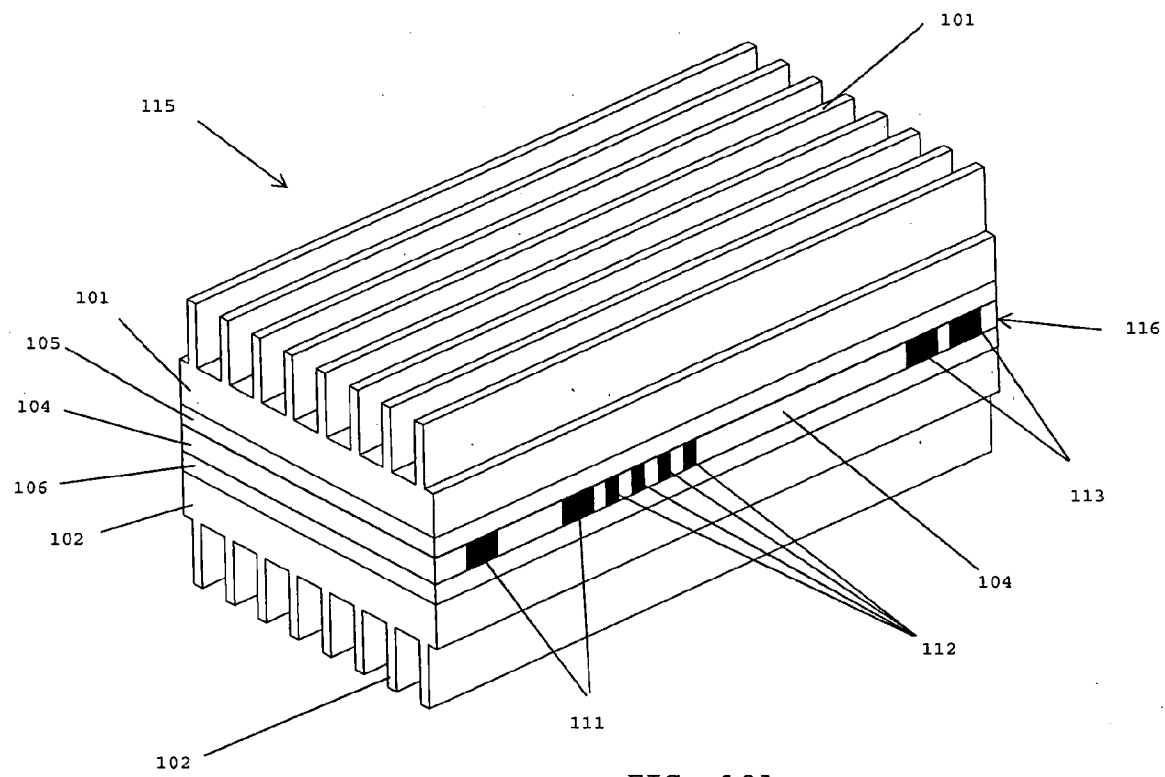


FIG. 10A

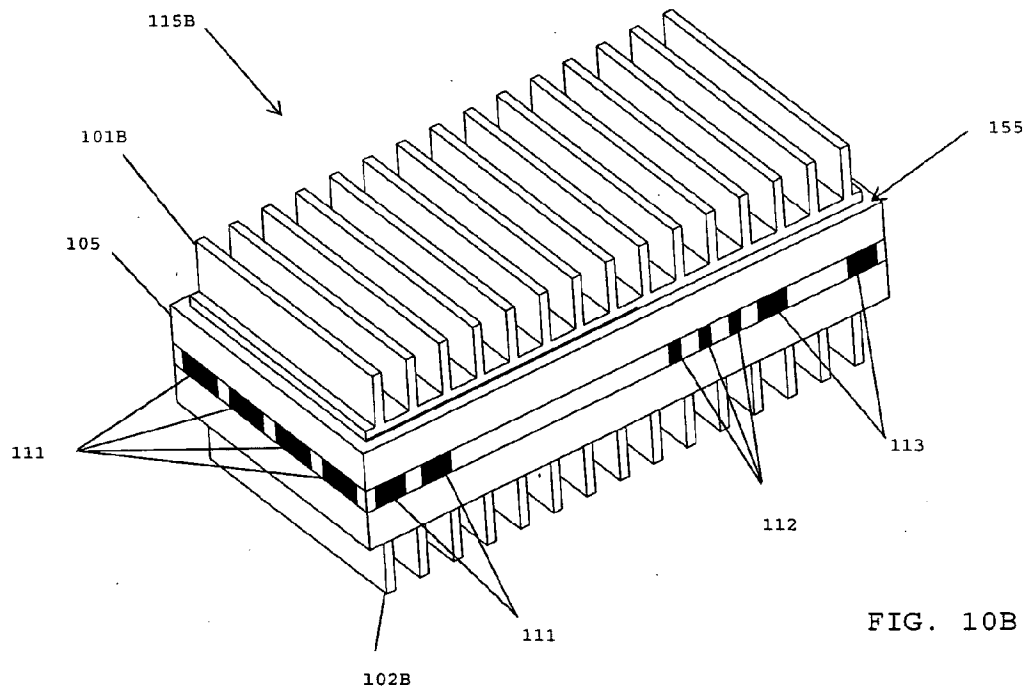
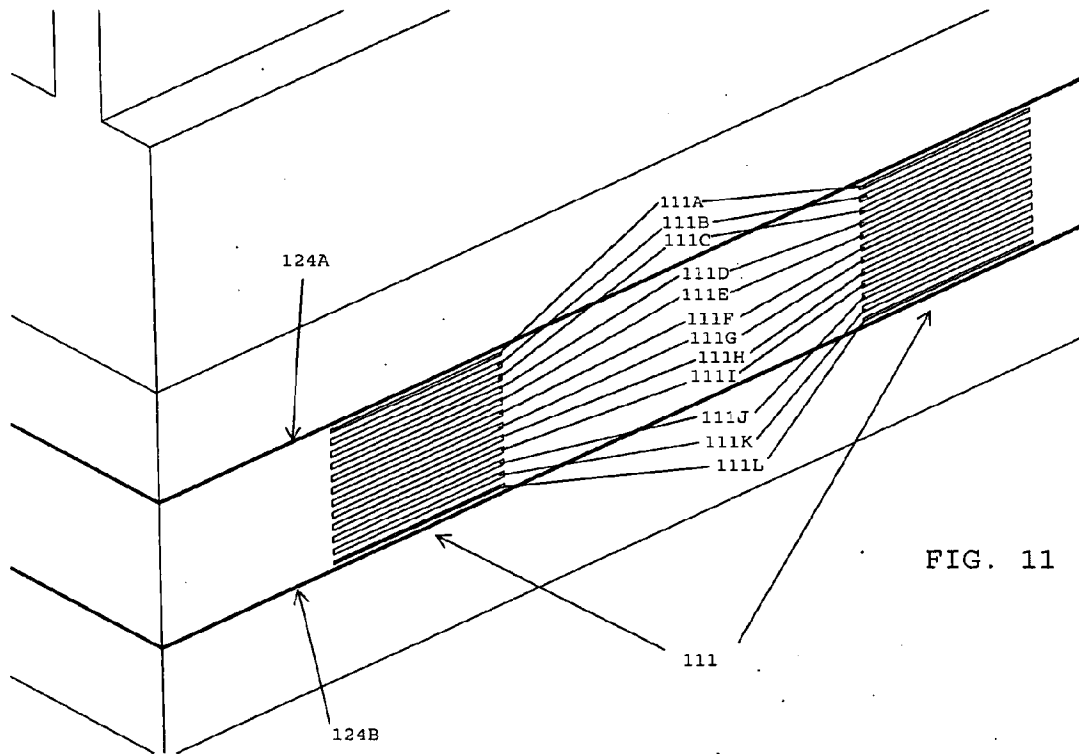


FIG. 10B



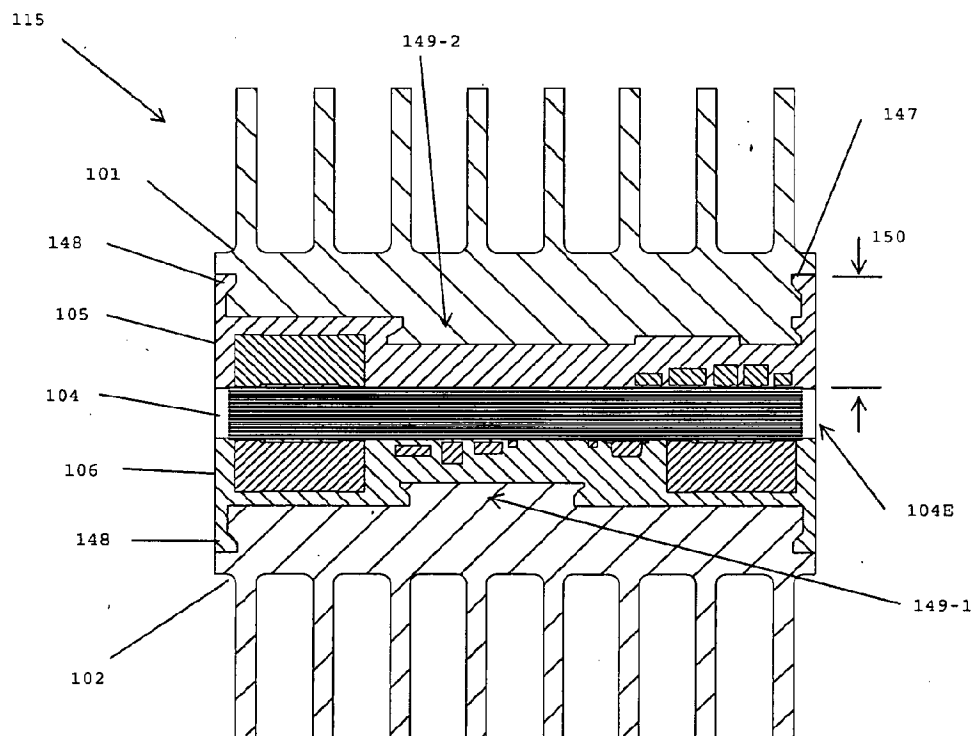


FIG. 12

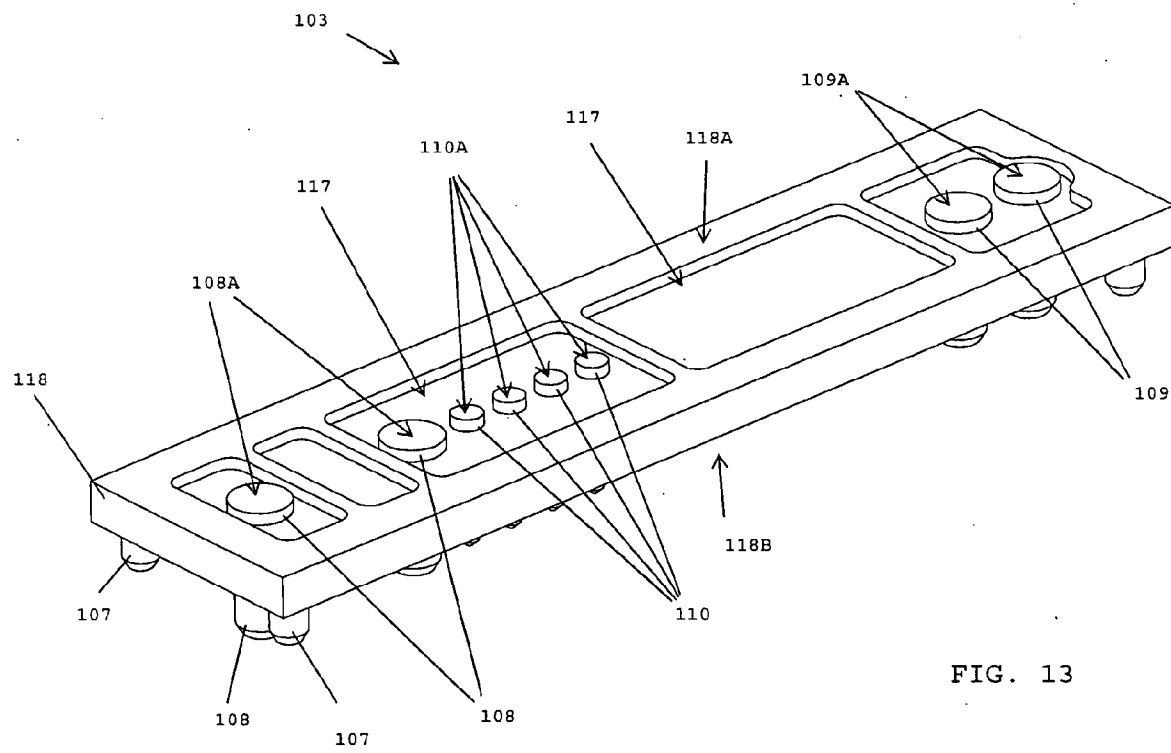


FIG. 13

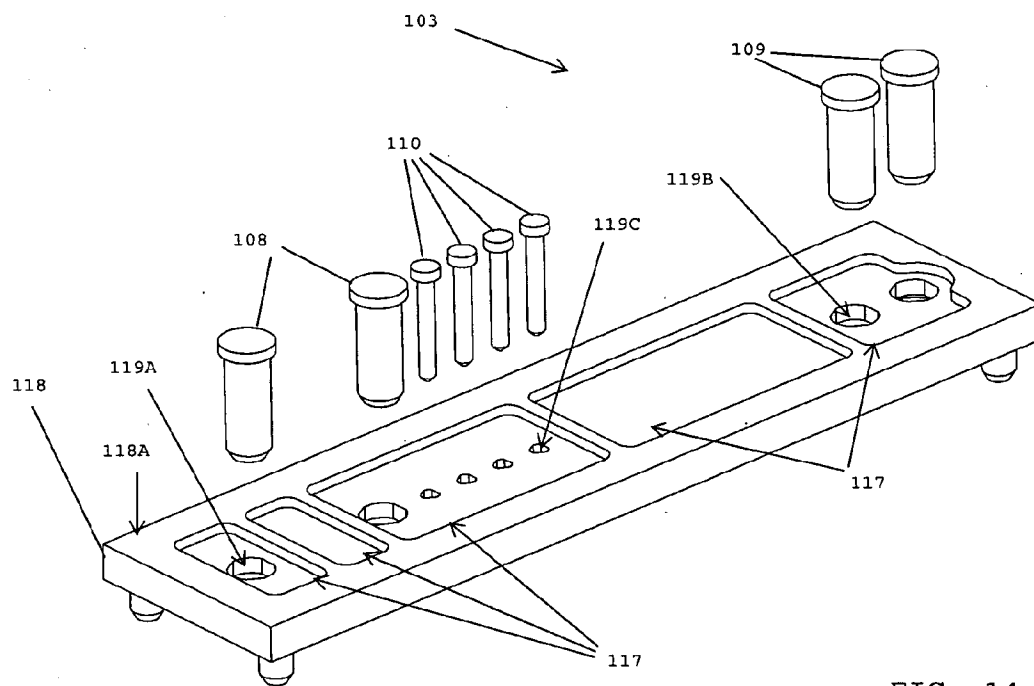


FIG. 14

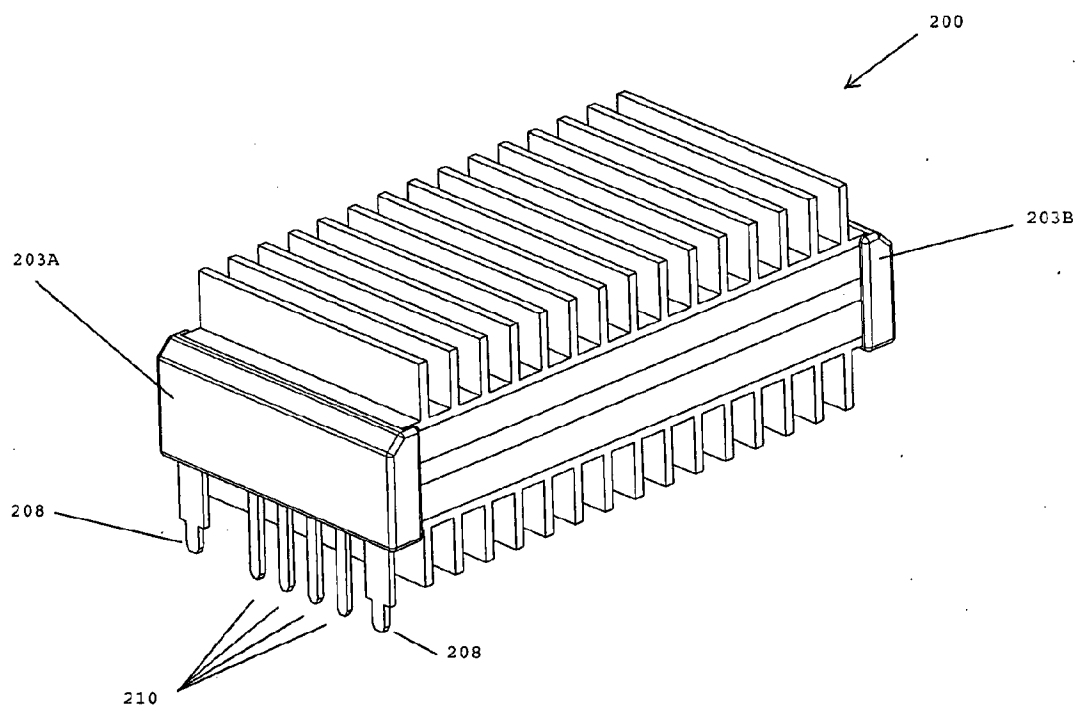


FIG. 15

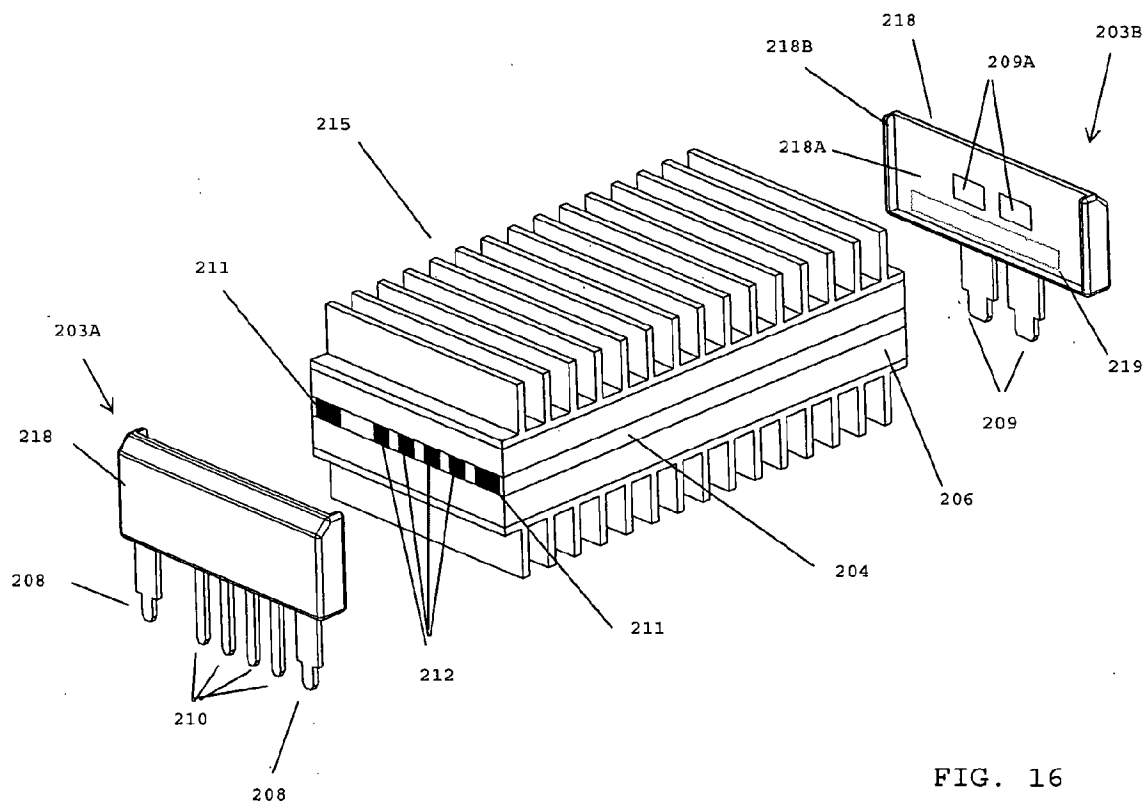


FIG. 16

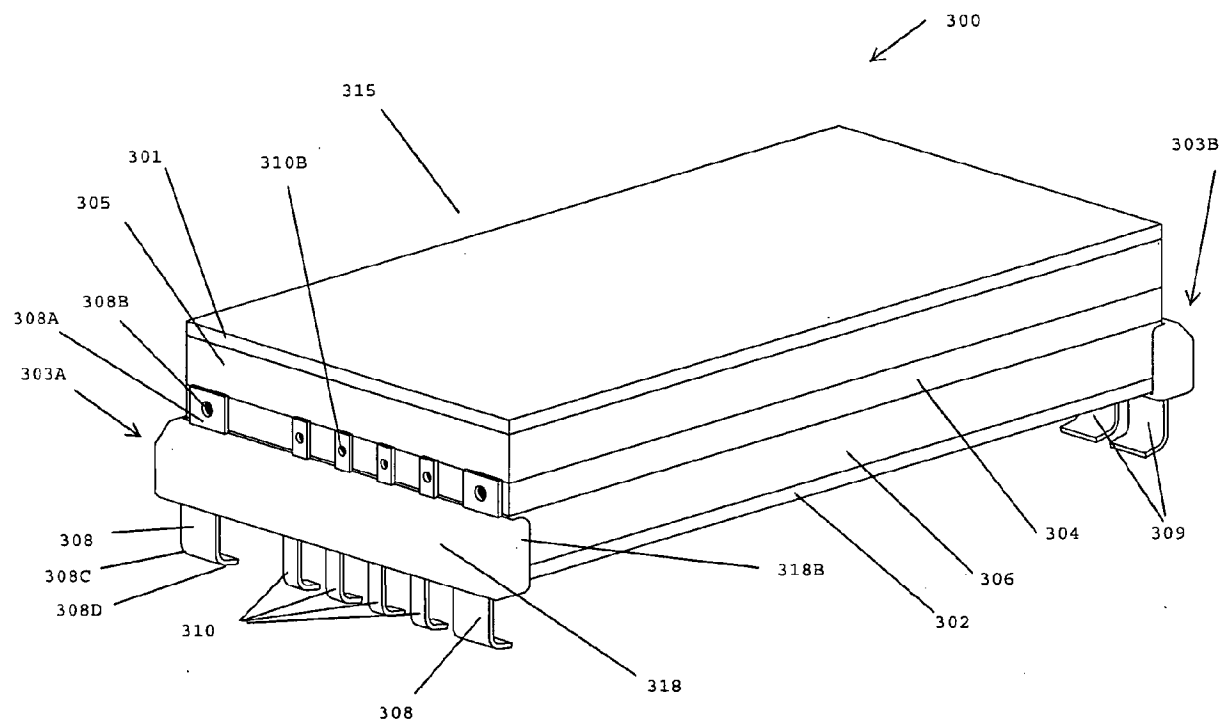


FIG. 17

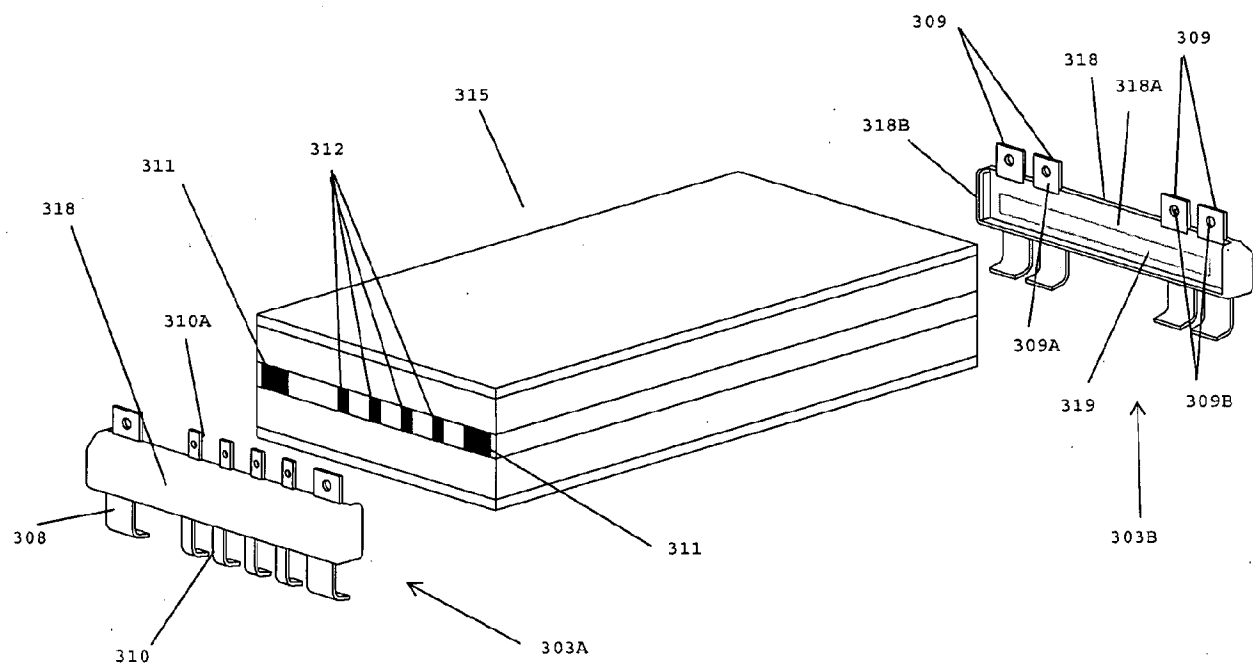


FIG. 18

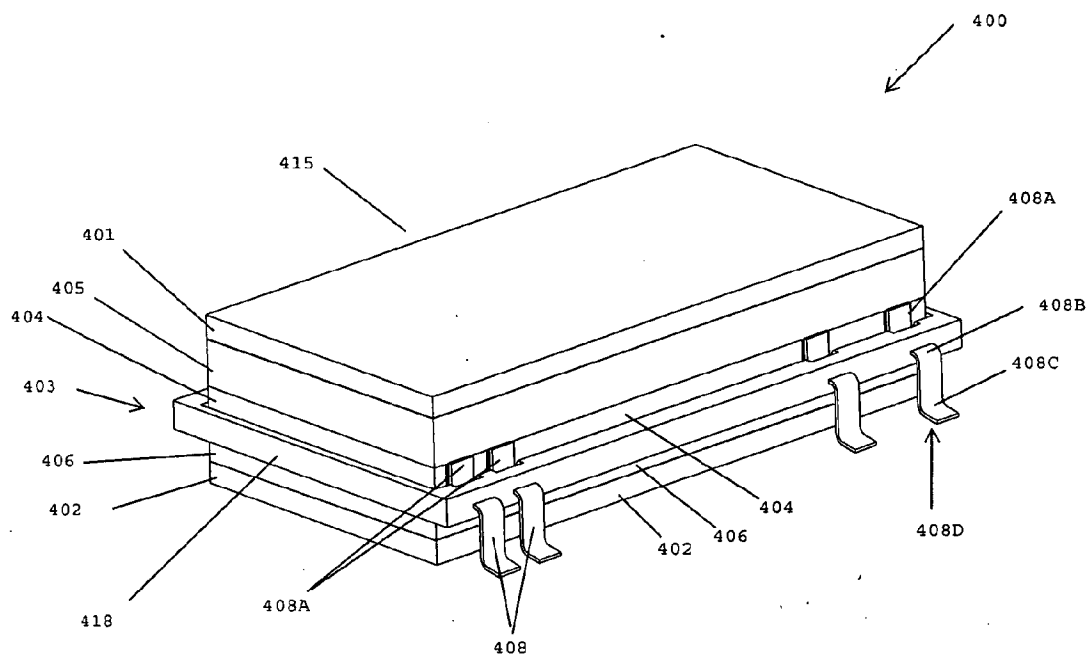


FIG. 19

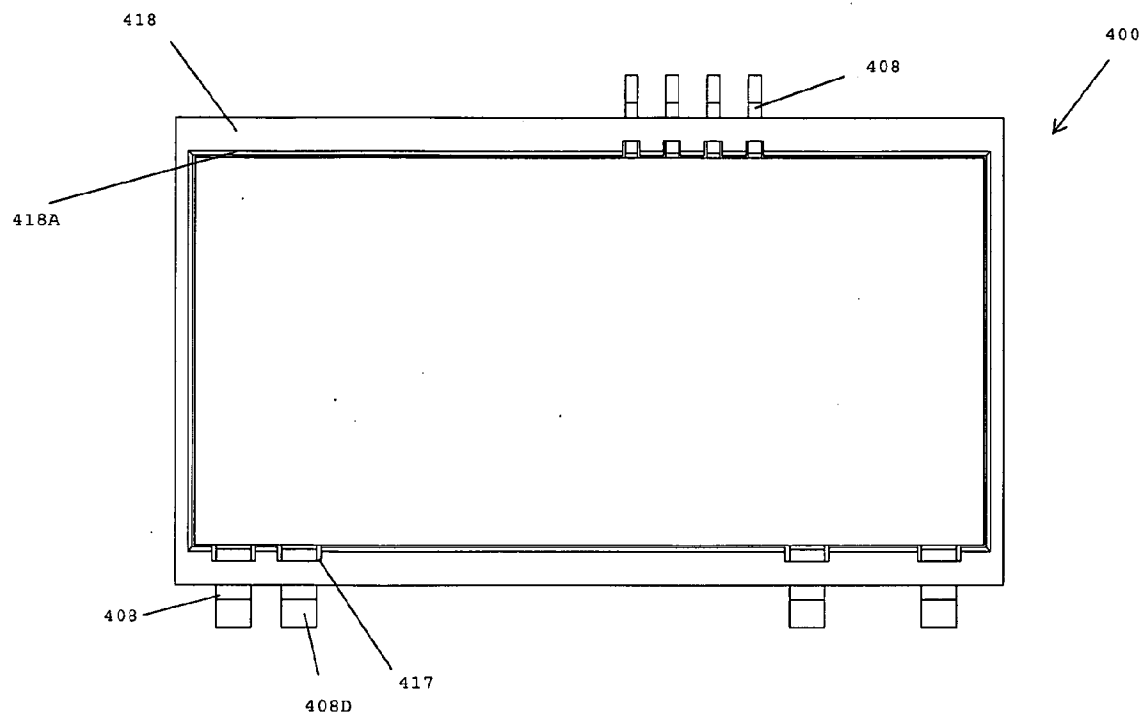


FIG. 20

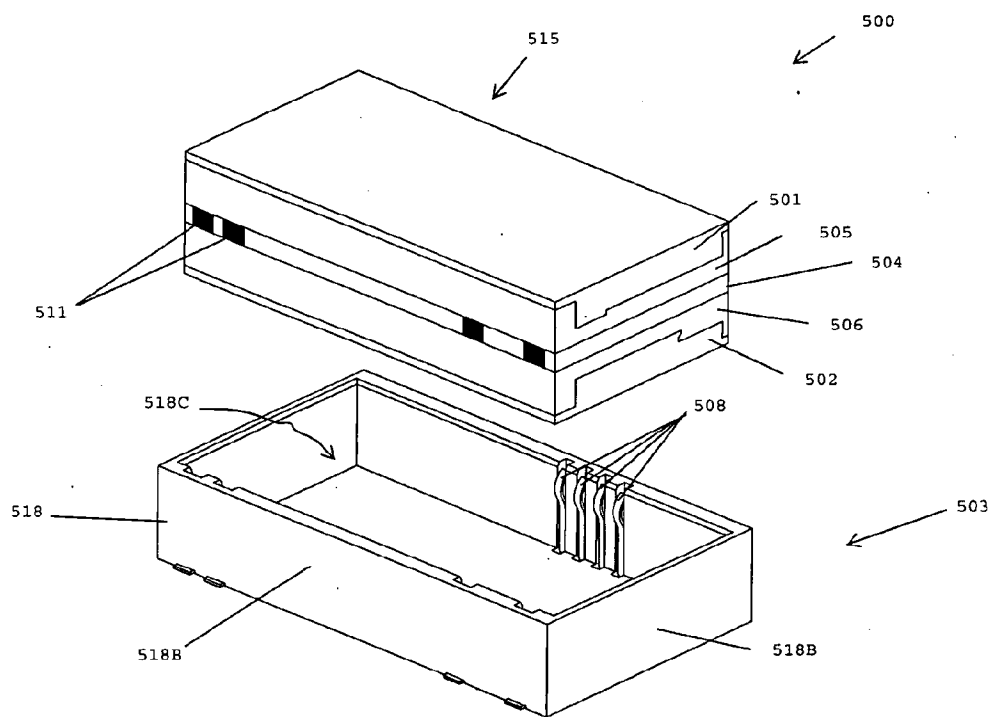


FIG. 21

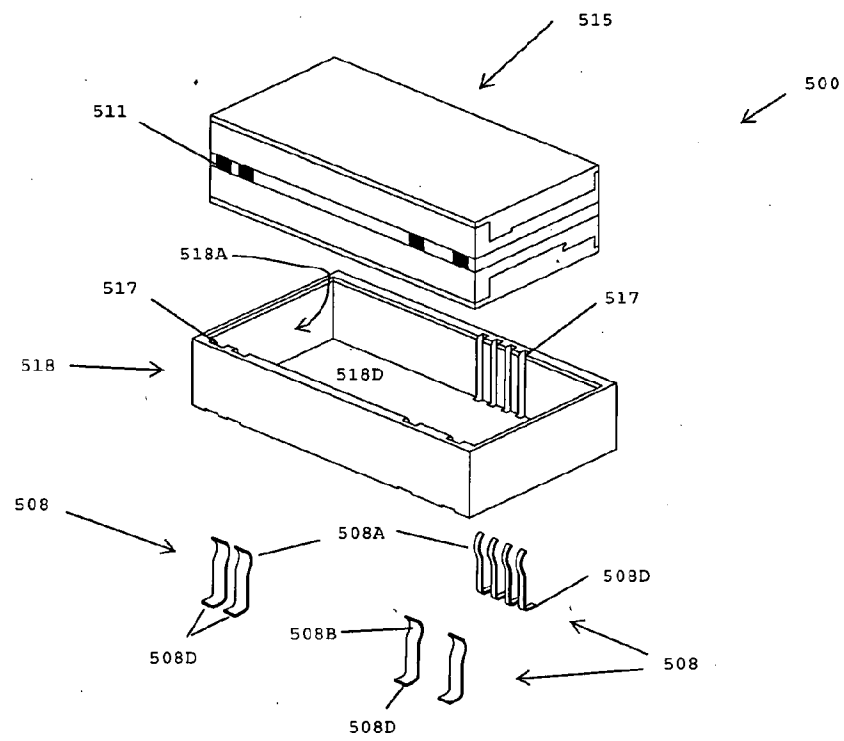


FIG. 22

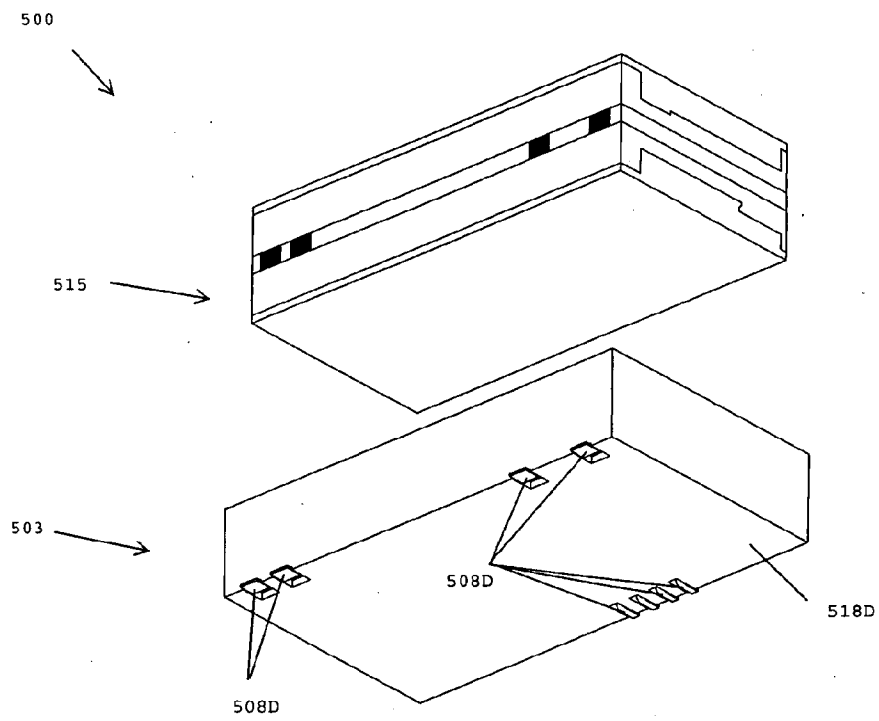


FIG. 23

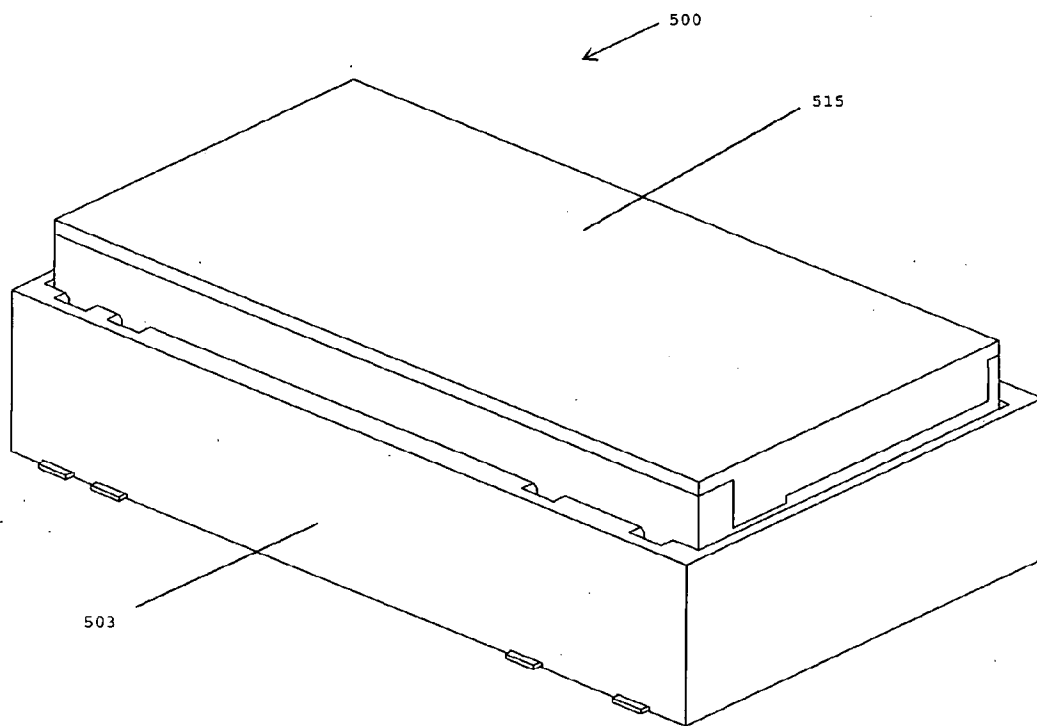


FIG. 24

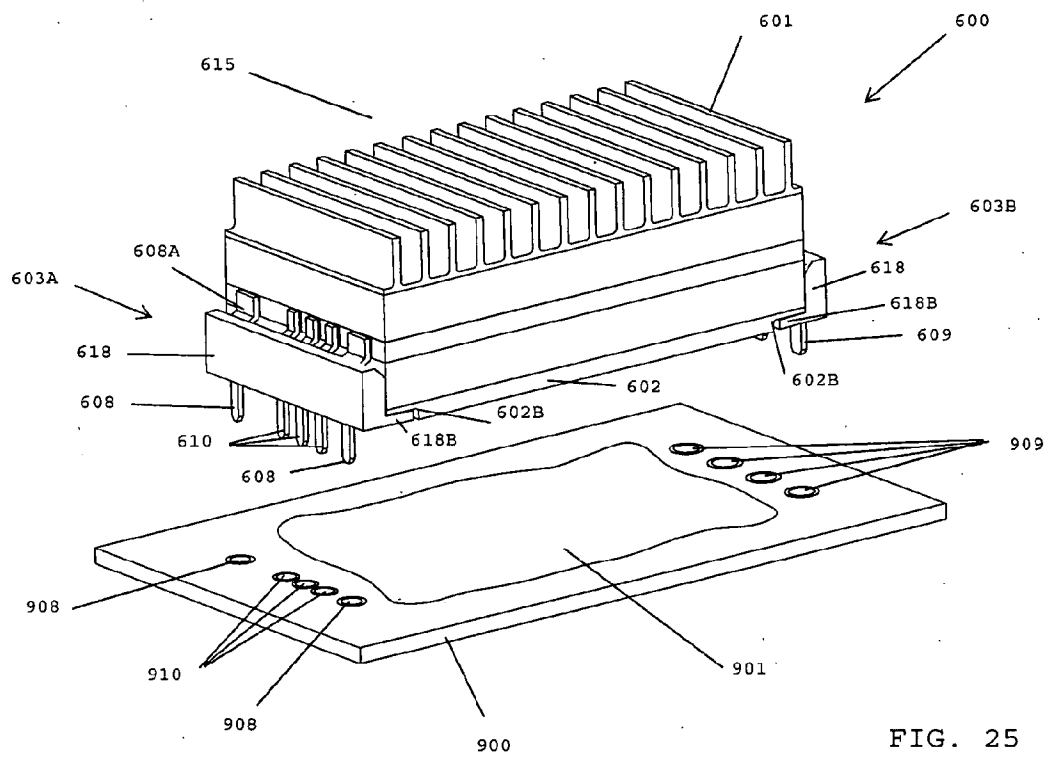


FIG. 25

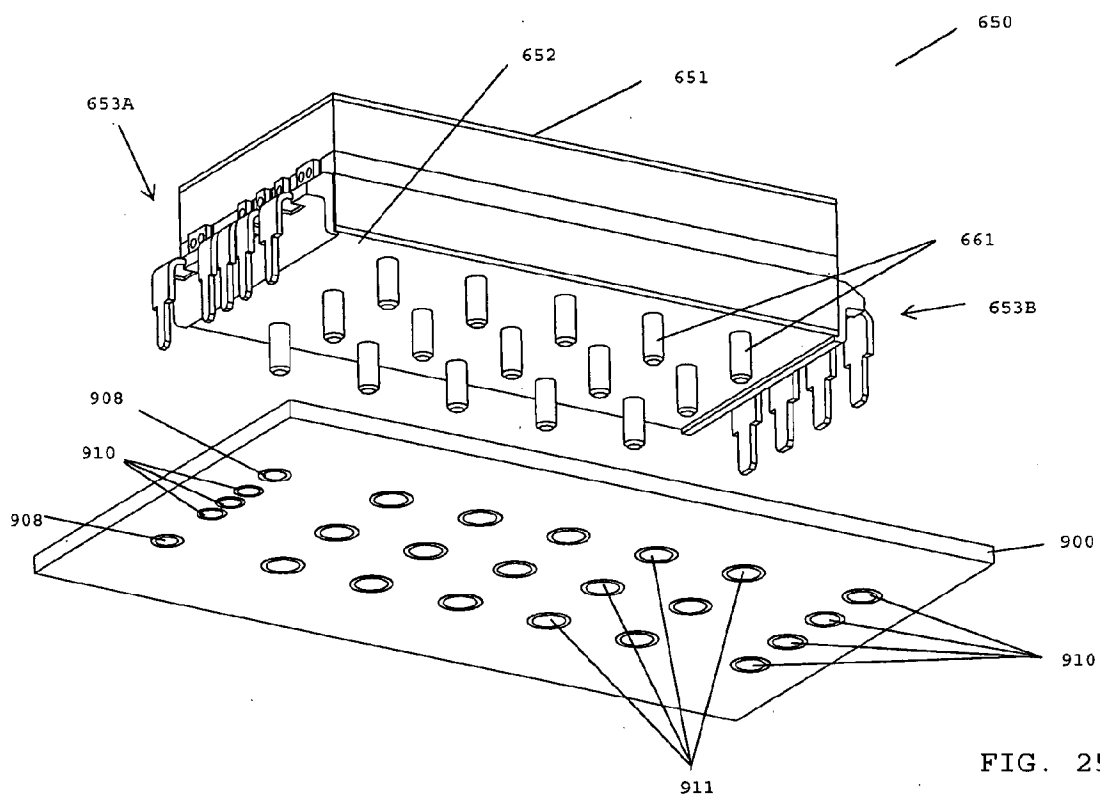
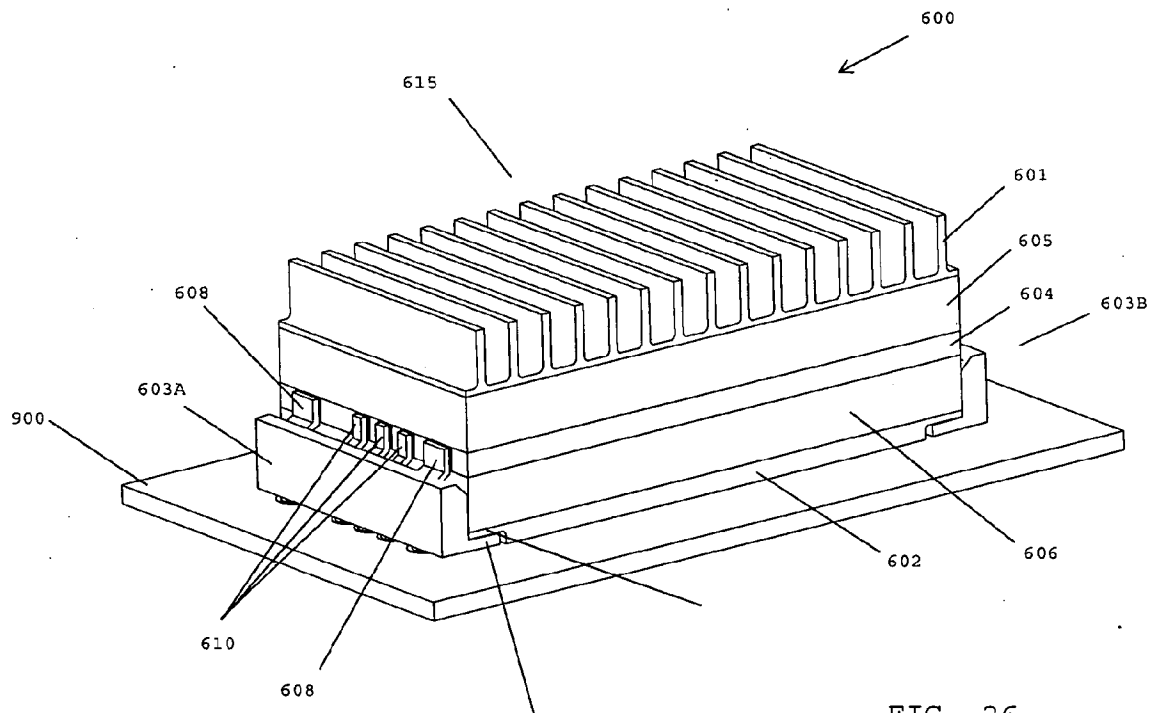
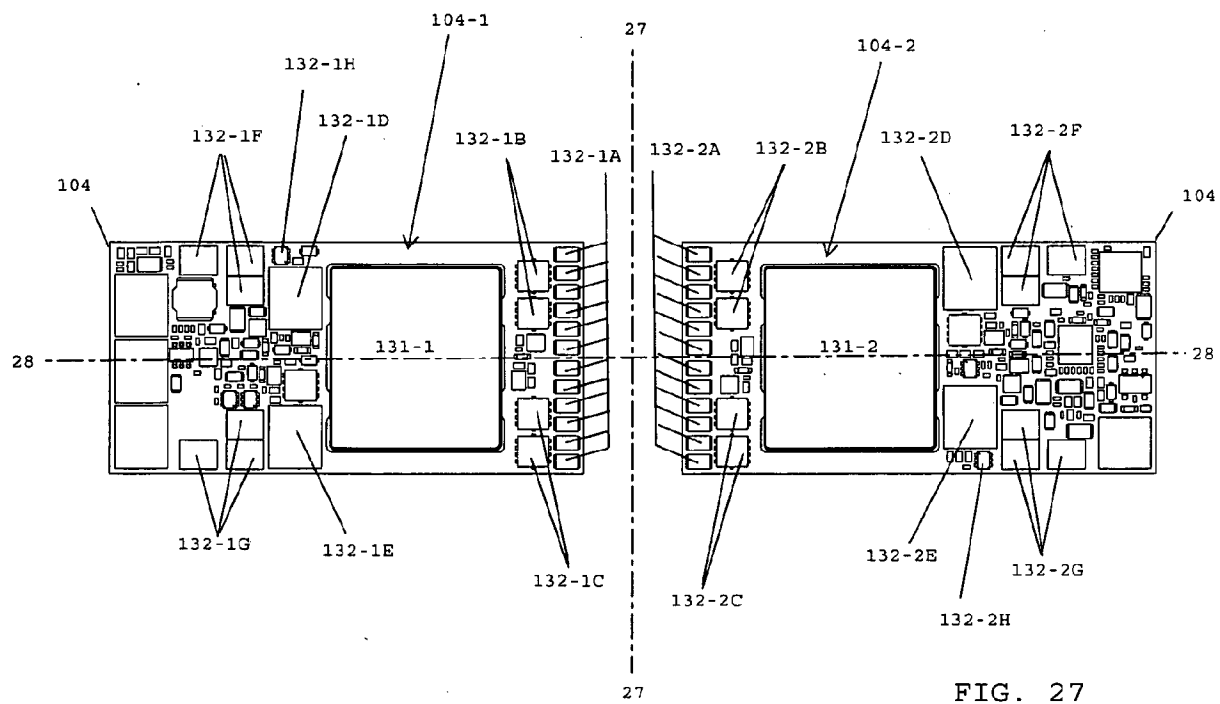


FIG. 25A





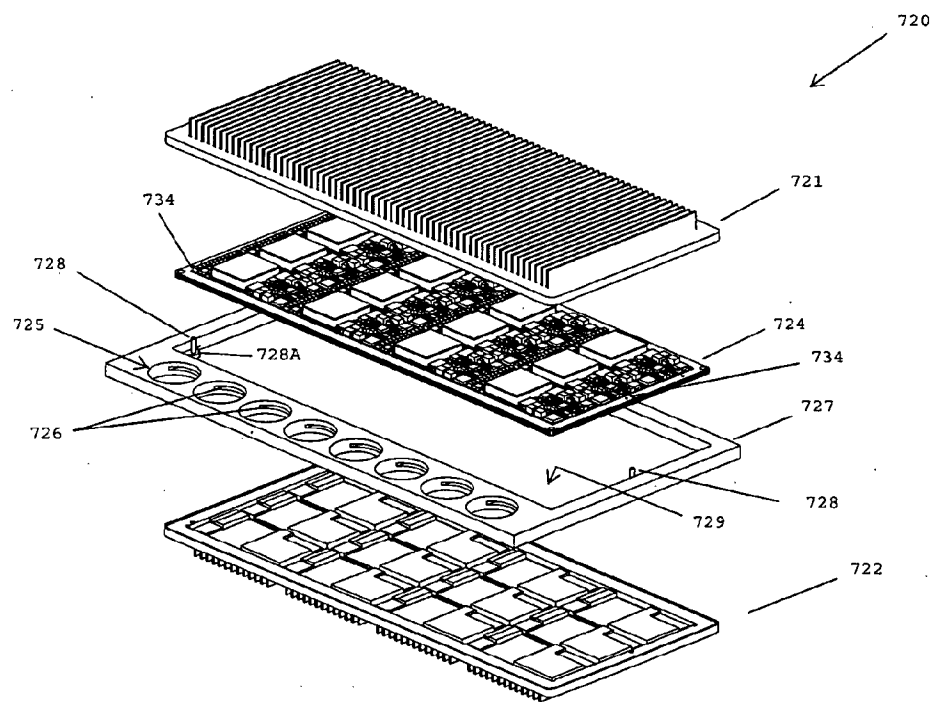


FIG. 28

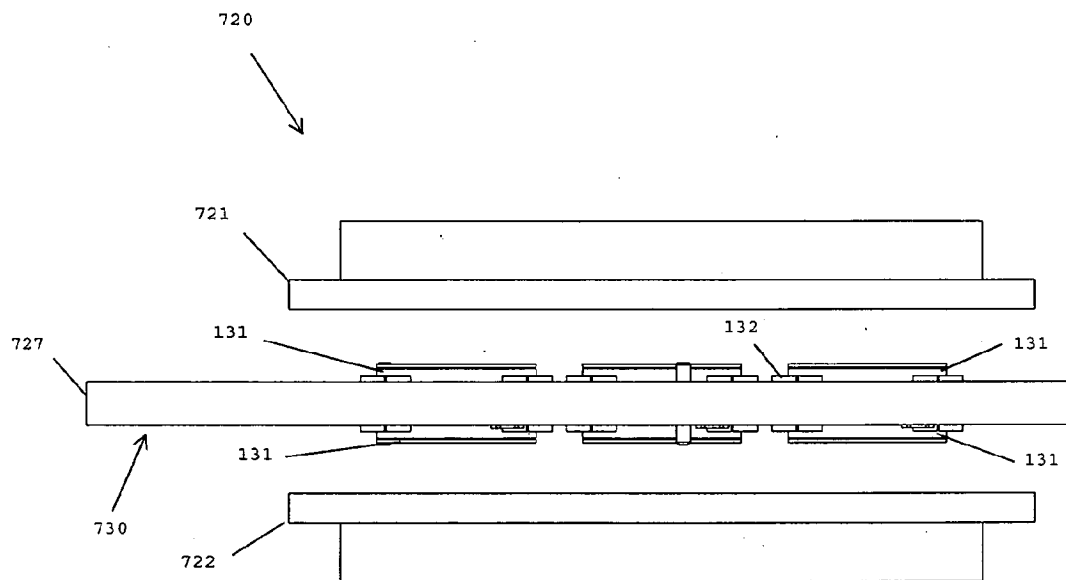
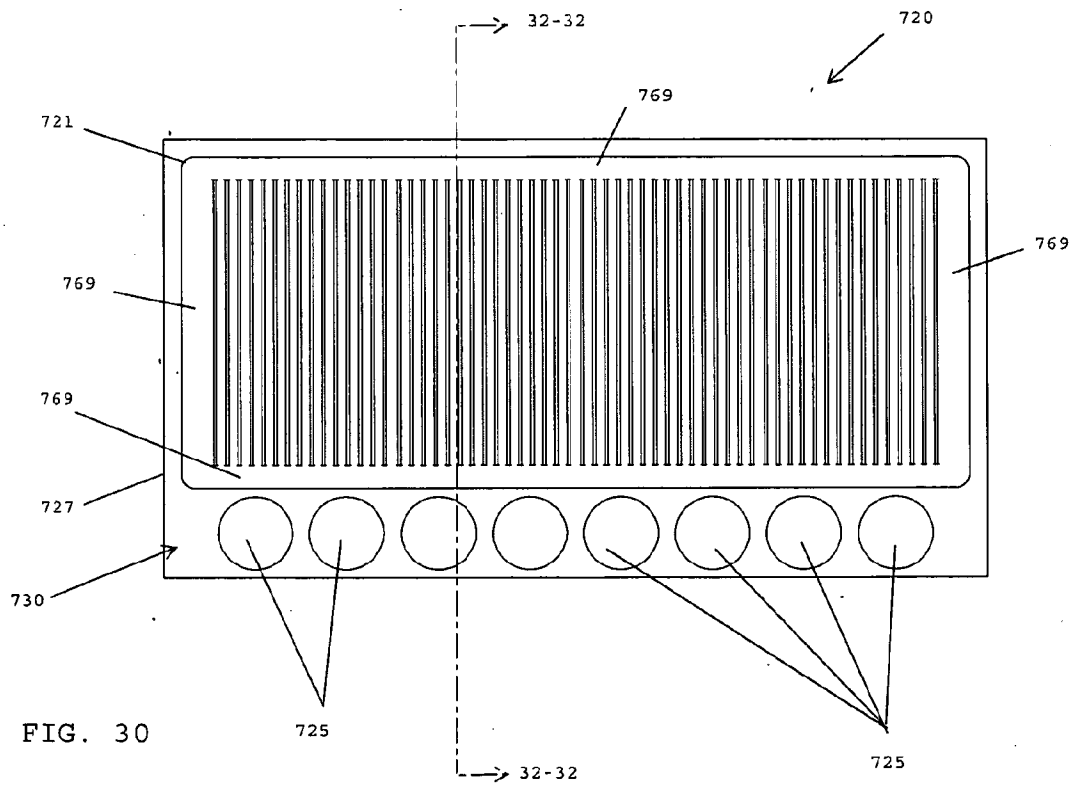


FIG. 29



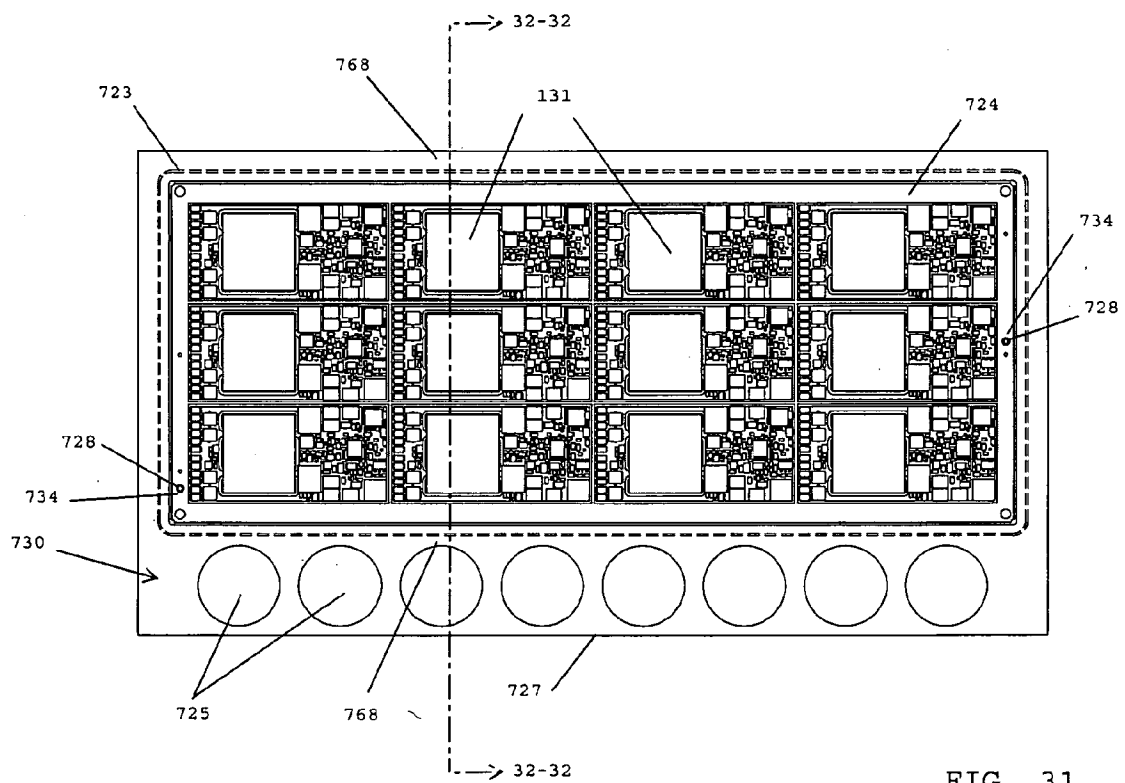


FIG. 31

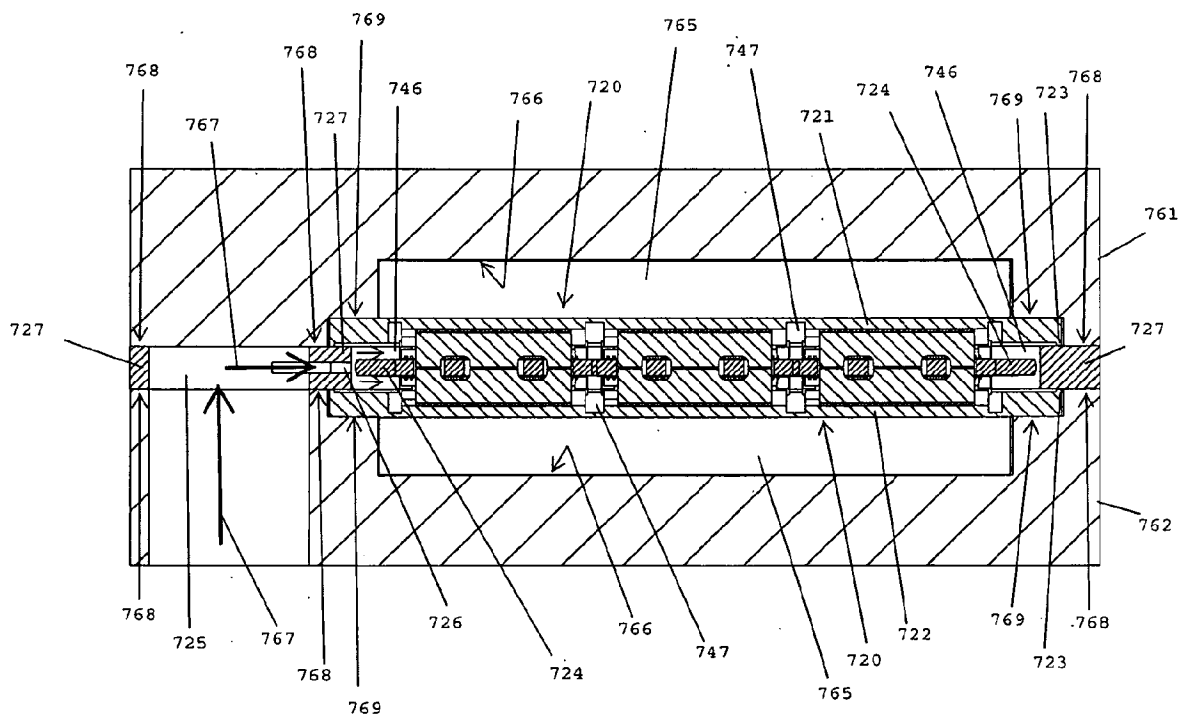


FIG. 32

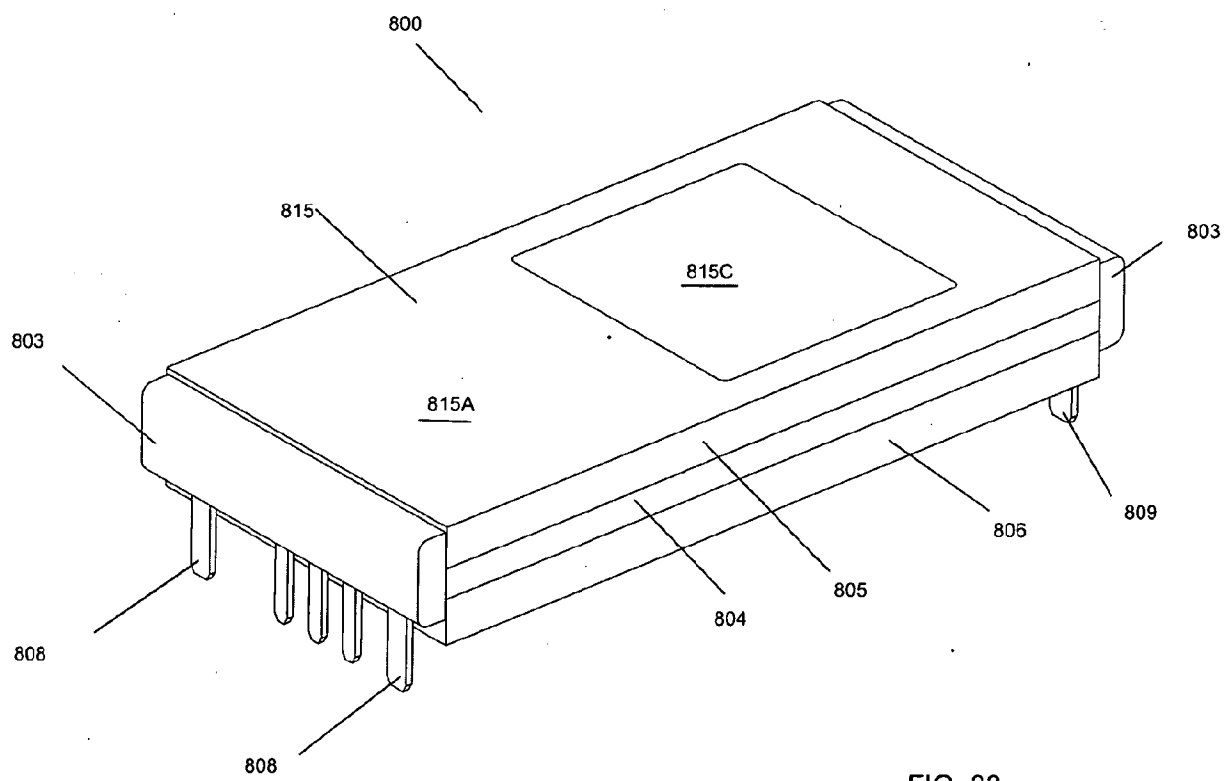


FIG. 33

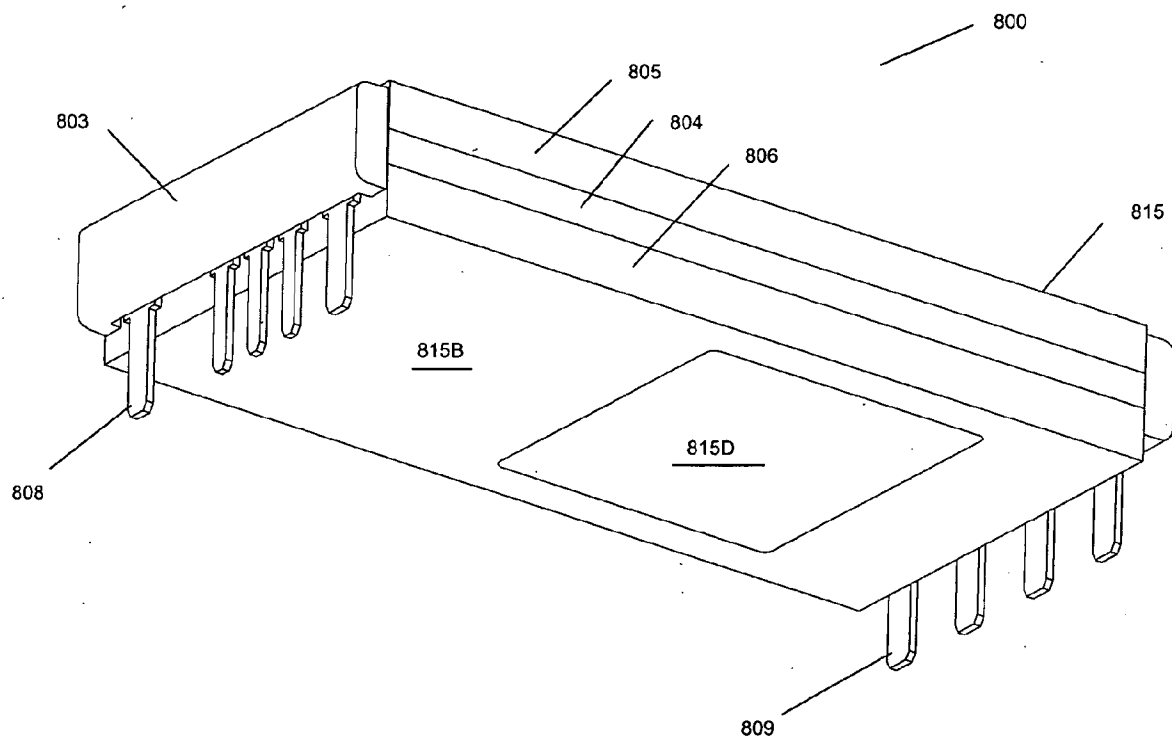


FIG. 34

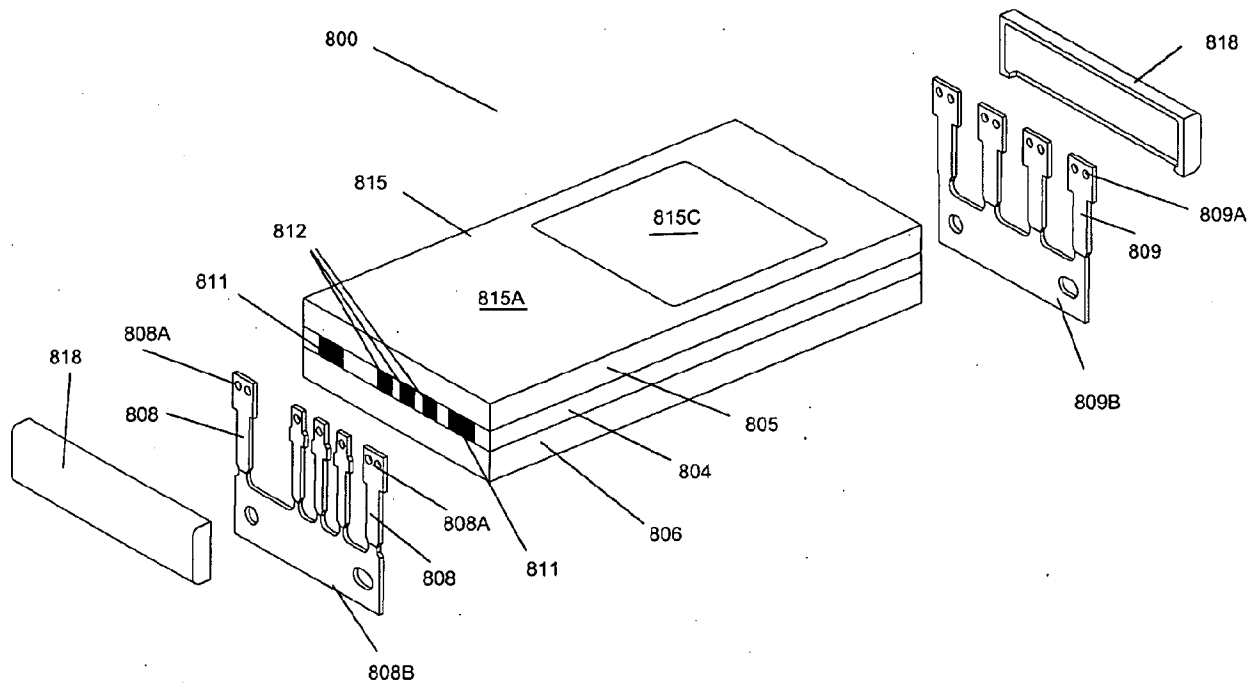


FIG. 35

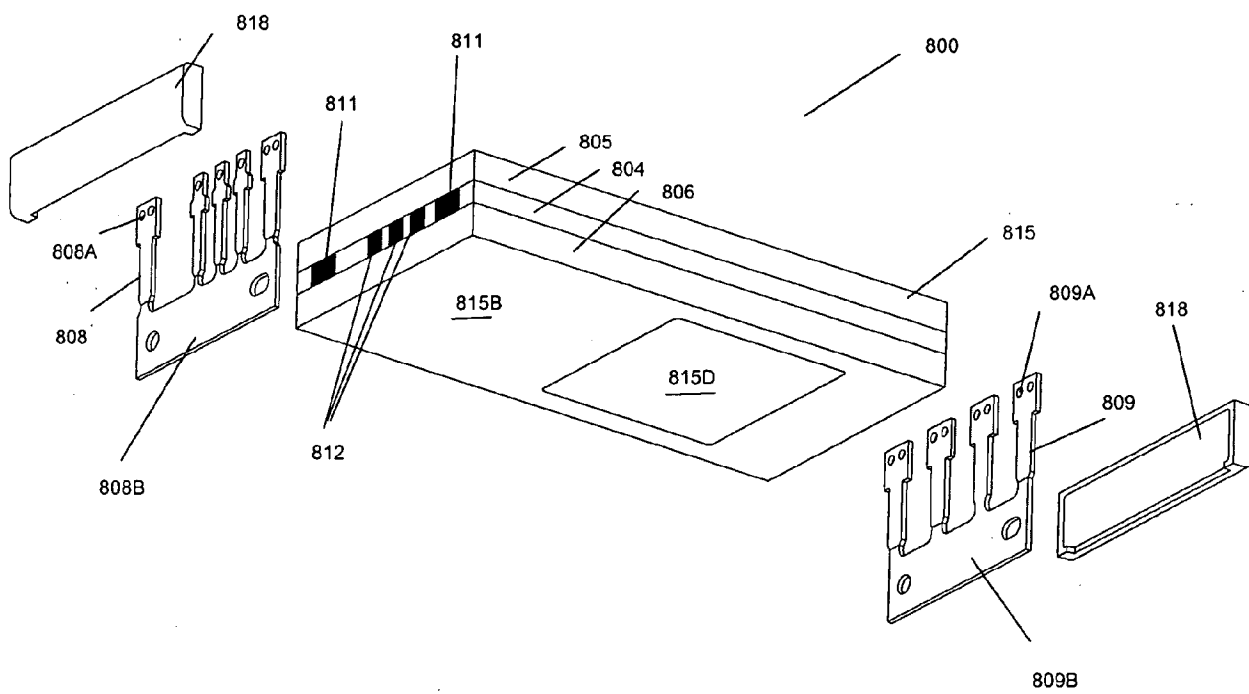


FIG. 36

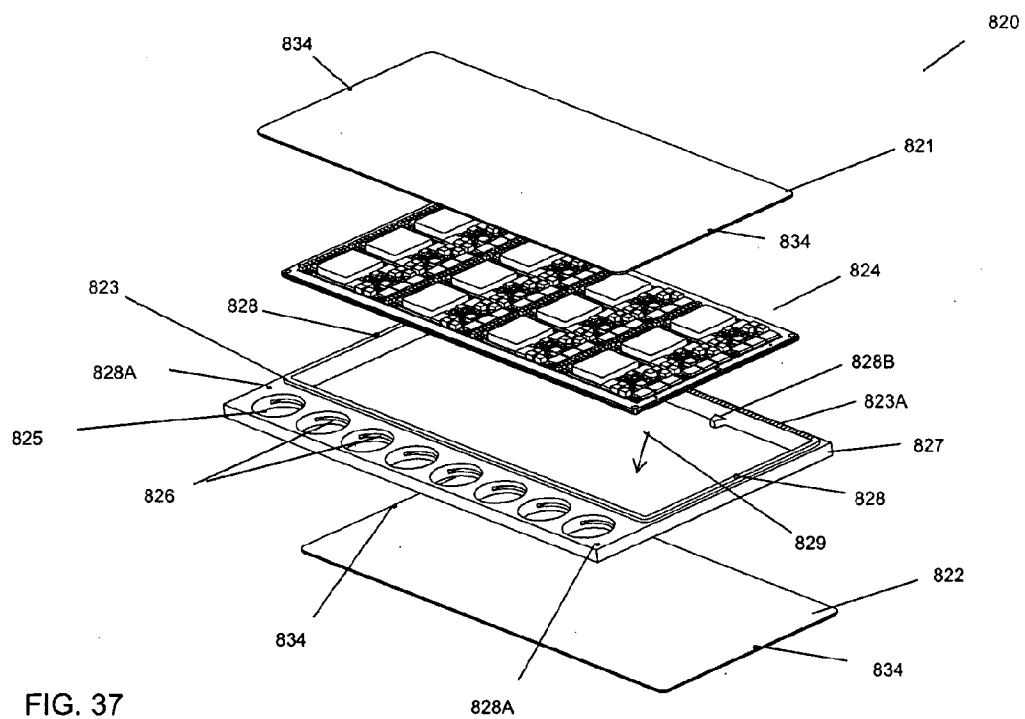


FIG. 37

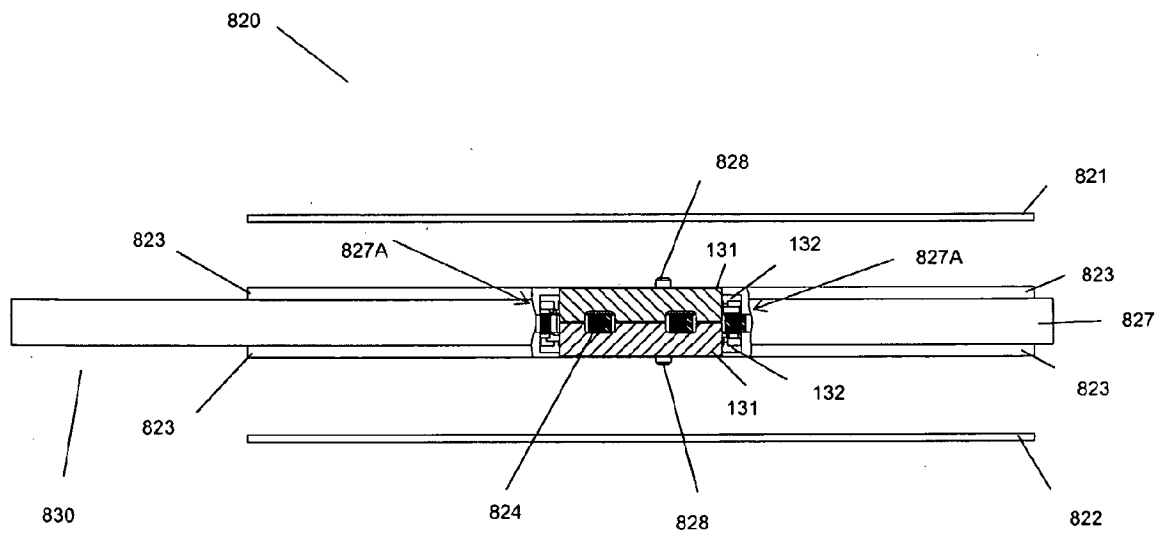


FIG.38

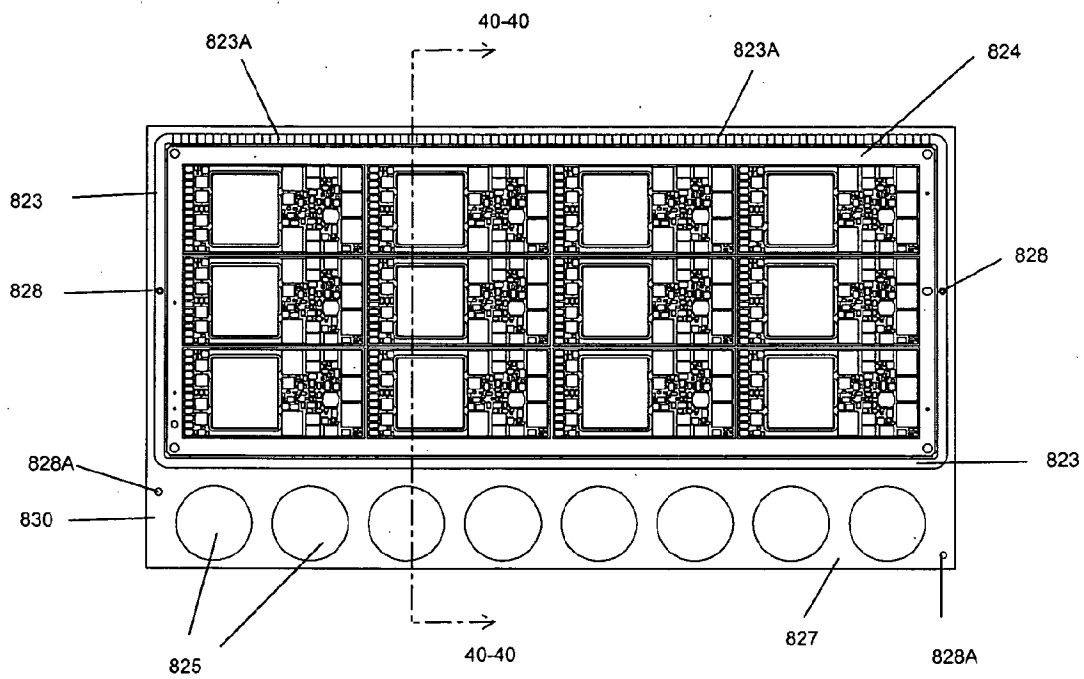


FIG.39

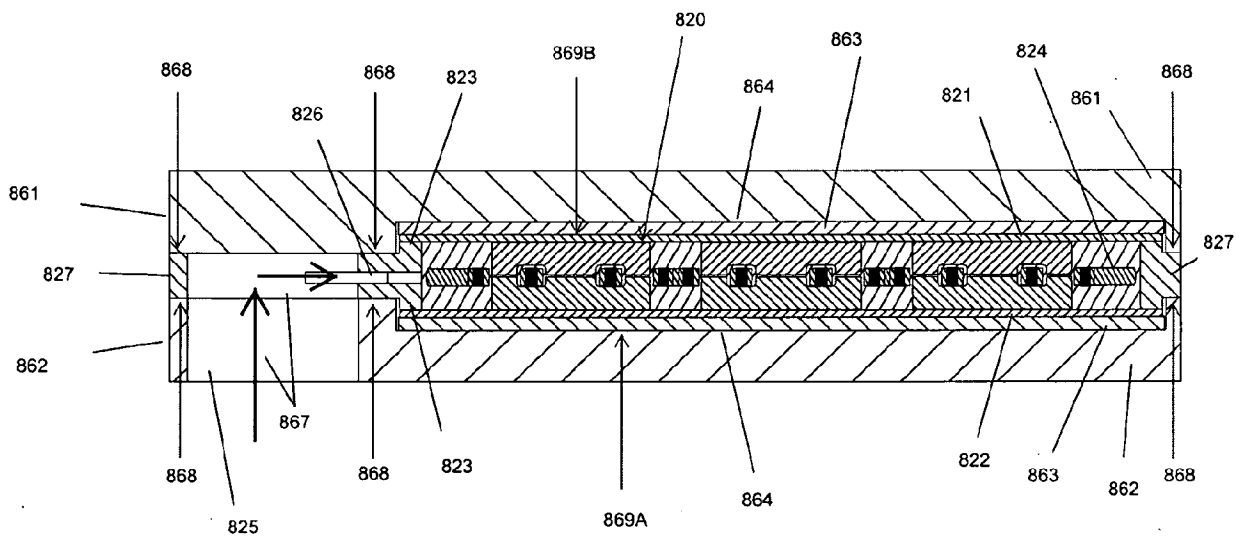


FIG. 40

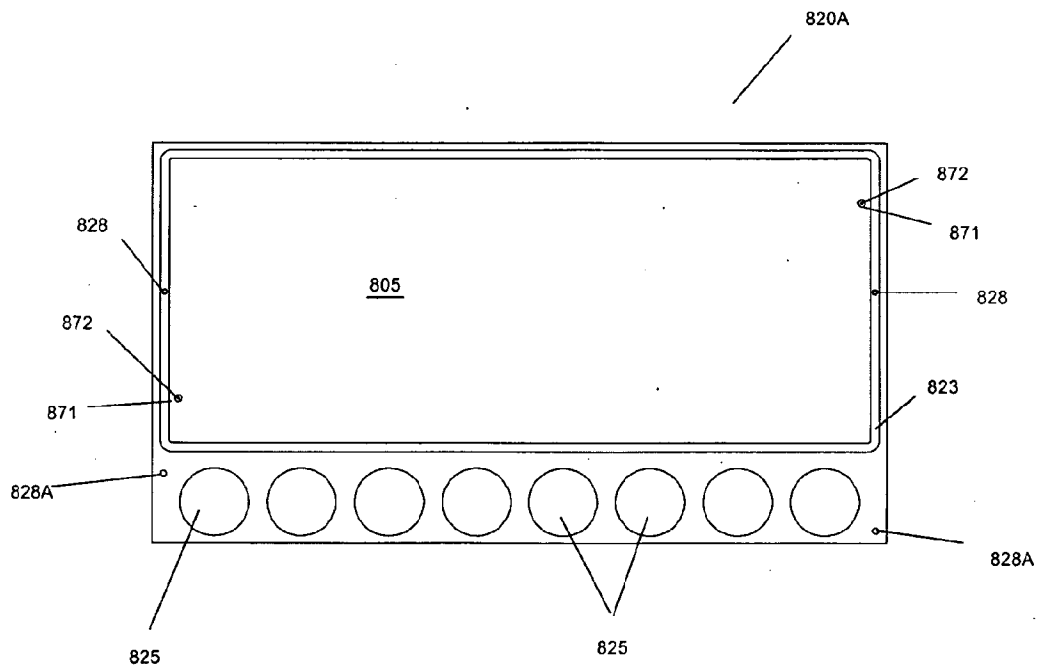


FIG. 41

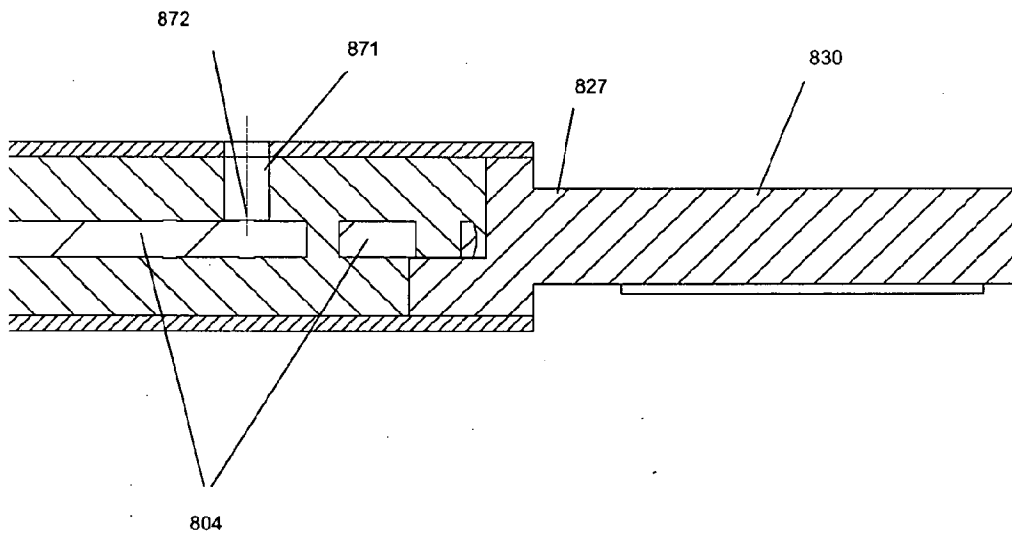


FIG. 42

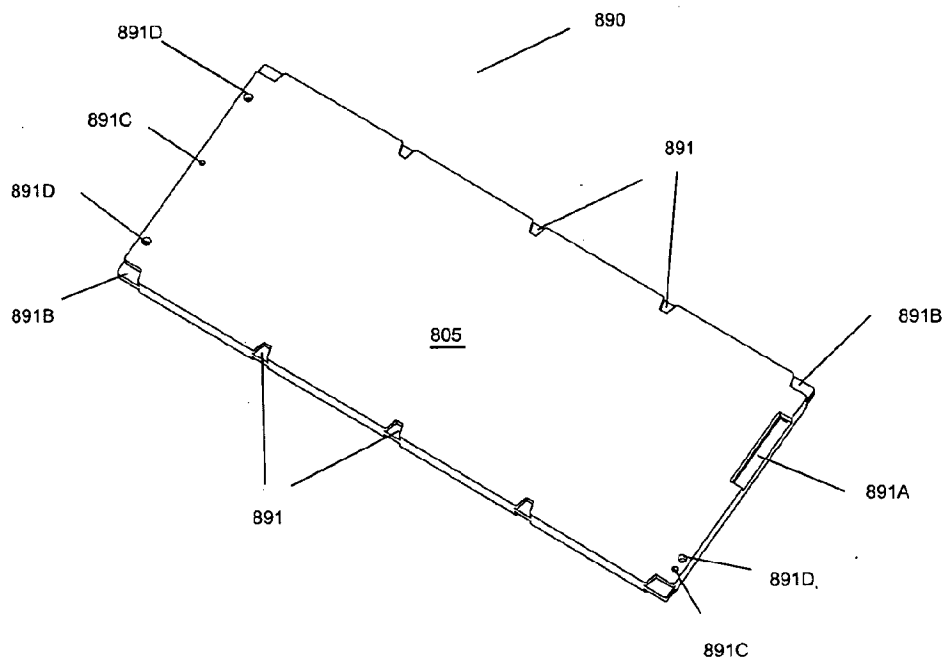


FIG. 43

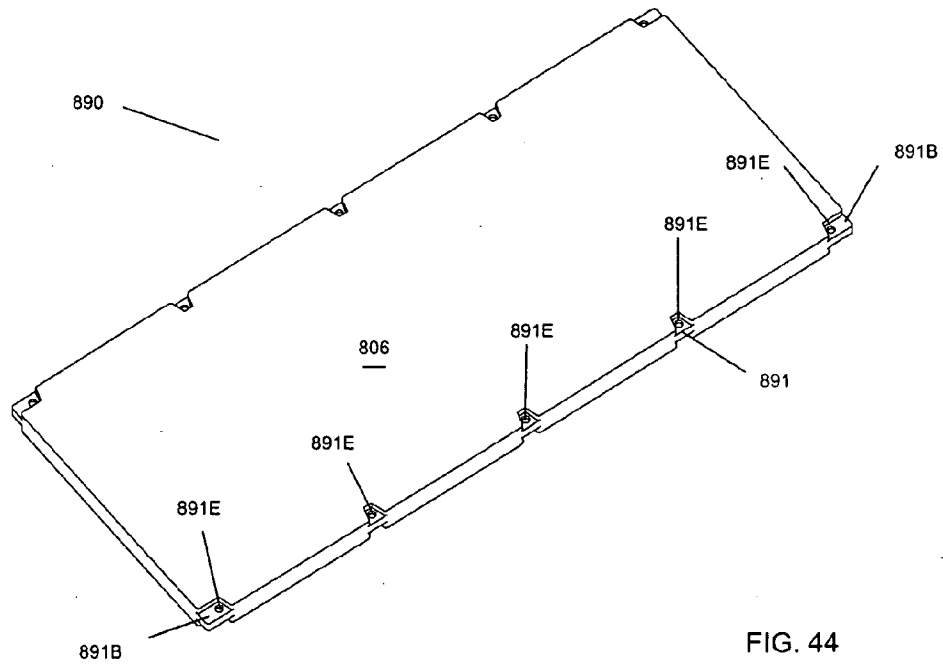


FIG. 44

Substitute Disclosure Form	U.S. Department of Commerce Patent and Trademark Office	Attorney Docket No. 00614-0183001	Application No. 16/218,418
Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))		Applicant VLT, Inc.	
		Filing Date December 12, 2018	Group Art Unit 2849

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Filing Date If Appropriate	Class	Subclass
	1.	10791645	9/29/2020	Vinciarelli			

Foreign Patent Documents or Published Foreign Patent Applications							
Examiner Initial	Desig. ID	Document Number	Publication Date	Country or Patent Office	Translation	Class	Subclass

Other Documents (include Author, Title, Date, and Place of Publication)		
Examiner Initial	Desig. ID	Document
	2.	Braun et al. "Opportunities of Wafer Level Embedded Technologies for MEMS Devices"; SEMI MEMS Tech Seminar; (possibly published on September 26, 2013); 22 pages
	3.	"Ideas for attaching / connecting / stacking one PCB onto another with no gap"; http://electronics.stackexchange.com/questions/45733/ideas-for-attaching-connecting-stacking-one-PCB-onto-another-with-no-gap ; accessed on June 3, 2015; 5 pages
	4.	PCT International Search Report and Written Opinion, PCT/US2012/37495, May 6, 2013, 18 pages
	5.	USPTO Office Action in U.S. Application No. 13/105,696, dated September 25, 2013, Examiner Arbes, 8 pages
	6.	USPTO Office Action in U.S. Application No. 14/116,642, dated February 9, 2016, Examiner Shrestha, 15 pages
	7.	USPTO Office Action in U.S. Application No. 14/635,420, dated October 22, 2015, Examiner Arbes, 6 pages

Examiner Signature	Date Considered
EXAMINER: Initials citation considered. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.	

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

First Named Inventor :	Patrizio Vinciarelli	Art Unit :	2849
Application No. :	16/218,418	Examiner :	Cassandra F. Cox
Filed :	December 12, 2018	Conf. No. :	1092
Title :	DELIVERING POWER TO SEMICONDUCTOR LOADS		

MAIL STOP AMENDMENT

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

INFORMATION DISCLOSURE STATEMENT

Please consider the documents listed on the disclosure form submitted with this statement. Copies of the documents are being submitted with this statement to the extent required under 37 C.F.R. § 1.98(a)(2).

This statement is being filed after a request for continued examination and before the mailing of a first action.

Apply any necessary charges or credits applicable under 37 C.F.R. § 1.17(p) to deposit account 06-1050, referencing the above attorney docket number.

Please contact the undersigned if there are any questions regarding this statement.

Respectfully submitted,

Date: November 10, 2020 _____

/Rex I. Huang/_____

Rex I. Huang
Reg. No. 57,661

Customer Number 26214
Fish & Richardson P.C.
Telephone: (617) 542-5070
Facsimile: (877) 769-7945

Electronic Acknowledgement Receipt

EFS ID:	41088635
Application Number:	16218418
International Application Number:	
Confirmation Number:	1092
Title of Invention:	DELIVERING POWER TO SEMICONDUCTOR LOADS
First Named Inventor/Applicant Name:	Patrizio Vinciarelli
Customer Number:	26214
Filer:	Rex I. Huang/Peter Wolff
Filer Authorized By:	Rex I. Huang
Attorney Docket Number:	00614-0183001
Receipt Date:	10-NOV-2020
Filing Date:	12-DEC-2018
Time Stamp:	17:59:07
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Non Patent Literature	Braun_OpportunitiesofWaferLevelEmbeddedTech.pdf	1723417	no	22
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Warnings:

Information:					
2	Non Patent Literature	IdeasAttachingConnectingStackingOnePCB.pdf	429157	no	7
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Warnings:					
Information:					
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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

Opportunities of Wafer Level Embedded Technologies for MEMS Devices

T. Braun ⁽¹⁾, K.-F. Becker ⁽¹⁾, R. Kahle ⁽²⁾, V. Bader ⁽¹⁾, S. Voges ⁽²⁾, T. Thomas ⁽²⁾,
R. Aschenbrenner ⁽¹⁾, K.-D. Lang ⁽²⁾

⁽¹⁾ Fraunhofer Institute for Reliability and Microintegration
Gustav-Meyer-Allee 25, 13355 Berlin, Germany
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phone: +49-30/464 03 244 fax.: +49-30/464 03 254
⁽²⁾ Technical University Berlin, Microperipheric Center

Tanja Braun, IHT

Outline

- Introduction – FOWLP
- Application Examples
 - MST SmartSense Multi-Sensor Package
 - Sensor Integration into Microfluidics

Fan-out Wafer Level Packaging (FOWLP)

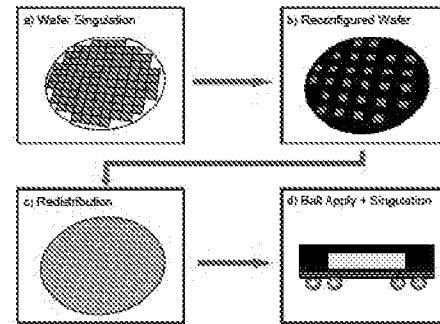
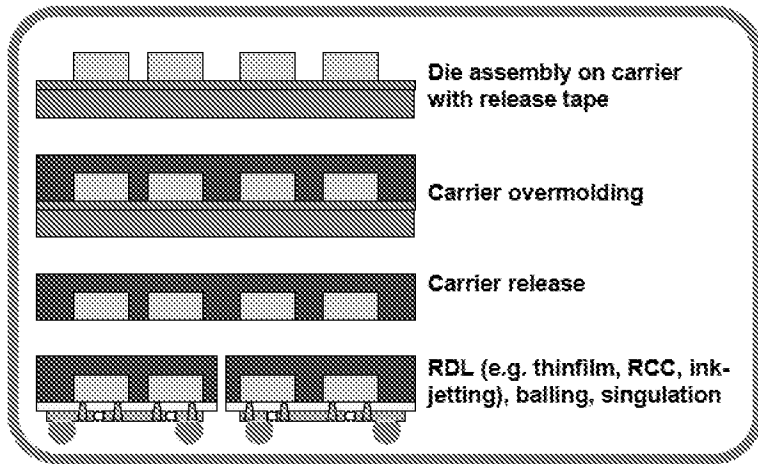


Figure 2a-d. Schematic process flow for a fan-out wafer-level package

“Molded Reconfigured Wafer”, Infineon

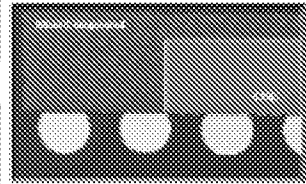
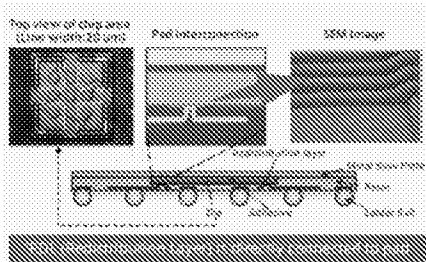


Fig. 3. Cross section of wLBP



„Wafer-Level Fan-Out Packaging, WFOP“, J-Devices

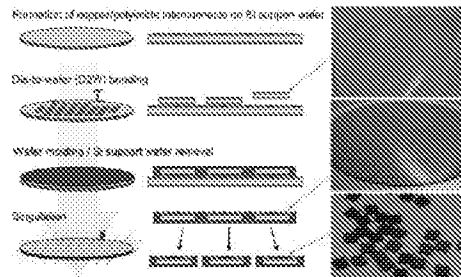


Fig. 2. Process flow for new wafer-level package with fan-out RDL

„Fan-Out Wafer-Level Packaging“, Renesas Electronics Corporation

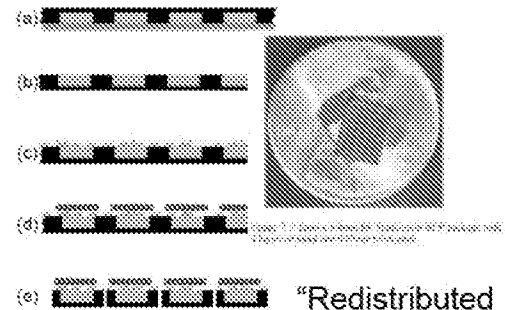


Figure 1. RCP Process Flow: (a) Place the active die from an external wafer, (b) remove substrate, (c) redistribution layer, (d) mold and protect (3-epoxy 96/3: 96/3: 3/3) and (e) singulate

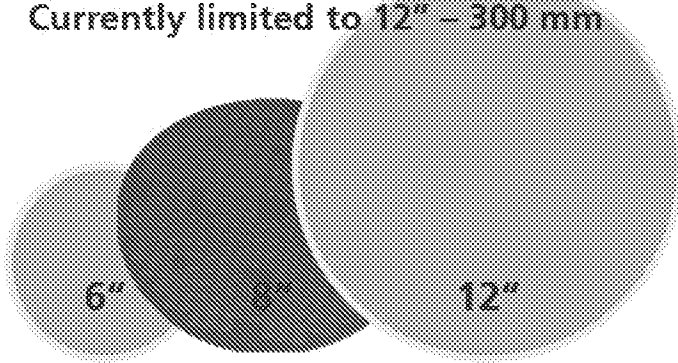
“Redistributed Chip Package”, Freescale

Tanja Braun, DHT

From Wafer to Panel Level Packaging

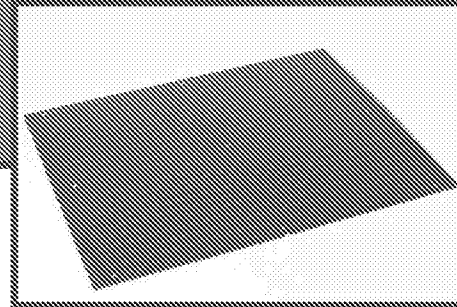
PCB Technologies

- Based on standard thin film technology equipment
- Tightest tolerances for fine pitch line/space (5/5 μm)
- Sensitive to substrate warpage
- Currently limited to 12" – 300 mm



Thin Film Technologies

- Based on standard PCB manufacturing equipment
- Intrinsic warpage compensation by lamination
- 3D and double sided routing are standard features for PCBs – Line/space down to 20/20 μm
- Full format/large area is standard



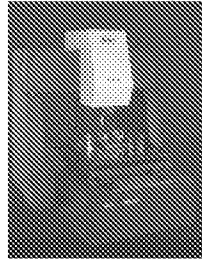
Tanja Braun, DHT

Chip Embedding - IZM Embedding & Substrate Line from Wafer Scale to Panel Scale 610 x 456 mm²

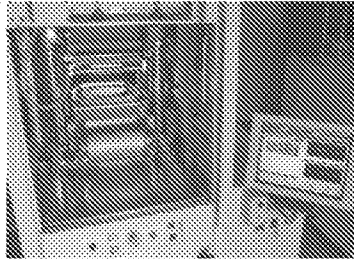
Placement ➡ Accuracy ➡ Molding ➡ Lamination ➡ Laser Drilling



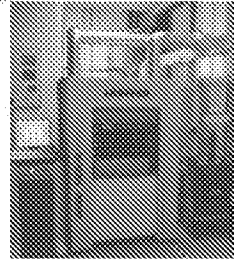
Datacon evo/
Siplace CA3



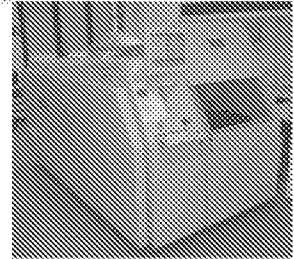
Mahr OMS 600



WL: Towa 120t
PL: NN – Q2 2014

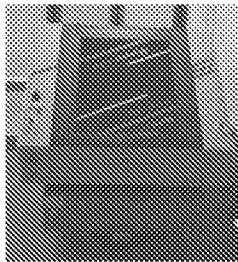


Lauffer/
Bürkle

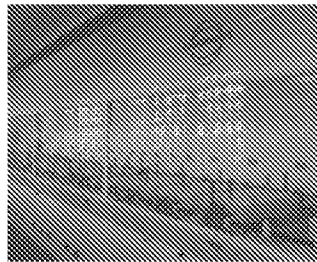


Siemens
Microbeam

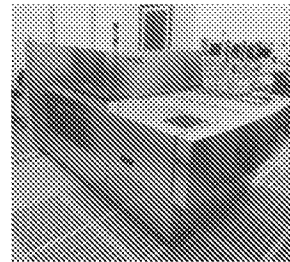
➡ Mech. Drilling ➡ Cu Plating ➡ Imaging ➡ Etching



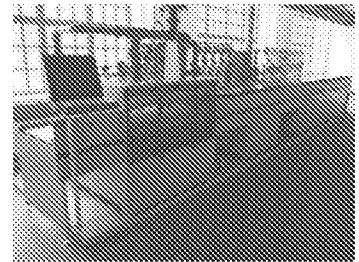
Schmoll MX1



Ramgraber auto-
matic plating line



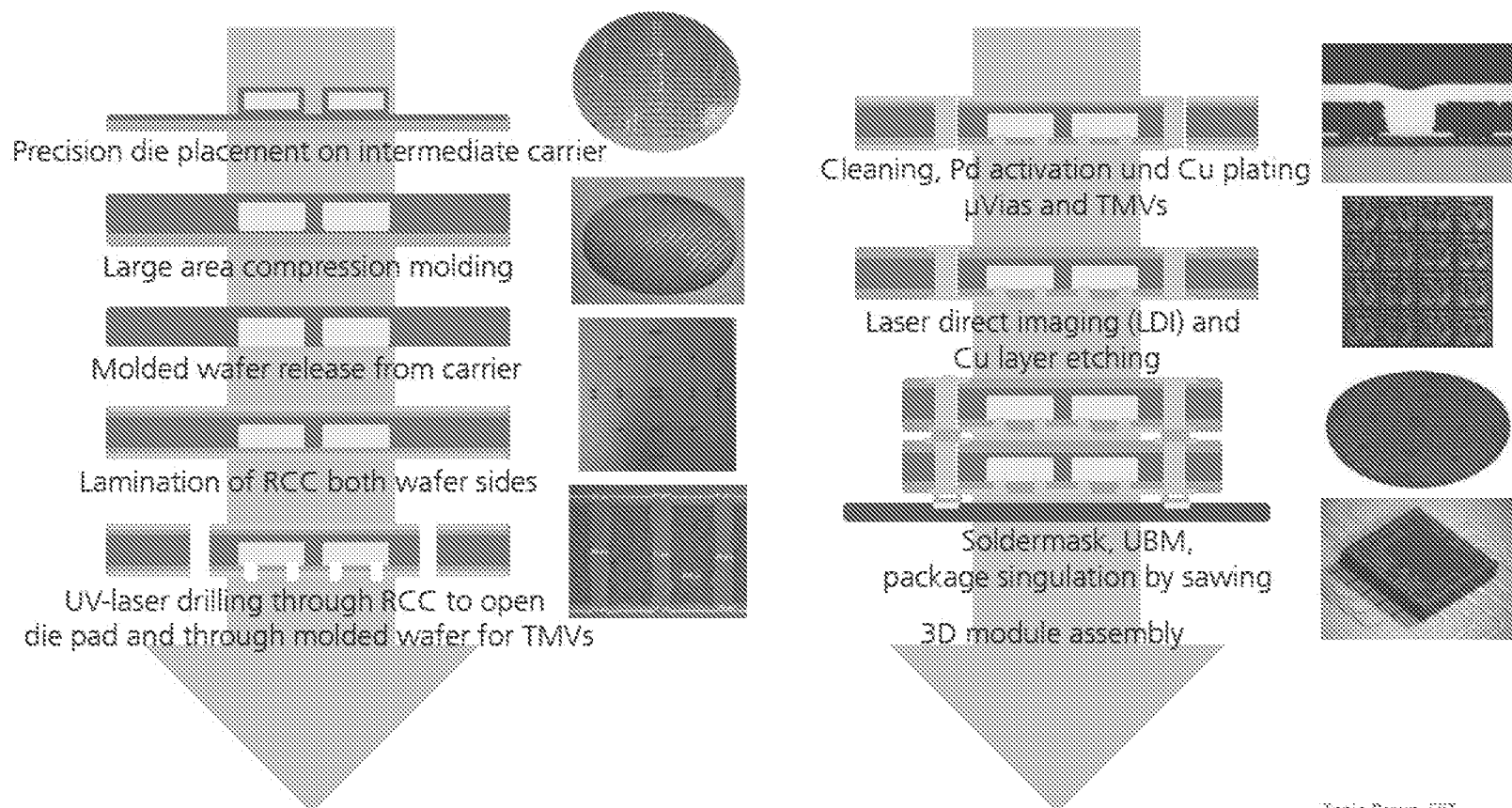
Orbotech
Paragon 9000



Schmid

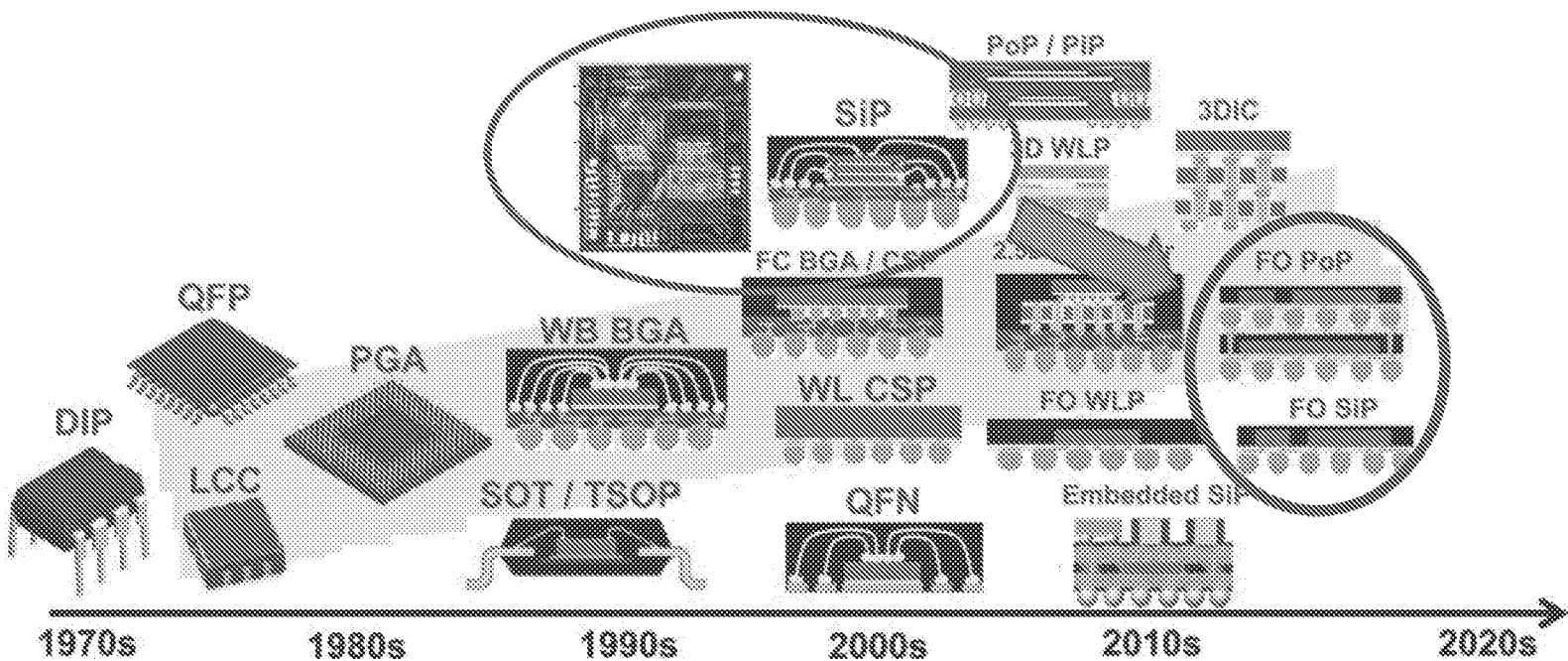
Tanja Braun, SHY

FOWLP with PCB based RDL and Through Mold Vias (TMV) Process Flow



Tanja Braun, DHT

Motivation SiP Wafer Level Fan-Out Embedding



Roadmap source: Yole

Tanja Braun, DHT

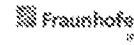
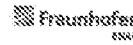
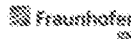
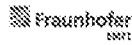
Application Example

MST SmartSense - Intelligent 3D MEMS Compass

Bosch Sensortec



SENSOTEC



BOSCH

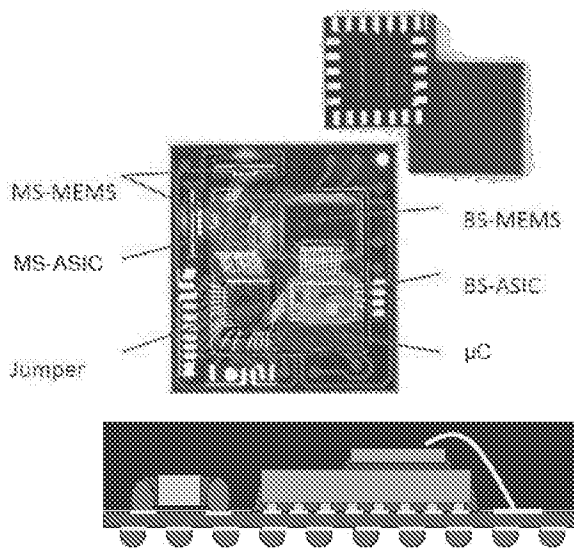
Tanja Braun, DHT

MST SmartSense - Intelligent 3D MEMS Compass

Technology Demonstrator

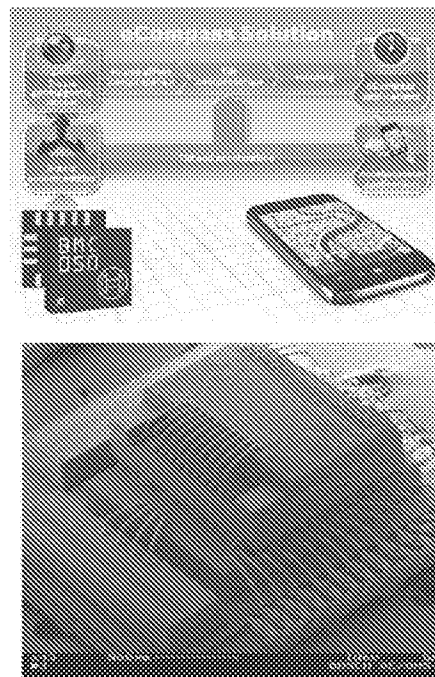
Commercial Product

Advanced Technology

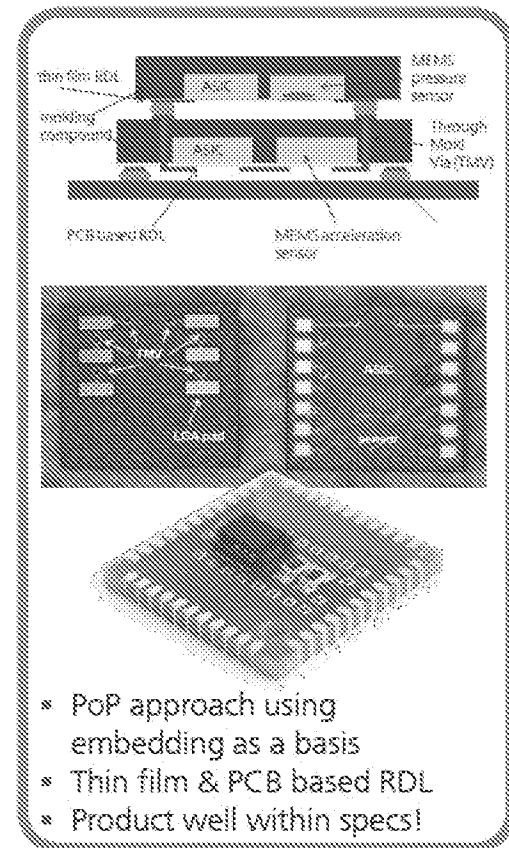


Heterogeneous Integration

- BGA Multi-Sensor Package
- Evaluation of Material Combinations
- Reliability Investigations



- Chip on Board technology
- Transfer molded LGA housing

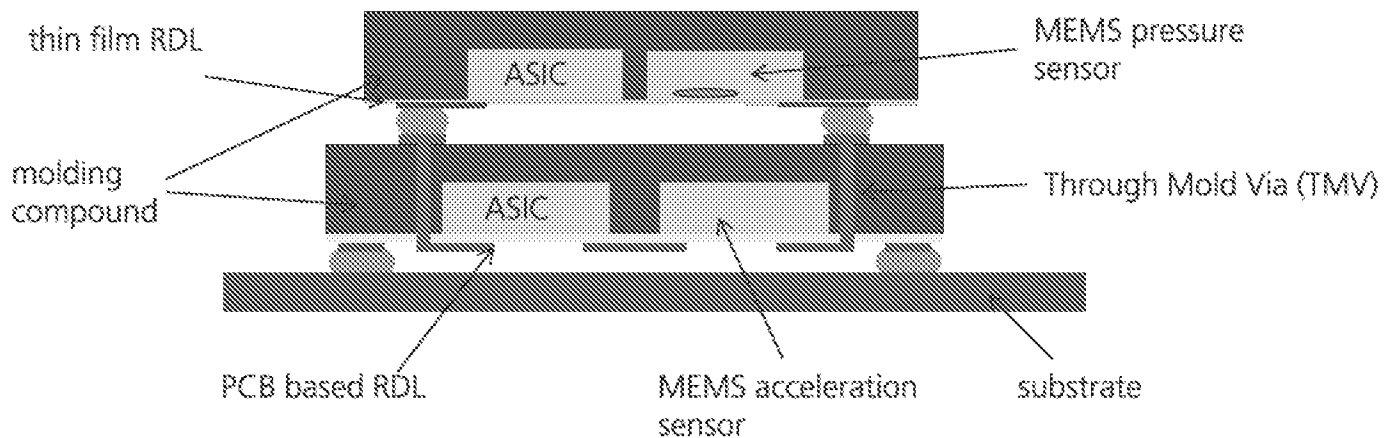


- PoP approach using embedding as a basis
- Thin film & PCB based RDL
- Product well within specs!

Tanja Braun, DHT

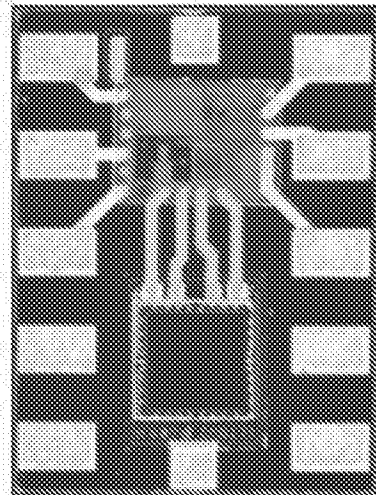
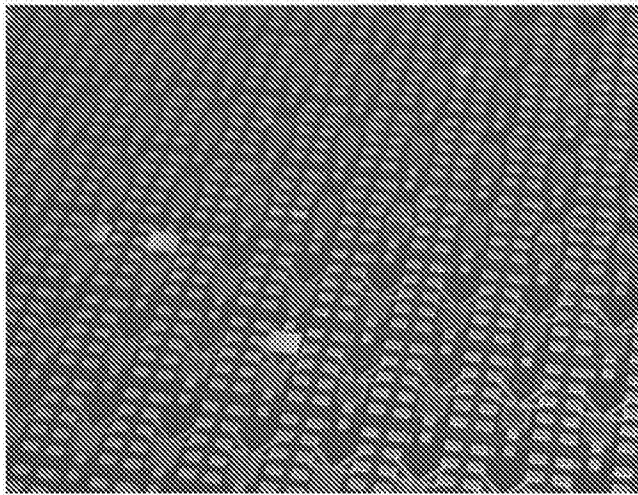
FOWLP Multi-Sensor Stack

- Manufacturing of functional demonstrators
- FOWLP Multi-Sensor Stack consisting of
 - Pressure Sensor/ASIC package with thin film RDL and RDL opening above sensor membrane
 - Acceleration sensor/ASIC package with PCB based RDL and Through Mold Vias (TMV) for package stacking



Tanja Braun, DHT

FOWLP – Pressure Sensor-ASIC Package



pressure sensor packages after wafer level molding and redistribution

- Thin film redistribution
- Open RDL layer above pressure sensor membrane
- Comparable sensor performance over the entire wafer

M. Bründel, U. Scholz, F. Haag, E. Graf, T. Braun, K.-F. Becker; Substrateless sensor packaging using wafer level fan-out technology; Proc. of EPTC 2012; Singapore.

Tanja Braun, SHY

Through Mold Via - Reliability

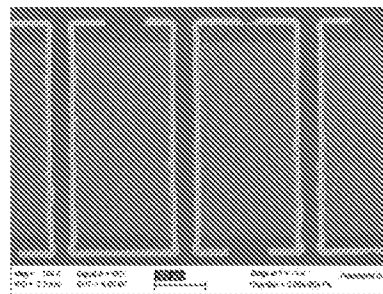
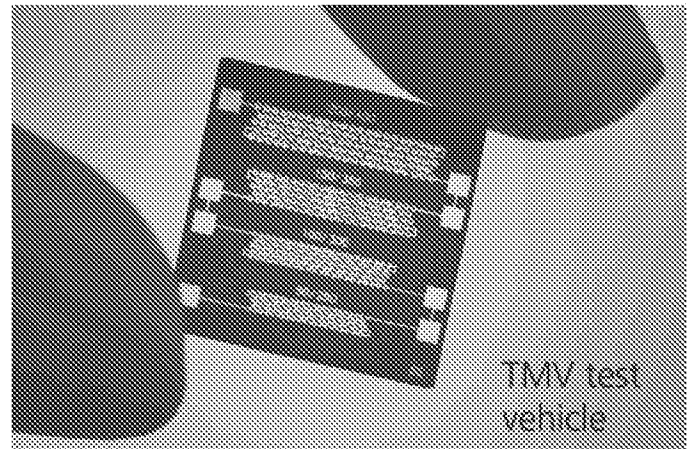
- Laser drilled Through Mold Vias:
100 up to 150 vias/s
- Direct metallization by Cu plating

Investigations of

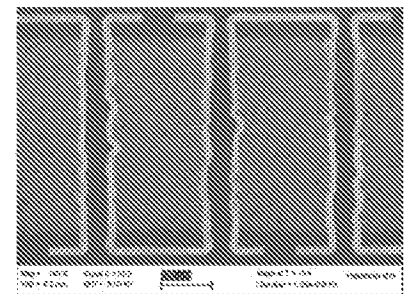
- 5 different epoxy molding compounds with max. filler sizes from 24 μm to 75 μm
- 4 different via diameters/pitches (200 μm /400 μm , 150 μm 350 μm , 100 μm /300 μm , **50 μm /250 μm**)

All samples passed:

- MSL 1
- 5000 temperature cycles
-55 °C/125 °C
- 5000 h humidity storage
85 °C/85 % r.h.
- without any electrical failures



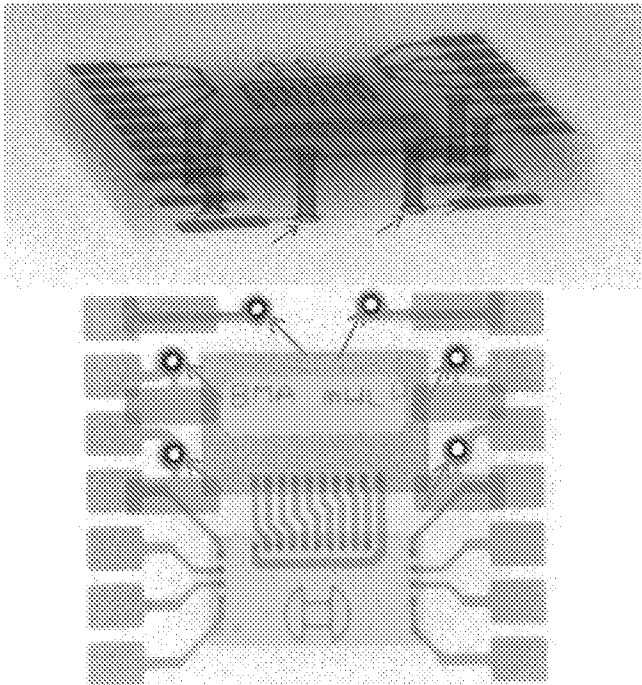
EMC with max. 24 μm filler size
50 μm via and 250 μm pitch



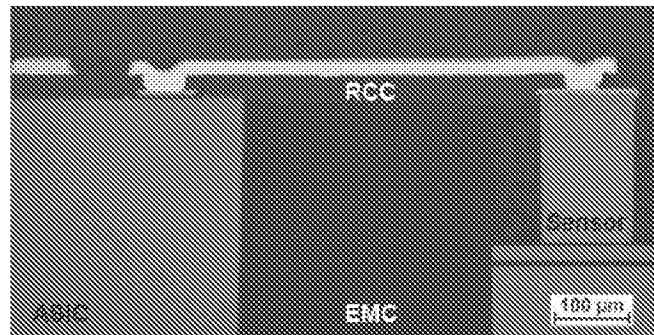
EMC with max. 75 μm filler size
50 μm via and 250 μm pitch

Tanja Braun, DHT

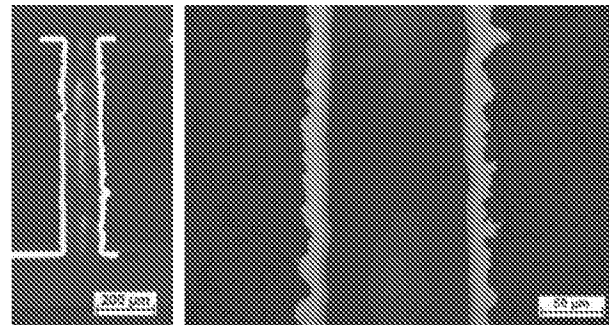
Sensor Package - Through Mold Via (TMV)



X-Ray image – acceleration sensor-ASIC package with through mold vias (red arrows)



Cross section – Through mold via interconnection between ASIC and sensor with PCB based RDL technology

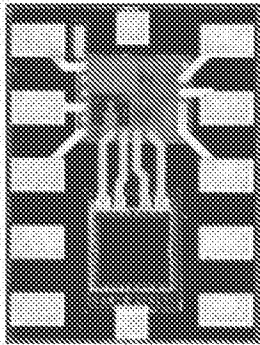


Cross section – Through mold via

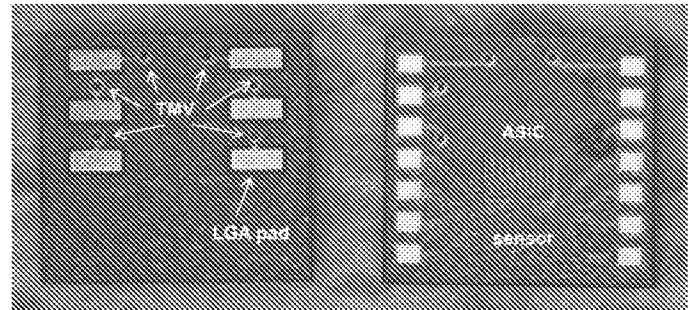
- Precisely laser drilled and homogeneously metalized through mold vias
- Well aligned and void-free Cu filled μ Via with 110 μ m pitch and structured conductor lines with 55 μ m lines and spaces

Tanja Braun, DHT

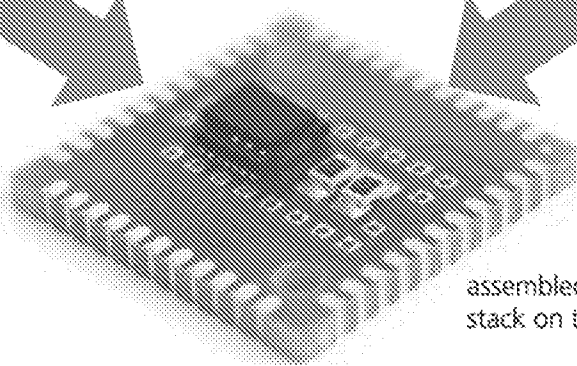
FOWLP Multi-Sensor Stack



pressure sensor/ASIC package with thin film RDL



Acceleration sensor/ASIC package with PCB based RDL and Through Mold Vias (TMV) for package stacking



assembled sensor stack on test board

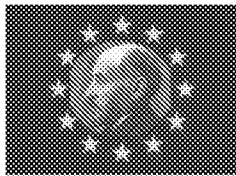
- Functional tests show sensor performance in specs

Tanja Braun, DHT

Application Example

Sensor Integration into Microfluidics

- ENIAC Cajal4EU



CAJAL4EU



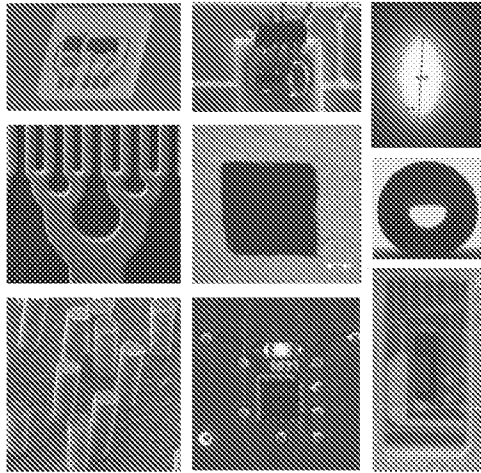
Fraunhofer
IZM

Tanja Braun, DHT

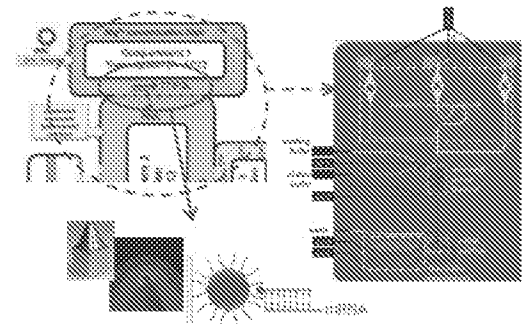
Sensor Integration into Microfluidics

- System design / simulation
- Fluidics
- World to chip interface

Concept and
CAD-Design



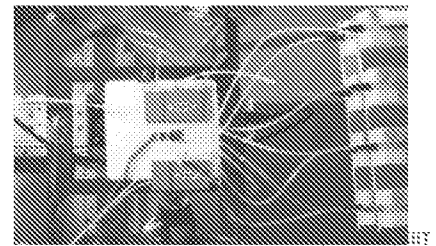
Fabrication



- Polymer / PDMS structuring & Bonding
- Surface modification
- **Integration of Microelectronics / MEMS / MOEMS...**
- Fusion of components to functional demonstrator

- Fluidic characterization
- Functional experiments
- System validation

Characterization



Example: Sensors and Actuators for Automated Cell-free Protein Production

Process automation:

- Phase interface and bubble detectors
- Micro actuators / pumps / valves

Control and regulation:

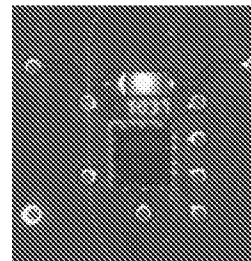
- Electrochemical pH-sensing
- Optical Mg-sensing by color change reaction
- Temperature spot control with micro peltiers
- Glucose measurement for energy regeneration system

Yield measurement:

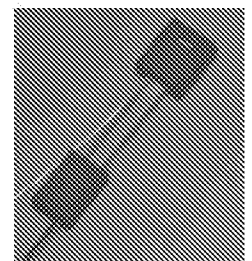
- Fluorescence measurement with μ PMT
- RNA purity control by UV-absorbance measurement



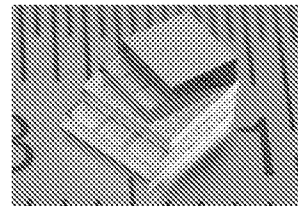
Bubble detector



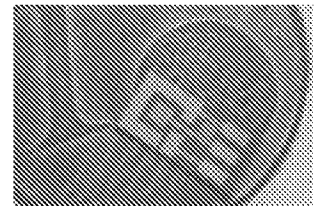
Integrated RGB-Sensor with LED



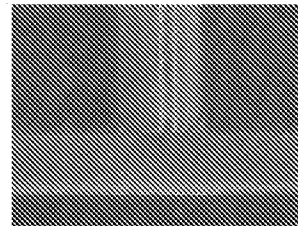
μ PMT (Hamamatsu)



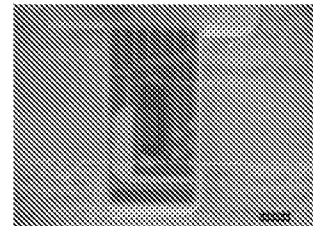
www.micropelt.com



Micro glucose sensor (Fh ISIT)

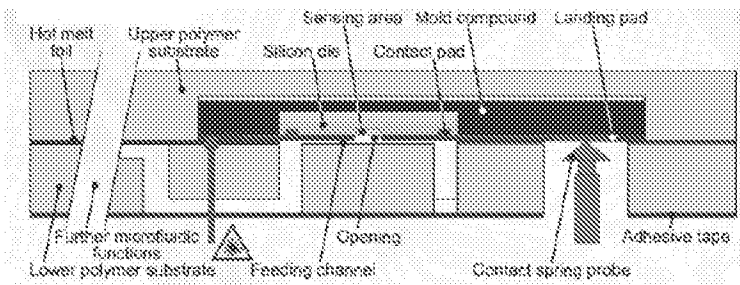


Integrated glass fiber



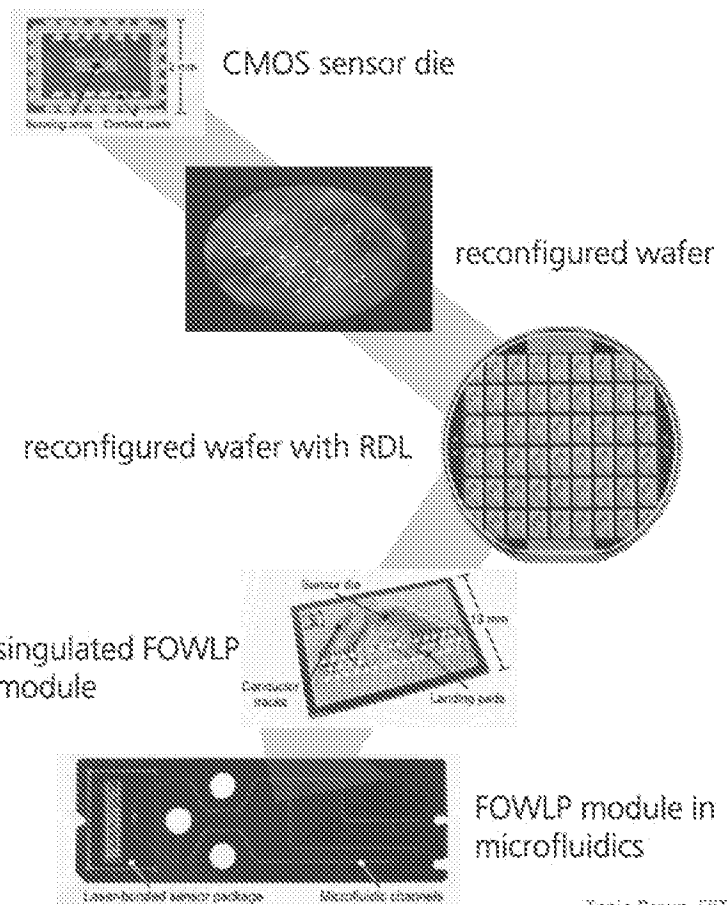
Integrated pH-Sensor (Fh ISIT)
Tanja Braun, DHT

Cajal4U: Integration of CMOS Biosensor into a Polymeric Lab-on-a-Chip Systems



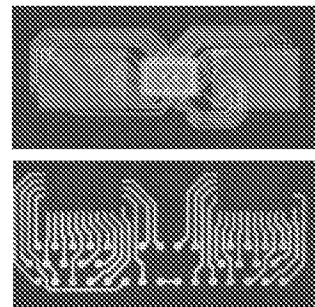
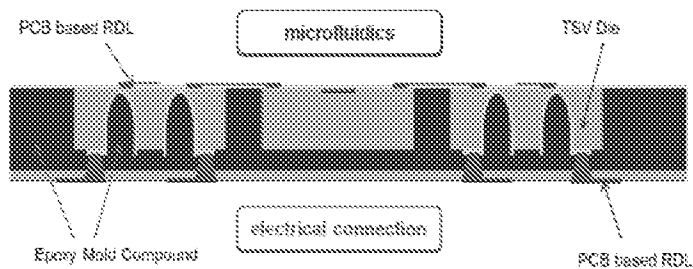
- Feasibility of packaging approach for silicon-into-polymer suitable for mass fabrication successfully shown
- Protection of sensing area during packaging allows application to wide range of biosensors
- Tightness high enough for most microfluidic applications

T. Brettschneider, C. Dörner, H. Suy, T. Braun, E. Jung, R. Hoofman, M. Bründel, R. Zengerle and F. Lärmer, "Integration of CMOS biosensor into a polymeric lab-on-a-chip systems", International Conference on Microfluidics and Nanofluidics, 2013, Venice, Italy.

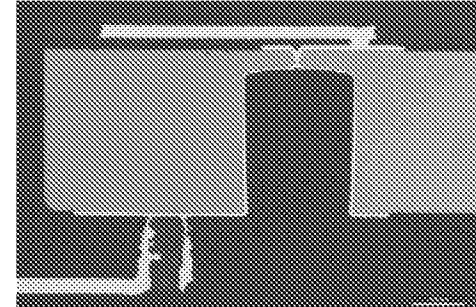


Tanja Braun, IZM

Cajal4U: Integration of CMOS Biosensor

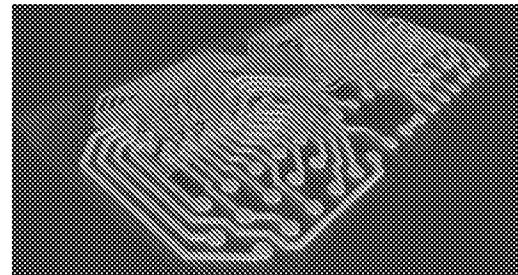


Photography of FOWLP

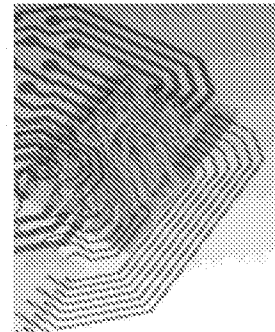


Cross section of connected TSV die

- Separation of "wet" microfluidics from "dry" electrical connection
- Use of TSV dies for 3D routing
- Backside TSV die connection by blind μ vias through molding compound



X-ray CT image of FOWLP



X-ray image of connected TSV die

Tanja Braun, DHT

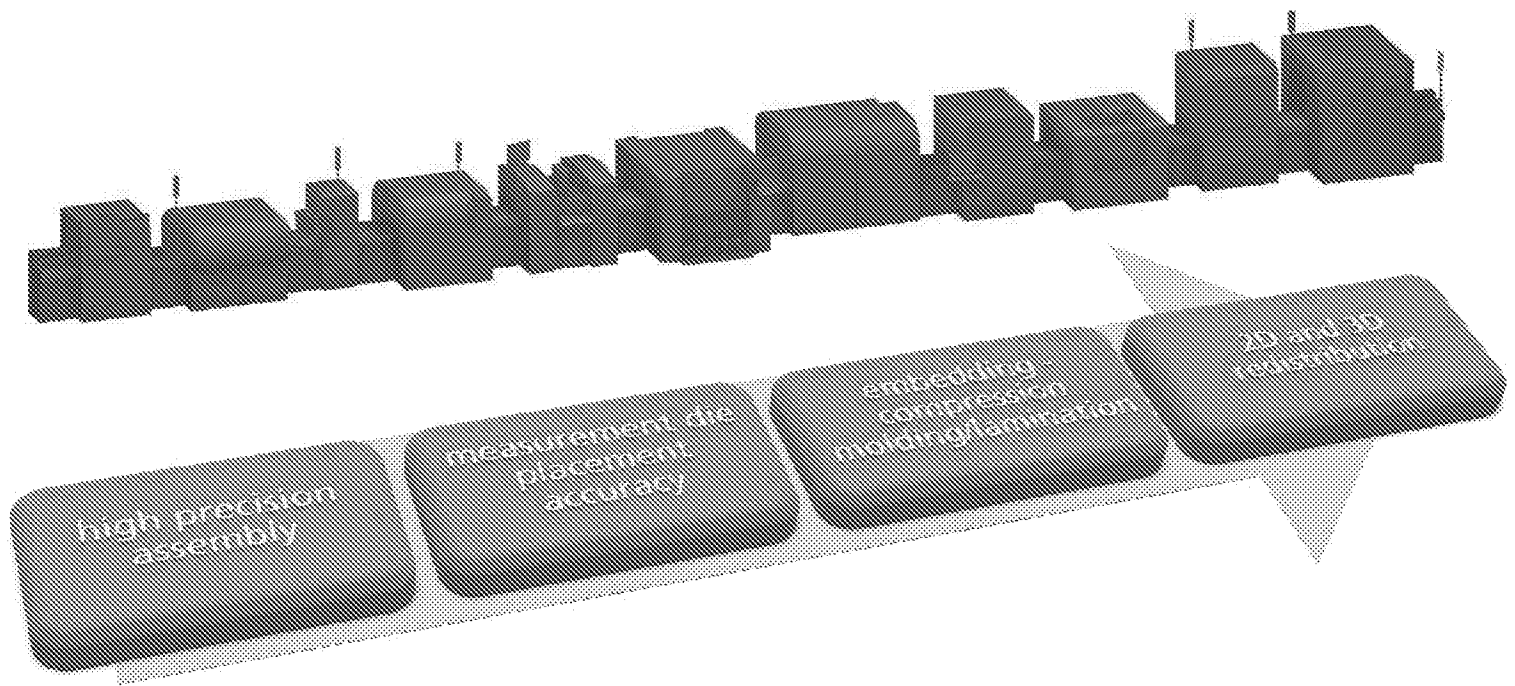
Conclusion

- » A technology has been successfully developed for stacking wafer level embedded packages by Through Mold Vias (TMV)
- » Technology based on large area compression molding and
 - PCB based redistribution technology with potential to full PCB format (610 x 457 mm²) or
 - Standard thin film redistribution technology
- » Laser drilled TMV technology has very high reliability potential – samples passed MSL1, 5000 TC -55 °C/125°C and 5000 h 85°C/85 % r.h. without failure
- » Technology was successfully demonstrated for
 - a functional ASIC- MEMS acceleration sensor package on which an ASIC – MEMS pressure sensor package is assembled
 - Sensor integration into microfluidics

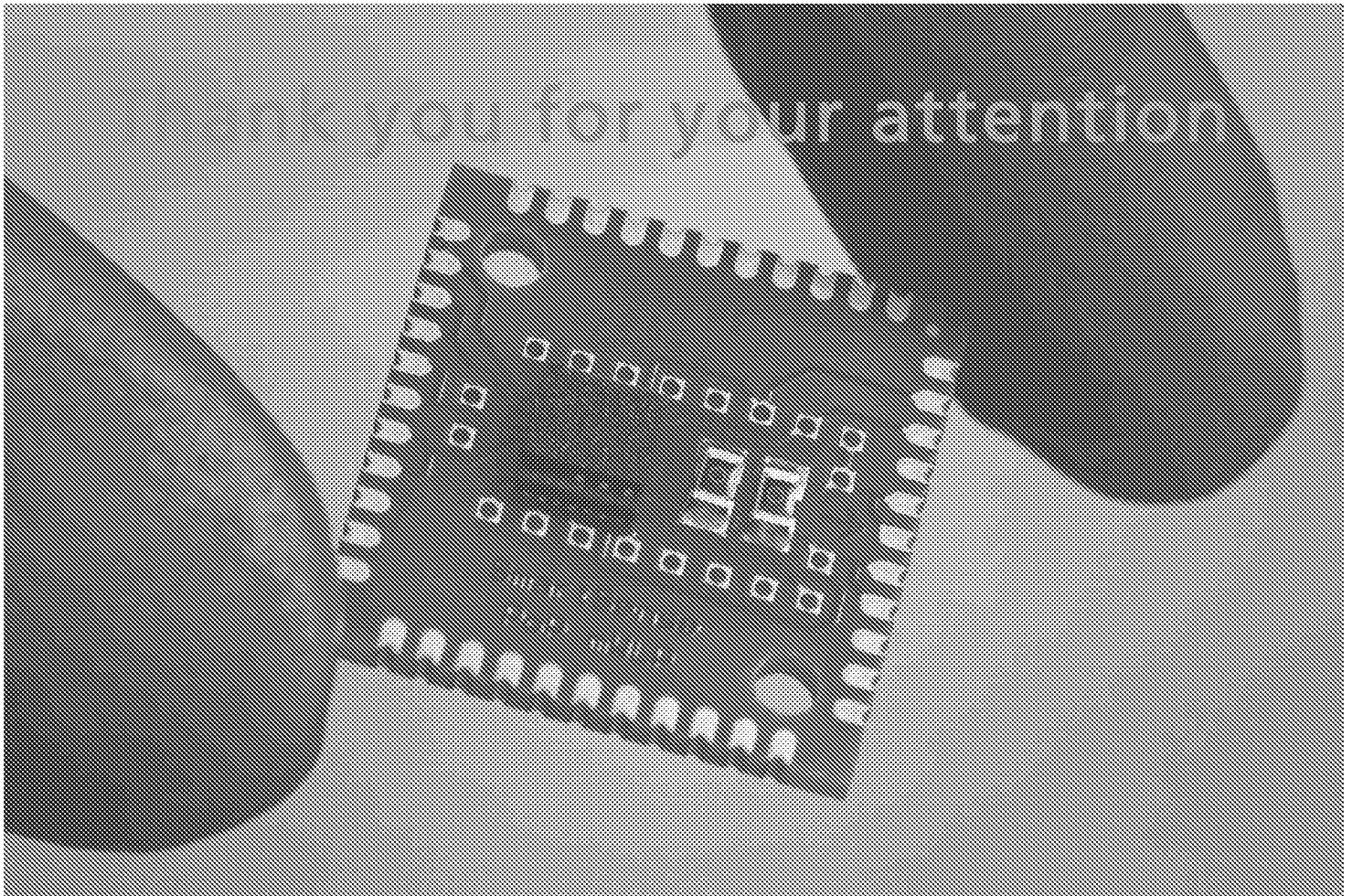
Tanja Braun, DHT

Outlook - Fan-out Panel Level Packaging (FOPLP)

- Continuous manufacturing line for FOPLP up to 24" x 18" / 610 x 457 mm²



Tanja Braun, DHT



Tanja Braun, DHT

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Sign up to join this community

Anybody can ask a question

Anybody can answer

The best answers are voted up and rise to the top



Ideas for attaching / connecting / stacking one PCB onto another with no gap

Asked 7 years ago Active 7 years ago Viewed 11k times



15



6

What methods might be feasible for **attaching/stacking one PCB immediately on top of another PCB**, with the following conditions:

- Zero spacing/gap between the two PCBs
- Electrical contacts are needed, not just physical attachment
- Assume that the top PCB is about a third the size of the bottom PCB

I'm at the early design stage of a project and am trying to survey the options first, so I'm open to recommendations of standard methods as well as any creative ideas.

Note: I'm already familiar with edge castellations (AKA "half vias"), so other suggestions would be of interest.

For instance, is it possible to design it such that the top PCB has pad-contacts only at the bottom (QFN/QFP style) which are somehow solderable onto pads on the bottom PCB?

EDIT: To answer @Andrew's question:

By using our site, you acknowledge that you have read and understand our [Cookie Policy](#), [Privacy Policy](#), and our [Terms of Service](#).

contacts it has), hence the goal of having one constant Base PCB with pads onto which I can attach a variable Top PCB.

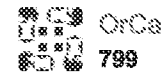
pcb

pcb-design

connector

edited Oct 29 '12 at 10:56

asked Oct 29 '12 at 10:16



OrCa

799 4 8 19

I have to ask: Why? assuming you must have the space on the main board to fit the daughtercard... whilst this may be technically possible, I'm concerned from a manufacturing/assembly viewpoint, especially with your "somehow solder" comment. -- Andrew Oct 29 '12 at 10:37

- 1 @Andrew: Added answer to your question above. And what is the concern from the assembly viewpoint? Very likely this is not an uncommon setup (?) -- OrCa Oct 29 '12 at 10:56 ✓
- 4 I'd say this is a very uncommon approach, usually people use connectors or castellation as you say. You could do what you are talking about like a QFN. A real QFN is essentially just a tiny pcb with a die on top and pads on the bottom. One major difficulty with something this size will be coplanarity. Your boards will have to be very flat in order to be assembled easily, and I assure you they are not flat enough by default ;) IPC class 2 allows for some bow/twist of the board, and during reflow they will flex and twist as well. The bigger they are the harder this will be to assemble. -- Some Hardware Guy Oct 29 '12 at 14:39

5 Answers

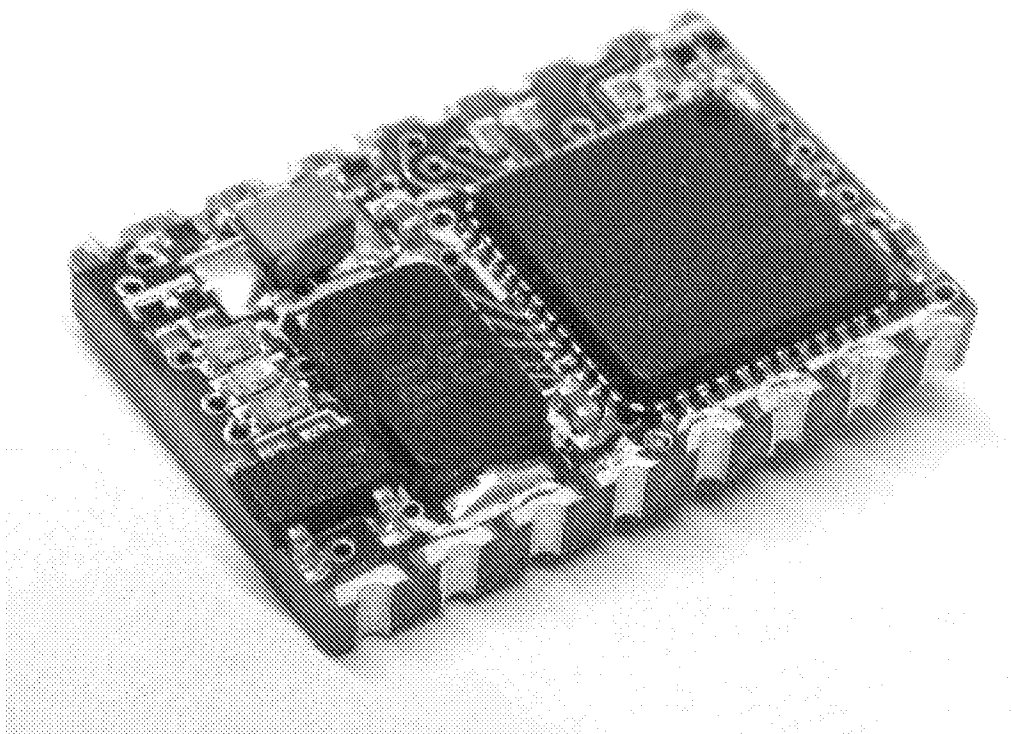


This is not a direct answer to your question, but I think it's quite relevant.

14

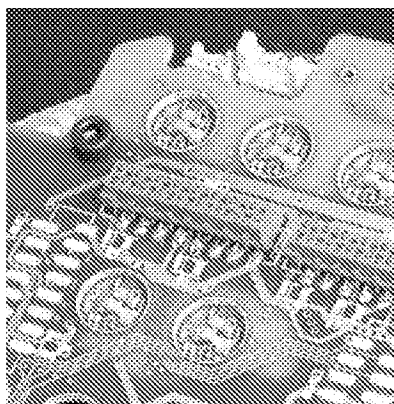
A few years ago, we did the same thing. We made little daughter boards that used edge castellations to solder it onto the mother board.





The difficulty was that we had components on the bottom side of the PCB. These were the vital decoupling capacitors needed by the chip.

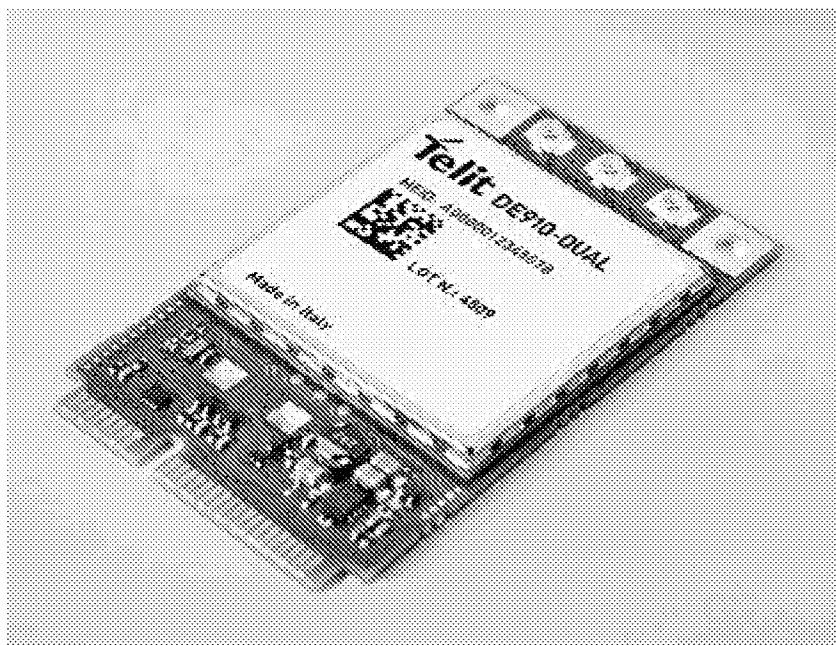
So the motherboard had very large vias to accommodate these components.



You can see several large round holes in the PCB. Through the holes you can see the capacitors on the flip side of the daughter boards. Since the holes are just large vias, they end up through-plated (our supplier doesn't offer unplated holes), so you have to be careful that the plating doesn't short any pads on the daughter board.

A few thoughts about using pads under the PCB. I assume you mean something like this Telit HE910 module:

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Which reflow solders directly onto a PCB. Notice that in the picture the gap between the module and main PCB is not zero, but certainly less than 1mm. Clearly this technique works. Whatever components are inside the module don't mind undergoing an extra reflow process. This is because components can usually survive at least two reflows (once for each side of the board). Since those modules only have components on one side of the PCB, they have almost certainly experienced only one reflow.

Instead of reflow, you might be tempted to use a hot plate to solder a module like this. This would enable you to solder the module down without getting the components inside the module too hot. However, I would advise against this method. At the moment the solder solidifies, the mother PCB will be much hotter than the daughter PCB. As the mother cools and shrinks, it will generate shear forces in the solder joints, and may warp.

edited Oct 29 '12 at 12:40

answered Oct 29 '12 at 12:15



Rocketmagnet

20.7k 12 73 145

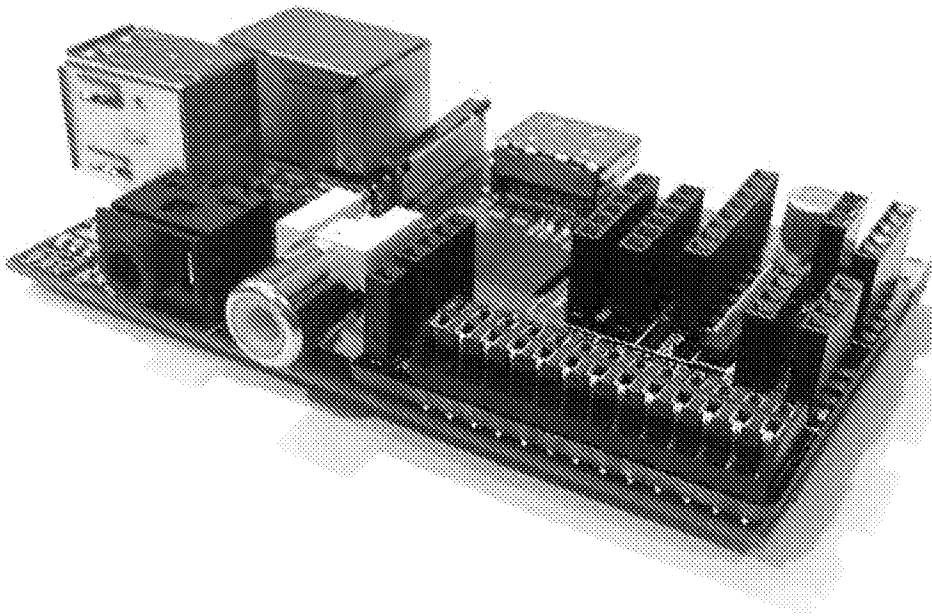
- 1 One potential solution to this is that he could solder the raw PCBs together first, then populate and reflow the boards. It doesn't add much in the way of quick mix-and-matching but it would help against having to require components to endure a second reflow cycle. – Toby Lawrence Oct 29 '12 at 14:10
- 7 I like that you drilled giant holes in the mother board to accommodate the decoupling caps, that's pretty awesome. – Some Hardware Guy Oct 29 '12 at 14:23
- 1 Well, we made castellations the bad way (use vias on the outline of the board). The problem is that, when the board is routed, it tears the plating from the vias. They were very unreliable. A nightmare. It's best to get your PCB manufacturer to make them properly for you. If you can't do that, then expand the board outline so that the vias are left intact, then use a belt sander to sand away half the vias. This reduces stress on them. – Rocketmagnet Oct 29 '12 at 21:52
- 1 They're probably still vulnerable to things like thermal and mechanical stress. You might consider: 1) strengthening the copper around the castellation with a couple of small vias. This helps to rivet the copper down. 2) have all castellations along one edge, and use some flexible glue along the other edge. This should

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- 2 @Sener - These are IDC wire-to-board connectors called Micro-Match by TE-Connectivity. – Rocketmagnet Nov 7 '18 at 11:08

Maybe not exactly what you are asking, but I suggest you to check out PiCrust for ideas. They use connectors by Hirose to achieve a compact stacked design on top of the Raspberry Pi board.

- 5 If the board should be replaceable without soldering, this sounds like a pretty simple solution to the problem.



answered Oct 29 '12 at 15:34

 Nakedible
286 3 15

In my (admittedly narrow experience) daughter cards are usually fitted on header connectors, not directly soldered.

- 3 In reply to a question about Very low stacking height PCB connector, @trygvis suggested this Molex connector

Maybe that's of use?

The problem with face-to-face soldering as you describe is that this will have to be a manual process (not pick-and-place with reflow) unless you want to reflow your PCBs. Also, you would need to be sure of the mechanical fixing - a few solder tags probably won't be enough - you will need mechanical fixing otherwise there is a serious risk of vibration fracture.

edited Nov 19 '17 at 17:30

answered Oct 29 '12 at 11:38

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Two principal things come to mind:

1

1) what is you describe could be used to allow describe a solder bump (BGA) type package that uses a FR-4 substrate. This is not an uncommon packaging option.

2) there used to be a type of tape that you could get that would preferentially enhance electrical connection through the thickness of the tape whilst minimizing lateral conduction. I used to be available from 3M, but I haven't seen it in years. And it's conductivity was probably insufficient for your use if you need to carry 100's of mA. This might give you an idea or two.

answered Oct 29 '12 at 15:13



placeholder

27.8k 8 54 98

Re #2: You're thinking of 3M Z-Axis tape. It only conducts vertically between boards, not horizontally between pads. – Navin May 17 at 6:03

1

You might consider using a combination of SMT receptacle and thru hole pin header, for example:

2.54mm DIL SMT socket [BG120](#)

2.54mm DIL Thru hole pin header [BG040](#)

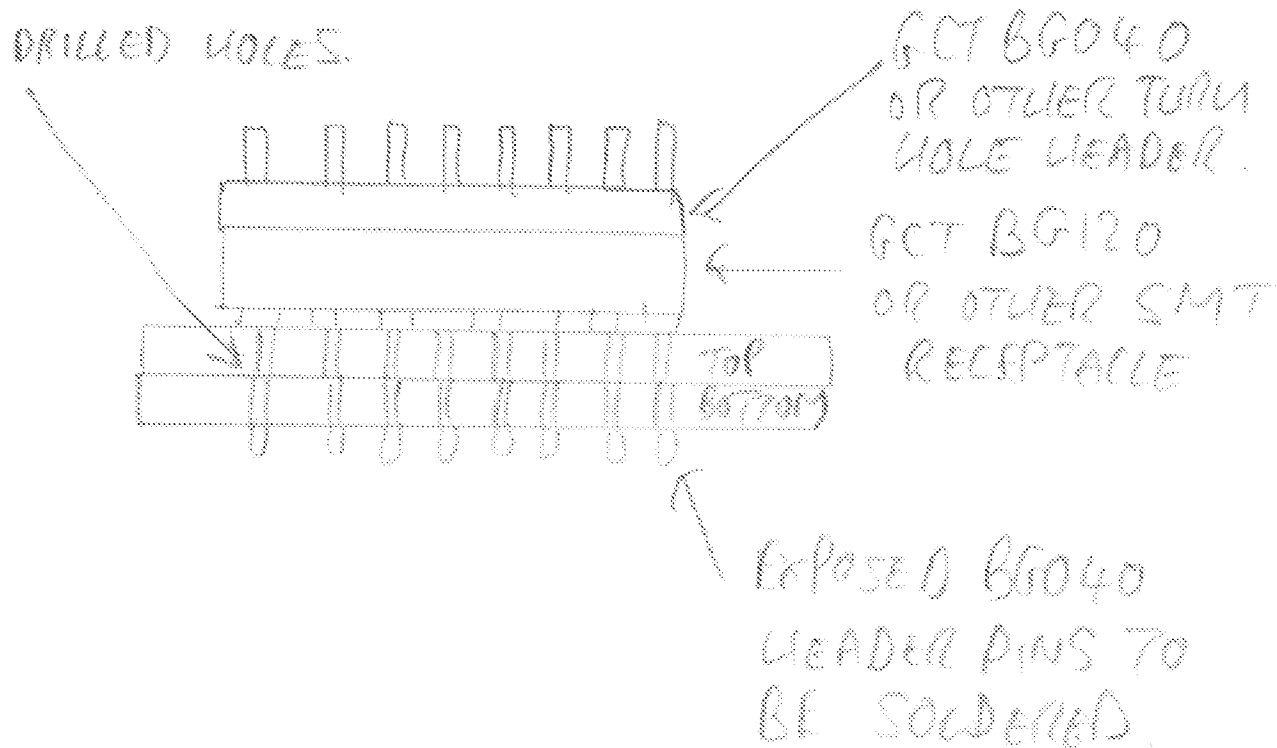
You could choose single row, GCT also offer finer pitches if required, other options [here](#).

-Mount the SMT socket on the top PCB.

-Plug a thru hole pin header (from above) all the way through both PCBs. Obviously the mating pin header pins should be long enough to pass through the SMT female receptacle, both PCB's and leave you enough space to hand solder.

-Hand solder the exposed header pins on the bottom side of the bottom PCB.

See sketch enclosed (excuse my awful drawing)



, I'm not sure if this will work for you, just an idea!

Note: GCT standard products available via Newark, any non standard pin lengths carry a higher MOQ (at least 1k pieces).

edited Oct 29 '12 at 15:37

 embedded.kyle
7,825 2 22 41

answered Oct 29 '12 at 11:49

 Laurence Hill
213 1 6

PATENT COOPERATION TREATY

From the INTERNATIONAL SEARCHING AUTHORITY

To:

REX HUANG
FISH & RICHARDSON P.C.
P.O. BOX 1022
BOSTON, MA 02210

PCT

NOTIFICATION OF TRANSMITTAL OF
THE INTERNATIONAL SEARCH REPORT AND
THE WRITTEN OPINION OF THE INTERNATIONAL
SEARCHING AUTHORITY, OR THE DECLARATION

(PCT Rule 44.1)

Date of mailing
(day/month/year)

06 MAY 2013

Applicant's or agent's file reference
006140169WO1

FOR FURTHER ACTION See paragraphs 1 and 4 below

International application No.
PCT/US 12/37495

International filing date
(day/month/year) 11 May 2012 (11.05.2012)

Applicant VLT, INC.

1. ☒ The applicant is hereby notified that the international search report and the written opinion of the International Searching Authority have been established and are transmitted herewith.

Filing of amendments and statement under Article 19:

The applicant is entitled, if he so wishes, to amend the claims of the international application (see Rule 46):

When? The time limit for filing such amendments is normally two months from the date of transmittal of the international search report.

Where? Directly to the International Bureau of WIPO, 34 chemin des Colombettes
1211 Geneva 20, Switzerland, Facsimile No.: +41 22 338 82 70

For more detailed instructions, see PCT Applicant's Guide, International Phase, paragraphs 9.004 – 9.011.

2. ☐ The applicant is hereby notified that no international search report will be established and that the declaration under Article 17(2)(a) to that effect and the written opinion of the International Searching Authority are transmitted herewith.
3. ☐ **With regard to any protest** against payment of (an) additional fee(s) under Rule 40.2, the applicant is notified that:
- ☐ the protest together with the decision thereon has been transmitted to the International Bureau together with any request to forward the texts of both the protest and the decision thereon to the designated Offices.
- ☐ no decision has been made yet on the protest: the applicant will be notified as soon as a decision is made.

4. **Reminders**

The applicant may submit comments on an informal basis on the written opinion of the International Searching Authority to the International Bureau. The International Bureau will send a copy of such comments to all designated Offices unless an international preliminary examination report has been or is to be established. Following the expiration of 30 months from the priority date, these comments will also be made available to the public.

Shortly after the expiration of **18 months** from the priority date, the international application will be published by the International Bureau. If the applicant wishes to avoid or postpone publication, a notice of withdrawal of the international application, or of the priority claim, must reach the International Bureau before the completion of the technical preparations for international publication (Rules 90bis.1 and 90bis.3).

Within **19 months** from the priority date, but only in respect of some designated Offices, a demand for international preliminary examination must be filed if the applicant wishes to postpone the entry into the national phase **until 30 months** from the priority date (in some Offices even later); otherwise, the applicant must, **within 20 months** from the priority date, perform the prescribed acts for entry into the national phase before those designated Offices.

In respect of other designated Offices, the time limit of **30 months** (or later) will apply even if no demand is filed within 19 months.

For details about the applicable time limits, Office by Office, see www.wipo.int/pct/en/texts/time_limits.html and the *PCT Applicant's Guide, National Chapters*.

Name and mailing address of the ISA/
Mail Stop PCT, Attn: ISA/US
Commissioner for Patents
P.O. Box 1450, Alexandria, Virginia 22313-1450
Facsimile No. 571-273-3201

Authorized officer

Lee W. Young

PCT Helpdesk: 571-272-4300

Telephone No. PCT OSP: 571-272-7774

PATENT COOPERATION TREATY

PCT

INTERNATIONAL SEARCH REPORT

(PCT Article 18 and Rules 43 and 44)

Applicant's or agent's file reference 006140169WO1	FOR FURTHER ACTION see Form PCT/ISA/220 as well as, where applicable, item 5 below.	
International application No. PCT/US 12/37495	International filing date (<i>day/month/year</i>) 11 May 2012 (11.05.2012)	(Earliest) Priority Date (<i>day/month/year</i>) 11 May 2011 (11.05.2011)
Applicant VLT, INC.		

This international search report has been prepared by this International Searching Authority and is transmitted to the applicant according to Article 18. A copy is being transmitted to the International Bureau.

This international search report consists of a total of 8 sheets.

☐ It is also accompanied by a copy of each prior art document cited in this report.

1. Basis of the report

a. With regard to the **language**, the international search was carried out on the basis of:

- ☒ the international application in the language in which it was filed.
☐ a translation of the international application into _____ which is the language of a translation furnished for the purposes of international search (Rules 12.3(a) and 23.1(b)).

b. ☐ This international search report has been established taking into account the **rectification of an obvious mistake** authorized by or notified to this Authority under Rule 91 (Rule 43.6bis(a)).

c. ☐ With regard to any **nucleotide and/or amino acid sequence** disclosed in the international application, see Box No. I.

2. ☐ **Certain claims were found unsearchable** (see Box No. II).

3. ☐ **Unity of invention is lacking** (see Box No. III).

4. With regard to the **title**,

- ☒ the text is approved as submitted by the applicant.
☐ the text has been established by this Authority to read as follows:

5. With regard to the **abstract**,

- ☐ the text is approved as submitted by the applicant.
☒ the text has been established, according to Rule 38.2, by this Authority as it appears in Box No. IV. The applicant may, within one month from the date of mailing of this international search report, submit comments to this Authority.

6. With regard to the **drawings**,

- a. the figure of the **drawings** to be published with the abstract is Figure No. 1
- ☒ as suggested by the applicant.
☐ as selected by this Authority, because the applicant failed to suggest a figure.
☐ as selected by this Authority, because this figure better characterizes the invention.
- b. ☐ none of the figures is to be published with the abstract.

INTERNATIONAL SEARCH REPORT

International application No.

PCT/US 12/37495

Box No. IV Text of the abstract (Continuation of item 5 of the first sheet)

A method of encapsulating a panel of electronic components such as power converters reduces wasted printed circuit board area. The panel, which may include a plurality of components, may be cut into one or more individual pieces after encapsulation. The mold may be used to form part of the finished product. Interconnection features provided along boundaries of individual circuits are exposed during the singulation process providing electrical connections to the components without wasting valuable PCB surface area. The molds may include various internal features. Wide cuts may be made in the molds after encapsulation reducing thermal stresses. Blank mold panels may be machined to provide some or all of the above features in an on-demand manufacturing system. Connection adapters may be provided to use the modules in vertical or horizontal mounting positions in connector, through-hole, surface-mount solder variations.

INTERNATIONAL SEARCH REPORT

International application No.

PCT/US 12/37495

A. CLASSIFICATION OF SUBJECT MATTER

IPC(8) - H05K 7/00 (2013.01)

USPC - 361/728

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

IPC(8): H05K 7/00 (2013.01)

USPC: 361/728

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

IPC(8): H05K 7/00 (2013.01) (keyword limited, see terms below)
361/728, 752, 783; 249/83; 29/825 (keyword limited, see terms below)

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

PatBase, Google Scholar, Google Patents. Search terms used: encapsulate, mold, insulate, cut, slice, break, separate, circuit, power, package, semiconductor, connector, conductor, expose, cut, remove, solder, centrifuge, magnet, heatsink, fin, footprint, metal, aluminum, cavity, recess, trace, cure, component, substrate, pcb, printed, circuit

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	US 2002/0096348 A1 (SAXELBY, JR et al.) 25 July 2002 (25.07.2002), entire document, especially: Fig 4, 5a, 7, 19, 20, para [0040]-[0054]	1-6, 15, 17, 18, 20-24, 26, 28, 49-62, 66-69, 73-85, 89-97, 109, 143-146, 149-152, 155, 158, 162-164

Y		7-14, 16, 19, 25, 27, 29-48, 63-65, 70-72, 86-88, 98-108, 110-142, 147, 148, 153, 154, 156, 157, 159-161
Y	US 2004/0100778 A1 (VINCIARELLI et al.) 27 May 2004 (27.05.2004), entire document, especially: Fig 3a, 5a, 5b, 8; para [0052], [0055], [0060], [0065]	7-14, 16, 19, 25, 30-48, 63-65, 70-72, 86-88, 98-108, 110-142, 147, 148, 153, 154, 156, 157, 159-161
Y	US 2010/0246152 A1 (LIN et al.) 30 September 2010 (30.09.2010), para [0194]	27, 29
A	WO 95/27308 A1 (PARTHASARATHI et al.) 12 October 1995 (12.10.1995), pg 4-5	1-164
A	US 2007/0241440 A1 (HOANG et al.) 18 October 2007 (18.10.2007), entire document	1-164



Further documents are listed in the continuation of Box C.



* Special categories of cited documents:

"A" document defining the general state of the art which is not considered to be of particular relevance

"E" earlier application or patent but published on or after the international filing date

"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)

"O" document referring to an oral disclosure, use, exhibition or other means

"P" document published prior to the international filing date but later than the priority date claimed

"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

"&" document member of the same patent family

Date of the actual completion of the international search

06 April 2013 (06.04.2013)

Date of mailing of the international search report

06 MAY 2013

Name and mailing address of the ISA/US

Mail Stop PCT, Attn: ISA/US, Commissioner for Patents
P.O. Box 1450, Alexandria, Virginia 22313-1450

Facsimile No. 571-273-3201

Authorized officer:

Lee W. Young

PCT Helpdesk: 571-272-4300
PCT OSP: 571-272-7774

PATENT COOPERATION TREATY

From the
INTERNATIONAL SEARCHING AUTHORITY

To: REX HUANG
FISH & RICHARDSON P.C.
P.O. BOX 1022
BOSTON, MA 02210

PCT

WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY

(PCT Rule 43bis.1)

Date of mailing
(day/month/year)

06 MAY 2013

Applicant's or agent's file reference
006140169WO1

FOR FURTHER ACTION

See paragraph 2 below

International application No.

PCT/US 12/37495

International filing date (day/month/year)

11 May 2012 (11.05.2012)

Priority date (day/month/year)

11 May 2011 (11.05.2011)

International Patent Classification (IPC) or both national classification and IPC

IPC(8) - H05K 7/00 (2013.01)

USPC - 361/728

Applicant VLT, INC.

1. This opinion contains indications relating to the following items:

- ☒ Box No. I Basis of the opinion
- ☐ Box No. II Priority
- ☐ Box No. III Non-establishment of opinion with regard to novelty, inventive step and industrial applicability
- ☐ Box No. IV Lack of unity of invention
- ☒ Box No. V Reasoned statement under Rule 43bis.1(a)(i) with regard to novelty, inventive step or industrial applicability: citations and explanations supporting such statement
- ☐ Box No. VI Certain documents cited
- ☐ Box No. VII Certain defects in the international application
- ☐ Box No. VIII Certain observations on the international application

2. FURTHER ACTION

If a demand for international preliminary examination is made, this opinion will be considered to be a written opinion of the International Preliminary Examining Authority ("IPEA") except that this does not apply where the applicant chooses an Authority other than this one to be the IPEA and the chosen IPEA has notified the International Bureau under Rule 66.1 *bis(b)* that written opinions of this International Searching Authority will not be so considered.

If this opinion is, as provided above, considered to be a written opinion of the IPEA, the applicant is invited to submit to the IPEA a written reply together, where appropriate, with amendments, before the expiration of 3 months from the date of mailing of Form PCT/ISA/220 or before the expiration of 22 months from the priority date, whichever expires later.

For further options, see Form PCT/ISA/220.

Name and mailing address of the ISA/US
Mail Stop PCT, Attn: ISA/US
Commissioner for Patents
P.O. Box 1450, Alexandria, Virginia 22313-1450
Facsimile No. 571-273-3201

Date of completion of this opinion

06 April 2013 (06.04.2013)

Authorized officer:

Lee W. Young

PCT Helpdesk: 571-272-4300
PCT OSP: 571-272-7774

WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY

International application No.

PCT/US 12/37495

Box No. I Basis of this opinion

1. With regard to the **language**, this opinion has been established on the basis of:

- ☒ the international application in the language in which it was filed.
☐ a translation of the international application into _____ which is the language of a translation furnished for the purposes of international search (Rules 12.3(a) and 23.1(b)).

2. ☐ This opinion has been established taking into account the **rectification of an obvious mistake** authorized by or notified to this Authority under Rule 91 (Rule 43*bis*.1(a))

3. With regard to any **nucleotide and/or amino acid sequence** disclosed in the international application, this opinion has been established on the basis of a sequence listing filed or furnished:

a. (means)

- ☐ on paper
☐ in electronic form

b. (time)

- ☐ in the international application as filed
☐ together with the international application in electronic form
☐ subsequently to this Authority for the purposes of search

4. ☐ In addition, in the case that more than one version or copy of a sequence listing has been filed or furnished, the required statements that the information in the subsequent or additional copies is identical to that in the application as filed or does not go beyond the application as filed, as appropriate, were furnished.

5. Additional comments:

**WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY**

International application No.

PCT/US 12/37495

Box No. V Reasoned statement under Rule 43bis.1(a)(i) with regard to novelty, inventive step or industrial applicability; citations and explanations supporting such statement

1. Statement

Novelty (N)	Claims	<u>see below</u>	YES
	Claims	<u>see below</u>	NO
Inventive step (IS)	Claims	<u>None</u>	YES
	Claims	<u>1-164</u>	NO
Industrial applicability (IA)	Claims	<u>1-164</u>	YES
	Claims	<u>None</u>	NO

2. Citations and explanations:

1. Statement

Novelty (N)	Claims: 2,3,7-14,16,19,25,27,29-32,34-48,58,63-65,70-72,85-88,98-108,110-142,147,148,153,154,156,157,159-161	YES
	Claims: 1,4-6,15,17,18,20-24,26,28,33,49-57,59-62,66-69,73-84,89-97,109,143-146,149-152,155,158,162-164	NO

Claims 1, 4-6, 15, 17, 18, 20-24, 26, 28, 33, 49-57, 59-62, 66-69, 73-84, 89-97, 109, 143-146, 149-152, 155, 158 and 162-164 lack novelty under PCT Article 33(2) as being anticipated by US 2002/0096348 A1 to SAXELBY, JR et al. (hereinafter 'Saxelby').

As per claim 1, Saxelby teaches a method of making a plurality of electronic devices (para [0053]), comprising: providing a plurality of mold panels (60a, 60b), which when assembled, form an internal chamber (68) (Fig 7, para [0043]); providing a substrate (20) having a plurality of conductive traces and a plurality of components (12, 14, 15, 16, 17, 18) electrically connected to form at least one circuit (Fig 4, para [0040]), the substrate having at least one contact (30) for making an electrical connection to the circuit (Fig 4, para [0040]); forming a panel assembly including the mold panels assembled with the substrate in the internal chamber and an encapsulant (35) filling spaces between the substrate and interior surfaces of the chamber (para [0043]-[0045], [0047]); curing the encapsulant (para [0051], [0052]); and cutting the panel assembly to expose at least a portion of the at least one contact and form a respective exposed contact (cutting line 116, Fig 19, para [0053], exposed conductive pads).

As per claim 4, Saxelby teaches the method of claim 1, and further teaches wherein the cutting divides the panel assembly producing at least one module having a plurality of layers including respective portions of each of the panel molds, the substrate, and the encapsulant, and the module contains the at least one circuit (Fig 20).

As per claim 5, Saxelby teaches the method of claim 4, and further teaches wherein the cutting defines at least one side of the module (cut line 116, Fig 19).

As per claim 6, Saxelby teaches the method of claim 4, and further teaches wherein the cutting defines two or more sides of the module (cut line 116, Fig 19).

As per claim 15, Saxelby teaches the method of claim 4, and further teaches wherein the mold panels are metal (para [0061]).

As per claim 17, Saxelby teaches the method of claim 1, and further teaches wherein the providing mold panels further comprises providing contours in an internal surface of one or more of the mold panels (Fig 20, para [0054]-[0055]).

As per claim 18, Saxelby teaches the method of claim 17, and further teaches wherein the providing contours further comprises matching depths of portions of the internal surface to heights of one or more selected components (Fig 20, para [0054]-[0055]).

As per claim 20, Saxelby teaches the method of claim 4, and further teaches wherein the traces and components are electrically connected to form a plurality of separable circuits arranged in a pattern on the substrate (Fig 19, para [0053]); and the cutting divides the panel assembly along spaces between the separable circuits into a plurality of modules each containing at least one respective circuit (Fig 19, para [0053]).

As per claim 21, Saxelby teaches the method of claim 20, and further teaches wherein the spaces between the separable circuits have dimensions approximately matching a width of cuts produced by equipment used to cut the panel assembly (Fig 19, para [0053]).

As per claim 22, Saxelby teaches the method of claim 21, and further teaches wherein the at least one contact comprises a plurality of contacts located in the spaces between the separable circuits (Fig 19, para [0053]).

As per claim 23, Saxelby teaches the method of claim 22, and further teaches wherein the plurality of contacts are formed in the substrate (Fig 19, para [0053]).

As per claim 24, Saxelby teaches the method of claim 23, and further teaches wherein the plurality of contacts are formed and buried below at least one surface of the substrate (para [0040]).

---(continued in Supplemental Boxes)---

WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY

International application No.

PCT/US 12/37495

Supplemental Box

In case the space in any of the preceding boxes is not sufficient.

Continuation of:
Box V.2 Citations and explanations:

As per claim 26, Saxelby teaches the method of claim 1, and further teaches wherein forming the panel assembly comprises: dispensing encapsulant into a bottom panel mold (para [0043]-[0048]); assembling the substrate into the bottom panel mold (para [0043]-[0048]); dispensing encapsulant onto a top of the substrate (para [0043]-[0048]); and assembling a top panel mold onto the substrate (para [0043]-[0048]).

As per claim 28, Saxelby teaches the method of claim 1, and further teaches wherein forming the panel assembly comprises: assembling the substrate with a first substrate surface facing into a first panel closing a second panel mold onto the substrate covering a second substrate surface (para [0043]-[0048]), providing one or more conduits to the internal chamber (para [0043]-[0048]), and forcing encapsulant through the one or more conduits into the chamber (para [0043]-[0048]).

As per claim 49, Saxelby teaches an apparatus comprising: a first mold panel (60b) comprising an exterior surface (Fig 7, para [0043]), an interior surface defining an internal cavity (68, Fig 7, para [0043]), a clamp region located at points along a circumference of the internal cavity (para [048]-[0049]), and an opening (84, Fig 7, para [0045]), the first mold panel being adapted to (a) be engaged by pressure in the clamp region (para [0048]-[0049]), (b) receive in the cavity a circuit panel containing a plurality of components (para [0040]) and mold compound to fill empty spaces in the internal cavity (para [0045]), and (c) be cut after curing of the mold compound (para [0053]).

As per claim 50, Saxelby teaches the apparatus of claim 49, and further teaches wherein the interior surface in the region of the internal cavity is adapted to adhere to the mold compound (para [0045], [0052]).

As per claim 51, Saxelby teaches the apparatus of claim 49, and further teaches further comprising a second mold panel (60a) comprising an exterior surface (Fig 7), an interior surface (Fig 7), and a clamp region (para [0048], [0049]), wherein the second mold panel is adapted to (a) close against and mate with the first mold panel (para [0048], [0049]), (b) be engaged by pressure in the clamp region (para [0048], [0049]), and (c) be cut after curing of the mold compound (para [0053]).

As per claim 52, Saxelby teaches the apparatus of claim 51, and further teaches wherein the interior surface of the second mold further defines a second internal cavity, the clamp region of the second mold panel is located at points along a circumference of the second internal cavity, and the internal cavity of the first mold panel is adapted to receive a first side of the circuit panel and the second internal cavity is adapted to receive a second opposite side of the circuit panel (para [0043]-[0045]).

As per claim 53, Saxelby teaches the apparatus of claim 52, and further teaches further comprising contours formed in the interior surface of at least one of the mold panels (Fig 20, para [0054]-[0055]).

As per claim 54, Saxelby teaches the apparatus of claim 53, and further teaches wherein the contours are adapted to match predetermined characteristics of selected ones of the plurality of components (Fig 20, para [0054]-[0055]).

As per claim 55, Saxelby teaches the apparatus of claim 53, and further teaches wherein the internal surface of the mold panel is adapted to adhere to the molding compound (para [0052]).

As per claim 56, Saxelby teaches the apparatus of claim 55, and further teaches wherein the contours form interlocking features with cured mold compound (para [0052]).

As per claim 57, Saxelby teaches the apparatus of claim 53, and further teaches wherein at least one of the mold panels comprises a non-ferrous metal (para [0061]).

As per claim 59, Saxelby teaches the apparatus of claim 51, and further teaches wherein at least one of the mold panels comprises a thermally conductive material (para [0061]).

As per claim 60, Saxelby teaches the apparatus of claim 53, and further teaches wherein at least one of the mold panels comprises a non-metallic substance (para [0047]).

As per claim 61, Saxelby teaches the apparatus of claim 51, and further teaches further comprising at least one channel connected to the internal cavity for allowing expansion of the mold compound (84, Fig 7, para [0045]).

As per claim 62, Saxelby teaches the apparatus of claim 51, and further teaches further comprising at least one channel connected to the internal cavity for injecting mold compound into the internal cavity (84, Fig 7, para [0045]).

As per claim 66, Saxelby teaches the apparatus of claim 49, and further teaches wherein the internal cavity includes features formed in the interior surface and arranged in a predetermined pattern. (Fig 20, para [0054]-[0055]).

As per claim 67, Saxelby teaches the apparatus of claim 49, and further teaches wherein the internal cavity includes features formed in the interior surface along lines through which the mold panel maybe cut to establish a setback from a cut edge of the mold panel (Fig 20, para [0054]-[0055]).

As per claim 68, Saxelby teaches the apparatus of claim 49, and further teaches wherein the mold panel further comprises at least one registration feature adapted to engage and establish a predetermined relationship with the circuit panel (116, Fig 19, para [0053]).

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As per claim 69, Saxelby teaches the apparatus of claim 49, and further teaches wherein one or more selected portions of the moldpanel are adapted for incorporation into one or more products, the products being formed by a process which uses the mold panel to contain the molding compound (Fig 19, para [0053]).

As per claim 73, Saxelby teaches an apparatus comprising: a panel assembly having external surfaces defined by a first mold panel (60a) and a second mold panel (60b) (Fig 7, para [0043]-[0044]); the first and second mold panels forming an internal cavity (68) enclosing an internal circuit board (20), the internal circuit board having a first surface and second surface and a plurality of components (12, 14, 15, 17, 18) in an active circuit area on at least one of the surfaces, the components being electrically connected to interconnects (30) contained within the internal cavity (Fig 7, para [0040], [0043]-[0044]); the internal cavity being filled with mold compound (35) in spaces unoccupied by the circuit board and components (Fig 7, para [0045]); the panel assembly being adapted to have first selected portions of the first and second mold panels cut away to expose the interconnects and to have second selected portions of the respective mold panels situated near the active circuit area remain attached to the assembly following the cut (Fig 19, para [0053]).

As per claim 74, Saxelby teaches the apparatus of claim 73, and further teaches wherein the panel assembly includes a clamp region located at points along a circumference of the first and second internal cavities, and the first selected portions comprise the clamp region (para [0048]-[0049]).

As per claim 75, Saxelby teaches an apparatus comprising: a panel assembly having external surfaces defined by a first mold panel (60a) and a second mold panel (60b) (fig 7, para [0040]-[0044]); the first and second mold panels forming an internal cavity (68) enclosing an internal circuit board (20) (Fig 7, para [0040]-[0044]), the internal circuit board having a first surface and second surface and a plurality of components electrically (12, 14, 15, 17, 18) connected to form a plurality of individual circuits (Fig 7, 19, para [0040]-[0044]), each individual circuit being electrically connected to respective interconnects (30) located along a respective circuit perimeter, the interconnects being contained within the cavity (Fig 7, 19, para [0040]-[0044], [0053]).

As per claim 76, Saxelby teaches the apparatus of claim 75, and further teaches wherein the panel assembly is adapted to be cut along the circuit perimeter separating the individual circuits, dividing the panel assembly into individual circuit modules, and exposing selected portions of the interconnects (para [0053]).

As per claim 77, Saxelby teaches the apparatus of claim 75, and further teaches wherein the panel assembly is constructed to retain an integral layered structure after being filled with mold compound (para [0045]) which is subsequently cured (para [0052]), the layered structure comprising a first layer including a portion of the first mold, a second layer including a portion of the mold compound, a third layer including a portion of the circuit board, a fourth layer including a portion of the mold compound, and a fifth layer including a portion of the second mold (Fig 7, 19, para [0043], [0044]).

As per claim 78, Saxelby teaches the apparatus of claim 75, and further teaches wherein the internal circuit board comprises a plurality of circuits having the same functionality (Fig 19, para [0053]).

As per claim 79, Saxelby teaches the apparatus of claim 75, and further teaches further comprising one or more conduits connecting the internal cavity to an external opening (Fig 7, para [0045]).

As per claim 80, Saxelby teaches the apparatus of claim 75, and further teaches wherein spaces unoccupied by the circuit board, components, and interconnects in the internal cavity are filled with mold compound (para [0045]).

As per claim 81, Saxelby teaches the apparatus of claim 80, and further teaches wherein the mold compound has been cured (para [0052]).

As per claim 82, Saxelby teaches the apparatus of claim 81, and further teaches wherein selected portions of at least one of the mold panels have been removed, the selected portions being located near the circuit perimeters (Fig 19, para [0053]).

As per claim 83, Saxelby teaches an apparatus comprising: a modular package (9, Fig 5a, para [0040]) having a first external surface, a second external surface opposite the first external surface (Fig 5a), and a side wall extending along the perimeter of and connecting with the first and second external surfaces (left most side of Fig 5a), the modular package including a first layer defining the first external surface and a second layer defining the second external surface, the first and second layers being separated by and in contact with cured mold compound (Fig 5a, para [0040]); and an electrical circuit (10) located between the first and second layers and within the cured mold compound and including at least one electrical component (12) electrically connected to a plurality of interconnects (30, fig 4, 7, para [0040]); the side wall comprising a strip formed by the first layer, a strip formed by the second layer, and a strip formed by the cured mold compound (Fig 5a, para [0040]-[0044]); and the interconnects being disposed within the side wall (Fig 5a, 19, para [0040]).

As per claim 84, Saxelby teaches the apparatus of claim 83, and further teaches wherein the first and second layers further comprise a non-ferrous metal (para [0061]).

As per claim 89, Saxelby teaches the apparatus of claim 83, and further teaches wherein the electrical circuit further comprises a circuit board and the side wall further comprises a strip formed by the circuit board and an additional strip formed by cured mold compound (Fig 5a, para [0040]).

As per claim 90, Saxelby teaches the apparatus of claim 89, and further teaches wherein the interconnects further comprise conductive features in the circuit board (para [0040]-[0041]).

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As per claim 91, Saxelby teaches the apparatus of claim 90, and further teaches wherein the circuit board comprises a multilayer printed circuit board ("PCB") and each of the interconnects comprises a plurality of conductive layers in the PCB (para [0040]-[0042]).

As per claim 92, Saxelby teaches the apparatus of claim 90, and further teaches wherein each of the interconnects comprises a plurality of conductive vias in the circuit board (para [0040]-[0043]).

As per claim 93 Saxelby teaches the apparatus of claim 83, and further teaches further comprising interlocking features comprising a contour formed in an interior surface of the first layer, the contour being filled with cured mold compound (Fig 20, para [0054]-[005]).

As per claim 94, Saxelby teaches the apparatus of claim 91, and further teaches wherein the circuit board includes a top surface and a bottom surface and the at least one electrical component comprises a set of top-side components mounted on the top surface and a set of bottom-side components mounted on the bottom surface (para [0040]).

As per claim 95, Saxelby teaches the apparatus of claim 94, and further teaches wherein the top-side components include a number, T, of large-footprint components, the bottom-side components include a number, B, of large-footprint components, and the number T is approximately equal to the number B (Fig 8b).

As per claim 96, Saxelby teaches the apparatus of claim 95, and further teaches wherein each of most of the top-side large-footprint components shares a respective set of conductive vias with a corresponding one of the bottom-side large-footprint components (para [0040]).

As per claim 97, Saxelby teaches the apparatus of claim 95, and further teaches wherein each of most of the top-side large-footprint components is located in a respective footprint shared by a corresponding one of the bottom-side large-footprint components (para [0040]).

As per claim 109, Saxelby teaches the apparatus of claim 83, and further teaches wherein the interconnects further comprise a layer of metal plating (para [0055]).

As per claim 143, Saxelby teaches a method of manufacturing a plurality of products comprising: inserting a plurality of components into a cavity (68) formed by one or more molds (60a, 60b) (Fig 7, para [0040], [0043]-[0044]); closing the one or more molds to form a seal around the cavity (para [0048]-[0049]); filling the cavity with mold compound (para [0045]); curing the mold compound in the cavity to secure the components, cured mold compound, and molds together into an assembly (para [0052]); and cutting the assembly to separate the plurality of products from the assembly (Fig 19, para [0053]), the products each including a respective section of the one or more molds which remains as an integral part of each respective product (Fig 19, para [0053]).

As per claim 144, Saxelby teaches the method of claim 143, and further teaches further comprising maintaining a predetermined alignment between the plurality of components and the one or more molds (para [0048]-[0050]).

As per claim 145, Saxelby teaches the method of claim 143, and further teaches wherein the plurality of components include a substrate having conductive features; and the cutting comprises exposing portions of the conductive features in each respective product(para[0055]).

As per claim 146, Saxelby teaches a method of making a plurality of electronic devices, comprising: providing a PCB panel (20) including a PCB having a plurality of conductive traces and a plurality of components (12, 14, 15, 17, 18) electrically connected to form a plurality of circuits (Fig 7, para [0040]), the PCB having a plurality of contacts for making an electrical connections (30) to the plurality of circuits (para [0040]-[0042]); enclosing the PCB panel, including the PCB, components, and contacts in a mold cavity (68) (fig 7, para [0041]-[0043]); filling the mold cavity with mold compound (para [0045]); curing the mold compound (para [0052]); removing the encapsulated PCB panel from the mold (para [0053]); and cutting the encapsulated PCB panel to expose at least a portion of the plurality of contacts and form respective exposed contacts (para [0053]).

As per claim 149, Saxelby teaches the method of claim 146, and Saxelby further teaches further comprising: providing one or more registration features for positioning the panel assembly in the mold cavity; creating voids in the mold compound with one or more of the registration features by displacing uncured mold compound for reference after the encapsulated PCB panel is removed from the mold (Fig 19, para [0053]).

As per claim 150, Saxelby teaches the method of claim 146, and further teaches further comprising: making holes in the cured mold compound to expose portions of the PCB (para [0053]).

As per claim 151, Saxelby teaches the method of claim 146, and further teaches further comprising removing a layer of material from a first surface of the encapsulated panel assembly (para [0053]).

As per claim 152, Saxelby teaches the method of claim 151, and further teaches further comprising removing a layer of material from a second surface of the encapsulated PCB panel (para [0053]).

As per claim 155, Saxelby teaches the method of claim 146, and further teaches wherein the cutting divides the substrate producing at least one module having a plurality of layers including respective portions of the substrate and the encapsulant on each side of the substrate, and the module contains at least one of the circuits (Fig 19, para [0053]).

As per claim 158, Saxelby teaches the method of claim 155, and further teaches wherein the traces and components are electrically connected to form a plurality of separable circuits arranged in a pattern on the PCB; and the cutting divides the PCB panel along spaces between the separable circuits into a plurality of modules each containing at least one respective circuit (Fig 19, para [0053]).

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As per claim 162, Saxelby teaches a method of manufacturing a plurality of products comprising: inserting a printed circuit board (20) including a plurality of components (12, 14, 15, 17, 18) into a mold (60) (Fig 4, 7, para [0040]-[0043]) closing the mold to form a seal around the cavity (para [0048]-[0050]); filling the cavity with mold compound (para [0045]); curing the mold compound in the cavity to form an encapsulated panel assembly (para [0052]); cutting the encapsulated panel assembly to separate the plurality of products from the encapsulated panel assembly, the products each including a respective section of the printed circuit board (Fig 19, para [0053]).

As per claim 163, Saxelby teaches an apparatus comprising: a panel assembly having external surfaces defined at least in part by cured mold compound (Fig 19, para [0053]); the panel assembly including an internal circuit board (20), the internal circuit board having a first surface and second surface and a plurality of components (12, 14, 15, 17, 18) electrically connected to form a plurality of individual circuits (Fig 4, para [0040]), each individual circuit being electrically connected to respective interconnects (30) located along a respective circuit perimeter, the interconnects being contained within the cured mold compound (Fig 7, 19, para [0040], [0053]).

As per claim 164, Saxelby teaches the apparatus of claim 163, and further teaches wherein the panel assembly is adapted to be cut along the circuit perimeter separating the individual circuits, dividing the panel assembly into individual circuit modules, and exposing selected portions of the interconnects (Fig 19, para [0053]).

Claims 2, 3, 58 and 85 lack an inventive step under PCT Article 33(3) as being obvious over Saxelby.

As per claim 2, Saxelby teaches the method of claim 1, but does not teach wherein the cutting comprises making a first cut in at least one of the mold panels, and in the first cut, making a second narrower cut through the panel assembly. However, it would have been obvious to one of ordinary skill in the art as a matter of routine experimentation to make a first cut in the mold panels followed by a second narrower cut in the panel assembly so as the mold panel is not cut during the second cut to prevent contamination of the circuit by cuttings of the panel material.

As per claim 3, Saxelby teaches the method of claim 2, but does not teach wherein the curing comprises raising the temperature of the panel assembly and the first cut is made before the panel assembly cools after the curing. However, it would have been obvious to one of ordinary skill in the art as a matter of routine experimentation to use heat to cure, because it is a very simple process and easy to regulate.

As per claim 58, Saxelby teaches the apparatus of claim 57, but does not teach wherein at least one of the mold panels comprises aluminum. However, it would have been obvious to one of ordinary skill in the art as a matter of design choice to use aluminum over other metals, because it has good thermal conductivity and is readily available.

As per claim 85, Saxelby teaches the apparatus of claim 84, but does not teach wherein the first and second layers further comprise aluminum. However, it would have been obvious to one of ordinary skill in the art as a matter of design choice to use aluminum over other metals, because it has good thermal conductivity and is readily available.

Claims 7-14, 16, 19, 25, 30-33, 34-48, 63-65, 70-72, 86-88, 98-108, 110-142, 147, 148, 153, 154, 156, 157 and 159-161 lack an inventive step under PCT Article 33(3) as being obvious over Saxelby in view of US 2004/0100778 A1 to VINCIARELLI et al. (hereinafter 'Vinciarelli').

As per claim 7, Saxelby teaches the method of claim 1, but does not teach further comprising treating the exposed contact to protect against oxidation. However, Vinciarelli does further comprising treating the exposed contact (para [0055], pre-applying solder). It would have been obvious to one of ordinary skill in the art to modify the method, as taught by Saxelby, so as to include treating the exposed contact, as taught by Vinciarelli, because protecting the exposed contact prevents any unwanted contaminants, including protection against oxidation, from degrading the quality of the contacts during the manufacturing process.

As per claim 8, Saxelby in view of Vinciarelli teaches the method of claim 7, and Vinciarelli further teaches wherein the treating comprises applying a removable conformal coating to the exposed contact (para [0055]).

As per claim 9, Saxelby in view of Vinciarelli teaches the method of claim 7, and Vinciarelli further teaches wherein the treating comprises applying a layer of metal to the exposed contact (para [0055]).

As per claim 10, Saxelby in view of Vinciarelli teaches the method of claim 9, and Vinciarelli further teaches wherein the applying a layer of metal comprises applying solder (para [0055]).

As per claim 11, Saxelby in view of Vinciarelli teaches the method of claim 9, and Vinciarelli further teaches wherein the applying a layer of metal comprises applying a precious metal (para [0055]).

As per claim 12, Saxelby in view of Vinciarelli teaches the method of claim 9, and Vinciarelli further teaches wherein the applying a layer of metal comprises plating (para [0055]).

As per claim 13, Saxelby teaches the method of claim 4, but does not teach further comprising providing an adapter having at least one electrical terminal; and attaching the at least one electrical terminal to the exposed contact. However, Vinciarelli does teach further comprising providing an adapter (extender, 50) having at least one electrical terminal (pins 60, Fig 5, para [0060]); and attaching the at least one electrical terminal to the exposed contact (Fig 5, para [0060]). It would have been obvious to one of ordinary skill in the art to modify the method, as taught by Saxelby, so as to include the use of an adapter, as taught by Vinciarelli, because the adapter allows for ease of connecting/disconnecting from the exposed contact. --- (continued on next page) ---

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As per claim 14, Saxelby in view of Vinciarelli teaches the method of claim 13, and Vinciarelli further teaches wherein the at least one contact comprises a plurality of contacts (33) (Fig 8, para [0060]); the module comprises a plurality of exposed contacts (Fig 8, para [0060]); the adapter has a plurality of electrical terminals arranged to match respective ones of the plurality of exposed contacts (Fig 8, para [0060]); and the adapter is mechanically secured to the module (Fig 8, para [0060]).

As per claim 16, Saxelby teaches the method of claim 15, but does not teach wherein the respective portions of the mold panels provide heat sink surfaces for the module. However, Vinciarelli does teach wherein the respective portions of the mold panels provide heat sink surfaces for the module (para [0052]). It would have been obvious to one of ordinary skill in the art to modify the method, so as to include the use of heat sink surfaces, because heat sinks improve the thermal dissipation of the method.

As per claim 19, Saxelby teaches the method of claim 18, and further teaches wherein the selected components include at least one semiconductor device (para [0041]). Saxelby does not teach and at least one magnetically permeable core. However, Vinciarelli does teach and at least one magnetically permeable core (422a, 422b, Fig 5a, para [0052]). It would have been obvious to one of ordinary skill in the art to modify the method, so as to include the use of a magnetically permeable core, because the magnetic core increases the number of different circuitry that can be created.

As per claim 25, Saxelby teaches the method of claim 15, but does not teach further comprising treating at least one exterior surface of at least one of the mold panels for solderability. However, Vinciarelli does teach further comprising treating at least one exterior surface of at least one of the mold panels for solderability (para [0055]). It would have been obvious to one of ordinary skill in the art to modify the method, so as to include treating for solderability, because solderability creates an increase in the number of connectors that can be created.

As per claim 30, Saxelby in view of Vinciarelli teaches the method of claim 9, and Vinciarelli further teaches further comprising providing a connector for removably mating with and providing electrical connection to the metal (para [0060]).

As per claim 31, Saxelby teaches the method of claim 1, but does not teach further comprising: providing a center plate having an opening to accommodate the substrate; and wherein the forming a panel assembly includes positioning the substrate in the opening of the center plate and closing the mold panels against the center plate. However, Vinciarelli does teach further comprising: providing a center plate (43) having an opening to accommodate the substrate (Fig 3a, para [0052], [0056]); and wherein the forming a panel assembly includes positioning the substrate in the opening of the center plate and closing the mold panels against the center plate (para [0052], [0056]). It would have been obvious to one of ordinary skill in the art to modify the method, so as to include the use of a center plate, because the center plate ensures the correct alignment of the substrate during the molding process.

As per claim 32, Saxelby in view of Vinciarelli teaches the method of claim 31, and Vinciarelli further teaches further comprising providing at least one opening in the center plate connected to the internal chamber by at least one conduit; and forcing the encapsulant through the at least one opening and at least one conduit into the internal chamber (Fig 17a, 17b, para [0078], [0079]).

As per claim 33, Saxelby in view of Vinciarelli teaches the method of claim 32, and further teaches wherein the forming a panel assembly further comprises securing the mold panels together prior to curing the encapsulant (para [0048]-[0051]).

As per claim 34, Saxelby teaches a method of forming an electrical contact (9) (Fig 7, para [0040]) comprising: assembling a panel including a substrate (20) having one or more conductive features (30) enclosed within the panel and unexposed to an exterior surface of the panel (Fig 7, 19, para [0040], [0053]), the one or more conductive features being located along a cut line (116) (Fig 19, para [0053]); cutting the panel along the cut line exposing portions of the one or more conductive features for use as electrical connections to the substrate (Fig 19, para [0053]). Saxelby does not teach treating the portions of the one or more conductive features exposed from the cutting for preservation as electrical connections. However, Vinciarelli does teach treating the portions of the one or more conductive features exposed from the cutting for preservation as electrical connections (para [0055], pre-applying solder). It would have been obvious to one of ordinary skill in the art to modify the method, so as to include treating the exposed contact, because protecting the exposed contact prevents any unwanted contaminants from degrading the quality of the contacts during the manufacturing process.

As per claim 35, Saxelby in view of Vinciarelli teaches the method of claim 34, and Vinciarelli further teaches wherein the treating further comprises applying solder to the portions of the one or more conductive features exposed from the cutting (para [0055]).

As per claim 36, Saxelby in view of Vinciarelli teaches the method of claim 34, and Vinciarelli further teaches wherein the treating further comprises applying a conformal coating to the portions of the one or more conductive features exposed from the cutting to protect against oxidation (para [0055]).

As per claim 37, Saxelby in view of Vinciarelli teaches the method of claim 34, and Vinciarelli further teaches wherein the treating further comprises applying a metal layer to the portions of the one or more conductive features exposed from the cutting (para [0055]).

As per claim 38, Saxelby in view of Vinciarelli teaches the method of claim 34, and Vinciarelli further teaches wherein the treating further comprises soldering a lead of an adapter to the portions of the one or more conductive features exposed from the cutting (para [0055]).

As per claim 39, Saxelby in view of Vinciarelli teaches the method of claim 34, and Saxelby further teaches further comprising covering at least one surface of the substrate in an area including the cut line and the conductive features prior to cutting the substrate (para [0053]).

As per claim 40, Saxelby in view of Vinciarelli teaches the method of claim 39, and Saxelby further teaches wherein the covering comprises encapsulating the substrate with a molding compound; and further comprising: providing a registration feature having a predetermined relationship to the substrate (116, Fig 19, para [0053]), and using the registration feature to align the cutting relative to the cut line (para [0054]).

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As per claim 41, Saxelby in view of Vinciarelli teaches the method of claim 34, and Saxelby further teaches further comprising establishing a pattern including at least one conductive layer in the substrate along the cut line to form the conductive features (30, Fig 19, para [0053]).

As per claim 42, Saxelby in view of Vinciarelli teaches the method of claim 41, and Saxelby further teaches wherein the substrate comprises a multilayer printed circuit board and the establishing a pattern includes establishing a plurality of conductive layers along the cut line to form the conductive features (para [0040], [0053]).

As per claim 43, Saxelby in view of Vinciarelli teaches the method of claim 34, and Saxelby further teaches further comprising establishing a pattern including at least one conductive connection in the substrate along the cut line to form the conductive features (para [0040], [0053]). Saxelby does not teach the conductive connection is a conductive via. However, Vinciarelli does teach that the conductive connection is a conductive via (para [0065]).

As per claim 44, Saxelby in view of Vinciarelli teaches the method of claim 43, and Vinciarelli further teaches wherein the conductive via is filled with a conductive material (para [0065]).

As per claim 45, Saxelby in view of Vinciarelli teaches the method of claim 43, and Vinciarelli further teaches wherein the conductive via is buried in the substrate (para [0065]).

As per claim 46, Saxelby in view of Vinciarelli teaches the method of claim 43, and Vinciarelli further teaches wherein the conductive via is a through hole contacting the surfaces of the substrate (para [0065]).

As per claim 47, Saxelby in view of Vinciarelli teaches the method of claim 46, and Vinciarelli further teaches wherein the through hole is filled with a conductive material (para [0065]).

As per claim 48, Saxelby in view of Vinciarelli teaches the method of claim 34, and Saxelby further teaches wherein the conductive features are covered at the surfaces of the substrate by an insulative layer (para [0052]).

As per claim 63, Saxelby teaches the apparatus of claim 59, but does not teach wherein the exterior surface of the mold panel comprises a plurality of fins. However, Vinciarelli does teach wherein the exterior surface of the mold panel comprises a plurality of fins (para [0052]). It would have been obvious to one of ordinary skill in the art to modify the apparatus, so as to include the use of fins, because fins increase the heat dissipation of the apparatus.

As per claim 64, Saxelby teaches the apparatus of claim 59, but does not teach wherein the exterior surface of the mold panel comprises a flat surface. However, Vinciarelli does teach wherein the exterior surface of the mold panel comprises a flat surface (Fig 8, para [0060]). It would have been obvious to one of ordinary skill in the art to modify the apparatus, so as to include a panel with a flat surface, because having a flat surface reduces the amount of space the apparatus takes up.

As per claim 65, Saxelby in view of Vinciarelli teaches the apparatus of claim 64, and Vinciarelli further teaches wherein the flat surface is adapted for a solder joint (Fig 8, para [0060]).

As per claim 70, Saxelby teaches the apparatus of claim 59, but does not teach wherein the exterior surface of the mold panel comprises a flat surface with one or more pins protruding from the flat surface. However, Vinciarelli does teach wherein the exterior surface of the mold panel comprises a flat surface with one or more pins protruding from the flat surface (Fig 8, para [0060]). It would have been obvious to one of ordinary skill in the art to modify the apparatus, so as to include the use of pins, because the pins provide a simple and reliable form of connector.

As per claim 71, Saxelby teaches the apparatus of claim 59, but does not teach wherein the exterior surface of the mold panel comprises a flat surface having one or more holes in the flat surface. However, Vinciarelli does teach wherein the exterior surface of the mold panel comprises a flat surface having one or more holes in the flat surface (Fig 8, para [0060]). It would have been obvious to one of ordinary skill in the art to modify the apparatus, so as to include the use of holes, because the holes provide a simple and reliable form of connector.

As per claim 72, Saxelby in view of Vinciarelli teaches the apparatus of claim 71, and Vinciarelli further teaches further comprising a pin inserted into a respective one of the one or more holes (Fig 8, para [0060]).

As per claim 86, Saxelby teaches the apparatus of claim 85, but does not teach wherein the first external surface further comprises a plurality of fins. However, Vinciarelli does teach wherein the first external surface further comprises a plurality of fins (para [0052]). It would have been obvious to one of ordinary skill in the art to modify the apparatus, so as to include the use of fins, because fins increase the heat dissipation of the apparatus.

As per claim 87, Saxelby teaches the apparatus of claim 85, but does not teach wherein the first external surface further comprises an essentially flat area. However, Vinciarelli does teach wherein the first external surface further comprises an essentially flat area. (Fig 8, para [0060]). It would have been obvious to one of ordinary skill in the art to modify the apparatus, so as to include a panel with a flat surface, because having a flat surface reduces the amount of space the apparatus takes up.

As per claim 88, Saxelby in view of Vinciarelli teaches the apparatus of claim 87, and Vinciarelli further teaches wherein the essentially flat area is adapted for a solder joint (Fig 8, para [0060]).

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As per claim 98, Saxelby teaches the apparatus of claim 89, but does not teach wherein the circuit board includes a top surface and a bottom surface, the at least one electrical component comprises a number, T, of largefootprint components mounted on the top surface, and most of the large-footprint components are distributed symmetrically in relation to an axis on the top surface. However, Vinciarelli does teach wherein the circuit board includes a top surface and a bottom surface, the at least one electrical component comprises a number, T, of largefootprint components mounted on the top surface, and most of the large-footprint components are distributed symmetrically in relation to an axis on the top surface (Fig 5b). It would have been obvious to one of ordinary skill in the art to modify the apparatus, so as to include a symmetrical component layout, because having a symmetrical layout simplifies the manufacturing process.

As per claim 99, Saxelby in view of Vinciarelli teaches the apparatus of claim 98, and further teaches wherein the axis is along a midline of the top surface (Fig 5b).

As per claim 100, Saxelby in view of Vinciarelli teaches the apparatus of claim 98, and further teaches wherein the axis is defined in relation to a predetermined component (Fig 5b).

As per claim 101, Saxelby teaches the apparatus of claim 83, but does not teach further comprising: an adapter for providing mechanical and electrical connections between the modular package and an external mounting surface, the adapter having a body and a plurality of electrical terminals supported by the body; and electrical connections formed between the adapter terminals and respective interconnects on the modular package; the adapter body being mechanically secured to the modular package; the adapter terminals being arranged to mate with the external mounting surface. However, Vinciarelli does teach further comprising: an adapter (extender 50) for providing mechanical and electrical connections between the modular package and an external mounting surface (Fig 8, para [0060]), the adapter having a body and a plurality of electrical terminals (pins 60) supported by the body (Fig 8, para [0060]); and electrical connections formed between the adapter terminals and respective interconnects on the modular package (Fig 8, para [0060]); the adapter body being mechanically secured to the modular package; the adapter terminals being arranged to mate with the external mounting surface (Fig 8, para [0060]). It would have been obvious to one of ordinary skill in the art to modify the apparatus, so as to include the use of an adapter, because the adapter allows for ease of connecting/disconnecting from the exposed contact.

As per claim 102, Saxelby in view of Vinciarelli teaches the apparatus of claim 101, and Vinciarelli teaches wherein the external mounting surface is a circuit board and the terminals comprise ends constructed and arranged to be inserted into conductive holes in the circuit board (Fig 8, para [0060]).

As per claim 103, Saxelby in view of Vinciarelli teaches the apparatus of claim 101, and Vinciarelli teaches wherein the external mounting surface is a circuit board and the terminals comprise ends constructed and arranged to be surface mount soldered to the circuit board (Fig 8, para [0060]).

As per claim 104, Saxelby in view of Vinciarelli teaches the apparatus of claim 101, and Vinciarelli teaches wherein the external mounting surface includes a connector and the terminals comprise ends constructed and arranged to mate with the connector (Fig 8, para [0060]).

As per claim 105, Saxelby in view of Vinciarelli teaches the apparatus of claim 101, and Vinciarelli further teaches wherein the interconnects are disposed along a long edge of the modular package, the adapter is secured to the long edge of the modular package, and the first and second layers of the modular package are oriented perpendicular to the mounting surface (Fig 8, para [0060]).

As per claim 106, Saxelby in view of Vinciarelli teaches the apparatus of claim 101, and Vinciarelli further teaches wherein the interconnects are disposed along opposite edges of the modular package, the adapter is secured to the opposite edges, and the first and second layers of the modular package are oriented essentially parallel to the mounting surface (Fig 8, para [0060]).

As per claim 107, Saxelby in view of Vinciarelli teaches the apparatus of claim 106, and Vinciarelli further teaches wherein the adapter is constructed and arranged to maintain one of the first or second layers in contact with the mounting surface (Fig 8, para [0060]).

As per claim 108, Saxelby in view of Vinciarelli teaches the apparatus of claim 83, and Vinciarelli further teaches wherein the interconnects further comprise a surface constructed and arranged for engagement with a connector terminal (Fig 8, para [0060]).

As per claim 110, Saxelby teaches an apparatus comprising: a modular package (9) (Fig 5a, para [0040]) having a first external surface, a second external surface opposite the first external surface (Fig 5a), and a side wall extending along the perimeter of and connecting with the first and second external surfaces (left most side Fig 5a), the modular package including a first layer of metal defining the first external surface and a second layer of metal defining the second external surface (para [0061]), the first and second layers being separated by and in contact with cured mold compound (Fig 5a); and an electrical circuit, including a printed circuit board ("PCB") (20) (Fig 4, para [0040]), a plurality of components (12, 14, 15, 17, 18) including semiconductors mounted to the PCB (Fig 4, para [0040]-[0041]), and a plurality of interconnects (30) electrically connected to the components (Fig 4, para [0040]-[0041]), the electrical circuit being located between the first and second layers and within the cured mold compound (Fig 7, para [0040], [0043]-[0044]); the side wall comprising a strip formed by the first layer, a strip formed by the second layer, a strip formed by the PCB, and one or more strips formed by the cured mold compound (Fig 5a, para [0040]); the interconnects being disposed within the side wall (Fig 5a, para [0040]). Saxelby does not teach an electrical circuit for converting power. However, Vinciarelli does teach an electrical circuit for converting power (para [0014]). It would have been obvious to one of ordinary skill in the art to modify the apparatus, so as to include the use of a power converter circuit, because utilizing a power converter circuit increases the number of situations in which an apparatus can be used.

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Box V.2 Citations and explanations:

As per claim 111, Saxelby in view of Vinciarelli teaches the apparatus of claim 110, and Vinciarelli further teaches and further teaches wherein the first and second layers further comprise aluminum, and the first external surface further comprises a plurality of fins (para [0052]).

As per claim 112, Saxelby in view of Vinciarelli teaches the apparatus of claim 110, and Vinciarelli further teaches wherein the first and second layers further comprise aluminum, and the first external surface comprises an essentially flat area (Fig 8, para [0060]).

As per claim 113, Saxelby in view of Vinciarelli teaches the apparatus of claim 112, and Vinciarelli further teaches wherein the essentially flat area is adapted for a solder joint (Fig 8, para [0060]).

As per claim 114, Saxelby in view of Vinciarelli teaches the apparatus of claim 110, and Saxelby further teaches wherein the PCB includes a top surface and a bottom surface, the semiconductors include large-footprint switches, a number, T, of top-side large-footprint switches are mounted on the top surface, a number, B, of bottom-side large-footprint switches are mounted on the bottom surface, and the number T is approximately equal to the number B (para [0040]).

As per claim 115, Saxelby in view of Vinciarelli teaches the apparatus of claim 114, and Saxelby further teaches wherein each large-footprint switch is connected to one or more other components by a respective set of conductive vias in the PCB, and each of a plurality of the top-side large-footprint switches shares its respective set of conductive vias with a corresponding one of the bottom-side large-footprint switches (para [0040]).

As per claim 116, Saxelby in view of Vinciarelli teaches the apparatus of claim 114, and Saxelby further teaches wherein each of most of the large-footprint switches is positioned on one surface in a location substantially overlapping a location on the other surface occupied by another large-footprint switch (para [0040]).

As per claim 117, Saxelby in view of Vinciarelli teaches the apparatus of claim 110, and Vinciarelli further teaches wherein the PCB includes a top surface and a bottom surface, the semiconductors include large-footprint switches mounted on the top surface, most of the large-footprint components are distributed symmetrically between a left side of the top surface and an opposite right side of the top surface (Fig 5b).

As per claim 118, Saxelby in view of Vinciarelli teaches the apparatus of claim 110, and Vinciarelli further teaches wherein the PCB includes a top surface and a bottom surface, the semiconductors include large-footprint switches mounted on the top surface, and most of the large-footprint components are distributed symmetrically between quadrants on the top surface (Fig 5b).

As per claim 119, Saxelby in view of Vinciarelli teaches the apparatus of claim 118, and Vinciarelli further teaches wherein the components further comprise a transformer core, and the quadrants surround the transformer core (422a, Fig 5b, para [0052]).

As per claim 120, Saxelby in view of Vinciarelli teaches the apparatus of claim 110, and Saxelby further teaches wherein the PCB includes a top surface and a bottom surface, the components include a set of high-profile components having similar heights, a number, T, of the high-profile components are mounted on the top surface, a number, B, of the high-profile components are mounted on the bottom surface, and the number T is approximately equal to the number B (Fig 8b).

As per claim 121, Saxelby in view of Vinciarelli teaches the apparatus of claim 120, and Saxelby further teaches wherein a spatial distribution of the high-profile components on the top surface roughly matches a spatial distribution of the high-profile components on the bottom surface (para [0040]).

As per claim 122, Saxelby in view of Vinciarelli teaches the apparatus of claim 110, and Vinciarelli further teaches further comprising: an adapter (50) for providing mechanical and electrical connections between the modular package and an external mounting surface (Fig 8, para [0060]), the adapter having a body and a plurality of electrical terminals (pins 60) supported by the body (Fig 8, para [0060]); and electrical connections formed between the adapter terminals and respective interconnects on the modular package, the adapter body being mechanically secured to the modular package (Fig 8, para [0060]); the adapter terminals being arranged to mate with the external mounting surface (Fig 8, para [0060]).

As per claim 123, Saxelby in view of Vinciarelli teaches the apparatus of claim 122, and Vinciarelli further teaches wherein the external mounting surface is a circuit board and the terminals comprise ends constructed and arranged to be inserted into conductive holes in the circuit board (Fig 8, para [0060]).

As per claim 124, Saxelby in view of Vinciarelli teaches the apparatus of claim 122, and Vinciarelli further teaches wherein the external mounting surface is a circuit board and the terminals comprise ends constructed and arranged to be surface mount soldered to the circuit board (Fig 8, para [0060]).

As per claim 125, Saxelby in view of Vinciarelli teaches the apparatus of claim 122, and Vinciarelli further teaches wherein the external mounting surface includes a connector and the terminals comprise ends constructed and arranged to mate with the connector (Fig 8, para [0060]).

As per claim 126, Saxelby in view of Vinciarelli teaches the apparatus of claim 122, and Vinciarelli further teaches wherein the interconnects are disposed along a long edge of the modular package, the adapter is secured to the long edge of the modular package, and the first and second layers of the modular package are oriented perpendicular to the mounting surface (Fig 8, para [0060]).

---Remaining Claims contained in attached document---

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As per claim 127, Saxelby in view of Vinciarelli teaches the apparatus of claim 122, and Vinciarelli further teaches wherein the interconnects are disposed along opposite edges of the modular package, the adapter is secured to the opposite edges, and the first and second layers of the modular package are oriented essentially parallel to the mounting surface (Fig 8, para [0060]).

As per claim 128, Saxelby in view of Vinciarelli teaches the apparatus of claim 127, and Vinciarelli further teaches wherein the adapter is constructed and arranged to maintain one of the first-or second layers in contact with the mounting surface (Fig 8, para [0060]).

As per claim 129, Saxelby in view of Vinciarelli teaches the apparatus of claim 110, and Vinciarelli further teaches wherein the interconnects further comprise a surface constructed and arranged for engagement with a connector terminal (Fig 8, para [0060]).

As per claim 130, Saxelby in view of Vinciarelli teaches the apparatus of claim 110, and Vinciarelli further teaches wherein the interconnects further comprise a layer of metal plating (para [0055]).

As per claim 131, Saxelby in view of Vinciarelli teaches the apparatus of claim 110, and Saxelby further teaches wherein the first layer further comprises contours formed in an interior surface of the first layer, the contours including a first feature having a shape and an elevation to accommodate a first component on the PCB, the first component having a height greater than or less than other components on the PCB (Fig 20, para [0054]-[0055]).

As per claim 132, Saxelby in view of Vinciarelli teaches the apparatus of claim 131, and Vinciarelli further teaches wherein the first component includes a magnetic core structure and the elevation is a recess in the interior surface (422a, Fig 5a, para [0052]).

As per claim 133, Saxelby in view of Vinciarelli teaches the apparatus of claim 131, and Saxelby further teaches wherein the first component includes a semiconductor switch and the elevation is a protrusion from the interior surface (para [0041]).

As per claim 134, Saxelby in view of Vinciarelli teaches the apparatus of claim 110, and Saxelby further teaches further comprising interlocking features comprising a contour formed in an interior surface of the first layer, the contour being filled with cured mold compound (Fig 20, para [0054]-[0055]).

As per claim 135, Saxelby in view of Vinciarelli teaches the apparatus of claim 110, and Saxelby further teaches wherein the modular package further comprises a recess formed in the first layer adjacent one or more of the interconnects providing a setback between the first layer and the one or more interconnects (Fig 20, para [0054]-[0055]).

As per claim 136, Saxelby teaches an apparatus comprising: an electrical circuit including a printed circuit board ("PCB") (20) comprising a plurality of conductive layers and having a top surface and a bottom surface (Fig 5a, para [0040]). Saxelby does not teach that the electrical circuit is a power converter including a magnetic core structure magnetically coupled to a winding formed by traces in one or more of the conductive layers in the PCB; and a plurality of power semiconductor devices (para [0041]), a first set of the power semiconductor devices being mounted on the top surface and electrically connected to dissipate power at a level, Pt, during operation of the converter, a second set of the power semiconductor devices being mounted on the bottom surface and electrically connected to dissipate power at a level, Pb, during operation of the converter; wherein the power semiconductor devices are distributed between the first and second sets to distribute heat generation during operation of the converter such that each level Pt, Pb is less than 150% of the other level Pb, Pt. However, Vinciarelli does teach that the electrical circuit is a power converter (para [0014]) including magnetic core structure magnetically coupled to a winding formed by traces in one or more of the conductive layers in the PCB (Fig 5b, para [0052]); and a plurality of power semiconductor devices (para [0041]), a first set of the power semiconductor devices being mounted on the top surface and electrically connected to dissipate power at a level, Pt, during operation of the converter, a second set of the power semiconductor devices being mounted on the bottom surface and electrically connected to dissipate power at a level, Pb, during operation of the converter (Fig 5b, para [0052]); wherein the power semiconductor devices are distributed between the first and second sets to distribute heat generation during operation of the converter such that each level Pt, Pb is less than 150% of the other level Pb, Pt (para [0051], [0052]). It would have been obvious to one of ordinary skill in the art to modify the apparatus so as to include the use of a power converter and magnetic core, because utilizing a power converter circuit increases the number of situations in which an apparatus can be used.

As per claim 137, Saxelby in view of Vinciarelli teaches the apparatus of claim 136, and Saxelby further teaches wherein a plurality of the power semiconductor devices in the first set are each positioned in a location on the top surface substantially overlapping a location on the bottom surface occupied by a power semiconductor device in the second set (Fig 7, para [0040]).

As per claim 138, Saxelby in view of Vinciarelli teaches the apparatus of claim 136, and Vinciarelli further teaches wherein the power semiconductor devices are electrically connected using a respective set of conductive vias in the PCB, and a plurality of the power semiconductor devices in the first set share their respective sets of conductive vias with corresponding power semiconductor devices in the second set (para [0065]).

As per claim 139, Saxelby in view of Vinciarelli teaches the apparatus of claim 137, and Vinciarelli further teaches wherein the power converter further comprises circuitry including a pair of cells having a common circuit topology and each including power semiconductor switches from each of the first and second sets; each cell having its respective components arranged in a pattern, wherein the pattern of components of one cell is substantially a mirror image of the pattern of components in the other cell (Fig 5b, para [0051]-[0052]).

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Box V.2 Citations and explanations:

As per claim 140, Saxelby in view of Vinciarelli teaches the apparatus of claim 139, and Vinciarelli further teaches wherein a component from one of the cells is located on an opposite surface of a respective component from the other one of the cells (Fig 5b, para [0051]-[0052]).

As per claim 141, Saxelby in view of Vinciarelli teaches the apparatus of claim 140, and Vinciarelli further teaches wherein the cells comprise input cells (para [0051]-[0052]).

As per claim 142, Saxelby in view of Vinciarelli teaches the apparatus of claim 138, and Vinciarelli further teaches wherein the power semiconductor devices comprise output switches (para [0051]-[0052]).

As per claim 147, Saxelby teaches the method of claim 146, but does not teach further comprising: mating the PCB panel with a center plate having an opening for receiving the PCB panel; covering the opening of the center plate with at least one plate; and wherein the enclosing comprises closing the mold against the center plate and exerting pressure to force the at least one plate against the center plate; and wherein the removing comprises removing the at least one plate from the encapsulated assembly. However, Vinciarelli does teach further comprising: mating the PCB panel with a center plate (43) having an opening for receiving the PCB panel (Fig 5a, para [0052], [0056]); covering the opening of the center plate with at least one plate (para [0052], [0056]); and wherein the enclosing comprises closing the mold against the center plate and exerting pressure to force the at least one plate against the center plate (para [0052], [0056]); and wherein the removing comprises removing the at least one plate from the encapsulated assembly (para [0052], [0056]). It would have been obvious to one of ordinary skill in the art to modify the method, so as to include the use of a center plate, because the center plate ensures the correct alignment of the substrate during the molding process.

As per claim 148, Saxelby teaches the method of claim 146, but does not teach further comprising: mating the PCB panel with a center plate; and wherein the enclosing comprises closing the mold against the center plate; wherein the curing bonds the PCB panel to the center plate; wherein the removing comprises removing the bonded centerplate and PCB panel. However, Vinciarelli does teach further comprising: mating the PCB panel with a center plate (para [0052], [0056]); and wherein the enclosing comprises closing the mold against the center plate (para [0052], [0056]); wherein the curing bonds the PCB panel to the center plate (para [0052], [0056]); wherein the removing comprises removing the bonded centerplate and PCB panel (para [0052], [0056]). It would have been obvious to one of ordinary skill in the art to modify the method, so as to include the use of a center plate, because the center plate ensures the correct alignment of the substrate during the molding process.

As per claim 153, Saxelby teaches the method of claim 151, but does not teach wherein the components include a magnetic core and the removing a layer of material exposes a surface of the magnetic core. However, Vinciarelli does teach wherein the components include a magnetic core and the removing a layer of material exposes a surface of the magnetic core (Fig 5b, para [0052]). It would have been obvious to one of ordinary skill in the art to modify the method, so as to include the use of a magnetically permeable core, because the magnetic core increases the number of different circuitry that can be created.

As per claim 154, Saxelby in view of Vinciarelli teaches the method of claim 153, and Vinciarelli further teaches wherein the removing a layer of material from the top surface of the encapsulated panel assembly provides finished surfaces and reduces the encapsulated panel assembly to a controlled thickness (Fig 5b, para [0052]).

As per claim 156, Saxelby in view of Vinciarelli teaches the method of claim 155, but does not teach further comprising providing an adapter having at least one electrical terminal; and attaching the at least one electrical terminal to the exposed contact. However, Vinciarelli further teaches further comprising providing an adapter (50) having at least one electrical terminal (60) (Fig 8, para [0060]); and attaching the at least one electrical terminal to the exposed contact (Fig 8, para [0060]). It would have been obvious to one of ordinary skill in the art to modify the method, as taught by Saxelby, so as to include the use of an adapter, as taught by Vinciarelli, because the adapter allows for ease of connecting/disconnecting from the exposed contact.

As per claim 157, Saxelby in view of Vinciarelli teaches the method of claim 156, and Vinciarelli further teaches wherein the at least one contact comprises a plurality of contacts; the module comprises a plurality of exposed contacts; the adapter has a plurality of electrical terminals arranged to match respective ones of the plurality of exposed contacts; and the adapter is mechanically secured to the module (Fig 8, para [0060]).

As per claim 159, Saxelby teaches an apparatus comprising: a modular package (9, Fig 5a, para [0040]) having a first external surface, a second external surface opposite the first external surface (Fig 5a), and a side wall extending along the perimeter of and connecting with the first and second external surfaces (Fig 5a), the first and second external surfaces each being defined at least in part by a respective layer of cured mold compound (Fig 5a); and an electrical circuit including a printed circuit board ("PCB") (20), a plurality of components including semiconductors (para [0040]-[0041]) mounted to the PCB and a plurality of interconnects (30) electrically connected to the components (Fig 7, para [0040], [0043]-[0044]), the electrical circuit being located within the cured mold compound (Fig 7, para [0040], [0043]-[0044]); the side wall comprising a strip formed by the PCB between strips formed by the layers of cured mold compound defining the first and second external surfaces; the interconnects being disposed within the side wall (Fig 5a, para [0040]). Saxelby does not teach an electrical circuit for converting power. However, Vinciarelli does teach an electrical circuit for converting power (para [0014]). It would have been obvious to one of ordinary skill in the art to modify the apparatus, so as to include the use of a power converter circuit, because utilizing a power converter circuit increases the number of situations in which an apparatus can be used.

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Box V.2 Citations and explanations:

As per claim 160, Saxelby in view of Vinciarelli teaches the apparatus of claim 159, and Vinciarelli further teaches wherein the electrical circuit further comprises a magnetic core, and at least one of the first and second external surfaces is also defined by a surface of the magnetic core (422a, Fig 5b, para [0052]).

As per claim 161, Saxelby in view of Vinciarelli teaches the apparatus of claim 160, and Vinciarelli further teaches further comprising: an adapter (50) for providing mechanical and electrical connections between the modular package and an external mounting substrate, the adapter having a body and a plurality of electrical terminals (60) (Fig 8, para [0060]); and electrical connections formed between the adapter terminals and respective interconnects on the modular package (Fig 8, para [0060]), the adapter body being mechanically secured to the modular package and covering the electrical connections (Fig 8, para [0060]); and the adapter terminals being arranged to mate with the external mounting substrate (Fig 8, para [0060]).

Claims 27 and 29 lack an inventive step under PCT Article 33(3) as being obvious over Saxelby in view of US 2010/0246152 A1 to LIN et al. (hereinafter 'Lin').

As per claim 27, Saxelby teaches the method of claim 26, but does not teach further comprising centrifuging the assembly before curing the encapsulant. However, Lin does teach further comprising spinning the assembly before curing the encapsulant (para [0194]). It would have been obvious to one of ordinary skill in the art to modify the method so as to include spinning the assembly before curing, because spinning the assembly ensures that there are no air bubbles in the encapsulant, thus improving the strength of the encapsulant.

As per claim 29, Saxelby teaches the method of claim 28, but does not teach further comprising centrifuging the assembly before curing the encapsulant. However, Lin does teach further comprising centrifuging the assembly before curing the encapsulant (para [0194]). It would have been obvious to one of ordinary skill in the art to modify the method so as to include spinning the assembly before curing, because spinning the assembly ensures that there are no air bubbles in the encapsulant, thus improving the strength of the encapsulant.

Claims 1-164 have industrial applicability as defined by PCT Article 33(4) because the subject matter can be made or used in industry.



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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13/105,696

05/11/2011

Patrizio Vinciarelli

00614-0168001

3765

26214 7590 09/25/2013

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EXAMINER

ARBES, CARL J

ART UNIT

PAPER NUMBER

3729

NOTIFICATION DATE

DELIVERY MODE

09/25/2013

ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

PATDOCTC@fr.com

Office Action Summary	Application No. 13/105,696	Applicant(s) VINCIARELLI ET AL.	
	Examiner CARL ARBES	Art Unit 3729	AIA (First Inventor to File) Status No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTH(S) OR THIRTY (30) DAYS, WHICHEVER IS LONGER, FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

- 1) ☒ Responsive to communication(s) filed on 03 June 2013.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.
- 2a) ☐ This action is **FINAL**. 2b) ☒ This action is non-final.
- 3) ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
- 4) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims

- 5) ☒ Claim(s) 34-48 and 146-152 is/are pending in the application.
5a) Of the above claim(s) _____ is/are withdrawn from consideration.
- 6) ☐ Claim(s) _____ is/are allowed.
- 7) ☒ Claim(s) 34-48 and 146-148 is/are rejected.
- 8) ☒ Claim(s) 149-152 is/are objected to.
- 9) ☐ Claim(s) _____ are subject to restriction and/or election requirement.

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

Application Papers

- 10) ☐ The specification is objected to by the Examiner.
- 11) ☒ The drawing(s) filed on 11 May 2011 is/are: a) ☒ accepted or b) ☐ objected to by the Examiner.
Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).
Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

- 12) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some * c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

* See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

- | | |
|---|---|
| 1) ☒ Notice of References Cited (PTO-892) | 3) ☐ Interview Summary (PTO-413)
Paper No(s)/Mail Date. _____ |
| 2) ☒ Information Disclosure Statement(s) (PTO/SB/08)
Paper No(s)/Mail Date <u>herein</u> . | 4) ☐ Other: _____ |

Art Unit: 3729

The following is a quotation of 35 U.S.C. 112(b):

(b) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:

The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

Claims 38, 39, 45 and 146 are rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor, or for pre-AIA the applicant regards as the invention. To the extent that the language ...attaching a lead of an adapter... is not understood claim 38 is held to be unclear, vague and indefinite. Where is "an adapter" defined? What is its purpose or function? What does it adapt? With respect claim 39 what does the language ... covering at least one surface of the substrate in the area... mean? Covering the surface with what? Is the surface covered with air? Is the surface covered with paint? If so what type? Claim 39 is held to be unclear, vague and indefinite. As applied to claim 45 it is not understood what Applicants intend when they recite...the conductive via is buried in the substrate.... By definition a via a conductive or non-conductive hole or region that is located in a substrate. That substantially all conductive vias are "buried in substrates". What is it that Applicants intend by this language? Claim 45 is held to be unclear, vague and indefinite.

Claims 34-47, 146 and 147 are rejected under pre-AIA 35 U.S.C. 103(a) as being unpatentable over Gilbert et al (Pat No. 4,551,747); hereinafter Gilbert et al.

Gilbert et al teach a carrier apparatus (40) for mounting logic components on the surface of a circuit board (41). The carrier includes bonding structures defining top and bottom surfaces and further defining cavity (50) in the bottom surface for receipt of a logic component (51). A recessed cover portion (56) is attached to the housing. Apparatus (40) includes means for mounting housing onto a circuit board such that cover (56) does not make contact with the surface of the circuit board (41). Housing also includes means for electrically interconnecting component (51) to circuit board (41). (Cf. Abstract). Additionally Gilbert et al teach substrate dielectric layer (46) positioned adjacent first dielectric layer (42). Layer (46) is provided with die cavity (50) for reception of logic components. Layer (46) also includes

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also includes elongated electroplated signal leads or traces (48) (Cf. Col. 7). Additionally Gilbert et al teach that substrate dielectric layer (46) have a plurality of spaced, vertically extending, semi-cylindrical indentations or castellations (60) along the edges thereof. (Cf. Col. 8). It would have been obvious to produce layers having vias with conductive material therein and which are integrated to make carrier (40) and then cutting these vias to produce a portion conductive castellations. As applied to claim 35 inasmuch as Gilbert et al teach applying solder paste (70) in cylindrical spaces (69) the limitation recited in said claim 35 is held to have been obvious in view of Gilbert et al. As applied to claim 36 inasmuch as Gilbert et al teach that castellations are gold-plated in order to prevent oxidation (Cf Col. 8) it is held that the limitation recited in said claim 36 would have been obvious in view of Gilbert et al. As applied to claim 37. Gilbert et al teach applying metal layer to one or more conductive features exposed to the cutting Cf e.g .Fig 2-5 and corresponding disclosure) and therefore the limitation recited in said 37 is held to have been obvious in view of Gilbert et al. As applied to claim 38 Gilbert et al teach solder paste (70) used to secure and electrically connect castellation (60) to signal traces (71) (Cf. Col. 9 and Fig.8) and therefore the limitation recited in said claim 38 is held to have been obvious. As applied to claim 40 before dicing (or cutting) the semiconductor wafer (20) there must be some means of determining where to cut so as to expose the conductive features after the cuts are made. A PHOSITA would have understood this without the use of undue skill in this art. Therefore it is held to have been obvious to provide a registration feature having a predetermined relationship to the substrate and using this registration feature to properly align the cutting relative to the cut line. As applied to claim 41-44 inasmuch as Gilbert et al teach providing conductive features in the layers of substrates along the cut lines and are filled with a conductive material e.g. (60, 70) (Cf. e.g. Figs 2,(Cf. e.g. Figs 2,,3, 15 and 17) it is held that the limitations recited in said claims 41-44 would have been obvious in view of Gilbert et al. As applied to claim 45-47 inasmuch as Gilbert et al teach providing vias that contact the surface of the substrate and are filled with conductive material (Cf. e.g. Figs. 1-3, 15 and 17) it would have been obvious to provide that conductive vias are buried, that they contact the surface of the substrate and are filled with a conductive material. As applied to claims 146 and 147 inasmuch as Gilbert et al teach a leadless chip carrier apparatus that can be

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connected to a circuit board (Cf. Col. 9 and Fig. 8) it is held that the limitation recited in said claims 146 and 147 would have been obvious in view of Gilbert et al.

The following is a quotation of pre-AIA 35 U.S.C. 103(a) which forms the basis for all obviousness rejections set forth in this Office action:

(a) A patent may not be obtained though the invention is not identically disclosed or described as set forth in section 102 of this title, if the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains. Patentability shall not be negated by the manner in which the invention was made.

Claims 34-39, 41 and 43-47 are further rejected under pre-AIA 35 U.S.C. 103(a) as being unpatentable over Estacio (Pat No. 6,830,959 B2); hereinafter Estacio.

Estacio teaches a semiconductor package with a semiconductor having a side electrical connection. Additionally Estacio teaches semiconductor (20) having a plurality of semiconductor dies (24) formed therein described by scribe lines (22). Each cavity (28) extends partially or entirely through semiconductor wafer (20). These cavities can be formed by means of a photolithographic and etching, well as wet or dry etching, by laser etching or by water-jet etching processes. (Cf. Col. 4). Additionally Estacio teaches dicing (using a saw or a laser) and bumping the semiconductor the castellated recesses (34) receive solder that is placed in the bottom of the recess. (Cf. Col. 5) If indeed Estacio does not specifically teach dicing the semiconductor wafer before the or cavities are diced (to provide castellations) a person of ordinary skill in this art i.e. a PHOSITA would have been able to fill the non-diced cavities with conductive material and then dice the cavities to provide the castellations. As applied to claim 35 Estacio teaches solder bumps i.e. 66(a), 66(b) wherein solder bump 66(b) couples with solder bump (34). (Cf Col. 7) and therefore it is held that the limitation recited in said claim 35 it would have been obvious in view of Estacio. As applied to claim 36 even if Estacion does not teach a conformal coating to one or more of the conductive features exposed from the cutting (dicing) nevertheless a PHOSITA with the use of ordinary skill would have been able to use such a coating in order to protect the conductive features from oxidation's harmful effects. As applied to claim 37 Estacio teaches solder joint (68) to castellations that are formed in the semiconductor die and this joint is connect to solder bump (32). (Cf. Col. 7). Hence it would have been obvious to apply a conductive layer to the portions of one or more

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conductive features exposed from the cutting and hence said claim 37 is held to have been made obvious. As applied to claim 40 before dicing (or cutting) the semiconductor wafer (20) there must be some means of determining where to cut so as to expose the conductive features after the cuts are made. A PHOSITA would have understood this without the use of undue skill in this art. Therefore it is held to have been obvious to provide a registration feature having a predetermined relationship to the substrate and using this registration feature to properly align the cutting relative to the cut line before dicing (or cutting) the semiconductor wafer (20) there had be some means of determining where to cut so as to expose the conductive features after the cuts are made. A PHOSITA would have understood this without the use of undue skill in this art. As applied to claims 39 and 41 instead of cutting the substrate to provide castellations first and then placing conductive material into the cavities or vias a PHOSITA using only ordinary skill in this art would have been able to introduce the conductive material into the vias first and then cut the substrates to provide conductive castellations on the edges of the substrate. Using only a limited number of attempts the PHOSITA would have been able to accomplish what is recited in said claims 39 and 41. As applied to claims 43-47 inasmuch as Estacio teaches providing conductive vias along the cut line and filling the vias with conductive material (Cf. e.g. Fig. 1(h) & 3(k) and related disclosure) it is held that the limitation recited in said claims 43-47 would have been obvious in view of Estacio.

Claims 48 and 148 are rejected under pre-AIA 35 U.S.C. 103(a) as being unpatentable over Gilbert et al in view of Mullen, III et al (Pat. No. 5,241,133); hereinafter Mullen, III et al.

The teaching of Gilbert et al has been provide above and is not repeated. Mullen, III et al teach a leadless pad array chip carrier including a circuit board, an array of solder pads on the bottom side of the circuit board Plastic cover is transfer molded about the semiconductor device.(Cf. Abstract). It would have been obvious to combine he two teachings and to substitute the molded material taught by Mullen, III et al for the lid 956) taught by Gilbert et al since encapsulation of a semiconductor is a patentable equivalent of providing a lid. That is each of these commonly used means is used to protect the semiconductor from exterior environmental conditions. As applied to claim 148 inasmuch as Mullen, III et

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al teach encapsulating the surface of the printed circuit board it is held that the limitation recited in said claim 148 would have been obvious in view of the cited prior art.

Claims 149-152 are objected to as being dependent upon a rejected base claim, but would be allowable if rewritten in independent form including all of the limitations of the base claim and any intervening claims.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to CARL ARBES whose telephone number is (571)272-4563. The examiner can normally be reached on M, T, R and F from 8 to 6.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, D. Banks, can be reached on (571) 272-4690. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://portal.uspto.gov/external/portal>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free).

/C. J. Arbes/

Primary Examiner, Art Unit 3729

Notice of References Cited	Application/Control No. 13/105,696		Applicant(s)/Patent Under Reexamination VINCIARELLI ET AL.	
	Examiner CARL ARBES		Art Unit 3729	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	Classification
*	A	US-4,366,342	12-1982	Breedlove, James G.	174/557
*	B	US-4,551,747	11-1985	Gilbert et al.	257/691
*	C	US-5,241,133	08-1993	Mullen et al.	174/534
*	D	US-5,864,092	01-1999	Gore et al.	174/51
*	E	US-6,830,959 B2	12-2004	Estacio, Maria Cristina B.	438/113
*	F	US-2010/0197150 A1	08-2010	Smejtek, Dalibor	439/61
*	G	US-7,972,143 B2	07-2011	Smejtek, Dalibor	439/61
	H	US-			
	I	US-			
	J	US-			
	K	US-			
	L	US-			
	M	US-			

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

Substitute Form PTO-1449 (Modified)	U.S. Department of Commerce Patent and Trademark Office	Attorney Docket No. 00614-0168001	Application No. 13/105,696
Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))		Applicant Patrizio Vinciarelli et al.	
		Filing Date May 11, 2011	Group Art Unit

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Class	Subclass	Filing Date If Appropriate
	1	5,728,600	03/17/1998	Saxelby, Jr. et al.	437	207	
	2	6,403,009	06/11/2002	Saxelby, Jr. et al.	264	272.15	
	3	6,940,013	09/06/2005	Vinciarelli et al.	174	52.1	
	4	7,361,844	04/22/2008	Vinciarelli et al.	174	260	
	5						
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Foreign Patent Documents or Published Foreign Patent Applications								
Examiner Initial	Desig. ID	Document Number	Publication Date	Country or Patent Office	Class	Subclass	Translation	
							Yes	No
	12							
	13							
	14							
	15							
	16							

Other Documents (include Author, Title, Date, and Place of Publication)		
Examiner Initial	Desig. ID	Document
	17	U.S. Patent Application Serial No. 12/493,773, entitled "Encapsulation Method and Apparatus for Electronic Modules", Application and Preliminary Amendment filed June 29, 2009, 63 pages
	18	
	19	
	20	

Examiner Signature /Carl Arbes/	Date Considered 9/20/13
EXAMINER: Initials citation considered. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.	

Substitute Disclosure Form (PTO-1449)

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /C.A./

Substitute Disclosure Form	U.S. Department of Commerce Patent and Trademark Office	Attorney Docket No. 00614-0168001	Application No. 13/105,696
Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))		Applicant Patrizio Vinciarelli et al.	
		Filing Date May 11, 2011	Group Art Unit 2835

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Class	Subclass	Filing Date If Appropriate
	1	2002/0096348	07/25/2002	Saxelby et al.			
	2	2004/0100778	05/27/2004	Vinciarelli et al.			
	3	2007/0241440	10/18/2007	Hoang et al.			
	4	2010/0246152	09/30/2010	Lin et al.			

Foreign Patent Documents or Published Foreign Patent Applications								
Examiner Initial	Desig. ID	Document Number	Publication Date	Country or Patent Office	Class	Subclass	Translation	
							Yes	No
	5	WO1995/027308	10/12/1995	WIPO				

Other Documents (include Author, Title, Date, and Place of Publication)		
Examiner Initial	Desig. ID	Document
	6	9/20/13 International Search Report and Written Opinion, PCT/US2012/37495, May 6, 2013, 18 pages.

Examiner Signature /Carl Arbes/	Date Considered
EXAMINER: Initials citation considered. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.	

Substitute Form PTO-1449 (Modified)	U.S. Department of Commerce Patent and Trademark Office	Attorney Docket No. 00614-0168001	Application No. 13/105,696
Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))		Applicant Patrizio Vinciarelli et al.	
		Filing Date May 11, 2011	Group Art Unit

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Class	Subclass	Filing Date If Appropriate
	1						
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Foreign Patent Documents or Published Foreign Patent Applications								
Examiner Initial	Desig. ID	Document Number	Publication Date	Country or Patent Office	Class	Subclass	Translation	
							Yes	No
	12							
	13							
	14							
	15							
	16							

Other Documents (include Author, Title, Date, and Place of Publication)		
Examiner Initial	Desig. ID	Document
	17	AFM Microelectronics Inc., "General Purpose Capacitors", www.afmmicroelectronics.com , 13 pages, published on or before July 26, 2010
	18	Kahn, "Technical Information: Multilayer Ceramic Capacitors – Materials and Manufacture", Microelectronics Inc., 9 pages, published on or before July 26, 2010
	19	
	20	

Examiner Signature /Carl Arbes/	Date Considered 9/20/13
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EXAMINER: Initials citation considered. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/116,642	07/24/2014	Patrizio Vinciarelli	00614-0169US1	5649

26214 7590 02/09/2016
FISH & RICHARDSON P.C. (VLT, INC)
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022

EXAMINER

SHRESTHA, SAGAR

ART UNIT	PAPER NUMBER
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2835

NOTIFICATION DATE	DELIVERY MODE
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02/09/2016

ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

PATDOCTC@fr.com

Office Action Summary	Application No. 14/116,642	Applicant(s) VINCIARELLI ET AL.	
	Examiner SAGAR SHRESTHA	Art Unit 2835	AIA (First Inventor to File) Status No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

- 1) ☒ Responsive to communication(s) filed on 11/19/2015.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.
- 2a) ☐ This action is **FINAL**. 2b) ☒ This action is non-final.
- 3) ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
- 4) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

- 5) ☒ Claim(s) 146-183 is/are pending in the application.
5a) Of the above claim(s) 146-158 and 162-181 is/are withdrawn from consideration.
- 6) ☐ Claim(s) _____ is/are allowed.
- 7) ☒ Claim(s) 159-161 and 182 is/are rejected.
- 8) ☒ Claim(s) 183 is/are objected to.
- 9) ☐ Claim(s) _____ are subject to restriction and/or election requirement.

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating Intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

Application Papers

- 10) ☐ The specification is objected to by the Examiner.
- 11) ☒ The drawing(s) filed on 11/08/2013 is/are: a) ☐ accepted or b) ☐ objected to by the Examiner.
Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).
Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

- 12) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some** c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

- | | |
|--|---|
| 1) ☒ Notice of References Cited (PTO-892) | 3) ☐ Interview Summary (PTO-413)
Paper No(s)/Mail Date. _____ |
| 2) ☒ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b)
Paper No(s)/Mail Date _____ | 4) ☐ Other: _____ |

DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application is being examined under the pre-AIA first to invent provisions.

Double Patenting

2. The nonstatutory double patenting rejection is based on a judicially created doctrine grounded in public policy (a policy reflected in the statute) so as to prevent the unjustified or improper timewise extension of the “right to exclude” granted by a patent and to prevent possible harassment by multiple assignees. A nonstatutory double patenting rejection is appropriate where the claims at issue are not identical, but at least one examined application claim is not patentably distinct from the reference claim(s) because the examined application claim is either anticipated by, or would have been obvious over, the reference claim(s). See, e.g., *In re Berg*, 140 F.3d 1428, 46 USPQ2d 1226 (Fed. Cir. 1998); *In re Goodman*, 11 F.3d 1046, 29 USPQ2d 2010 (Fed. Cir. 1993); *In re Longi*, 759 F.2d 887, 225 USPQ 645 (Fed. Cir. 1985); *In re Van Ornum*, 686 F.2d 937, 214 USPQ 761 (CCPA 1982); *In re Vogel*, 422 F.2d 438, 164 USPQ 619 (CCPA 1970); and *In re Thorington*, 418 F.2d 528, 163 USPQ 644 (CCPA 1969).

A timely filed terminal disclaimer in compliance with 37 CFR 1.321(c) or 1.321(d) may be used to overcome an actual or provisional rejection based on a nonstatutory double patenting ground provided the reference application or patent either is shown to be commonly owned with this application, or claims an invention made as a result of activities undertaken within the scope of a joint research agreement. See MPEP §

717.02 for applications subject to examination under the first inventor to file provisions of the AIA as explained in MPEP § 2159. See MPEP §§ 706.02(I)(1) - 706.02(I)(3) for applications not subject to examination under the first inventor to file provisions of the AIA. A terminal disclaimer must be signed in compliance with 37 CFR 1.321(b).

The USPTO Internet website contains terminal disclaimer forms which may be used. Please visit www.uspto.gov/forms/. The filing date of the application in which the form is filed determines what form (e.g., PTO/SB/25, PTO/SB/26, PTO/AIA/25, or PTO/AIA/26) should be used. A web-based eTerminal Disclaimer may be filled out completely online using web-screens. An eTerminal Disclaimer that meets all requirements is auto-processed and approved immediately upon submission. For more information about eTerminal Disclaimers, refer to <http://www.uspto.gov/patents/process/file/efs/guidance/eTD-info-I.jsp>.

3. Claim 159, 160 and 161 are provisionally rejected on the ground of nonstatutory double patenting as being unpatentable over claims 110, 118, 119 and 122 of copending Application No. 14/635,467 (reference application). Although the claims at issue are not identical, they are not patentably distinct from each other because the copending application teaches all the limitations

Regarding claim 159, claim 110 of co-pending application No. 14/635,467 discloses all the limitations recited in claim 159 of the instant application.

Regarding claim 160 and 161, claims 118, 119 and 122 of co-pending application No. 14/635,467 discloses all the limitations recited in claim 160 and 161 of the instant application.

This is a provisional nonstatutory double patenting rejection because the patentably indistinct claims have not in fact been patented.

Claim Rejections - 35 USC § 103

4. The following is a quotation of pre-AIA 35 U.S.C. 103(a) which forms the basis for all obviousness rejections set forth in this Office action:

(a) A patent may not be obtained though the invention is not identically disclosed or described as set forth in section 102 of this title, if the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains. Patentability shall not be negated by the manner in which the invention was made.

5. Claims 159, 160, 182 are rejected under pre-AIA 35 U.S.C. 103(a) as being unpatentable over Ozawa (US 5,471,366; Ozawa hereinafter) in view of Vinciarelli et al. (US 2004/0100778; Vinciarelli hereinafter).

Regarding claim 159, Ozawa (figs. 2-6) discloses an apparatus comprising:
a modular package (34, fig. 3) having a first external surface (top external surface of 34), a second external surface (bottom external surface of 34) opposite the first external surface, and a side wall (side wall of 34) extending along the perimeter of and connecting with the first and second external surfaces, the first and second external surfaces each being defined at least in part by a respective layer of cured mold compound (Figs. 2, 3); and

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an electrical circuit, including a printed circuit board ("PCB") (31, figs. 2-5), a plurality of components including semiconductors (32-1 - 32-4, figs. 2, 3, Col. 4: lines 45-54) mounted to the PCB and a plurality of interconnects electrically connected to the components, the electrical circuit being located within the cured mold compound (figs. 2, 3, col. 4: lines 45-64 and col.5: lines 6-23);

the side wall comprising a strip formed by the PCB between strips formed by the layers of cured mold compound defining the first and second external surfaces (the side wall comprises of PCB strips in between strips formed by the first external surface and second external surface, Figs. 2, 3) ;

the interconnects being disposed within the side wall (figs. 2, 3, col. 4: lines 45-64 and col.5: lines 6-23).

Ozawa does not explicitly disclose that an electrical circuit for power converting.

Vinciarelli teaches an electrical circuit for converting power (Par. [0014]).

It would have been obvious to one of ordinary skill in the art before the invention was filed to include the power converting circuit of Vinciarelli into the circuit of the apparatus of Ozawa in order to increase the usage of the apparatus. The modified apparatus could also be used as a power converter.

Regarding claim 160, Ozawa discloses the apparatus as set forth in claim 159.

Ozawa does not explicitly disclose the electrical circuit further comprises a magnetic core, and at least one of the first and second external surfaces is also defined by a surface of the magnetic core.

Vinciarelli (fig. 5B) teaches the electrical circuit further comprises a magnetic core, and the first external surface is defined by a surface of the magnetic core (fig. 5B, Par. [0052]).

It would have been obvious to one of ordinary skill in the art before the invention was filed to include the magnetic core of power converting circuit of Vinciarelli into the circuit of the apparatus of Ozawa in order to increase the usage of the apparatus. The modified apparatus could also be used as a power converter.

Regarding claim 182, Ozawa (figs. 2-5) discloses the apparatus of claim 159, comprising a portion of a mold panel, in which a surface of the portion of the mold panel forms at least a portion of the first external surface (figs. 2-5, Col. 5: lines 6-23 and Col.6: lines 63-67).

6. Claim 161 is rejected under pre-AIA 35 U.S.C. 103(a) as being unpatentable over Ozawa and Vinciarelli as applied to claim 160, and in further view of Brancalone (US 4,326,765; Brancalone hereinafter).

Regarding claim 161, the modified apparatus of Ozawa discloses the apparatus as set forth in claim 160.

The modified apparatus of Ozawa does not disclose the apparatus of claim 160 further comprising: an adapter for providing mechanical and electrical connections between the modular package and an external mounting substrate, the adapter having a body and a plurality of electrical terminals; and electrical connections formed between the adapter terminals and respective interconnects on the modular package, the

adapter body being mechanically secured to the modular package and covering the electrical connections; the adapter terminals being arranged to mate with the external mounting substrate.

Brancaleone (figs. 1, 2) teaches a modular package comprises an adapter (16) for providing mechanical and electrical connections between the modular package and an external mounting substrate (12), the adapter having a body (fig. 1) and a plurality of electrical terminals (30); and electrical connections formed between the adapter terminals and respective interconnects on the modular package (figs. 1, 2), the adapter body being mechanically secured to the modular package and covering the electrical connections; the adapter terminals being arranged to mate with the external mounting substrate (figs. 1, 2).

It would have been obvious to one of ordinary skill in the art before the claimed invention was filed to incorporate the adapter of Brancaleone onto the modified apparatus of Ozawa in order to make the connecting and disconnecting mechanism of the apparatus on the external circuit easier and convenient, and provide easy replacement of the apparatus in the external circuit as suggested by Brancaleone (Col. 4: lines 34-48).

7. Claims 159, 160 and 182 are rejected under pre-AIA 35 U.S.C. 103(a) as being unpatentable over Saxelby (US 2002/0096348; Saxelby hereinafter) in view of Vinciarelli et al. (US 2004/0100778; Vinciarelli hereinafter).

Regarding claim 159, Saxelby (figs 1-21) disclose an apparatus comprising:

a modular package (9, figs. 5a,5b) having a first external surface, a second external surface opposite the first external surface (modular package has top external surface and bottom external surface opposite to the first external surface, figs. 5a, 5b) , and a side wall extending along the perimeter of and connecting with the first and second external surfaces (modular package 9 has side wall around the perimeter of the modular package 9), the first and second external surfaces each being defined at least in part by a respective layer of cured mold compound (figs. 2a-5b, par. [0040]); and

an electrical circuit, including a printed circuit board ("PCB") (20, figs. 2a-5b), a plurality of components (12, 14, 15, 17, 18) including semiconductors mounted to the PCB (20) and a plurality of interconnects electrically connected to the components (figs. 4-6a, par. [0040] ~ [0041]), the electrical circuit being located within the cured mold compound (figs. 4-6a, par. [0040] ~ [0041]);

the side wall comprising a strip formed by the PCB between strips formed by the layers of cured mold compound defining the first and second external surfaces (the sidewall comprises PCB strips formed between the strips formed by the first and the second external surfaces, fig. 5a, 5b, Par. [0040]);

the interconnects being disposed within the side wall (fig. 5a, 5b, Par. [0040]).

Saxelby does not explicitly disclose that an electrical circuit for power converting.

Vinciarelli teaches an electrical circuit for converting power (Par. [0014]).

It would have been obvious to one of ordinary skill in the art before the invention was filed to include the power converting circuit of Vinciarelli into the circuit of the

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apparatus of Saxelby in order to increase the usage of the apparatus. The modified apparatus could also be used as a power converter.

Regarding claim 160, Saxelby discloses the apparatus as set forth in claim 159.

Saxelby does not explicitly disclose the electrical circuit further comprises a magnetic core, and at least one of the first and second external surfaces is also defined by a surface of the magnetic core.

Vinciarelli (fig. 5B) teaches the electrical circuit further comprises a magnetic core, and the first external surface is defined by a surface of the magnetic core (fig. 5B, Par. [0052]).

It would have been obvious to one of ordinary skill in the art before the invention was filed to include the magnetic core of power converting circuit of Vinciarelli into the circuit of the apparatus of Saxelby in order to increase the usage of the apparatus. The modified apparatus could also be used as a power converter.

Regarding claim 182, Saxelby (figs. 2a-5b) disclose that the apparatus of claim 159, comprising a portion of a mold panel (figs. 2a-5b), in which a surface of the portion of the mold panel forms at least a portion of the first external surface (the portion of the mold panel above the circuit board 20 formed a portion of the first external surface, figs. 2a-5b).

8. Claim 161 is rejected under pre-AIA 35 U.S.C. 103(a) as being unpatentable over Saxelby and Vinciarelli as applied to claim 160, and in further view of Brancalone (US 4,326,765; Brancalone hereinafter).

Regarding claim 161, the modified apparatus of Saxelby discloses the apparatus as set forth in claim 160.

The modified apparatus of Saxelby does not disclose the apparatus of claim 160 further comprising: an adapter for providing mechanical and electrical connections between the modular package and an external mounting substrate, the adapter having a body and a plurality of electrical terminals; and electrical connections formed between the adapter terminals and respective interconnects on the modular package, the adapter body being mechanically secured to the modular package and covering the electrical connections; the adapter terminals being arranged to mate with the external mounting substrate.

Brancaleone (figs. 1, 2) teaches a modular package comprises an adapter (16) for providing mechanical and electrical connections between the modular package and an external mounting substrate (12), the adapter having a body (fig. 1) and a plurality of electrical terminals (30); and electrical connections formed between the adapter terminals and respective interconnects on the modular package (figs. 1, 2), the adapter body being mechanically secured to the modular package and covering the electrical connections; the adapter terminals being arranged to mate with the external mounting substrate (figs. 1, 2).

It would have been obvious to one of ordinary skill in the art before the claimed invention was filed to incorporate the adapter of Brancaleone onto the modified apparatus of Saxelby in order to make the connecting and disconnecting mechanism of the apparatus on the external circuit easier and convenient, and provide easy

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replacement of the apparatus in the external circuit as suggested by Brancalone (Col. 4: lines 34-48).

Allowable Subject Matter

9. Claim 183 is objected to as being dependent upon a rejected base claim, but would be allowable if rewritten in independent form including all of the limitations of the base claim and any intervening claims.

The following is a statement of reasons for the indication of allowable subject matter:

Regarding dependent claim 183, the prior art of record, taken alone or in combination, fails to teach or fairly suggest, in combining with other limitations recited within the claim and in the independent claim 159, a combination of limitations that a first set of the power semiconductor devices being mounted on the top surface and electrically connected to dissipate power at a level, P_t , during operation of the converter, a second set of the power semiconductor devices being mounted on the bottom surface and electrically connected to dissipate power at a level, P_b , during operation of the converter, wherein the power semiconductor devices are distributed between the first and second sets to distribute heat generation during operation of the converter such that each level P_t , P_b is less than 150% of the other level P_b , P_t . None of the reference art of record discloses or renders obvious such a combination.

Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably

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accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

Conclusion

10. The prior art made of record and not relied upon is considered pertinent to applicant's disclosure are listed in the form 892.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to SAGAR SHRESTHA whose telephone number is (571)270-1236. The examiner can normally be reached on mon-fri: 9:00am to 5:30 pm EST.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Jinhee Lee can be reached on 5712721977. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

Application/Control Number: 14/116,642
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/S. S/
Examiner, Art Unit 2835

/HOA C NGUYEN/
Primary Examiner, Art Unit 2847

Notice of References Cited	Application/Control No. 14/116,642		Applicant(s)/Patent Under Reexamination VINCIARELLI ET AL.	
	Examiner SAGAR SHRESTHA		Art Unit 2835	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	CPC Classification	US Classification
*	A	US-5,471,366 A	11-1995	Ozawa; Takashi	H01L23/3677	174/521
*	B	US-2006/0097831 A1	05-2006	Lotfi; Ashraf W.	H01F27/027	336/84C
*	C	US-2003/0011054 A1	01-2003	Jeun, Gi-young	H01L21/565	257/678
*	D	US-6,184,585 B1	02-2001	Martinez; Roberto	H01L23/49575	257/666
*	E	US-7,554,181 B2	06-2009	Satou; Yukihiro	H01L23/49575	257/676
*	F	US-2010/0259909 A1	10-2010	Ho; Dominique	H01F19/08	361/767
*	G	US-4,326,765 A	04-1982	Brancaleone; Salvatore T.	H05K7/1092	439/225
	H	US-				
	I	US-				
	J	US-				
	K	US-				
	L	US-				
	M	US-				

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	CPC Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

Substitute Disclosure Form U.S. Department of Commerce Patent and Trademark Office Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))	Attorney Docket No. 00614-0169US1	Application No. 14/116,642
	Applicant Patrizio Vinciarelli et al.	
	Filing Date July 24, 2014	Group Art Unit 2835

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Class	Subclass	Filing Date If Appropriate
/S.S./	1.	5,557,142	09/1996	Gilmore			
/S.S./	2.	6,734,552	05/2004	Combs			
/S.S./	3.	7,015,587	03/2006	Poddar			
/S.S./	4.	7,049,682	05/2006	Mathews			
/S.S./	5.	7,198,987	04/2007	Warren			
/S.S./	6.	8,138,584	03/2012	Wang			

Foreign Patent Documents or Published Foreign Patent Applications								
Examiner Initial	Desig. ID	Document Number	Publication Date	Country or Patent Office	Class	Subclass	Translation	
							Yes	No
	7.							

Other Documents (include Author, Title, Date, and Place of Publication)		
Examiner Initial	Desig. ID	Document
	8.	

Examiner Signature /Sagar Shrestha/	Date Considered 02/02/2016
EXAMINER: Initials citation considered. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.	

Substitute Disclosure Form U.S. Department of Commerce Patent and Trademark Office Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))	Attorney Docket No. 00614-0169US1	Application No. 14/116,642
	First Named Inventor Patrizio Vinciarelli	
	Filing Date November 8, 2013	Group Art Unit

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Class	Subclass	Filing Date If Appropriate
/S.S./	1	2010/0197150	08/2010	Smejtek			
/S.S./	2	2012/0287582	11/15/2012	Vinciarelli et al.			
/S.S./	3	4,366,342	12/1982	Breedlove			
/S.S./	4	4,551,747	11/1985	Gilbert et al.			
/S.S./	5	5,241,133	08/1993	Mullen et al.			
/S.S./	6	5,728,600	03/17/1998	Saxelby, Jr. et al.			
/S.S./	7	5,864,092	01/1999	Gore et al.			
/S.S./	8	6,403,009	06/11/2002	Saxelby, Jr. et al.			
/S.S./	9	6,830,959	12/2004	Estacio			
/S.S./	10	6,940,013	09/06/2005	Vinciarelli et al.			
/S.S./	11	7,361,844	04/22/2008	Vinciarelli et al.			
/S.S./	12	7,972,143	07/2011	Smejtek			
/S.S./	13	8,427,269	04/23/2013	Vinciarelli			

Foreign Patent Documents or Published Foreign Patent Applications								
Examiner Initial	Desig. ID	Document Number	Publication Date	Country or Patent Office	Class	Subclass	Translation	
							Yes	No
	14							

Other Documents (include Author, Title, Date, and Place of Publication)		
Examiner Initial	Desig. ID	Document
/S.S./	15	AFM Microelectronics Inc., "General Purpose Capacitors", www.afmmicroelectronics.com , 13 pages, published on or before July 26, 2010.
/S.S./	16	Kahn, "Technical Information: Multilayer Ceramic Capacitors – Materials and Manufacture", Microelectronics Inc., 9 pages, published on or before July 26, 2010.

Examiner Signature /Sagar Shrestha/	Date Considered /Sagar Shrestha/
EXAMINER: Initials citation considered. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.	

Substitute Disclosure Form

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /S.S./



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/635,420	03/02/2015	Patrizio Vinciarelli	00614-0168002	6226

26214 7590 10/22/2015
FISH & RICHARDSON P.C.
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022

EXAMINER

ARBES, CARL J

ART UNIT	PAPER NUMBER
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3729

NOTIFICATION DATE	DELIVERY MODE
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10/22/2015

ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

PATDOCTC@fr.com

Office Action Summary	Application No. 14/635,420	Applicant(s) VINCIARELLI ET AL.	
	Examiner CARL ARBES	Art Unit 3729	AIA (First Inventor to File) Status No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

- 1) ☒ Responsive to communication(s) filed on 02 March 2015.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.
- 2a) ☐ This action is **FINAL**. 2b) ☒ This action is non-final.
- 3) ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
- 4) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

- 5) ☒ Claim(s) 1-33 is/are pending in the application.
5a) Of the above claim(s) _____ is/are withdrawn from consideration.
- 6) ☐ Claim(s) _____ is/are allowed.
- 7) ☒ Claim(s) 1-33 is/are rejected.
- 8) ☐ Claim(s) _____ is/are objected to.
- 9) ☐ Claim(s) _____ are subject to restriction and/or election requirement.

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating Intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

Application Papers

- 10) ☐ The specification is objected to by the Examiner.
- 11) ☒ The drawing(s) filed on 02 March 2015 is/are: a) ☒ accepted or b) ☐ objected to by the Examiner.
Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).
Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

- 12) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some** c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

- | | |
|--|---|
| 1) ☒ Notice of References Cited (PTO-892) | 3) ☐ Interview Summary (PTO-413)
Paper No(s)/Mail Date. _____ |
| 2) ☒ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b)
Paper No(s)/Mail Date <u>herein</u> . | 4) ☐ Other: _____ |

Art Unit: 3729

The present application is being examined under the pre-AIA first to invent provisions.

The following is a quotation of pre-AIA 35 U.S.C. 103(a) which forms the basis for all obviousness rejections set forth in this Office action:

(a) A patent may not be obtained though the invention is not identically disclosed or described as set forth in section 102 of this title, if the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains. Patentability shall not be negated by the manner in which the invention was made.

The following is a quotation of 35 U.S.C. 112(b):

(b) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:

The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

Claims 1-33 are further rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor, or for pre-AIA the applicant regards as the invention. .To the extent that Applicant 's recitation of the language ...providing a plurality of mold panels which when assembled, form an internal chamber,.... (Cf. Claim 1) is not understood, the Claims are held to unclear, vague and indefinite. What is a plurality of mold panels? What is a "panel assembly"? What is a "panel"? What is an internal chamber?

Claims 1 and 5-12, assuming that these Claims are based on and enabling disclosure and are furthermore clear and definite, are further rejected under pre-AIA 35 U.S.C. 103(a) as being unpatentable over Mahadevan et al. (Pat. No. 7,030,469 B2); hereinafter Mahadevan et al.

Mahadevan et al teach a method of forming a semiconductor package by forming a conductive layer (42, 64) over a mold encapsulant (35, 62). Semiconductor package (2, 50) includes a semiconductor electrically connected to a lead frame (10,52).by wire-bonding two device portions (2, 4, 6, 8) of lead frame (10) together and cutting wire-bond (32) by forming groove (40). In overlying encapsulant (35) to form two wires (33). Conductive layer (42) is electrically coupled to each of the two